Comps Mega Review

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1 About

As part of the mathematics PhD program at Stony Brook University, graduate students are required to pass seven core classes (Algebra I and II, Real Analysis I and II, Differential Geometry I and II, and Complex Analysis). Furthermore, students are required to pass a comprehensive qualifying exam based on the syllabi of each of these courses.
To that end, this document is a compilation of the notes, exercises/problems, and general work I have completed to prepare for these exams (along with a few miscellaneous notes I've added since I didn't wish to put them in a separate document).
As always, the aim is to contain no errors, especially since these are comps problems. However, it is entirely possible that an original writeup in this document might be incorrect since I could've forgotten to update the solutions after fixing my errors. Any errors in these pages is entirely my responsibility.

1.1 Organization

The problems are divided by subjects. The problem labeling scheme is as follows:

(1) Any problem that has the format "`{20--}-{ }-{ }-{ }`" or "`{19--}-{ }-{ }-{ }`" are questions from past comprehensive exams. The labels indicate (1) in what year, (2) in what month, (3) in which part, and (4) in what the order the problem was given, respectively.
(2) Any problem that has the format "`LSM-{ }-{ }`" was taken from Lee's *Introduction to Smooth Manifolds* (2nd Ed.)
(3) Any problem that the format "`D&F-{ }-{ }`" was taken from Dummit& Foote's *Abstract Algebra* (3rd ed.)
(4) Any problem that has the format "`F-{ }-{ }-{ }`" was taken from Folland's *Real Analysis*.
(5) Any problem that has the format "`SS-{ }-{ }`" was taken from Stein and Shakarchi's *Functional Analysis*.

Some resources I found indispensable while studying for this exam are:

(1) Math StackExchange
(2) Wikipedia
(3) The SBU Past Comps website

2 Differential Geometry and Topology

2.1 Frobenius Theorem

Problem 2025-A-II-2. Consider the plane distribution in $\mathbb{R}^3$ spanned by the two vector fields

$$V = \partial_x + 2xy\partial_z, \qquad W = x\partial_x + \partial_y + (2x^2y + x^2 - 2y)\partial_z.$$

(i) Show that this distribution is integrable.
(ii) Does the pair of vector fields V and W generate a coordinate system on integral surfaces? If not, find a pair that can play this role for the local integral surfaces passing through the point $(0,0,z_0)$.

(i) Let D denote the distribution spanned by the two vector fields V and W. By the Frobenius Theorem, D is integrable if and only if $[V,W]$ is a smooth section of D. Given the vector fields V and W, we first compute

$$\begin{aligned} V(W) &= \partial_x + (4xy + 2x)\partial_z. \\ W(V) &= (2xy + 2x)\partial_z. \end{aligned}$$

Thus $[V,W] = V(W) - W(V) = \partial_x + 2xy\partial_z = V$. Since V is a smooth section of D, we conclude that D is integrable.

(ii) We recall that a pair of vecor fields generates a coordinate system on integral surfaces if and only if the Lie bracket of the vector fields is identically zero. However, in (i), we showed that the Lie bracket of V and W is not identically zero. Thus, V and W cannot generate a coordinate system on integral surfaces. However, we can easily show that the pair of vector fields $(\widetilde{V}, \widetilde{W})$ given by $\widetilde{V} = V$ and $\widetilde{W} = W - xV$ has identically zero Lie bracket. Moreover, at the point $(0,0,z_0)$, $\widetilde{V}$ and $\widetilde{W}$ are linearly independent. Thus, this pair of vector fields generates a coordinate system on local integral surfaces passing through the point $(0,0,z_0)$.

Problem 2024-A-II-2. On $\mathbb{R}^5$, equipped with standard coordinates (v,w,x,y,z), consider the 1-form

$$\theta = dz + v\,dx + w\,dy.$$

Are there two smooth functions $f,g : \mathbb{R}^5 \to \mathbb{R}$ such that $\theta = f\,dg$? Justify your answer by means of concrete calculations.

No. Assume to the contrary that there exist two smooth functions $f,g : \mathbb{R}^5 \to \mathbb{R}$ so that $\theta = f\,dg$. Then we compute,

$$\theta \wedge d\theta = (f\,dg) \wedge (df \wedge dg) = 0.$$

On the other hand,

$$\begin{aligned} (dz + v\,dx + w\,dy) \wedge d(dz + v\,dx + w\,dy) &= (dz + v\,dx + w\,dy) \wedge (dv \wedge dx + dw \wedge dy) \\ &= dz \wedge dv \wedge dx + dz \wedge dw \wedge dy + v\,dx \wedge dw \wedge dy \\ &\quad + w\,dy \wedge dv \wedge dx, \end{aligned}$$

which is not identically zero because of the first two terms. Thus, the proof concludes by contradiction.

Problem 2024-J-II-6. Let M be a smooth n-manifold, and let φ be a differential k-form on M which is closed, in the sense that $d\varphi = 0$. At each point $p \in M$, define

$$D_p = \{v \in T_pM : v \lrcorner \varphi = 0\},$$

where $\lrcorner$ denotes the interior product. Assume $\ell \coloneqq \dim D_p$ is independent of p, so that $D \subset TM$ is a rank-ℓ vector subbundle of the tangent bundle of M. Prove that D is an integrable distribution of ℓ-planes, in the sense of the Frobenius Theorem.

Let D denote the integrable distribution of ℓ-planes defined locally as stated above. Let V, W be smooth sections (i.e., smooth vector fields) of D. Then by the Frobenius theorem, it suffices to show that their Lie bracket $[V, W]$ is also a smooth section of D. First, one has

$$[V, W] \lrcorner \varphi = \mathcal{L}_V(W \lrcorner \varphi) - V \lrcorner (\mathcal{L}_W \varphi).$$

Then by Cartan's Formula,

$$\mathcal{L}_W \varphi = W \lrcorner d\varphi + d(W \lrcorner \varphi).$$

Thus,

$$[V, W] \lrcorner \varphi = \mathcal{L}_V(W \lrcorner \varphi) - V \lrcorner [W \lrcorner d\varphi + d(W \lrcorner \varphi)].$$

Since W is a smooth section of D, by definition of D, $W \lrcorner \varphi = 0$. Next since φ is closed, $d\varphi = 0$. Since $\mathcal{L}_V(0) = 0$, and $V \lrcorner 0 = 0$, we conclude that $[V, W] \lrcorner \varphi$ is identically zero. Thus $[V, W]$ is a smooth section of D, whence we conclude that D is an integrable distribution in the sense of the Frobenius theorem.

Problem 2023-A-II-5. Let (t, x, y, z) be the standard coordinate system on $\mathbb{R}^4$, and let ϕ be the nonzero smooth 1-form on $\mathbb{R}^4$ defined by

$$\phi = dt + y\,dx + z\,dy.$$

Let D be the 3-plane field on $\mathbb{R}^4$ that consists of tangent vectors V such that $\phi(V) = 0$. Is D Frobenius integrable? Support your answer with a proof.

Let (t, x, y, z) be the standard coordinate system on $\mathbb{R}^4$, and let ϕ be the given nonzero smooth 1-form on $\mathbb{R}^4$. Let D be the 3-plane field on $\mathbb{R}^4$ that consists of tangent vectors V such that $\phi(V) = 0$. By the Frobenius theorem, D is integrable if and only if $\phi \wedge d\phi = 0$. We compute,

$$d\phi = dy \wedge dx + dz \wedge dy \implies \phi \wedge d\phi = dt \wedge dy \wedge dx + y\,dx \wedge dz \wedge dy + dt \wedge dz \wedge dy,$$

which contains the nonzero term $dt \wedge dy \wedge dx$. Thus, $\phi \wedge d\phi$ is not identically zero, which implies that D is not integrable.

Problem 2019-J-II-6. Let X and Y be vector fields on $\mathbb{R}^3$, defined by

$$X = \frac{\partial}{\partial x} + x\frac{\partial}{\partial y} + y\frac{\partial}{\partial z} \quad \text{and} \quad Y = y\frac{\partial}{\partial x} + z\frac{\partial}{\partial y} + \frac{\partial}{\partial z}.$$

Is there a coordinate chart $\varphi = (x_1, x_2, x_3) : U \to \mathbb{R}^3$ of the neighborhood of the origin $0 \in \mathbb{R}^3$ such that

$$X|_U = \frac{\partial}{\partial x_1} \quad \text{and} \qquad Y|_U = \frac{\partial}{\partial x_2}?$$

Let X and Y be the vector fields on $\mathbb{R}^3$ given above. We compute,

$$\begin{aligned}
[V, W] &= \left(\frac{\partial}{\partial x} + x\frac{\partial}{\partial y} + y\frac{\partial}{\partial z}\right)\left(y\frac{\partial}{\partial x} + z\frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right) \\
&\quad - \left(y\frac{\partial}{\partial x} + z\frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)\left(\frac{\partial}{\partial x} + x\frac{\partial}{\partial y} + y\frac{\partial}{\partial z}\right) \\
&= \left(x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y}\right) - \left(y\frac{\partial}{\partial y} + z\frac{\partial}{\partial z}\right) = x\frac{\partial}{\partial x} - z\frac{\partial}{\partial z}.
\end{aligned}$$

Since $[V, W]$ is not identically zero in any neighborhood about the origin in $\mathbb{R}^3$, we conclude that there does not exist a coordinate chart $\varphi : (x_1, x_2, x_3) : U \to \mathbb{R}^3$ of a neighborhood of the origin $0 \in \mathbb{R}^3$ that satisfies the condition above.

Problem 2015-J-II-3. Let ω be a closed differential k-form on a smooth manifold M. At each point $p \in M$, notice that ω defines a linear mapping

$$T_pM \to \bigwedge_p^{k-1} T_p^*(M): \qquad V \mapsto \omega(V, -, \ldots, -).$$

Let D_p be the kernel of this linear mapping, and suppose that ω has the special property that the dimension $n = \dim D_p$ is independent of the point $p \in M$. Prove that, through any point of M, one can find an embedded submanifold $N \subseteq M$, such that $T_pN = D_p$ for every $p \in N$.

To prove that for any point of M, one can find an embedded submanifold $N \subseteq M$ such that $T_pN = D_p$ for every $p \in M$, we must show that D is an integrable distribution, where D is the distribution defined locally by D_p. By the Frobenius theorem, it suffices to show that if $V, W \in D_p$, then $[V, W] \in D_p$. We recall Koszul-Cartan's formula which states that if ω is a k-form, then

$$d\omega(X_0, \ldots, X_k) = \sum_{i=0}^{k} (-1)^i X_i(\omega(X_0, \ldots, \hat{X}_i, \ldots, X_k)) + \sum_{0 \le i < j \le k} (-1)^{i+j} \omega([X_i, X_j], X_0, \ldots, \hat{X}_i, \ldots, \hat{X}_j, \ldots, X_k).$$

Let $X_0 = V$ and $X_1 = W$. Then let $X_2, \ldots, X_k$ be arbitrary smooth vector fields on M. Since ω is a closed differential k-form, one has

$$d\omega(X_0, \ldots, X_k) = 0.$$

Next, by definition of X_0 and X_1,

$$\sum_{i=0}^{k} (-1)^i X_i(\omega(X_0, \ldots, \hat{X}_i, \ldots, X_k)) = X_0(\omega(\hat{X}_0, X_1, \ldots, X_k)) + X_1(\omega(X_0, \hat{X}_1, \ldots, X_k)).$$

Likewise,

$$\sum_{0 \le i < j \le k} (-1)^{i+j} \omega([X_i, X_j], X_0, \ldots, \hat{X}_i, \ldots, \hat{X}_j, \ldots, X_k) = -\omega([X_0, X_1], \hat{X}_0, \hat{X}_1, X_2, \ldots, X_k).$$

Therefore, we must have

$$\omega([X_0, X_1], \hat{X}_0, \hat{X}_1, X_2, \ldots, X_k) = X_0(\omega(\hat{X}_0, X_1, \ldots, X_k)) + X_1(\omega(X_0, \hat{X}_1, \ldots, X_k)).$$

By definition of X_0, the second term on the right side must be zero. On the other hand,

$$\omega(\hat{X}_0, X_1, \ldots, X_k) = -\omega(X_1, \hat{X}_0, \ldots, X_k) = 0 \implies X_0(\omega(\hat{X}_0, X_1, \ldots, X_k)) = 0.$$

Thus, $\omega([X_0, X_1], X_2, \ldots, X_k) = 0$. Since $X_2, \ldots, X_k$ were arbitrary smooth vector fields on M, we conclude by the definition of D that $[V, W]$ is a smooth section of D. Thus, by the Frobenius distribution, D is integrable, which means that for each $p \in M$, we can find an embedded submanifold $N \subseteq M$ that meets the desired properties. Note that this was a longer version of using Cartan's formula.

Problem 2013-A-II-4. Let θ be a smooth 1-form on a manifold M such that $\theta \neq 0$ everywhere. Let $D \subset TM$ be the vector subbundle defined by

$$D = \ker \theta = \{v \in TM : \theta(v) = 0\}.$$

Prove that D is Frobenius integrable if and only if $\theta \wedge d\theta = 0$ everywhere.

[!! Complete Later !!]

Problem 2012-J-I-4. Let (x, y, z) be the usual coordinates on $\mathbb{R}^3$. Consider the 1-form on $\mathbb{R}^3$ given by

$$\varphi = dx + y\,dz.$$

Is it possible to find smooth functions u and v on $\mathbb{R}^3$ such that $\varphi = u\,dv$? Why?

We claim that it is not possible to find smooth functions u, v on $\mathbb{R}^3$ so that $\varphi = u\,dv$. We first compute,

$$\varphi \wedge d\varphi = (dx + y\,dz) \wedge (dx + dy \wedge dz) = dx \wedge dy \wedge dz + y\,dz \wedge dx,$$

which is not identically zero as it contains the nonzero term $dx \wedge dy \wedge dz$. On the other hand,

$$(u\,dv) \wedge d(u\,dv) = (u\,dv) \wedge (du \wedge dv),$$

is identically zero. Thus the proof concludes.

Problem 2011-A-II-5. On $\mathbb{R}^4$, equipped with standard coordinates (x, y, z, t), let X and Y be the vector fields given by

$$X = \frac{\partial}{\partial x} + z\frac{\partial}{\partial y} \qquad \text{and} \qquad Y = x\frac{\partial}{\partial z} + \frac{\partial}{\partial t}.$$

If $f : \mathbb{R}^4 \to \mathbb{R}$ is a smooth function satisfying $Xf = Yf = 0$, show that f is constant.

Let (x, y, z, t) be the standard coordinates on $\mathbb{R}^4$, let X, Y be the vector fields given above, and let $f : \mathbb{R}^4 \to \mathbb{R}$ be a smooth function so that $Xf = Yf = 0$ for all $p \in \mathbb{R}^4$. First, this means that $[X,Y]f = 0$ since $[X,Y]f = X(Yf) - Y(Xf)$. We compute,

$$[X,Y] = \left(\frac{\partial}{\partial z}\right) - \left(x\frac{\partial}{\partial y}\right) = \frac{\partial}{\partial z} - x\frac{\partial}{\partial y}.$$

[!! Complete Later !!]

Problem 2010-J-I-2. Consider the differential 1-form $\varphi = dx^1 + x^2\,dx^3$ on $\mathbb{R}^3$. Is it possible to find a smooth coordinate system (y^1, y^2, y^3) on a neighborhood of the origin such that $\varphi = f\,dy^1$ in these new coordinates, for some smooth function $f(y^1, y^2, y^3)$? Support your answer with a proof.

We begin by computing,

$$\varphi \wedge d\varphi = (dx^1 + x^2\,dx^3) \wedge (dx^2 \wedge dx^3) = dx^1 \wedge dx^2 \wedge dx^3,$$

which is not identically zero. On the other hand, let $f : \mathbb{R}^3$ be a smooth function. Then,

$$(f\,dy^1) \wedge d(f\,dy^1) = (f\,dy^1) \wedge \left(\partial_{y^2} f\,dy^2 \wedge dy^1 + \partial_{y^3} f\,dy^3 \wedge dy^1\right)$$

is identically zero. Thus, we conclude that it is not possible to find a smooth coordinate system (y^1, y^2, y^3) on a neighborhod of the origin such that $\varphi = f\,dy^1$ in these new coordinates, for some smooth function $f(y^1, y^2, y^3)$.

Problem 1997-J-I-4. Let N be a smooth manifold and let ω be a smooth closed 2-form on N. For $x \in N$, let

$$D_x = \{v \in T_xN : \omega(v, w) = 0, \forall w \in T_xN\},$$

and suppose that $m = \dim D_x$ is independent of $x \in N$. Show that there exist m-dimensional submanifolds $M \subset N$ such that $T_xM = D_x$ for all $x \in M$.

Let $D : p \mapsto D_p$ be the distribution defined locally by D_p. Since $m = \dim D_x$ is independent of $x \in N$, by the Frobenius theorem, to show that there exist m-dimensional submanifolds $M \subseteq N$ such that

$T_x M = D_x$ for all $x \in M$, it suffices to show that D is an integrable distribution. To do so, we will show that if V, W are smooth sections of D, then their Lie bracket $[V, W]$ is also a smooth section of D. Then

$$\mathcal{L}_{[V,W]}\omega = \mathcal{L}_V(\mathcal{L}_W\omega) - \mathcal{L}_W(\mathcal{L}_V\omega).$$

Cartan's formula tells us that

$$\begin{aligned}
\mathcal{L}_V\omega &= V \lrcorner\, d\omega + d(V \lrcorner\, \omega) \\
&= 0 + d(0) = 0, \\
\mathcal{L}_W\omega &= W \lrcorner\, d\omega + d(W \lrcorner\, \omega) \\
&= 0 + d(0) = 0,
\end{aligned}$$

where in both lines, the first terms on the right sides are zero since ω is closed, and the second terms are zero since V and W are smooth sections of D. Thus, we conclude that $\mathcal{L}_{[V,W]}\omega = 0$. But this means that $[V, W]$ must be a smooth section of D. Thus, by the Frobenius Theorem, we conclude that D is integrable, which means that there exist m-dimensional submanifolds $M \subseteq N$ so that $T_x M = D_x$ for all $x \in M$.

2.2 Stokes' Theorem

Problem 2025-J-I-3. Let M be an orientable connected and compact smooth n-manifold with boundary. Show that there is no smooth retraction to the boundary; that is, there does not exist a smooth map $f : M \twoheadrightarrow \partial M$ such that $f(x) = x$ when $x \in \partial M$.

Let M be an orientable connected and compact smooth n-manifold with boundary. Note that this implies that ∂M is an orientable $(n-1)$-smooth manifold, so that there exists a nowhere vanishing top form ω on ∂M. Assume to the contrary that there exists a smooth retraction $r : M \twoheadrightarrow \partial M$. By Stokes' Theorem,

$$\int_{\partial M} r^*\omega = \int_M d(r^*\omega) = \int_M r^*(d\omega) = 0,$$

where the final equality follows from the fact that $d\omega = 0$ on ∂M since ω is a top form on ∂M. But since r is the identity on ∂M,

$$\int_{\partial M} r^*\omega = \int_{\partial M} \omega \neq 0,$$

since ω is an orientation form on ∂M. Thus, we have arrived at a contradiction. Therefore, we conclude there is no smooth retraction $r : M \twoheadrightarrow \partial M$.

Problem 2023-J-II-3. Consider the form $\omega = (x^2 + x + y)\ dy \wedge dz$ on $\mathbb{R}^3$. Let $S^2 \subset \mathbb{R}^3$ be the unit sphere, and $\iota : S^2 \to \mathbb{R}^3$ be the inclusion map.

(a) Calculate $\int_{S^2} \iota^*\omega$.

(b) Construct a closed form α on $\mathbb{R}^3$ such that $\iota^*\alpha = \iota^*\omega$, or show that such a form α does not exist.

(a) Let ω be the given 2-form on $\mathbb{R}^3$. We note that $S^2 = \partial B^3$, where B^3 is the unit ball in $\mathbb{R}^3$. Thus, we must have

$$\int_{S^2} \iota^*\omega = \int_{B^3} d(\iota^*\omega) = \int_{B^3} \iota^*(d\omega).$$

We compute

$$d\omega = (2x+1)\ dx \wedge dy \wedge dz.$$

Thus,

$$\int_{B^3} i^*(d\omega) = \int_{B^3} (2x+1)\ dxdydz = \int_{B^3} dxdydz = \Gamma(B^3) = \frac{4}{3}\pi,$$

where

$$\int_{B^3} 2x\ dxdydz = 0,$$

by symmetry and oddness of the function x.

(b) We will assume that such a form α does not exist. If such a form exists, then $d(i^*\omega) = d(i^*\alpha) = i^*(d\alpha) = i^*(0) = 0$. This would imply that $\int_{S^2} i^*\omega = 0$, which contradicts our finding from (a).

Problem 2016-J-I-5. Let M be a smooth compact connected n-manifold without boundary, and let ω be a smooth n-form on M such that $\omega \neq 0$ at every point. Prove that there does not exist a smooth vector field X on M such that $\omega = \mathcal{L}_X\omega$, where $\mathcal{L}_X$ denotes the Lie derivative with respect to X.

Let M be a smooth compact connected n-manifold without boundary, and let ω be a smooth n-form on M so that $\omega \neq 0$ at every point. I.e., ω is an orientation form on M. Assume to the contrary that there exists a vector field X on M so that $\omega = \mathcal{L}_X\omega$. By Cartan's formula, one has

$$\mathcal{L}_X\omega = X \lrcorner\, d\omega + d(X \lrcorner\, \omega) = d(X \lrcorner\, \omega),$$

where the first term is zero since $d\omega = 0$, which follows from the fact that ω is a top form on M. This means that $\omega = d(X \lrcorner \omega)$ is an exact form. Since M is compact connected oriented without boundary, it follows that (e.g., by Corollary 16.13 of Lee *Smooth Manifolds*) that

$$\int_M d(X \lrcorner \omega) = 0.$$

But this contradicts the fact that ω is an orientation form on M so that $\int_M \omega \neq 0$. Thus by contradiction, it follows that no such vector field X exists on M.

Problem 2009-J-II-3. A sympletic form $\omega \in \Omega^2(M^{2n})$ is a closed 2-form such that

$$\omega^{\wedge n} = \underbrace{\omega \wedge \cdots \wedge \omega}_{n \text{ times}} \neq 0$$

at all points, where $n \neq 0$. If M is compact, then show that $[\omega] \neq 0 \in H^2(M)$.

Let ω be a sympletic form on M^{2n}, where $n \neq 0$. Let M be compact, and assume to the contrary that $[\omega] = 0 \in H^2(M)$ (i.e., ω is exact). Let α be a 1-form such that $\omega = d\alpha$. Since ω is closed, one has

$$\omega^{\wedge n} = d(\alpha \wedge \omega^{\wedge n-1}).$$

Thus, $\int_M \omega^{\wedge n} = 0$ on a compact manifold without boundary. But since $\omega^{\wedge n}$ is nowhere vanishing, it is a volume form so that $\int_M \omega^{\wedge n} \neq 0$. Thus, by contradiction, the proof concludes.

Problem 2004-J-II-1. Let X denote the product of the 2-dimensional sphere with the circle; let Y denote the 3-dimensional torus; and let $f : X \to Y$ denote an infinitely differentiable map. Show that for any 3-form ω on Y we have that

$$\int_X f^* \omega = 0.$$

Problem 2002-J-I-6. Define forms θ_1 and θ_2 on the torus $T^2 = S^1 \times S^1$ by pulling back area forms from the two projections onto circles. Let M be a compact oriented 3-manifold with boundary T^2. Show that $\int_{T^2} \theta_1 \wedge \theta_2$ is non-zero, and as a result that θ_1 and θ_2 cannot both extend to closed forms on M.

Let $T^2 = S^1 \times S^1$ denote the 2-torus, and let θ_1, θ_2 be two 1-forms on T^2 obtained by pulling back area forms from the two projections onto circles. Let M be a compact oriented 3-manifold with boundary T^2. Then by Stokes' Theorem, one has

$$\int_{T^2} \theta_1 \wedge \theta_2 = \int_{\partial M} \theta_1 \wedge \theta_2 = \int_M d\theta_1 \wedge \theta_2 - \int_M \theta_1 \wedge d\theta_2.$$

Problem 1999-A-I-4. Let M be a smooth compact oriented manifold with boundary. Use Stokes' Theorem to show that there is no smooth map $\Phi : M \to \partial M$ such that $\Phi|_{\partial M}$ is the identity.

Let M be a smooth compact oriented manifold with boundary, and assume to the contrary that there exists a smooth retraction $\Phi : M \twoheadrightarrow \partial M$. Since M is orientable, ∂M is a smooth orientable $(n-1)$-manifold. Thus there exists a nowhere vanishing $(n-1)$-form ω on ∂M that defines an orientation on ∂M; equip ∂M with this orientation. Then by Stokes' Theorem,

$$\int_{\partial M} \Phi^* \omega = \int_M d(\Phi^* \omega) = \int_M \Phi^*(d\omega) = 0,$$

where the final equality follows from the fact that $d\omega = 0$ since ω is a top form on ∂M. On the other hand, since Φ is the identity map on ∂M,

$$\int_{\partial M} \Phi^* \omega = \int_{\partial M} \omega > 0.$$

This produces a contradiction. Therefore, we conclude that no such retraction Φ can exist.

Problem 1998-J-I-5. Show that a nowhere vanishing 2-form on a compact orientable 2-dimensional manifold cannot be exact.

Let M denote a compact orientable 2-dimensional manifold without boundary, and let ω be a nowhere vanishing 2-form on M. Assume to the contrary that ω is exact; i.e., assume there exists some 1-form α so that $\omega = d\alpha$. Since ω is nowhere vanishing, it is a volume form on M so that if M is equipped with the orientation induced

$$\int_M \omega > 0.$$

On the other hand, since M is without boundary, by Stokes' Theorem,

$$\int_M \omega = \int_M d\alpha = 0,$$

which is a contradiction. Thus, by contradiction, we conclude that ω cannot be exact.

2.3 Submersions and Immersions

Problem Comps Lemma (Submersion). Let M, N be smooth connected n-manifolds, and let $f : M \to N$ be a smooth submersion. If M is compact and nonempty, then N is compact and f is a smooth covering map.

Let M and N be smooth connected n-manifolds, and let $f : M \to N$ be a smooth submersion. Assume M is compact and nonempty. Since f is a submersion, $df_p : T_pM \to T_{f(p)}N$ is surjective for all $p \in M$. Since $\dim T_pM = \dim M = \dim N = \dim T_{f(p)}N$ for all $p \in M$, it follows that df_p must also be injective. Therefore, the differential of f at each $p \in M$ is a bijective map. By the inverse function theorem, it follows that f is a local diffeomorphism. Since local diffeomorphisms are open maps, $f(M)$ is a nonempty open subset of N. Next since f is continuous, and the image of compact spaces under continuous maps is compact, $f(M)$ is compact in N (and hence, closed in N since N is Hausdorff). Since N is connected and $f(M)$ is nonempty, $f(M) = N$ so that N is compact and f is surjective. Now let $q \in N$, and consider $f^{-1}(q) \subset M$. Since f is closed and $\{q\} \subset N$ is closed, $f^{-1}(q)$ is a closed subset of M. Moreover, $f^{-1}(q)$ must be discrete: assume not. Then there exists some $p \in f^{-1}(q)$ and a neighborhood U of p so that $f : U \to f(U)$ is a diffeomorphism. Moreover, there exists some $x \in (f^{-1}(q) \cap U) \setminus \{p\}$ so that $f(x) = q$. But this contradicts the fact that $f : U \to f(U)$ is injective. Thus by contradiction, every $p \in f^{-1}(q)$ is isolated and therefore, $f^{-1}(q)$ is discrete. But since every closed discrete subset of a compact Hausdorff space is finite, $f^{-1}(q)$ is finite. Fix some $q \in N$, and let $\{x_1, \ldots, x_s\} = f^{-1}(q)$. For each x_j, let U_j be a neighborhood of x_j so that $f : U_j \to f(U_j)$ is a diffeomorphism. We can shrink each U_j so that $f : U_j \to f(U_j)$ remains a diffeomorphism and the U_j are pairwise disjoint. Set $V = \bigcap_1^s f(U_j)$. Then V is an evenly covered neighborhood of $q \in N$. Since $q \in N$ was arbitrary, we conclude that f is a smooth covering map.

Problem Comps Lemma (Local Homeomorphism). Let M, N be smooth connected n-manifolds, and let $f : M \to N$ be a local homeomorphism. If M is compact and nonempty, then N is compact and f is a covering map.

Let M, N be smooth connected n-manifolds, and let $f : M \to N$ be a local homeomorphism. Assume M is compact and nonempty. Since the image of compact sets under continuous maps is compact, $f(M)$ is compact and nonempty in N. Since compact subspaces of Hausdorff spaces are closed, and N is Hausdorff, $f(M)$ is closed and nonempty in N. Next, since local homeomorphisms are open maps, $f(M)$ is also open in N. Since N is connected, and $f(M)$ is a closed, open, nonempty subset of N, we conclude that $f(M) = N$. Thus N is compact and f is surjective. The rest of the proof follows identically to the proof given above for the Comps Lemma (Submersions) statement. Thus, f is a finite-sheeted covering map.

Problem 2025-J-II-4. Let Σ_2 be a compact oriented surface of genus 2. Is there a submersion

$$f : \Sigma_2 \to S^1 \times S^1,$$

where S^1 denotes the unit circle?

Let Σ_2 be a compact oriented surface of genus 2. Suppose there exists a submersion $f : \Sigma_2 \to S^1 \times S^1$, where S^1 denotes the unit circle. By the Comps Lemma it follows that f is a smooth covering map. Thus, the induced map $f_* : \pi_1(\Sigma_2) \to \pi_1(S^1 \times S^1) \cong \pi_1(S^1) \times \pi_1(S^1) \cong \mathbb{Z} \times \mathbb{Z}$ must be injective. However, $\pi_1(\Sigma_2)$ is nonabelian, while $\mathbb{Z} \times \mathbb{Z}$ is abelian. Thus, by contradiction, there cannot be any submersion from $\Sigma_2 \to S^1 \times S^1$.

Problem 2024-A-I-5. Determine whether or not the complex number $i = \sqrt{-1}$ is in the field $\mathbb{Q}(\alpha)$, where α is any complex number subject to the relation $\alpha^3 + \alpha + 1$. Justify your answer.

We claim that i is not in the field $\mathbb{Q}(\alpha)$ for any root α of the monic polynomial $x^3 + x + 1$. Indeed, by real analytic arguments, we find that the polynomial has a unique real root, α_1, and a pair of conjugate complex roots α_2, α_3. Since α_1 is real, $\mathbb{Q}(\alpha) \hookrightarrow \mathbb{R}$. Thus $i \notin \mathbb{Q}(\alpha)$. By the rational root test, the monic polynomial is irreducible over $\mathbb{Q}$. Thus the minimal polynomials of α_2 and α_3 are of degree 3, which means that $\mathbb{Q}(\alpha_2), \mathbb{Q}(\alpha_3)$ are both degree 3 field extensions. Assume $i \in \mathbb{Q}(\alpha_2)$. Then $\mathbb{Q}(\alpha_2)$ must contain the degree 2 field extension $\mathbb{Q}(i)$, which then implies that $2 \mid 3$, a contradiction. Thus $\mathbb{Q}(\alpha_2)$ cannot contain i. Likewise, $\mathbb{Q}(\alpha_3)$ cannot contain i. The proof concludes.

Problem 2024-A-II-1. Recall that S^n denotes the unit sphere in $\mathbb{R}^{n+1}$. Also recall that a smooth map is called a smooth submersion if its differential is everywhere surjective. Prove or disprove each of the following statements:

(a) There is a smooth submersion $F: S^3 \to S^1$.

(b) There is a smooth submersion $F: S^3 \to S^2$.

(a) We claim there is no smooth submersion from $F: S^3 \to S^1$. Assume to the contrary that there exists such a smooth submersion, and consider the nowhere vanishing 1-form $d\theta$ on S^1. Then since F is a submersion, by definition of the pullback, $F^* d\theta$ is a nowhere vanishing closed 1-form on S^3. Since $H^1_{\mathrm{dR}}(S^3) = 0$, $F^* d\theta$ ought to be exact; i.e., there exists some smooth function $f: S^3 \to \mathbb{R}$ so that $F^* d\theta = df$. But by the extreme value theorem, f has an extrema at some point $p \in M$, which means that $df_p = 0$. Thus, $F^* d\theta$ vanishes at $p \in M$, which is a contradiction. Thus, by contradiction, F cannot be a submersion.

(b) There does exist a smooth submersion $F: S^3 \to S^2$. In particular, the Hopf map $H: S^3 \subset \mathbb{C}^2 \to \mathbb{CP}^1 \cong S^2$, given by $H(z_1, z_2) = [z_1 : z_2]$ is smooth and has a surjective differential at every point. Equivalently, since its fibers are the free S^1 orbits $(z_1, z_2) \mapsto (e^{it} z_1, e^{it} z_2)$, it is a smooth submersion.

Problem 2023-A-I-2. Let $f: T^2 \to S^2$ be a smooth map from the 2-torus to the 2-sphere. Can f be an immersion? If the answer is yes, give an explicit example. If the answer is no, then give a proof.

Let T^2 and S^2 denote the 2-torus, and 2-sphere. Assume to the contrary that there exists a smooth immersion $f: T^2 \to S^2$. Then by the Comps Lemma, f is a covering map, which means that the induced map $f_*: \pi_1(T^2, e_0) \to \pi_1(S^2, x_0)$ is injective. Since S^2 is simply connected, $\pi_1(S^2, x_0) = \{e\}$. On the other hand, since $T^2 \cong S^1 \times S^1$, $\pi_1(T^2, e_0) \cong \mathbb{Z} \times \mathbb{Z}$. Thus we have shown that $\mathbb{Z} \times \mathbb{Z} \hookrightarrow \{e\}$, which is a contradiction! Thus by contradiction, there exists no immersion from the 2-torus to the 2-sphere.

Problem 2023-A-I-5. Let T be the 2-torus $S^1 \times S^1$ with an open 2-disk removed. Show that there is no continuous retraction r onto its boundary (i.e., no continuous map $r: T \to \partial T$ satisfying $r|_{\partial T} = \mathrm{id}_{\partial T}$).

Let T be the 2-torus with an open 2-disk removed (i.e., so that $\partial T \cong S^1$). Assume to the contrary that there is a continuous retraction r onto its boundary. Let $\iota: \partial T \hookrightarrow T$ denote the natural inclusion map. Then $r_* \circ \iota_*: \pi_1(\partial T) \to \pi_1(\partial T)$ must be the identity map. We note that $\pi_1(T) \cong \mathbb{Z} * \mathbb{Z}$, the free product group on two generators, where the boundary curve represents the commutator $aba^{-1}b^{-1}$. Since $\pi_1(\partial T) \cong \pi_1(S^{-1}) \cong \mathbb{Z}$ is abelian, every homomorphism from $\mathbb{Z} * \mathbb{Z}$ must map commutators to zero. But this means that $r_* \circ \iota_*$ cannot be the identity map, which is a contradiction! Thus there cannot be a continuous retraction of T onto its boundary.

Problem 2019-A-I-4. Let $f: \mathbb{RP}^3 \to \mathbb{T}^3 = S^1 \times S^1 \times S^1$ be a smooth map. Show that f is not an immersion.

Let $f: \mathbb{RP}^3 \to \mathbb{T}^3$ be a smooth map. Assume to the contrary that f is an immersion. Since $\mathbb{RP}^3$ is compact and nonempty, aand then by the Comps Lemma, f is a covering map. Therefore, the induced map $f_*: \pi_1(\mathbb{RP}^3) \to \pi_1(S^1 \times \cdots \times S^1)$ is injective. Therefore, $\pi_1(\mathbb{RP}^3)$ is isomorphic to a subgroup of $\pi_1(S^1 \times S^1 \times S^1)$. But this is a contradiction: $\pi_1(\mathbb{RP}^3) \cong \mathbb{Z}/2\mathbb{Z}$, while $\pi_1(S^1 \times S^1) \cong \mathbb{Z}^3$ is torsion-free. Thus, f cannot be an immersion.

Problem 2017-A-I-1. Let M be a smooth compact connected n-manifold (without boundary), and let $F : M \to \mathbb{R}^n$ be a smooth map. Does F necessarily have a critical point?

Let M be a smooth compact connected n-manifold (without boundary), and let $F : M \to \mathbb{R}^n$ be a smooth map. Then F necessarily has a critical point. Recall that a point $p \in M$ is said to be a ***regular point*** of F if and only if $dF_p : T_pM \to T_{F(p)}N$ is surjective, and ***critical*** otherwise. Let $F = (f_1, \ldots, f_n)$, where for each $1 \leq j \leq n$, $f_j : M \to \mathbb{R}^n$ is a smooth function. Since M is compact, by the extreme value theorem, each f_j has an extrema at some point $p_j \in M$, which means that $df_j|_{p_j} = 0$. But this means that dF_{p_j} (which can be represented by a square $n \times n$ matrix) does not always have full rank n, and therefore, is not surjective. Thus, each p_j is a critical point of F.

Problem 2017-J-I-1. Let Σ_1 be a torus and let Σ_2 be a genus 2 surface. Show that there is no submersion from Σ_2 to Σ_1.

Let Σ_1 be a torus and let Σ_2 be a genus 2 surface. Assume to the contrary that there exists a submersion from $f : \Sigma_2 \to \Sigma_1$. Then by the Comps Lemma, f must be a k-sheeted covering map for some finite integer $k > 0$. Thus, if $\chi(\Sigma_j)$ denotes the Euler characteristic of Σ_j, we must have $\chi(\Sigma_2) = k \cdot \chi(\Sigma_1)$. We compute,

$$\chi(\Sigma_1) = 2 - 2(1) = 0 \qquad \text{and} \qquad \chi(\Sigma_2) = 2 - 2(2) = 2 - 4 = -2.$$

Since there exists no positive integer k such that $-2 = 0(k) = 0$, we have arrived at a contradiction. Thus, there exist no submersions from Σ_2 to Σ_1.

Problem 2015-A-II-3. Suppose $f : \Sigma_2 \to \Sigma_1$ is a continuous map from a compact orientable surface Σ_2 of genus 2 to a torus Σ_1 (a compact orientable surface of genus 1). Show that f is not a local homeomorphism (i.e., for some $x \in \Sigma_2$, there exists no neighborhood on which f is a homeomorphism onto its image).

Let $f : \Sigma_2 \to \Sigma_1$ be a continuous map. Assume to the contrary that f is a local homeomorphism. Then by the Comps Lemma, f is a k-sheeted covering map for some finite positive integer k. Since f is a covering map, we must have $\chi(\Sigma_2) = k \cdot \chi(\Sigma_1)$. But

$$\chi(\Sigma_2) = 2 - 2(2) = 2 - 4 = -2 \qquad \text{and} \qquad \chi(\Sigma_1) = 2 - 2(1) = 0.$$

Since $k \cdot 0 = 0 \neq -2$, we have reached a contradiction. Thus by contradiction, there cannot be a local homeomorphism from Σ_2 to Σ_1.

Problem 2011-J-I-3. Let Σ be the double torus (i.e., a compact oriented surface of genus 2) and $T = S^1 \times S^1$ the torus, where S^1 is the circle. Suppose that $f : \Sigma \to T$ is continuous. Show that f is not a local homeomorphism.

Let $f : \Sigma_2 \to T$ be a continuous map. Assume to the contrary that f is a local homeomorphism. Then by the Comps Lemma, f is a k-sheeted covering map for some finite positive integer k. Since f is a covering map, we must have $\chi(\Sigma_2) = k \cdot \chi(T)$. But

$$\chi(\Sigma_2) = 2 - 2(2) = 2 - 4 = -2 \qquad \text{and} \qquad \chi(T) = 2 - 2(1) = 0.$$

Since $k \cdot 0 = 0 \neq -2$, we have reached a contradiction. Thus by contradiction, there cannot be a local homeomorphism from Σ_2 to T.

Problem 2010-J-I-3. Let $f : S^1 \times S^1 \to \mathbb{RP}^2$ be a smooth map from the 2-torus to the projective plane. Prove that f cannot be an immersion.

Let $f : S^1 \times S^1 \to \mathbb{RP}^2$ be a smooth map from the 2-torus to the projective plane. Assume to the contrary that f is an immersion. Then from the Comps Lemma, f must be a covering map, so that the induced map $f_* : \pi_1(S^1 \times S^1) \to \pi_1(\mathbb{RP}^2)$ is injective, which means that the first fundamental group of the two torus would have to be isomorphic to a subgroup of the first fundamental group of the projective plane. But $\pi_1(S^1 \times S^1) \cong \mathbb{Z} \times \mathbb{Z}$ (which is an infinite group), while $\pi_1(\mathbb{RP}^2) \cong \mathbb{Z}/2\mathbb{Z}$ (a finite group). Thus, the proof concludes by contradiction.

Problem 2002-A-I-1. Let $f : S^3 \to \mathbb{R}^3$ be a $\mathcal{C}^\infty$-map. Show that there must exist at least one point where f does not locally have a $\mathcal{C}^\infty$ inverse.

Let $f : S^3 \to \mathbb{R}^3$ be a smooth map, and assume to the contrary that f has a local smooth inverse at every $p \in S^3$. By definition, it follows that f is a local diffeomorphism. Since diffeomorphisms are open maps, $f(S^3)$ is a nonempty open subset of the manifold $\mathbb{R}^3$. On the other hand, since the image of compact spaces under continuous maps is compact, $f(S^3)$ is compact in $\mathbb{R}^3$. Since $\mathbb{R}^3$ is Hausdorff and compact spaces of Hausdorff spaces are closed, $f(S^3)$ is closed in $\mathbb{R}^3$. Thus, $f(S^3)$ is a nonempty closed and open subset of $\mathbb{R}^3$. Since $\mathbb{R}^3$ is connected, it follows that $f(S^3) = \mathbb{R}^3$. But then, $\mathbb{R}^3$ must be a compact subspace of itself, which is a contradiction. Thus, by contradiction, there must exist some point $p \in S^3$ so that f does not have a smooth inverse in some neighborhood of p.

Problem 2002-J-II-1. Let M and N be smooth, connected, n-dimensional manifolds, and let $f : M \to N$ be an immersion (i.e., assume that the derivative of f always sends nonzero tangent vectors to non-zero tangent vectors). If M is compact and nonempty, show that N is compact, and that f is a covering map.

Note that this is essentially the same as the Comps Lemma from above. Let M, N be smooth, connected, n-dimensional manifolds, and let $f : M \to N$ be an immersion. Since the image of compact spaces under continuous maps is compact, $f(M)$ is a nonempty, compact subset of N; since N is Hausdorff, and compact subspaces of Hausdorff spaces are closed, $f(M)$ is a nonempty closed subset of N. Now since f is an immersion, $df_p : T_pM \to T_{f(p)}N$ is injective for all $p \in M$. Since $\dim T_pM = \dim M = \dim N = \dim T_{f(p)}N$, it follows that df_p must also be surjective. Thus df_p is bijective. By the Inverse Function Theorem, it follows that f is a local diffeomorphism. Since local diffeomorphisms are open maps, $f(M)$ is an open subset of N. Thus $f(M)$ is a nonempty open and closed subset of N. Since N is connected, it follows that $f(M) = N$. And since $f(M)$ was shown to be compact in N, N is compact.
Now let $q \in N$, and consider $f^{-1}(q)$. Since $\{q\}$ is closed in N and f is continuous, $f^{-1}(q)$ is closed in M. Furthermore, we claim that $f^{-1}(q) \subset M$ is a discrete set. To see this, let $x \in f^{-1}(q)$ and assume to the contrary that $f^{-1}(q)$ is not discrete. Since f is a local diffeomorphism, there exists some neighborhood U of x in M so that $f : U \to f(U)$ is a diffeomorphism (in particular, is injective). Since x is not isolated, there exists some distinct $y \in f^{-1}(q) \cap U$ such that $f(y) = q$. But then $f(x) = q = f(y)$ and $x \neq y$, which contradicts the claim that f is injective. Thus by contradiction, x must be isolated. Since x was arbitrary, we conclude that every point in $f^{-1}(q)$ is isolated so that $f^{-1}(q)$ is discrete. Since every closed discrete subset of a compact space is finite, we conclude that $f^{-1}(q)$ is finite. Let $q \in N$ be fixed, and assume $\{x_1, x_2, \ldots, x_s\} = f^{-1}(q)$. For each $1 \leq j \leq s$, let $\widetilde{U}_j$ be a neighborhood of x_j in M so that $f : \widetilde{U}_j \to f(\widetilde{U}_j)$ is a diffeomorphism. Without loss of generality, we can assume that the $\widetilde{U}_j$ are disjoint. If they are not, then we can shrink each U_j so that they are disjoint and f remains a diffeomorphism on each shrunken neighborhood. Let $V = \bigcap_1^s f(\widetilde{U}_j)$, and define $U_j = f^{-1}(V) \cap \widetilde{U}_j$. Thus V is an evenly covered neighborhood of q in N. This conclude the proof that f is a covering map.

2.4 de Rham Cohomology

Problem 2026-J-II-6. Let M be the closed 3-manifold obtained from the 3-torus by removing two small disjoint 3-balls and identifying their boundary 2-spheres by an orientation-reversing diffeomorphism. Compute the de Rham cohomology of M.

Let M be the closed 3-manifold obtained by the construction given above. We recall that this procedure is called the *autosum* of M and denotes the connected sum of the 3-torus, T^3, with the handle $S^2 \times S^1$. Thus, for every p, $H^p_{\mathrm{dR}}(M) \cong H^p_{\mathrm{dR}}(T^3 \# (S^2 \times S^1))$. Since $\dim M = 3$, $H^k_{\mathrm{dR}}(M) \cong 0$ for all $k \geq 4$. Since M is compact connected and orientable, $H^3_{\mathrm{dR}}(M) \cong \mathbb{R}$. Likewise, since M is connected, $H^0_{\mathrm{dR}}(M) \cong \mathbb{R}$. Since T^3 and $S^2 \times S^1$ are connected and orientable, by Problem LSM-17-7, one has $H^p_{\mathrm{dR}}(M) \cong H^p_{\mathrm{dR}}(T^3) \oplus H^p_{\mathrm{dR}}(S^2 \times S^1)$ for $p = 1, 2$. We begin by computing the following:

$$\begin{aligned}
H^1_{\mathrm{dR}}(T^3) &= H^1_{\mathrm{dR}}(S^1 \times T^2) = \bigoplus_{i+j=1} H^i_{\mathrm{dR}}(S^1) \otimes H^j_{\mathrm{dR}}(T^2) \\
&= H^0_{\mathrm{dR}}(S^1) \otimes H^1_{\mathrm{dR}}(T^2) \oplus H^1_{\mathrm{dR}}(S^1) \otimes H^0_{\mathrm{dR}}(T^2) \\
&= \mathbb{R} \otimes \mathbb{R}^2 \oplus \mathbb{R} \otimes \mathbb{R} \cong \mathbb{R}^2 \oplus \mathbb{R} \cong \mathbb{R}^3. \\
H^1_{\mathrm{dR}}(S^2 \times S^1) &= \bigoplus_{i+j=1} H^i_{\mathrm{dR}}(S^2) \otimes H^j_{\mathrm{dR}}(S^1) \\
&= H^0_{\mathrm{dR}}(S^2) \otimes H^1_{\mathrm{dR}}(S^1) \oplus H^1_{\mathrm{dR}}(S^2) \otimes H^0_{\mathrm{dR}}(S^1) \\
&= \mathbb{R} \otimes \mathbb{R} \oplus 0 \otimes \mathbb{R} \cong \mathbb{R} \oplus 0 \cong \mathbb{R}. \\
H^2_{\mathrm{dR}}(T^3) &= H^2_{\mathrm{dR}}(S^1 \times T^2) = \bigoplus_{i+j=2} H^i_{\mathrm{dR}}(S^1) \otimes H^j_{\mathrm{dR}}(T^2) \\
&= H^0_{\mathrm{dR}}(S^1) \otimes H^2_{\mathrm{dR}}(T^2) \oplus H^1_{\mathrm{dR}}(S^1) \otimes H^1_{\mathrm{dR}}(T^2) \oplus H^2_{\mathrm{dR}}(S^1) \otimes H^0_{\mathrm{dR}}(T^2) \\
&= \mathbb{R} \otimes \mathbb{R} \oplus \mathbb{R} \otimes \mathbb{R}^2 \oplus 0 \otimes \mathbb{R} \\
&= \mathbb{R} \oplus \mathbb{R}^2 \cong \mathbb{R}^3. \\
H^2_{\mathrm{dR}}(S^2 \times S^1) &= \bigoplus_{i+j=2} H^i_{\mathrm{dR}}(S^2) \otimes H^j_{\mathrm{dR}}(S^1) \\
&= H^0_{\mathrm{dR}}(S^2) \otimes H^2_{\mathrm{dR}}(S^1) \oplus H^1_{\mathrm{dR}}(S^2) \otimes H^1_{\mathrm{dR}}(S^1) \oplus H^2_{\mathrm{dR}}(S^2) \otimes H^0_{\mathrm{dR}}(S^1) \\
&= \mathbb{R} \otimes 0 \oplus 0 \otimes \mathbb{R} \oplus \mathbb{R} \otimes \mathbb{R} \cong \mathbb{R}.
\end{aligned}$$

Therefore, using the above information, we conclude that

$$H^k_{\mathrm{dR}}(M) = \begin{cases} \mathbb{R}, & k = 0, 3 \\ \mathbb{R}^4, & k = 1, 2 \\ 0, & k \geq 4 \end{cases}$$

Problem 2026-J-II-6 (Modified). Let M be the closed 3-manifold obtained by removing two small 3-balls from two 3-tori and identifying their boundary 2-spheres by an orientation-reversing diffeomorphism. Compute the de Rham cohomology groups $H^k_{\mathrm{dR}}(M, \mathbb{R})$.

Let M be the closed 3-manifold obtained from the 3-torus by removing two small disjoint 3-balls from 2 3-tori and identifying their boundary 2-spheres by an orientation-reversing diffeomorphism. I.e., let $M = T^3 \# T^3$ be the connected sum of two 3-tori. Since M is connected, $H^0_{\mathrm{dR}}(M) \cong \mathbb{R}$. Since $\dim M = 3$, $H^k_{\mathrm{dR}}(M) \cong 0$ for all $k \geq 4$. On the other hand, since T^3 is compact and orientable, by Problem LSM-17-7,

$$H^1_{\mathrm{dR}}(M) \cong H^1_{\mathrm{dR}}(T^3) \oplus H^1_{\mathrm{dR}}(T^3) \cong \mathbb{R}^3 \oplus \mathbb{R}^3 \cong \mathbb{R}^6.$$

Likewise, by Problem LSM-17-7,

$$H^2_{\mathrm{dR}}(M) \cong H^2_{\mathrm{dR}}(T^3) \oplus H^2_{\mathrm{dR}}(T^3) \cong \mathbb{R}^3 \oplus \mathbb{R}^3 \cong \mathbb{R}^6.$$

Therefore, we conclude that

$$H^k_{\mathrm{dR}}(M) = \begin{cases} \mathbb{R}, & k = 0, 3 \\ \mathbb{R}^6, & k = 1, 2 \\ 0, & k \geq 4. \end{cases}$$

Problem 2024-A-I-1. Let M be a smooth compact manifold without boundary, and let φ be a smooth closed 1-form on M that has the property that $\varphi \neq 0$ at every point of M. Prove that the first de Rham cohomology $H^1_{\mathrm{dR}}(M)$ of the given manifold is non-zero.

Let M be a smooth compact manifold without boundary, and let φ be a smooth closed 1-form on M so that $\varphi \neq 0$ at every point of M. Assume to the contrary that $H^1_{\mathrm{dR}}(M) = 0$, which means that φ must be a closed 1-form; i.e., there exists a smooth function $f : M \to \mathbb{R}$ so that $\varphi = df$. Since M is compact and f is a smooth function, f has an extrema at some point $p \in M$. But this necessarily means that p is a critical point of f, which implies $df_p = 0$. I.e., φ is zero at $p \in M$, which is a contradiction. Thus, by contradiction, φ is not exact, which means that $H^1_{\mathrm{dR}}(M) \neq 0$ since there exists a closed 1-form that is not exact.

Problem 2024-J-I-5. Let α be a closed 1-form on $\mathbb{RP}^n$, $n > 1$. Show that if $f : [0,1] \to \mathbb{RP}^n$ is a smooth function such that $f(0) = f(1)$, then

$$\int_{[0,1]} f^*\alpha = 0.$$

Include all calculations that are relevant to your solution.

Let α be a closed 1-form on $\mathbb{RP}^n$, $n > 1$, and let $f : [0,1] \to \mathbb{RP}^n$ be a smooth function so that $f(0) = f(1)$. To calculate the integral $\int_{[0,1]} f^*\alpha$, we need to know the de Rham cohomology groups of $\mathbb{RP}^n$; in particular, we must know the first de Rham cohomology group of $\mathbb{RP}^n$. [!! Show explicit computation of the cohomology groups !!] Since $H^1_{\mathrm{dR}}(\mathbb{RP}^n) = 0$, α must also be an exact 1-form on $\mathbb{RP}^n$. I.e., there exists a smooth function $\beta : \mathbb{RP}^n \to \mathbb{R}$ so that $\alpha = d\beta$. Then one has

$$\int_{[0,1]} f^*\alpha = \int_{[0,1]} f^*d\beta = \beta(f(1)) - \beta(f(0)) = \beta(f(0)) - \beta(f(0)) = 0.$$

Problem 2021-A-II-1. Let M be a compact manifold (without boundary) and $\pi : M \to S^1$ a submersion onto the circle. Show that the de Rham group $H^1_{\mathrm{dR}}(M) \neq 0$.

Let M be a compact smooth manifold without boundary, and let $\pi : M \to S^1$ be a submersion onto the circle. Assume to the contrary that $H^1_{\mathrm{dR}}(M) = 0$, which means that every closed form on M is an exact form. The cohomology class of the closed 1-form $d\theta$ is a generator of $H^1_{\mathrm{dR}}(S^1) \cong \mathbb{R}$. Then $\pi^*d\theta$ is a closed 1-form on M. Since $H^1_{\mathrm{dR}}(M) = 0$, there exists a smooth function $f : M \to \mathbb{R}$ so that $\pi^*d\theta = df$. In particular, by the extreme value theorem, there exists some $p \in M$ so that f has an extrema at p. But this means that $df_p = 0$, so that $\pi^*d\theta$ vanishes at p. We will show that this is a contradiction. Since π is a submersion, its differential $d\pi_x$ is surjective at every $x \in M$. Then since $d\theta$ is nowhere vanishing, for all $x \in M$, $d\theta_{\pi(x)} \neq 0$. But then $(\pi^*d\theta)_x := d\theta_{\pi(x)} \circ d\pi_x \neq 0$, directly contradicting our above assumption. Thus by contradiction, we conclude that $H^1_{\mathrm{dR}}(M) \neq 0$.

Problem 2020-J-II-2. Let φ be a smooth closed 4-form on a compact 12-manifold M with smooth boundary. If $\varphi \wedge \varphi \wedge \varphi \neq 0$ everywhere on M, prove that the de Rham cohomologies $H^4_{\mathrm{dR}}(M)$ and $H^8_{\mathrm{dR}}(M)$ are both non-zero.

Let φ be a smooth closed 4-form on a compact 12-manifold M with smooth boundary. Assume also that $\varphi \wedge \varphi \wedge \varphi =: \varphi^3$ is nowhere vanishing on M; i.e., φ^3 defines an orientation form on M. Then $\int_M \varphi^3 \neq 0$.

Now assume to the contrary that $H^4_{\text{dR}}(M) = 0$. Then φ must be an exact form so that $\varphi = d\eta$ for some smooth 3-form η. Because $d\varphi = 0$, one can write

$$\varphi \wedge \varphi \wedge \varphi = d(\eta \wedge \varphi \wedge \varphi).$$

But then, by Stokes' Theorem,

$$\int_M \varphi^3 = \int_M d(\eta \wedge \varphi \wedge \varphi) = \int_{\partial M} \eta \wedge \varphi \wedge \varphi = 0,$$

since $\partial M = \varnothing$. This is a contradiction. Thus, $H^4_{\text{dR}}(M) \neq 0$. Likewise, assume $H^8_{\text{dR}}(M) = 0$. This means that φ^2 is exact; there exists some smooth 7-form η on M so that $\varphi^2 = d\eta$. Thus since $d\varphi = 0$, we can write $\varphi^3 = d(\eta \wedge \varphi)$. Then by Stokes' Theorem,

$$0 \neq \int_M \varphi^3 = \int_M d(\eta \wedge \varphi) = \int_{\partial M} \eta \wedge \varphi = 0,$$

since $\partial M = \varnothing$. By contradiction, we conclude that $H^8_{\text{dR}}(M) \neq 0$.

Problem 2019-A-II-4. Let M^n be a compact, connected, oriented n-manifold, and assume that n is odd. Compute the Euler characteristic $\chi(M^n \setminus \{\text{pt}\})$.

Let M^n be a compact, connected, oriented n-manifold, and assume that n is odd. Let us fix some point $p \in M^n$, and let B be a small n-ball in M centered at p. Then B deformation retracts onto $\{p\}$, and $M \setminus \{p\}$ deformation retracts onto $M \setminus B$. Moreover, $B \cap (M \setminus B) = S^{n-1}$. Thus by additivity of the Euler characteristic, one has

$$\chi(M) = \chi(M \setminus B) + \chi(B) - \chi(S^{n-1}).$$

Since n is odd, $n-1$ is even so that $\chi(S^{n-1}) = 2$. On the other hand, by Poincaré duality, one has $\chi(M) = 0$, and $\chi(B) = 1$. Thus, we conclude that $\chi(M \setminus B) = \chi(M \setminus \{p\}) = 1$.

Problem 2019-J-I-3. Let M and N be connected, compact, oriented n-manifolds. Prove that if $H^k_{\text{dR}}(N) \neq 0$ and $H^k_{\text{dR}}(M) = 0$ for some $k \neq 0$, then any smooth map $f : M \to N$ has degree 0.

Let M and N be connected, compact, oriented n-manifolds so that $H^k_{\text{dR}}(N) \neq 0$ and $H^k_{\text{dR}}(M) = 0$ for some $k \neq 0$. Further, let $f : M \to N$ be a smooth map. The map f induces a smooth pullback map $f^* : H^k_{\text{dR}}(N) \to H^k_{\text{dR}}(M)$. But since $H^k_{\text{dR}}(M) = 0$, f^* must be the zero map. Let α be a closed k-form on N so that $[\alpha] \neq 0$. Since f^* is the zero map, one must have $[f^*\alpha] = 0 \in H^k_{\text{dR}}(M)$ so that $f^*\alpha$ is an exact form. Let β be a $(k-1)$-form on M so that $f^*\alpha = d\beta$. Then by the Poincaré duality for de Rham cohomology, there exists a closed $(n-k)$-form γ on N so that

$$\int_N \alpha \wedge \gamma \neq 0.$$

But this implies that

$$\int_M f^*(\alpha \wedge \gamma) = \int_M f^*\alpha \wedge f^*\gamma = \int_M d\beta \wedge f^*\gamma = \int_M d(\beta \wedge f^*\gamma),$$

where in the final equality, we used the fact that $f^*\gamma$ is closed. Since M is without boundary, Stokes' Theorem tells us that the above integral must be zero. Thus,

$$\int_M f^*(\alpha \wedge \gamma) = 0 \cdot \int_N (\alpha \wedge \gamma).$$

By the definition of the degree of a smooth map, we conclude that $\deg f = 0$.

Problem 2015-J-I-5. Let X be a connected smooth manifold, and let U and V be connected open subsets such that $X = U \cup V$. Show that if $U \cap V$ is not connected, then the de Rham cohomology group $H^1_{\mathrm{dR}}(X, \mathbb{R})$ must be nonzero.

Let X be a connected smooth manifold, and let U and V be connected open subsets such that $X = U \cup V$. Further assume that $U \cap V$ is not connected; let n denote the number of connected components of $U \cap V$. Then, by the Mayer-Vietoris theorem, we have the exact sequence,

$$0 \to H^0_{\mathrm{dR}}(X) \xrightarrow{k^* \oplus l^*} H^0_{\mathrm{dR}}(U) \oplus H^0_{\mathrm{dR}}(V) \xrightarrow{i^* - j^*} H^0_{\mathrm{dR}}(U \cap V) \xrightarrow{\delta} H^1_{\mathrm{dR}}(X)$$
$$\xrightarrow{k^* \oplus l^*} H^1_{\mathrm{dR}}(U) \oplus H^1_{\mathrm{dR}}(V) \xrightarrow{i^* - j^*} H^1_{\mathrm{dR}}(U \cap V) \xrightarrow{\delta} \cdots,$$

where $i : U \cap V \hookrightarrow U$, $j : U \cap V \hookrightarrow V$, $k : U \hookrightarrow X$, and $l : V \hookrightarrow X$. Since X, U, V are connected and $U \cap V$ has n connected components, the exact sequence becomes,

$$0 \to \mathbb{R} \xrightarrow{k^* \oplus l^*} \mathbb{R}^2 \xrightarrow{i^* - j^*} \mathbb{R}^n \xrightarrow{\delta} H^1_{\mathrm{dR}}(X) \xrightarrow{k^* \oplus l^*} H^1_{\mathrm{dR}}(U) \oplus H^1_{\mathrm{dR}}(V) \xrightarrow{i^* - j^*} H^1_{\mathrm{dR}}(U \cap V) \xrightarrow{\delta} \cdots,$$

To show that $H^1_{\mathrm{dR}}(X)$ is nonzero, it suffices to show that the map δ is nonzero. We recall that each equivalence class in $H^0_{\mathrm{dR}}(U)$ and $H^0_{\mathrm{dR}}(V)$ is represented by a constant function on U and V, respectively. Since each such function restricts to a constant on each connected component W_j of $U \cap V$, we have that on each connected component W_j of $U \cap V$, $(i^* - j^*)|_{W_j}(c, d) = c - d$. Thus under the identification $H^0_{\mathrm{dR}}(U \cap V) \cong \mathbb{R}^n$, $i^* - j^*$ is given by the map $(c, d) \mapsto (c - d, \ldots, c - d)$. The image of this map is the set $\{(t, \ldots, t) \in \mathbb{R}^n : t \in \mathbb{R}\} \cong \mathbb{R}$ so that $\operatorname{rank}(i^* - j^*) = 1$. By exactness, $\operatorname{nullity}(\delta) = 1$. Thus by the rank-nullity theorem, we must have

$$\operatorname{rank}(\delta) + \operatorname{nullity}(\delta) = \dim(\mathbb{R}^n) = n \implies \operatorname{rank}(\delta) = n - 1.$$

Since we assumed $U \cap V$ was not connected, $n > 1$ (thus, $n - 1 > 0$). Therefore, since the rank of δ is positive, δ cannot be the zero map. Hence, $H^1_{\mathrm{dR}}(X)$ cannot be zero. The proof concludes.

Problem 2013-A-II-6. Let M be a smooth compact oriented n-manifold with boundary. Assume that ∂M is connected, and let $j : \partial M \hookrightarrow M$ denote the inclusion map. Prove that the induced map $j^* : H^{n-1}_{\mathrm{dR}}(M) \to H^{n-1}_{\mathrm{dR}}(\partial M)$ of de Rham cohomology is zero. Then use this to show there is no smooth retraction $F : M \to \partial M$.

Let M be a smooth compact oriented n-manifold with boundary, assume that ∂M is connected, and let $j : \partial M \hookrightarrow M$ denote the inclusion map. Consider the induced map $j^* : H^{n-1}_{\mathrm{dR}}(M) \to H^{n-1}_{\mathrm{dR}}(\partial M)$ of de Rham cohomology groups. Let $[\beta] \in H^{n-1}_{\mathrm{dR}}(M) \setminus \{0\}$ (i.e., let β be a closed, non-exact $(n-1)$-form on M). Assume to the contrary that $j^*\beta$ is not an exact $(n-1)$-form on N. Then since ∂M is also compact and oriented, one must have

$$\int_{\partial M} j^*\beta \neq 0.$$

But by Stokes' Theorem,

$$\int_{\partial M} j^*\beta = \int_M d\beta = 0,$$

since $d\beta$ is an exact n-form on M and M is compact. This is a contradiction. Thus, by contradiction $j^*\beta$ is exact. Since β was arbitrary, j^* maps every closed $(n-1)$-form on M to an exact $(n-1)$-form on ∂M so that j^* is the zero map on de Rham cohomology groups. Now assume to the contrary that there exists a smooth retraction $F : M \twoheadrightarrow \partial M$. Since ∂M is oriented, let ω be a smooth nowhere vanishing $(n-1)$-form on ∂M (i.e., a volume form). Let ∂M be equipped with the orientation induced by ω so that

$$\int_{\partial M} \omega > 0.$$

Note that ω is a closed differential form. Then, $F^*\omega$ is a closed $(n-1)$-form on M. By our above work, $j^*F^*\omega$ must be an exact $(n-1)$-form on ∂M so that since ∂M is compact, one has $\int_{\partial M} j^*F^*\omega = 0$. But $(F \circ j) = \mathrm{id}_{\partial M}$ so that $j^* \circ F^* = (F \circ j)^* = \mathrm{id}_{\partial M}^*$. This means that

$$\int_{\partial M} j^*F^*\omega = \int_{\partial M} \omega > 0,$$

which contradicts our previous finding. Thus, by contradiction, there cannot exist a smooth retraction $F : M \to \partial M$.

Problem 2013-J-II-6. Let M be a smooth compact manifold, and suppose that there is a smooth map $F : M \to S^1$ whose derivative is non-zero at every point. Prove that the de Rham cohomology space $H^1_{\mathrm{dR}}(M)$ is non-zero.

Let M be a smooth compact manifold, and suppose there is a smooth map $F : M \to S^1$ whose derivative is non-zero at every point $p \in M$. Assume to the contrary that $H^1_{\mathrm{dR}}(M) = 0$. Since $H^1_{\mathrm{dR}}(S^1) \cong \mathbb{R}$, this cohomology space is generated by the cohomology class of the nowhere vanishing closed 1-form $d\theta$ on S^1. Then $F^*d\theta$ is a closed 1-form on M. Since $H^1_{\mathrm{dR}}(M) = 0$, there exists a smooth function $f : M \to \mathbb{R}$ so that $F^*d\theta = df$. In particular, by the extreme value theorem, there exists some $p \in M$ so that f has an extrema at p. But this means that $df_p = 0$, so that $F^*d\theta$ vanishes at p. We will show that this is a contradiction. For any $p \in M$, $d\theta_{F(p)} \neq 0$. Moreover, by the given hypothesis, one has $dF_p \neq 0$. But this implies that

$$(F^*d\theta)_p \coloneqq d\theta_{F(p)} \circ dF_p \neq 0,$$

directly contradicting our result from above. Thus by contradiction, we conclude that $H^1_{\mathrm{dR}}(M) \neq 0$.

Problem 2002-J-II-5. Let θ be a closed 1-form on a compact manifold M without boundary. Further suppose that $\theta \neq 0$ at each point of M. Prove that $H^1_{\mathrm{dR}}(M) \neq 0$.

Let θ be a closed 1-form on a compact manifold M without boundary, and suppose that $\theta \neq 0$ at each point of M. Assume to the contrary that $H^1_{\mathrm{dR}}(M) = 0$. This means that θ is an exact 1-form, so that there exists some smooth 0-form f satisfying $\theta = df$ (i.e., $f : M \to \mathbb{R}$ is a smooth function). But since M is compact, f has an extrema at some point $p \in M$. This means that $df_p = 0$, and thus by extension, $\theta_p = 0$. But this contradicts our hypothesis on θ. Thus, by contradiction, θ cannot be an exact 1-form. Since we have shown the existence of a closed 1-form that is not exact, $H^1_{\mathrm{dR}}(M) \neq 0$.

Problem 1997-A-I-5. For which values of p is the de Rham cohomology $H^p_{\mathrm{dR}}(S^1 \times S^2)$ non-trivial? Explain.

First since $S^1 \times S^2$ is a smooth 3-manifold, $H^p_{\mathrm{dR}}(S^1 \times S^2) = 0$ for all $p \geq 4$. Moreover, since $S^1 \times S^2$ is connected, $H^0_{\mathrm{dR}}(S^1 \times S^2) \cong \mathbb{R}$ is non-trivial. Thus, it suffices to examine the case for $p = 1, 2, 3$. We recall Künneth's Formula, which states that for smooth manifolds M and N,

$$H^k_{\mathrm{dR}}(M \times N) \cong \bigoplus_{i+j=k} H^i_{\mathrm{dR}}(M) \otimes H^j_{\mathrm{dR}}(N).$$

Thus, we have

$$H^1_{\mathrm{dR}}(S^1 \times S^2) \cong \left(H^0_{\mathrm{dR}}(S^1) \otimes H^1_{\mathrm{dR}}(S^2)\right) \oplus \left(H^1_{\mathrm{dR}}(S^1) \otimes H^0_{\mathrm{dR}}(S^2)\right) \cong \mathbb{R} \otimes 0 \oplus \mathbb{R} \otimes \mathbb{R} \cong 0 \oplus \mathbb{R} \cong \mathbb{R}.$$

Next, we have

$$\begin{aligned} H^2_{\mathrm{dR}}(S^1 \times S^2) &\cong (H^2_{\mathrm{dR}}(S^1) \otimes H^0_{\mathrm{dR}}(S^2)) \oplus (H^1_{\mathrm{dR}}(S^1) \otimes H^1_{\mathrm{dR}}(S^2)) \oplus (H^0_{\mathrm{dR}}(S^1) \otimes H^2_{\mathrm{dR}}(S^2)) \\ &\cong (0 \otimes \mathbb{R}) \oplus (\mathbb{R} \otimes 0) \oplus (\mathbb{R} \otimes \mathbb{R}) \cong \mathbb{R}. \end{aligned}$$

Finally,

$$H^3_{\mathrm{dR}}(S^1 \times S^2) \cong (H^1_{\mathrm{dR}}(S^1) \otimes H^2_{\mathrm{dR}}(S^2)) \cong \mathbb{R}.$$

Thus, $H^p_{\mathrm{dR}}(S^1 \times S^2)$ is non-trivial for $p = 0, 1, 2, 3$.

Problem 2001-A-II-3. We say that a map $f : M \to N$ between two n-dimensional orientable manifolds is of "degree 1" if it induces an isomorphism $f_* : H^n_{\text{dR}}(N) \to H^n_{\text{dR}}(M)$. Prove that there is no map of degree 1 from the 2-sphere S^2 to the 2-dimensional torus T^2.

Assume to the contrary that there exists a map $f : S^2 \to T^2$ of degree 1, so that it induces an isomorphism $f_* : H^n_{\text{dR}}(T^2) \to H^n_{\text{dR}}(S^2)$ of de Rham cohomology groups. [!! Complete Later !!]

Problem 2007-A-II-3. Calculate the de Rham cohomology groups $H^k_{\text{dR}}(\mathbb{T}^2\#\mathbb{P}^2, \mathbb{R})$, where $\mathbb{T}^2$ is the torus and $\mathbb{P}^2$ is the projective plane.

Let $\mathbb{T}^2$ and $\mathbb{P}^2$ denote 2-torus and the projective plane, respectively. Since both of these are connected, $\mathbb{T}^2\#\mathbb{P}^2$ is also connected. Thus,

$$H^0_{\text{dR}}(\mathbb{T}^2\#\mathbb{P}^2) \cong \mathbb{R}.$$

Since $\mathbb{T}^2, \mathbb{P}^2$ are both compact, $\mathbb{T}^2\#\mathbb{P}^2$ is compact. However, since $\mathbb{P}^2$ is nonorientable, $\mathbb{T}^2\#\mathbb{P}^2$ is non-orientable. Thus since the connected sum is a smooth compact, non-orientable, and connected 2-manifold, $H^2_{\text{dR}}(\mathbb{T}^2\#\mathbb{P}^2) = 0$; $H^k_{\text{dR}}(\mathbb{T}^2\#\mathbb{P}^2) \cong 0$ for all $k \geq 3$ since the connected sum is a 2-manifold. Thus, we just need to compute the first de Rham cohomology group. We can solve this using the Euler characteristic. We recall that

$$\chi(M) = \sum_{0}^{\dim M} (-1)^i H^i_{\text{dR}}(M).$$

On the other hand, $\chi(M_1\#M_2) = \chi(M_1) + \chi(M_2) - 2$. Therefore,

$$-1 = 0 + 1 - 2 = 0 - \dim H^1_{\text{dR}}(\mathbb{T}^2\#\mathbb{P}^2) + 1 = 1 - \dim H^1_{\text{dR}}(\mathbb{T}^2\#\mathbb{P}^2).$$

Thus, $\dim H^1_{\text{dR}}(\mathbb{T}^2\#\mathbb{P}^2) = 2$ so that $H^1_{\text{dR}}(\mathbb{T}^2\#\mathbb{P}^2) = \mathbb{R}^2$. Therefore, we have the following:

$$H^k_{\text{dR}}(\mathbb{T}^2\#\mathbb{P}^2) = \begin{cases} \mathbb{R}, & k = 0 \\ \mathbb{R}^2, & k = 1 \\ 0, & k \geq 2 \end{cases}.$$

2.5 Fundamental Groups

Problem 2023-J-II-4. Prove that $S^2 \times S^2$ is not diffeomorphic to $M_1 \times M_2 \times M_3$, where M_1, M_2, M_3 are smooth manifolds of nonzero dimension.

Assume to the contrary that $f : S^2 \times S^2 \to M_1 \times M_2 \times M_3$ is a diffeomorphism. Since $\dim(M_1 \times M_2 \times M_3) = \sum \dim(M_j) = 4$, we have (up to relabeling of indices), $\dim M_1, \dim M_2 = 1$, and $\dim M_3 = 2$. Since $S^2 \times S^2$ is a compact manifold, f is continuous, and surjective, it follows that $M_1 \times M_2 \times M_3$ is compact. Thus, each M_j is compact. By the classification of manifolds of dimension 1, there exists exactly one smooth compact 1-manifold (up to diffeomorphism), which is the unit circle S^1. Thus, $M_1, M_2 \cong S^1$. This means that $\pi_1(M_1 \times M_2 \times M_3) \cong \mathbb{Z} \times \mathbb{Z}$ is nontrivial. But this is a contradiction: one has $\pi_1(S^2 \times S^2) \cong \pi_1(S^2) \times \pi_1(S^2) \cong 0$, and $f_* : \pi_1(S^2 \times S^2) \to \pi_1(M_1 \times M_2 \times M_3)$ is an isomorphism since f is a diffeomorphism implies that $0 \cong \mathbb{Z} \times \mathbb{Z}$. Thus, by contradiction, $S^2 \times S^2$ is not diffeomorphic to $M_1 \times M_2 \times M_3$.

Problem 2003-J-II-5. Is the unit sphere $\mathbb{S}^3$ diffeomorphic to the product of two other smooth manifolds of dimension > 0?

Assume that the unit sphere $\mathbb{S}^3$ is diffeomorphic to the product of two other smooth manifolds, M_1, M_2, both of nonzero dimension. Since diffeomorphisms preserve the total dimension, $\dim S^3 = \dim M_1 \times M_2 = \dim M_1 + \dim M_2$. Thus, up to relabeling of indices, $\dim M_1 = 1$ and $\dim M_2 = 2$. Moreover, since S^3 is compact, $M_1 \times M_2$ must also be compact. Since the product of manifolds is compact if and only if each factor is compact, M_1, M_2 are both compact. Since there exists only one compact smooth 1-manifold up to diffeomorphism, namely the unit circle S^1, $M_1 \cong S^1$. Diffeomorphic manifolds have isomorphic fundamental groups so that $\pi_1(S^3) \cong \pi_1(M_1 \times M_2) \cong \pi_1(M_1) \times \pi_1(M_2) \cong \pi_1(S^1) \times \pi_1(M_2)$. Since S^3 is simply connected, $\pi_1(S^3)$ is trivial. On the other hand, since $\pi_1(S^1) \cong \mathbb{Z}$, $\pi_1(M_1 \times M_2)$ must be non-trivial, which is a contradiction. Thus by contradiction, the unit sphere S^3 cannot be diffeomorphic to the product of two other smooth manifolds of nonzero dimensions.

2.6 Vector Fields

Problem 2020-J-I-4. Let θ be a closed smooth 1-form on a compact C^∞ manifold M with empty boundary, and let v be a smooth vector field on M. Prove that the Lie derivative $\mathcal{L}_v\theta$ vanishes at some point of M.

Let θ be a closed smooth 1-form on a smooth manifold M without boundary, and let V be a smooth vector field on M. By Cartan's formula,

$$\mathcal{L}_V\theta = V \lrcorner\, d\theta + d(V \lrcorner\, \theta) = d(V \lrcorner\, \theta),$$

where the first term is identically zero since θ is closed. Since θ is a smooth 1-form, $V \lrcorner\, \theta$ is a smooth 0-form on M; i.e., $V \lrcorner\, \theta$ is a smooth function from M to $\mathbb{R}$. Then by the extreme value theorem, $V \lrcorner\, \theta$ must have an extrema at some point $p \in M$, which means that $d(V \lrcorner\, \theta) = 0$ at p. But this means that $\mathcal{L}_V\theta$ vanishes at $p \in M$. Thus, the proof concludes.

Problem 2017-J-II-1. Let $f : M \to \mathbb{R}$ be a smooth function on a manifold M. In any arbitrary smooth local coordinate chart $x : U \to \mathbb{R}^n$ of M, define

$$\mathcal{D}f := \sum_{i=1}^{n} \frac{\partial f}{\partial x^i} \frac{\partial}{\partial x^i}.$$

Does $\mathcal{D}f$ give a well-defined vector field on M?

Let $f : M \to \mathbb{R}$ be a smooth function on a manifold M, and define $\mathcal{D}f$ as given above. We claim that $\mathcal{D}f$ does *not* give a well-defined vector field on M. Let $(U, (x^i)), (V, (\widetilde{x}^j))$ be overlapping coordinate charts. At any $p \in U \cap V$, we have

$$\begin{aligned} \mathcal{D}f &= \frac{\partial f}{\partial x^i} \frac{\partial}{\partial x^i} \\ &= \left(\frac{\partial f}{\partial \widetilde{x}^j} \frac{\partial \widetilde{x}^j}{\partial x^i} \right) \cdot \left(\frac{\partial}{\partial \widetilde{x}^j} \frac{\partial \widetilde{x}^j}{\partial x^i} \right) \\ &= \left(\frac{\partial \widetilde{x}^j}{\partial x^i} \right)^2 \cdot \left(\frac{\partial f}{\partial \widetilde{x}^j} \frac{\partial}{\partial \widetilde{x}^j} \right). \end{aligned}$$

I.e., $\mathcal{D}f$ does not follow the correct transformation law for tangent vector fields, so that the proof concludes.

2.7 Covering Spaces

Problem 2021-A-I-6. What connected spaces can be finitely-sheeted covering spaces of a sphere with three handles?

Let Σ_3 denote the compact connected orientable surface of genus 3, which is diffeomorphic to the sphere with three handles. Further, we will show that for every positive integer $k \geq 1$, Σ_3 has a k-sheeted covering space. Consider the map $\phi : \pi_1(\Sigma_3) \to \mathbb{Z}/k\mathbb{Z}$ defined by

$$\phi(a_1) = 1 \qquad \text{and} \qquad \phi(a_2), \phi(a_3), \phi(b_1), \ldots, \phi(b_3) = 0,$$

where

$$\pi_1(\Sigma_3) = \langle a_1, b_1, \ldots, a_3, b_3 : [a_1, b_1][a_2, b_2][a_3, b_3] = 1 \rangle.$$

Since 1 is a generator of the additive group $\mathbb{Z}/k\mathbb{Z}$, it follows that ϕ is surjective. It is then straightforward to check that ϕ is a well-defined group homomorphism. Let H denote the kernel of ϕ, which is a normal subgroup of $\pi_1(\Sigma_3)$. Thus by the First Isomorphism Theorem, one has that $\pi_1(\Sigma_3)/H \cong \mathbb{Z}/k\mathbb{Z}$.

Since $|\mathbb{Z}/k\mathbb{Z}| = k$, H is a subgroup of $\pi_1(\Sigma_3)$ of index k. Therefore, by the Galois correspondence for covering spaces, Σ_3 has a k-sheeted covering space. Now let $\rho : \widetilde{M} \to \Sigma_3$ be a k-sheeted covering space. Then we must have

$$\chi(\widetilde{M}) = k \cdot \chi(\Sigma_3) \iff 2 - 2g = -4k \iff g = 2k + 1.$$

I.e., $\widetilde{M}$ must be a surface of genus $2k+1$. Therefore, we conclude that the only connected finite k-sheeted covering spaces of a sphere with three handles are the surfaces Σ_{2k+1}, $k \geq 1$.

Problem 2016-J-I-2. Let $X = S^1 \times S^1$ be the 2-torus, and let $Y = X \setminus \{p\}$ be the complement of one point in the 2-torus.

(a) Prove that there is no covering map $X \to Y$.

(b) Prove that there is no covering map $Y \to X$.

Let $X = S^1 \times S^1$ denote the 2-torus, and let $Y = X \setminus \{p\}$ be the complement of a point in the 2-torus.

(a) Assume to the contrary that there exists a covering map $p : X \to Y$. We first note that $\chi(X) = 2 - 2(1) = 0$. On the other hand, if B is an open ball in X centered at p, then by the additivity of the Euler characteristic, one has

$$0 = \chi(X) = \chi(X \setminus B) + \chi(B) - \chi(S^1) = \chi(X \setminus B) + \chi(B) = \chi(Y) + \chi(\{p\}) = \chi(Y) + 1,$$

so that $\chi(Y) = -1$. Since X is assumed to be a compact covering space, the covering must be finitely sheeted. Thus, there exists a finite positive integer d so that $0 = d \cdot (-1) = -d$, which is a contradiction. Thus by contradiction, there cannot exist such a covering map.

(b) Now assume to the contrary that there is a covering map $p : Y \to X$. Then the induced map $p_* : \pi_1(Y) \to \pi_1(X)$ is an injective group homomorphism so that $\pi_1(Y)$ is isomorphic to some subgroup of $\pi_1(X)$. But since $\pi_1(X)$ is abelian, every one of its subgroups is abelian, while $\pi_1(Y)$ (the free group on two generators) is nonabelian. Thus by contradiction, Y cannot be a covering space of X.

Problem 2013-J-II-5. If M and N are compact 2-manifolds, their connected sum $M\#N$ is obtained by removing an open disk from each, and then identifying the resulting circle boundaries. It is a theorem that any smooth compact non-orientable surface without boundary is homeomorphic to an iterated connected sum

$$\Sigma_k := \underbrace{\mathbb{RP}^2\#\cdots\#\mathbb{RP}^2}_{k}$$

for some positive integer k. For which values of k is there a covering map $\Sigma_k \to \Sigma_3$?

Let Σ_3 denote the smooth compact non-orientable manifold homeomorphic to $\mathbb{RP}^2\#\cdots\#\mathbb{RP}^2$. Since we are looking for covering spaces Σ_k (which are compact), it suffices to restrict the search to finite d-sheeted covering spaces (infinitely-sheeted covering spaces are necessarily noncompact). If Σ_k is a covering space of Σ_3, then we must have

$$\begin{aligned} -d = d(3(1) - 2(2)) &= d(3\chi(\mathbb{RP}^2 - 2\chi(S^2)) \\ &= d \cdot \chi(\Sigma_3) \\ &= \chi(\Sigma_k) = k\chi(\mathbb{RP}^2) + (k-1)\chi(S^2) \\ &= 2 - k, \end{aligned}$$

so that $k = d + 2$. Since $d \geq 1$, we must have $k \geq 3$. On the other hand, suppose $d \geq 1$. Consider the map $\phi : \pi_1(\Sigma_3) \to \mathbb{Z}/d\mathbb{Z}$, where $\pi_1(\Sigma_3) \cong \langle a, b, c \mid a^2b^2c^2 = 1\rangle$, defined by

$$\phi(a) = 1 \qquad \phi(b) = d - 1 \qquad \phi(c) = 0.$$

Then it is clear that ϕ is surjective, and that $\phi(a^2b^2c^2) = 0$ so that ϕ is a well-defined homomorphism. Thus by the Isomorphism Theorems, $\pi_1(\Sigma_3)/\ker\varphi \cong \mathbb{Z}/d\mathbb{Z}$; i.e., $\ker\varphi$ is an index d subgroup of Σ_3. By the Galois correspondence of covering spaces, we conclude that Σ_3 must have a d-sheeted covering space of genus $g = -d$ so that this covering space is indeed Σ_{d+2}. Thus, the values of k for which there is a covering map $\Sigma_k \to \Sigma_3$ is precisely $k \geq 3$.

Problem 2013-J-II-5 (Modified). If M and N are compact 2-manifolds, their connected sum $M\#N$ is obtained by removing an open disk from each, and then identifying the resulting circle boundaries. It is a theorem that any smooth compact non-orientable surface without boundary is homeomorphic to an iterated connected sum

$$\Sigma_k := \underbrace{\mathbb{RP}^2\#\cdots\#\mathbb{RP}^2}_{k}$$

for some positive integer k. For which values of k is there a covering map $\Sigma_k \to \Sigma_4$?

As above, if Σ_k is a d-sheeted covering space ($d \geq 1$) of Σ_4, then we must have

$$\begin{aligned} -2d &= d \cdot \chi(\Sigma_4) \\ &= \chi(\Sigma_k) = 2 - k, \end{aligned}$$

so that $k = 2(d+1)$. I.e., k is an even integer that is at least 4. On the other hand, for every $d \geq 1$, there exists a surjective group homomorphism $\varphi : \pi_1(\Sigma_4) \cong \langle a, b, c, f \mid a^2b^2\cdots f^2 = 1\rangle \twoheadrightarrow \mathbb{Z}/d\mathbb{Z}$ given by $\varphi(a) = 1$, $\varphi(b) = d-1$, and $\varphi(c), \varphi(d) = 0$. Since the kernel of this group homomorphism is an index d subgroup of $\pi_1(\Sigma_4)$ (by the First Isomorphism Theorem), we conclude by the Galois correspondence of covering spaces that Σ_4 must have a d-sheeted covering space of genus $g = -2d$ so that this covering space is indeed $\Sigma_{2(d+1)}$. Thus, the values of k for which there is a covering map $\Sigma_k \to \Sigma_3$ is precisely $\{k \in \mathbb{N} \mid k \text{ even and } k \geq 4\}$.

Problem 2010-A-II-4. Let p be a prime number. Let M be the two-dimensional torus $S^1 \times S^1$. How many distinct, p-sheeted covering spaces $\varpi : \widetilde{M} \to M$ are there such that $\widetilde{M}$ is path connected? Count these up to isomorphism of covering spaces of M; i.e., $\varpi_1 : \widetilde{M}_1 \to M$ is isomorphic to $\varpi_2 : \widetilde{M}_2 \to M$ if there exists a homeomorphism $f : \widetilde{M}_1 \to \widetilde{M}_2$ such that $\varpi_2 \circ f = \varpi_1$.

Let p be a fixed prime number, and let $M = S^1 \times S^1$ denote the 2-torus. We note that M is connected and locally simply connected. Then by the Galois correspondence of covering spaces, the isomorphism classes of p-sheeted connected covering spaces of M are in one-to-one correspondence with the index p-subgroups of $\pi_1(M)$. Since $\pi_1(S^1) \cong \mathbb{Z}$, we have $\pi_1(M) \cong \mathbb{Z}\times\mathbb{Z}$. In turn, the number of such subgroups is equivalent to the number of kernels of surjective group homomorphisms: $\phi : \pi_1(M) \twoheadrightarrow \mathbb{Z}/p\mathbb{Z}$. We note that $a = (1,0)$ and $b = (0,1)$ are the generators of $\pi_1(M)$. Then each homomorphism $\phi : \pi_1(M) \to \mathbb{Z}/p\mathbb{Z}$ is uniquely determined by where ϕ maps a and b. Since $\mathbb{Z}/p\mathbb{Z}$ has p elements, and a, b can be mapped to any one of these elements, it follows that there are a total of p^2 homomorphisms from $\pi_1(M)$ to $\mathbb{Z}/p\mathbb{Z}$. However, a given homomorphism ϕ is not surjective if and only if neither $\phi(a)$ nor $\phi(b)$ is a generator of $\mathbb{Z}/p\mathbb{Z}$. Since p is prime, this means that the only non-surjective group homomorphism is the trivial homomorphism. Therefore, there are exactly $p^2 - 1$ surjective group homomorphisms from $\pi_1(M)$ onto $\mathbb{Z}/p\mathbb{Z}$. However, we must also account for the fact that two kernels may be the same up to multiplication by a constant (in which case, the corresponding connected p-sheeted covering spaces are isomorphic). Thus, we conclude that, up to isomorphism, there are exactly $(p^2-1)/(p-1) = p+1$ p-sheeted connected covering spaces of M.

Problem 2004-A-I-5. In this problem, the term "surface" means a compact orientable 2-manifold without boundary.

(a) Does a surface of genus 1 have a covering space which is homeomorphic to a surface of genus 2? Explain.

(b) Does a surface of genus 2 have a covering space which is homeomorphic to a surface of genus 3? Explain.

(a) Let M denote a surface of genus 1. We claim that there is no covering space of M that is homeomorphic to a surface of genus 2. Assume to the contrary that there exists a space $\widetilde{M}$ homeomorphic to a surface of genus 2 that is a k-sheeted covering space of M for some finite k. Then we must have $\chi_{\widetilde{M}} = k \cdot \chi_M$, where $\chi.$ denotes the Euler characteristic. Since $\chi_M = 2 - 2(1) = 0$, $\chi_{\widetilde{M}} = 2 - 2(2) = -2$, and there exists no finite positive integer k so that $-2 = k \cdot 0$, we conclude that M has no finitely-sheeted covering spaces homeomorphic to a surface 2 genus surface. On the other hand, any infinitely-sheeted covering space of M has to be noncompact, and thus cannot be homeomorphic to a (compact) surface of genus 2.

(b) Now let M denote a surface of genus 2. As above, it suffices to check if there exists a finitely-sheeted covering space of M with genus 3. By looking at the Euler characteristics, *if* M has a finitely-sheeted covering space of genus 3, then it must be a 2-sheeted covering space. But by replicating the work from Problem 2021-A-I-6, we see that by Galois correspondence of covering spaces, M must have a 2-sheeted covering space. Thus, we conclude that M has a covering space which is homeomorphic to a surface of genus 2.

3 Algebra

3.1 Linear Algebra

Problem 2022-A-I-6. Denote by $\mathrm{Mat}(m, \mathbb{C})$ and $\mathrm{Mat}(n, m, \mathbb{C})$ the space of $m \times m$ and $n \times m$ matrices with complex coefficients. Let $A \in \mathrm{Mat}(n, \mathbb{C}), B \in \mathrm{Mat}(m, \mathbb{C})$ be given and suppose A and B have no common eigenvalues. Show that the only solution of

$$AX = XB, \qquad X \in \mathrm{Mat}(n, m, \mathbb{C}),$$

is the trivial solution $X = 0$.

Assume all of the given hypotheses. Observe that $A^2 X = A(AX) = (AX)B = XB^2$. Hence, by induction, it follows that for all k,

$$A^k X = XB^k.$$

In particular, $p(A)X = Xp(B)$ for any $p \in \mathbb{C}[z]$. Let p be the characteristic polynomial $p_B(z)$ of B. By the Cayley-Hamilton theorem, one obtains,

$$p_B(A)X = Xp_B(B) = 0.$$

We can write

$$p_B(z) = \prod_{i=1}^{k} (z - \lambda_k)^{m_k},$$

where each λ_k is an eigenvalue of B, and m_k is its multiplicity. By the hypothesis, λ_k is not an eigenvalue of A. Hence, $(A - I\lambda_k)^{m_k}$ is invertible. Therefore, since each factor is invertible, $p_B(A)$ is invertible. Hence,

$$p_B(A)X = 0 \implies p_B^{-1}(A)p_B(A)X = 0 \implies X = 0.$$

Problem 2019-J-I-1. Let A and B be $n \times n$ invertible matrices over complex numbers, satisfying

$$AB = \lambda BA \qquad \text{for some } \lambda \in \mathbb{C}.$$

Prove that A^n and B commute.

Let A and B be $n \times n$ invertible matrices over the complex numbers, and $\lambda \in \mathbb{C}$ so that $AB = \lambda BA$. First we will show that λ^n is a root of unity. Taking the determinant on both sides and using the fact that $\det : \mathrm{GL}(n, \mathbb{C}) \to \mathbb{C}^\times$ is a group homomorphism, one gets,

$$\det A \det B = \det AB = \det(\lambda BA) = \lambda^n \det A \det B.$$

Since A, B are invertible, $\det A, \det B \neq 0$, so that dividing through by these quantities gets us $\lambda^n = 1$. Now we observe the following:

$$\begin{aligned} AB &= \lambda BA. \\ A^2 B &= \lambda(AB)A = \lambda^2 BA^2. \\ A^3 B &= \lambda^2 (AB)A^2 = \lambda^3 BA^3. \end{aligned}$$

Repeating this procedure inductively, one finally obtains $A^n B = \lambda^n BA^n$. But we showed above that $\lambda^n = 1$. Thus, one gets $A^n B = BA^n$, which concludes the proof that A^n and B commute.

3.2 Abelian Groups

Problem 2020-A-I-4. Let G be a finite group, and denote by $Z(G)$ its center. Assume that $G/Z(G)$ is abelian of order pq, where p, q are distinct primes. Prove that G is abelian.

Let G be a finite group, $Z(G)$ its center, and assume that $G/Z(G)$ is abelian of order pq, where p, q are distinct primes. Without loss of generality, assume $p < q$. To prove that G is abelian, it suffices to prove that $G/Z(G)$ is cyclic. By the classification of groups of order pq, there exist precisely two groups of order pq: (1) the abelian group $\mathbb{Z}/(pq)\mathbb{Z}$, and (2) the non-abelian group $\mathbb{Z}/p\mathbb{Z} \rtimes_{\varphi} \mathbb{Z}/q\mathbb{Z}$, where $\varphi : \mathbb{Z}/q\mathbb{Z} \to \operatorname{Aut}(\mathbb{Z}/q\mathbb{Z})$ is any non-trivial group homomorphism. Since $G/Z(G)$ is abelian, it follows that $G/Z(G) \cong \mathbb{Z}/(pq)\mathbb{Z}$ is cyclic, so that G is abelian.

Problem 2020-J-I-1. Let G be a finite non-abelian group, and let $Z(G)$ denote its center. Prove that $|Z(G)| \leq \frac{1}{4}|G|$, and give an example where equality holds.

Let G be a finite non-abelian group, and let $Z(G)$ denote its center. Assume to the contrary that $4|Z(G)| > |G| \implies 4 > |G|/|Z(G)| = |G/Z(G)|$, where the last equality follows from the fact that the center of a group is normal in the group. Thus, we must have $|G/Z(G)| = 1, 2, 3$. We know $|G/Z(G)| \neq 1$ since if this were the case, then $G = Z(G)$, contradicting the fact that G is abelian. If $|G/Z(G)| = 2$, then since up to isomorphism there exists exactly one group of order 2, namey $\mathbb{Z}/2\mathbb{Z}$, $G/Z(G) \cong \mathbb{Z}/2\mathbb{Z}$ is cyclic, implying that G is abelian (a contradiction). Likewise, if $|G/Z(G)| = 3$, then $G/Z(G) \cong \mathbb{Z}/3\mathbb{Z}$ is cyclic, and thus G is abelian (a contradiction). Thus we have reached a contradiction, which concludes the proof that $|Z(G)| \leq \frac{1}{4}|G|$.

Problem 2012-J-I-6. Let G be a group of order 315 whose center contains a subgroup N of order 9. Prove that G is abelian.

Let G be a group of order 315, and let $Z(G)$ be the center of the group. By Lagrange's Theorem,

$$|Z(G)| \in \{1, 3, 5, 7, 9, 15, 21, 35, 45, 63, 105, 315\} \cap \{0 \bmod 9\},$$

where the second set follows from the fact that $Z(G)$ contains a subgroup of order 9. Thus we have narrowed down our choices to $|Z(G)| \in \{9, 45, 63315\}$. If $|Z(G)| = 315$, we are done since $G = Z(G)$. If $|Z(G)| = 63$, then $|G/Z(G)| = |G|/|Z(G)| = 5$ so that $G/Z(G) \cong \mathbb{Z}/5\mathbb{Z}$ is cyclic. But this implies that G is abelian, contradicting the fact that $|Z(G)| \nleq 315$. Likewise, $|Z(G)| \neq 45$, since if not, then $|G/Z(G)| = 7$ so that $G/Z(G) \cong \mathbb{Z}/7\mathbb{Z}$ is cyclic and G is abelian. Suppose now that $|Z(G)| = 9$ so that $|G/Z(G)| = 35$. By Sylow's Theorems, $G/Z(G)$ must contain a normal Sylow 7-subgroup K and a normal Sylow 5-subgroup N, and these subgroups intersect trivially (by Lagrange's Theorem). Thus since K contains precisely 6 nonidentity elements of order 7 and N contains precisely 4 nonidentity elements of order 5, there are $35 - (1 + 6 + 4) = 24$ elements whose order is not $1, 5$, or 7. By Lagrange's Theorem, this elements must have order 35. Thus since $G/Z(G)$ contains an element of order $|G/Z(G)|$, G must be abelian (so that $|Z(G)| = 315$), which is a contradiction. Thus, we conclude that $G = Z(G)$ so that G is abelian.

3.3 Ring Theory

Problem 2025-J-I-1. Let R be a UFD (unique factorization domain). Let F be its quotient field. Let

$$p(x) = x^n + b_{n-1}x^{n-1} + \cdots + b_0 \in F[x]$$

be a monic polynomial with coefficients in R admitting a root $a \in F$. Prove that $a \in R$.

Let R be a UFD, and let F denote its quotient field. Let $p(x)$ be the aforementioned monic polynomial in $F[x]$. Furthermore, assume $a = b/c$, with $b, c \in R$, $c \neq 0$, and $\gcd(b, c) = 1$ (the last assumption is well-defined since (1) for any two elements in a UFD, the greatest common divisor is defined, and (2) if $\gcd(b,c) \neq 1$, then we can reduce the quotient repeatedly until this property is satisfied). Then plugging in $a = b/c$ into $p(x)$, one obtains:

$$\frac{b^n}{c^n} + b_{n-1}\frac{b^{n-1}}{c^{n-1}} + \cdots + b_1\frac{b}{c} + b_0 = 0,$$

which then implies that

$$b^n + b_{n-1}b^{n-1}c + \cdots + b_1bc^{n-1} + b_0c^n = 0 \implies b^n = -c(b_{n-1}b^{n-1} + b_{n-2}b^{n-2}c^2 + \cdots + b_0c^{n-1}),$$

where $(b_{n-1}b^{n-1} + b_{n-2}b^{n-2}c^2 + \cdots + b_0c^{n-1}) \in R$. Thus, it follows that $c \mid b^n$. Since $\gcd(b,c) = 1$, $\gcd(b^n, c) = 1$ since raising b to the nth power does not add any new prime divisors. Thus since $c \mid b^n$ and $\gcd(b^n, c) = 1$, $c = \pm 1$. This implies that $a = \pm b \in R$. Hence, the proof concludes.

3.4 Galois Theory

Problem Galois Lemma I. Let F be a field, and $p(x) \in F[x]$ be a separable polynomial. Then the Galois group action on the set of roots of $p(x)$ is transitive if and only if $p(x)$ is irreducible.

Problem 2025-J-II-6. The quaternion group is the 8-element subset $Q = \{\pm 1, \pm i, \pm j, \pm k\}$ of the quaternions, with multiplication as the group operation. Let $E/\mathbb{Q}$ be a Galois extension whose Galois group is isomorphic to Q. Show that E is not the splitting field of any degree-4 polynomial.

Assume to the contrary that E is the splitting field of some degree-4 polynomial $p(x)$. Further assume that $\mathrm{Gal}(E/\mathbb{Q}) \cong Q$. Since the Galois group action on the roots of $p(x)$ is faithful[1], it follows that the Galois group must be isomorphic to a subgroup of S_4. Each of the non-identity elements in Q has order 4, and the square of each element is identically -1. On the other hand, the elements of order 4 in S_4 are the 4-cycles, and the square of each such 4-cycle is a distinct product of transpositions. Thus, this leads to a contradiction, concluding the proof that E cannot be the splitting field of a quartic polynomial in $\mathbb{Q}[x]$.

Problem 2024-A-II-5. Let $\alpha \in \mathbb{C}$ be a complex number such that the field $\mathbb{Q}(\alpha)$ is a finite Galois extension of the field $\mathbb{Q}$. Let $L \supset \mathbb{Q}$ be an arbitrary finite extension. Show that $L(\alpha) \supset L$ is also a finite Galois extension.

Let $\alpha \in \mathbb{C}$ be a complex number such that $\mathbb{Q}(\alpha)$ is a finite Galois extension of the field $\mathbb{Q}$, and let $L \supset \mathbb{Q}$ be an arbitrary finite extension. [!! Complete Later !!]

Problem 2024-J-II-4. Let $\alpha = \sqrt{2+\sqrt{3}} \in \mathbb{C}$. Let K be the smallest Galois extension of $\mathbb{Q}$ which contains α. Describe the Galois group $\mathrm{Gal}(K/\mathbb{Q})$.

Let $\alpha = \sqrt{2+\sqrt{3}} \in \mathbb{C}$, and let K be the splitting field of α over $\mathbb{Q}$. First, we must determine the degree of this field extension. We first compute:

$$\alpha^2 - 2 = \sqrt{3} \implies \alpha^4 - 4\alpha^2 + 1 = 0.$$

I.e., the polynomial $m_\alpha(x) = x^4 - 4x^2 + 1 \in \mathbb{Q}[x]$ annihilates α. The roots of $m_\alpha(x)$ are $\pm\alpha, \pm\beta$, where

$$\beta = \sqrt{2-\sqrt{3}}.$$

[1] Note that if $p(x)$ is *any* quartic polynomial, then the group action is at most faithful. *If* $p(x)$ is also irreducible, then the group action is transitive. See above lemma.

It is straightforward to check that $m_\alpha(x)$ is irreducible over $\mathbb{Q}$, which means that K is a degree-4 extension. Now, since K is the splitting field of α, it must contain all four roots $\pm\alpha, \pm\beta$ of the minimal polynomial, and therefore the finite field extension $\mathbb{Q}(\alpha)$. On the other hand, $\mathbb{Q}(\alpha)$ contains all four roots since $\beta = \alpha^{-1}$. Thus, $K = \mathbb{Q}(\alpha)$. The Galois group $\mathrm{Gal}(K/\mathbb{Q})$ consists of all of the automorphisms σ on $K = \mathbb{Q}(\alpha)$ that preserve the algebraic relations between the four roots, and fix the base field $\mathbb{Q}$. Thus, we have the four automorphisms:

$$\begin{aligned} \mathrm{id} &: \alpha \mapsto \alpha, & \sigma &: \alpha \mapsto -\alpha; \\ \tau &: \alpha \mapsto \beta, & \rho &: \alpha \mapsto -\beta. \end{aligned}$$

For these four automorphisms, we compute the multiplication table:

$$\begin{array}{c|cccc} & \mathrm{id} & \sigma & \tau & \rho \\ \mathrm{id} & \mathrm{id} & \sigma & \tau & \rho \\ \sigma & \sigma & \mathrm{id} & \rho & \tau \\ \tau & \tau & \rho & \mathrm{id} & \sigma \\ \rho & \rho & \tau & \sigma & \mathrm{id} \end{array}.$$

Thus, we conclude that $\mathrm{Gal}(K/\mathbb{Q}) \cong \mathrm{V}_4$, the Klein-4 group.

Problem 2023-A-II-3. Let $K/\mathbb{Q}$ be a Galois extension whose Galois group is the quaternion group Q_8. (Recall that Q_8 is the group $\pm1, \pm i, \pm j, \pm k$ with $i^2 = j^2 = k^2 = ijk = -1$.) Show that K is not the splitting field of a quartic polynomial in $\mathbb{Q}[x]$.

Let $K/\mathbb{Q}$ be a Galois extension whose Galois group G is the quaternion group Q_8. Assume to the contrary that K is the splitting field of a quartic polynomial $p(x) \in \mathbb{Q}[x]$. Since the Galois group action on the roots of $p(x)$ is faithful, G must be isomorphic to a subgroup of S_4. We note that all of the elements of $Q_8 \setminus \{\pm 1\}$ are of order 4, and have the same square, -1. On the other hand, the elements of order 4 in S_4 are the 4-cycles, and the square of each 4-cycle is a distinct product of transpositions. Thus, we have a contradiction, concluding the proof that K cannot be the splitting field of a quartic polynomial in $\mathbb{Q}[x]$.

Problem 2023-J-I-5. Consider the following irreducible polynomial over $\mathbb{Q}$: $p(x) = x^4 - 3x^2 - 1$.
(a) Describe the splitting field of $p(x)$.
(b) Consider the Galois group of $p(x)$. Compute its order and determine if it is abelian.

(a) Let $p(x) = x^4 - 3x^2 - 1 \in \mathbb{Q}[x]$. By the rational root test, it is straightforward to check that $p(x)$ has no roots over $\mathbb{Q}$, and therefore cannot split into a product containing a linear factor. Now we will check if $p(x) = (x^2 + ax + b)(x^2 + cx + d)$ for some $a, b, c, d \in \mathbb{Q}[x]$. Expanding,

$$(x^2 + ax + b)(x^2 + cx + d) = x^4 + (a + c)x^3 + (b + ac + d)x^2 + (bc + ad)x + bd.$$

Hence, we have the following system of equations:

$$\begin{cases} a + c = 0, \\ b + ac + d = -3, \\ bc + ad = 0, \\ bd = -1. \end{cases}$$

Solving this system of equations, we notice there exist no $(a, b, c, d) \in \mathbb{Q}^4$ that solves the desired equations. Thus, we conclude that $p(x)$ is irreducible. Let $u = x^2$. Then

$$p(u) = u^2 - 3u - 1 \implies u = \frac{3 \pm \sqrt{9 + 4}}{2} = \frac{3 \pm \sqrt{13}}{2} \implies x = \pm\alpha, \pm\beta,$$

where
$$\alpha \coloneqq \sqrt{\frac{3+\sqrt{13}}{2}}, \qquad \beta \coloneqq \sqrt{\frac{3-\sqrt{13}}{2}}.$$
Here, we notice that $\beta = i/\alpha$. Let K be the splitting field of $p(x)$. Then K must contain all four roots of $p(x)$, and therefore, the field $\mathbb{Q}(i,\alpha)$. On the other hand, since $\beta = i/\alpha$, $\mathbb{Q}(i,\alpha)$ already contains all four roots of $p(x)$. Thus, $K = \mathbb{Q}(i,\alpha)$.

(b) Since $\mathbb{Q}(i,\alpha)$ is the splitting field of an irreducible polynomial over $\mathbb{Q}$, it follows that $\mathbb{Q}(i,\alpha)/\mathbb{Q})$ is Galois. Since the minimal polynomial of α over $\mathbb{Q}$ is of degree 4, $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 4$. Moreover, since $\mathbb{Q}(\alpha) \subset \mathbb{R}$, $i \notin \mathbb{Q}(\alpha)$ so that $[\mathbb{Q}(i,\alpha) : \mathbb{Q}(\alpha)] = 2$ (the minimal polynomial of i over $\mathbb{Q}(\alpha)$ being x^2+1). Thus, $[\mathbb{Q}(i,\alpha) : \mathbb{Q}] = 8$. The elements of the Galois group are the automorphisms on $\mathbb{Q}(i,\alpha)$ that preserve the algebraic relations between the generators i and α, and fix the base field $\mathbb{Q}$. In particular, since $p(x)$ is irreducible, the Galois group action on the roots of $p(x)$ must be transitive so that the Galois group corresponds to a transitive subgroup of S_4. Since the only transitive subgroup of S_4 up to isomorphism of order 8 is D_8, we conclude that the Galois group is D_8, and therefore, not abelian.

Problem 2022-A-I-2. Let $f(x) \in \mathbb{Q}[x]$ be an irreducible polynomial of odd degree greater than 1 and having only one real root $u \in \mathbb{R}$. Show that $\mathbb{Q}(u)/\mathbb{Q}$ is not a normal extension.

Let $f(x) \in \mathbb{Q}[x]$ be an irreducible polynomial of odd degree greater than 1 and having only one real root $u \in \mathbb{R}$. Assume to the contrary that $\mathbb{Q}(u)/\mathbb{Q}$ is a normal extension; by definition, this implies that $f(x)$ splits into the product of linear factors:
$$f(x) = (x-u)\cdot \prod_{j=1}^{n-1}(x-u_j),$$
where each $u_j \in \mathbb{Q}(u)$. But since u is real, $\mathbb{Q}(u)$ embeds into $\mathbb{R}$, which means that every u_j is real. This contradicts our hypothesis that $f(x)$ has exactly one real root. Thus, by contradiction, $\mathbb{Q}(u)/\mathbb{Q}$ is not a normal extension.

Problem 2021-A-I-4. If β is a nonzero algebraic number find the coefficients of the minimal polynomial over $\mathbb{Q}$ for β^{-1} in terms of coefficients of the minimal polynomial over $\mathbb{Q}$ for β.

Let β be a nonzero algebraic number, and let $p(x)$ denote the minimal polynomial over $\mathbb{Q}$ for β. First assume that $\beta \in \mathbb{Q}$. Then the minimal polynomial of β is $x-\beta$. Since $\beta^{-1} \in \mathbb{Q}$ (by virtue of the fact that $\beta \neq 0$), then the minimal polynomial of β^{-1} is $x - \beta^{-1}$. In other words, the constant term $\widetilde{a}_0$ of the minimal polynomial of β^{-1} over $\mathbb{Q}$ is a_0^{-1}, where $a_0 = -\beta$ is the constant term of the minimal polynomial of β over $\mathbb{Q}$.
Now assume that $\beta \notin \mathbb{Q}$ so that the minimal polynomial of β is given by the degree $n \geq 2$ polynomial $p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ (note that $a_0 \neq 0$ since $p(x)$ is irreducible by virtue of being a minimal polynomial of a nonzero algebraic number). It is straightforward to see that the corresponding polynomial,
$$q(x) = x^n + a_0^{-1}a_1 x^{n-1} + a_0^{-1}a_2 x^{n-2} + \cdots + a_0^{-1}a_{n-1}x + a_0^{-1}$$
annihilates β^{-1} over $\mathbb{Q}$. First we show that there cannot be a monic polynomial of smaller degree that annihilates β^{-1}. If there were such a polynomial $\widetilde{q}(x)$ with degree $m < n$, then the polynomial $x^m\widetilde{q}(x^{-1})$ (which also has degree m) must annihilate β. But this contradicts the fact that the minimal polynomial of β is a degree $n > m$ polynomial. Now we need to show that $q(x)$ is irreducible. Assume to the contrary that $q(x)$ reduces as the product of $f(x)g(x)$ of irreducible polynomials with $\deg f + \deg g = n$. Then we can write
$$p(x) = a_0 x^n f(x^{-1})g(x^{-1}) = a_0[x^{n-\deg f}f(x)]\cdot[x^{n-\deg g}g(x)]$$

as the product of smaller-degree polynomials, which means $p(x)$ is reducible (a contradiction!) Thus, $q(x)$ must be irreducible. Hence, we conclude that $q(x)$ is the minimal polynomial of β^{-1} over $\mathbb{Q}$. In particular, denoting $q(x) = x^n + b_{n-1}x^{n-1} + \cdots + b_1x + b_0$, we find that

$$b_j = \begin{cases} a_0^{-1}a_{n-j}, & 1 \le j \le n-1 \\ a_0^{-1}, & j = 0. \end{cases}$$

Problem 2021-J-I-5. L denotes the splitting field over the rational numbers $\mathbb{Q}$ for a polynomial $f(x) \in \mathbb{Q}[x]$ of degree 5. If the Galois group for the extension $\mathbb{Q} \subset L$ has order greater than 24, then show that $f(x)$ is irreducible.

Let L denote the splitting field over the rational numbers $\mathbb{Q}$ for a polynomial $f(x) \in \mathbb{Q}[x]$ of degree 5, and assume that the Galois group for the extension $\mathbb{Q} \subset L$ has order greater than 24. By definition, the Galois group action on the set of roots of $f(x)$ is faithful, and hence, induces an embedding $\mathrm{Gal}(L/\mathbb{Q}) \hookrightarrow S_5$. The only subgroups of S_5 of order greater than 24 are A_5 and S_5, both of which are transitive subgroups. Thus, the Galois group action on the set of roots is transitive. Since every polynomial in $\mathbb{Q}[x]$ is separable, and a separable polynomial is irreducible iff the Galois group action on the roots is transitive, we have shown that $f(x)$ is irreducible.

Problem 2019-A-I-6. Let $\Theta = \sqrt{2-\sqrt{3}}$. Prove that $\mathbb{Q}(\Theta)/\mathbb{Q}$ is a Galois extension, and compute its Galois group. Hint: $(2-\sqrt{3})(2+\sqrt{3}) = 1$.

Let $\Theta = \sqrt{2-\sqrt{3}}$, and let $\beta = \sqrt{2+\sqrt{3}}$. To show that $\mathbb{Q}(\Theta)$ is a Galois extension, it suffices to show that it is the splitting field of an irreducible polynomial in $\mathbb{Q}[x]$. Indeed, we first compute the following:

$$\Theta^2 - 2 = -\sqrt{3} \implies \Theta^4 - 4\Theta^2 + 1 = 0.$$

I.e., the polynomial $m_\Theta(x) \coloneqq x^4 - 4x^2 + 1$ annihilates Θ. By the rational root test, $m_\Theta(x)$ has no rational roots, and therefore no linear/cubic factors. Moreover, we can also show that $m_\Theta(x)$ is irreducible over $\mathbb{Q}$ as it cannot split into the product of two irreducible quadratics. Now, we will find the splitting field of $p(x)$ over $\mathbb{Q}$. Let $u = x^2$. Then we obtain

$$u^2 - 4u + 1 = 0 \implies u = \frac{4 \pm \sqrt{12}}{2} = 2 \pm \sqrt{3} \implies x = \pm\Theta, \pm\beta.$$

As suggested by the hint, one has $\Theta\beta = 1 \implies \beta = 1/\Theta \in \mathbb{Q}(\Theta)$. By definition, K must contain all of the roots of $m_\Theta(x)$, and thus, must contain the field $\mathbb{Q}(\Theta)$. On the other hand, the field $\mathbb{Q}(\Theta)$ already contains all of the roots since $\beta = \Theta^{-1}$. Thus, $K = \mathbb{Q}(\Theta)$, and therefore, $\mathbb{Q}(\Theta)$ is Galois. Since the minimal polynomial of Θ is off degree 4, the Galois group must be of order 4; the elements of the Galois group are the automorphisms of the splitting field $\mathbb{Q}(\Theta)$ that preserve the algebraic relations between the roots of $m_\Theta(x)$ and fix the base field $\mathbb{Q}$, and thus are uniquely determined by their actions on Θ. Thus, we have the four maps:

$$\begin{array}{ll} \mathrm{id} : \Theta \mapsto \Theta, & \sigma : \Theta \mapsto -\Theta; \\ \tau : \Theta \mapsto \beta, & , \quad \rho : \Theta \mapsto -\beta. \end{array}$$

Writing out the multiplication table, we deduce that the Galois group is isomorphic to V_4.

Problem 2017-A-II-3. Let K denote the splitting field of $f(x) = x^4 + x^2 + 1$ over $\mathbb{Q}$. Compute the Galois group of $\mathrm{Gal}(K/\mathbb{Q})$.

Let $f(x) = x^4 + x^2 + 1 \in \mathbb{Q}[x]$. By the rational root test, it follows that $f(x)$ has no rational roots, and therefore cannot split into a product consisting of a linear factor. We turn to check if $f(x)$ can be expanded into the product of two irreducible quadratics, i.e., if $f(x) = (x^2 + ax + b)(x^2 + cx + d)$ for

$a, b, c, d \in \mathbb{Q}$. Examining the following system of equations, we deduce that $f(x)$ is reducible as the product $(x^2 + x + 1)(x^2 - x + 1)$. By the rational root test, each of these factors is irreducible. Thus, $f(x)$ is the product of irreducible quadratics. We will look at the product of their discriminants. The discriminant of the first quadratic is simply,

$$D_1 = b^2 - 4ac = 1 - 4(1)(1) = 1 - 4 = -3,$$

while the discriminant of the second is simply,

$$D_2 = b^2 - 4ac = (-1)^2 - 4(1)(1) = 1 - 4 = -3.$$

Since their product $D_1 D_2 = 9$ is a square in $\mathbb{Q}$, we conclude that by the classification of Galois groups of quartics, the splitting field is $\mathbb{Q}(\sqrt{D_1}, \sqrt{D_2}) = \mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\sqrt{3}, i)$. The Galois group must then be of order 4, and consists of the automorphisms of the splitting field that fix the base field. Thus, we have the four automorphisms,

$$\begin{aligned} i &: \sqrt{3} \mapsto \sqrt{3}, i \mapsto i, & \sigma &: \sqrt{3} \mapsto -\sqrt{3}, i \mapsto i \\ \tau &: \sqrt{3} \mapsto \sqrt{3}, i \mapsto i, & \rho &: \sqrt{3} \mapsto -\sqrt{3}, i \mapsto -i. \end{aligned}$$

Writing out the multiplication table, we conclude that the corresponding Galois group must be the Klein 4-group, V_4.

Problem 2017-J-I-3. Let $\mathbb{F}_p$ be a prime field of order p and let $a \in \mathbb{F}_p \setminus \{0\}$. Show that the Galois group of $x^p - x - a$ is cyclic of order p and construct an explicit Galois automorphism generating this group.

Let $\mathbb{F}_p$ be a prime field of order p and let $a \in \mathbb{F}_p \setminus \{0\}$. Let $p(x) = x^p - x - a$. First, we note that $p(x)$ has no roots in $\mathbb{F}_p$: assume to the contrary that b was a root. Then since $t^p = t$ for all $t \in \mathbb{F}_p$, one has, $p(b) = b^p - b - a = -a \neq 0$ – a contradiction. In particular, this means that $x^p - x - a$, *if* reducible, cannot split into a product containing a linear factor. We will now show that $p(x)$ is indeed irreducible. Suppose α is some root of $p(x)$, and let $1 \in \mathbb{F}_p$. Then

$$p(\alpha + 1) = (\alpha + 1)^p - (\alpha + 1) - a = (\alpha^p - \alpha - a) + (1 - 1) = 0 + 0 = 0.$$

Then by induction, we can show that $\alpha + j$ is a root of $p(x)$ for all $j \in \mathbb{F}_p$. Thus since $\deg p(x) = p$, and there are precisely p elements in $\mathbb{F}_p$, these must be all of the roots of $p(x)$. [!! Complete Later !!]

Problem 2016-J-II-2. Find the Galois group of the polynomial $p(x) = x^3 - 2$ over the field $\mathbb{Z}_{11} := \mathbb{Z}/11\mathbb{Z}$.

Let $p(x) = x^3 - 2$, and let $\mathbb{Z}_{11} := \mathbb{Z}/11\mathbb{Z}$. First, we will check if $p(x)$ is irreducible. By plugging in the elements of this field, we find that $x = 7$ is a root of $p(x)$. This means that $p(x)$ factors as the product, $(x - 7)(x^2 + 7x + 5)$, where the latter quadratic is irreducible (if it were irreducible, then it must split as the product of two further linear factors, which would force $p(x)$ to have three roots in $\mathbb{Z}_{11}$ – a contradiction to our earlier finding). Thus, the Galois group of the quadratic, and hence $p(x)$, over $\mathbb{Z}_{11}$ must be $\mathbb{Z}/2\mathbb{Z}$.

Problem 2015-A-II-5. Find the splitting field and the Galois group of the polynomial $x^4 - 5x^2 + 5$ over $\mathbb{Q}$.

Let $p(x) = x^4 - 5x^2 + 5 \in \mathbb{Q}[x]$. By the rational root test, it follows that $p(x)$ has no rational roots, and thus cannot factor as the product of a linear and cubic factor. Therefore, we check to see if $p(x)$ can be decomposed into a product of the form $(x^2 + ax + b)(x^2 + cx + d)$. Expanding and matching coefficients, we obtain the following system of equations:

$$\begin{cases} a + c = 0 \\ ac + b + d = -5 \\ ad + bc = 0 \\ bd = 5. \end{cases}$$

From the first equation, one gets $a = -c$. Thus making this substitution, one obtains the following system of equations:

$$\begin{cases} b + d - c^2 = -5 \\ c(b-d) = 0 \\ bd = 5. \end{cases}$$

Since $\mathbb{Q}$ is an integral domain, either $c = 0$ or $b - d = 0$. If $c = 0$, then one has $b + d = -5$ and $bd = 5$. From this, one obtains $b = 5/d$. Plugging this into the first equation, one gets

$$-5 = b + d = \frac{5}{d} + d \implies -5d = 5 + d^2 \implies d^2 + 5d + 5 = 0.$$

But there exist no rational solutions to $d^2 + 5d + 5 = 0$. Now suppose $b - d = 0$ so that $b = d$. Then one has the system of equations

$$\begin{aligned} 2b - c^2 &= -5 \\ b^2 &= 5. \end{aligned}$$

Then second equation implies $b = \pm\sqrt{5}$ so that

$$\pm 2\sqrt{5} - c^2 = -5 \implies c^2 = \pm 2\sqrt{5} + 5 \implies c = \{\pm\sqrt{2\sqrt{5}+5}, \pm\sqrt{-2\sqrt{5}+5}\},$$

neither of which are rational. Thus, we conclude that $p(x)$ is irreducible over $\mathbb{Q}$. To find the splitting field and the Galois group, we will compute the roots of $p(x)$. Setting $u = x^2$, one has $0 = p(u) = u^2 - 5u + 5$ so that

$$u = \frac{5 \pm \sqrt{25 - 20}}{2} = \frac{5 \pm \sqrt{5}}{2} \implies x = \pm\alpha, \pm\beta,$$

where

$$\alpha \coloneqq \sqrt{\frac{5+\sqrt{5}}{2}}, \beta \coloneqq \sqrt{\frac{5-\sqrt{5}}{2}}.$$

Let K denote the splitting field of $p(x)$ over $\mathbb{Q}$. Then K must contain both α and β (and thus the field extension $\mathbb{Q}(\alpha, \sqrt{5})$ since $\alpha\beta = \sqrt{5}$). On the other hand, $\mathbb{Q}(\alpha, \sqrt{5})$ contains all of the roots of $p(x)$. Then the splitting field is $\mathbb{Q}(\alpha, \sqrt{5})$. Since in fact, $\sqrt{5} \in \mathbb{Q}(\alpha)$, we conclude that the splitting field is actually just $\mathbb{Q}(\alpha)$. Since $\mathbb{Q}(\alpha)$ is the splitting field of an irreducible monic polynomial, $\mathbb{Q}(\alpha)/\mathbb{Q}$ is Galois. The Galois group of the polynomial consists of the automorphisms on $\mathbb{Q}(\alpha)$ that preserve the base field $\mathbb{Q}$ and the algebraic relations between the roots. Thus we have the four automorphisms:

$$\begin{aligned} &\alpha \mapsto \alpha \qquad &&\alpha \mapsto -\alpha \\ &\alpha \mapsto \beta \qquad &&\alpha \mapsto -\beta. \end{aligned}$$

The first automorphism is trivial; the second automorphism has order 2, the third automorphism has order 4, and the fourth automorphism has order 4. Thus, we conclude that the Galois group is isomorphic to $\mathbb{Z}/4\mathbb{Z}$.

Problem 2014-A-II-1. Determine the Galois group of the polynomial $x^4 + 4$ over $\mathbb{Q}$.

Let $p(x) = x^4 + 4$. First, we begin by checking if $p(x)$ is irreducible over $\mathbb{Q}$. It is straightforward to see that p has no linear factors since it has no rational roots. Instead, we look for a decomposition of the form $(x^2 + ax + b)(x^2 + cx + d)$. Doing so, we obtain the decomposition,

$$x^4 + 4 = (x^2 - 2x + 2)(x^2 + 2x + 2).$$

Each of these quadratic are irreducible as they have no rational (nor real) roots. Solving these quadratics, we find the four roots of $p(x)$ to be $1 \pm i, -1 \pm i$. Thus, it is clear that the splitting field of $p(x)$ over $\mathbb{Q}$ must be $\mathbb{Q}(i)$. Since the degree of this field extension is 2, we conclude that the Galois group must be $\mathbb{Z}/2\mathbb{Z}$.

Problem 2014-J-I-5. Let K denote the splitting field for $(x^5-1)(x^3-2)$ over the rational numbers $\mathbb{Q}$. Compute the cardinality of the Galois group G for the extension $\mathbb{Q} \subset K$, and show that G is not abelian.

Let K denote the splitting field for $(x^5-1)(x^3-2)$ over the rational numbers $\mathbb{Q}$. The roots of this polynomial are ζ_5^j $(j = 0, \dots, 4)$, and $\sqrt[3]{2}\zeta_3^k$ $(k = 0, \dots, 2)$. The splitting field K must contain all of these roots, and thus the field $\mathbb{Q}(\zeta_3, \zeta_5, \sqrt[3]{2}) = \mathbb{Q}(\zeta_{15}, \sqrt[3]{2})$, where the equality stems from the fact that 3 and 5 are coprime. On the other hand, we also observe that $\mathbb{Q}(\zeta_{15}, \sqrt[3]{2})$ contains all of the roots of the polynomial, and thus must contain K. Therefore, the splitting field is $\mathbb{Q}(\zeta_{15}, \sqrt[3]{2})$. Since the minimal polynomial of $\sqrt[3]{2}$ over $\mathbb{Q}$ is x^3-2, $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$. On the other hand, since the minimal polynomial of ζ_{15} over $\mathbb{Q}(\sqrt[3]{2})$ is the 15th cyclotomic polynomial $\Phi_{15}(x)$, which has degree 8, we conclude that $[\mathbb{Q}(\zeta_{15}, \sqrt[3]{2}) : \mathbb{Q}] = 8 \cdot 3 = 24$. Thus, if G denotes the Galois group for the extension $K/\mathbb{Q}$, $|G| = 24$. Since the splitting roots of each irreducible factor ($\mathbb{Q}(\zeta_5)$ and $\mathbb{Q}(\zeta_3, \sqrt[3]{2})$. Since these fields are linearly disjoint over $\mathbb{Q}$, we conclude that the Galois group of the polynomial is the direct product of the Galois group of each factor.

The minimal polynomial of ζ_5 is of order 4, which means that the corresponding Galois group must also be of order 4. This group consists of the automorphisms of $\mathbb{Q}(\zeta_5)$ that fix $\mathbb{Q}$. Thus, it follows that the Galois group is $\mathbb{Z}/4\mathbb{Z}$. Next, since x^3-2 is irreducible over $\mathbb{Q}$, and its discriminant is not a square in $\mathbb{Q}$, we conclude that its Galois group is S_3. Therefore, $G = \mathbb{Z}/4\mathbb{Z} \times S_3$. Since S_3 is nonabelian, we conclude that G is nonabelian.

Problem 2013-A-I-2. Find the Galois group of x^3-2 over the field $\mathbb{Z}_5$.

Let $p(x) = x^3 - 2$. We will first look for root of $p(x)$ over the field $\mathbb{Z}/5\mathbb{Z}$. We note that $x = 3$ is a root of $p(x)$ over $\mathbb{Z}/5\mathbb{Z}$. Thus, in this field, $p(x) = (x-3)(x^2+3x+4)$. The quadratic has no roots in this field, and therefore, is irreducible over $\mathbb{Z}/5\mathbb{Z}$, and therefore, the splitting field of $p(x)$ is isomorphic to the splitting field of the irreducible quadratic. We find the roots of the quadratic in $\mathbb{C}$ to be

$$x = \frac{-3 \pm i\sqrt{7}}{2} \equiv 1 \pm 3i\sqrt{7} \pmod 5.$$

Let K denote the splitting field of the quadratic over $\mathbb{Z}/5\mathbb{Z}$. Since it must contain all of the roots, it must contain $i, \sqrt{7}$, and therefore, the field extension $\mathbb{Z}_5(i, \sqrt{7})$. On the other hand, $\mathbb{Z}_5(i, \sqrt{7})$ already contains all of the roots of the quadratic (and thus of the original cubic). Hence, $K = \mathbb{Z}_5(i, \sqrt{7})$. First, we note that

$$(\sqrt{7})^2 = 7 \equiv 2 \pmod 5,$$

so that the polynomial $m_1(x) = x^2 - 2$ annihilates $\sqrt{7}$ over $\mathbb{Z}_5$. This quadratic is irreducible as it has no roots in the field, which means that $\mathbb{Z}_5(\sqrt{7})$ is a degree 2 field extension. On the other hand, i is annihilated by the polynomial x^2+1 in $\mathbb{Z}_5$; this quadratic is reducible since 2 and 3 are roots. Therefore, $[\mathbb{Z}_5(i, \sqrt{7}) : \mathbb{Z}_5(\sqrt{7})] = 1$ and the splitting field of $p(x)$ is actually just $\mathbb{Z}_5(\sqrt{7})$. Therefore, since $\mathbb{Z}_5(\sqrt{7})$ is Galois of order 2, we conclude that the Galois group of $p(x)$ over $\mathbb{Z}_5$ is simply $\mathbb{Z}/2\mathbb{Z}$.

Problem 2011-A-II-1. Determine the Galois group of $x^4 - 25$ over $\mathbb{Q}$.

Let $p(x) = x^4 - 25$. First, we note that $p(x)$ is reducible: $p(x) = (x^2+5)(x^2-5)$, where each of the quadratic factors in this expression is irreducible as it has no rational roots. Indeed, the roots of $p(x)$ are $\pm\sqrt{5}$, $\pm i\sqrt{5}$. Let K denote the splitting field of $p(x)$ over $\mathbb{Q}$. Then K must contain all four roots, and thus the field $\mathbb{Q}(i, \sqrt{5})$. On the other hand, $\mathbb{Q}(i, \sqrt{5})$ already contains all of the roots of $p(x)$. Thus, $K = \mathbb{Q}(i, \sqrt{5})$. The minimal polynomial of $\sqrt{5}$ over $\mathbb{Q}$ is x^2-5, which is of degree 2. Therefore, $[\mathbb{Q}(\sqrt{5}) : \mathbb{Q}] = 2$. On the other hand, since $\sqrt{5} \in \mathbb{R}$, $\mathbb{Q}(\sqrt{5}) \hookrightarrow \mathbb{R}$, which means that $i \notin \mathbb{Q}(\sqrt{5})$. Thus, since the minimal polynomial of i over $\mathbb{Q}(\sqrt{5})$ is x^2+1 (degree 2), we conclude that $[\mathbb{Q}(i, \sqrt{5}) : \mathbb{Q}] = 4$. Since $\mathbb{Q}(i, \sqrt{5})$ is Galois over $\mathbb{Q}$, the elements of the Galois group are the automorphisms on $\mathbb{Q}(i, \sqrt{5})$ that preserve the relations between the generators, and fix the base field $\mathbb{Q}$. Thus, we have the four maps:

$$\begin{aligned} &\mathrm{id} : i \mapsto i, \sqrt{5} \mapsto \sqrt{5}, && \sigma : i \mapsto -i, \sqrt{5} \mapsto \sqrt{5}; \\ &\tau : i \mapsto i, \sqrt{5} \mapsto -\sqrt{5}, && \rho : i \mapsto -i, \sqrt{5} \mapsto -\sqrt{5}. \end{aligned}$$

Each of the nonidentity elements have order 2. Thus, the Galois group is V_4.

Problem 2010-A-II-2. Find a rational number c such that the splitting field over $\mathbb{Q}$ of the cubic polynomial $x^3 + cx - 1$ has a nonabelian Galois group over $\mathbb{Q}$. For your value of c, compute the isomorphism type of the Galois group.

Let $p(x) = x^3 + cx - 1$, where c is a rational number. Since the Galois group of polynomials must permute the roots of $p(x)$, we know that the galois Group, G, must be isomorphic to a subgroup of S_3 - the symmetry group on 3 letters. Up to conjugacy, there exist precisely 4 such subgroups: (1) the trivial subgroup, (2) the order 2 subgroup generated by a transposition (which is isomorphic to $\mathbb{Z}/2\mathbb{Z}$), (3) the order 3 subgroup A_3, and (4) the whole group S_3. Of these four choices, only S_3 is nonabelian. We recall that the Galois group of a cubic over $\mathbb{Q}$ iff it is irreducible and $\sqrt{D} \notin \mathbb{Q}$, where D is its discriminant. Since p is depressed, its discriminant is

$$D = -4p^3 - 27q^2 = -4c^3 - 27(-1)^2 = -4c^3 - 27.$$

If $c = 1$, then $D = -31$, which is not a square in $\mathbb{Q}$. Moreover, by the rational root test, $x^3 + x - 1$ has no rational roots (and thus is irreducible). Therefore, $c = 1$ is a good choice.

Problem 2010-J-I-5. Consider the following irreducible polynomial over $\mathbb{Q}$: $p(x) = x^4 - 3x^2 - 1$.
(a) Describe the splitting field of $p(x)$.
(b) Consider the Galois group of $p(x)$. Compute its order and if it is abelian.

(a) We begin by checking if $p(x)$ is irreducible. By the rational root test, $p(x)$ has no rational roots, and thus, cannot split into a product containing a linear factor. Now we will check if $p(x)$ can be broken into the product $(x^2 + ax + b)(x^2 + cx + d)$, where $a, b, c, d \in \mathbb{Q}$. Repeating work similar to that done in Problem 2015-A-II-5, we deduce that $p(x)$ is irreducible over $\mathbb{Q}$. Thus we now look for roots of $p(x)$. Set $u = x^2$. Then $0 = p(u) = u^2 - 3u - 1$ yields the solutions

$$u = \frac{3 \pm \sqrt{13}}{2} \implies x = \pm\alpha, \pm\beta,$$

where

$$\alpha \coloneqq \sqrt{\frac{3 + \sqrt{13}}{2}} \qquad \text{and} \qquad \beta \coloneqq \sqrt{\frac{3 - \sqrt{13}}{2}}.$$

We compute

$$\alpha\beta = i.$$

Let K denote the splitting field of $p(x)$ over $\mathbb{Q}$. Then K must contain $\pm\alpha, \pm\beta$ (and thus i). This implies $K \supseteq \mathbb{Q}(\alpha, i)$. On the other hand, $\mathbb{Q}(\alpha, i)$ already contains all of the roots of $p(x)$, and thus must contain K. Therefore, we conclude that the splitting field of $p(x)$ over $\mathbb{Q}$ is $\mathbb{Q}(\alpha, i)$. Since the minimal polynomial of α is of order 4, $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 4$. Since $\alpha \in \mathbb{R}$, $\mathbb{Q}(\alpha) \hookrightarrow \mathbb{R}$, and thus, the minimal polynomial of i over $\mathbb{Q}(\alpha)$ is $x^2 + 1$ (degree 2). Therefore, $[\mathbb{Q}(i, \alpha) : \mathbb{Q}] = 8$.

(b) Since $\mathbb{Q}(i, \alpha)$ is the splitting field of an irreducible polynomial over $\mathbb{Q}$, $\mathbb{Q}(i, \alpha)/\mathbb{Q}$ is a Galois field extension of degree 8; thus the Galois group of $p(x)$ is of order 8. The Galois group of $p(x)$ corresponds to the automorphism group on $\mathbb{Q}(i, \alpha)$ that permutes the roots of $p(x)$ while fixing the base field $\mathbb{Q}$. Thus, the Galois group must be isomorphic to a transitive subgroup of S_4, the symmetry group on four letters; the only transitive subgroup of S_4 of degree 8 is the nonabelian dihedral group D_8.

Problem 2005-J-I-4. Find the Galois group of the polynomial group $p(x) = x^3 - 3x + 1$ over $\mathbb{Q}$.

Let $p(x) = x^3 - 3x + 1 \in \mathbb{Q}[x]$. The possible candidates for rational roots of $p(x)$ are $x = \pm 1$. But since $p(1) = -1 \neq 0$ and $p(-1) = -3 \neq 0$, we conclude that $p(x)$ has no rational roots. Thus since $p(x)$ has degree 3, it must be irreducible over $\mathbb{Q}$. Let D denote the discriminant of $p(x)$. Since $p(x)$ is a depressed cubic, $D = -4p^3 - 27q^2 = 81$, which is a square in $\mathbb{Q}$. Therefore, by the classification of Galois groups of cubics over $\mathbb{Q}$, we conclude that the Galois group of $p(x)$ over $\mathbb{Q}$ is the alternating group A_3.

3.5 Sylow's Theorems

Problem 2022-J-I-3. Show that a group of order 1,000,000 contains a proper normal subgroup (i.e., is not simple).

Let G denote a group of order $10^6 = 2^6 \cdot 5^6$, and let n_5 denote the number of Sylow 5-subgroups of G. Then by Sylow's Theorems,

$$n_5 \in \{1, 2, 4, 8, 16, 32, 64\} \cap \{1, 6, 11, 16, 21, \ldots\} = \{1, 16\}.$$

If $n_5 = 1$, then we are done. So assume $n_5 = 16$. Let G act on the set $\mathrm{Syl}_5(G)$ of Sylow 5-subgroups by conjugation. This action induces a homomorphism $\varphi : G \to S_{16}$. Let $K \coloneqq \ker \varphi \trianglelefteq G$. If $K = \{e\}$, then $\varphi : G \hookrightarrow S_{16}$, which is a contradiction since $|G| = 10^6 \nmid 16! = |S_{16}|$. So K is nontrivial. If $K = G$, then the action is trivial, which means that for every $g \in G$ and every $P \in \mathrm{Syl}_5(G)$, $gPg^{-1} = P$. I.e., we must have $n_5 = 1$, contradicting our hypothesis. Thus, we conclude that K is a nontrivial, proper, normal subgroup of G.

Problem 2022-A-II-4. Let G be a finite group in which $(ab)^p = a^p b^p$ for every $a, b \in G$, where p is a prime dividing $|G|$. Prove that the p-Sylow subgroup of G is normal in G (and is in fact unique).

Let G be a finite group in which $(ab)^p = a^p b^p$ for every $a, b \in G$, where p is a prime dividing $|G|$. Consider the map $\varphi : G \to G$ defined by $\varphi(g) = g^p$. This map is a homomorphism since for any $g, h \in G$,

$$\varphi(gh) = (gh)^p = g^p h^p = \varphi(g)\varphi(h),$$

where the second equality follows from the hypothesis. Consider the map

$$\varphi^k \coloneqq \underbrace{\varphi \circ \cdots \circ \varphi}_{k \text{ copies}},$$

which must also be a homomorphism since the composition of homomorphisms is a homomorphism. The kernel of φ^k consists exactly of those elements $x \in G$ whose order is a power of p (i.e., $x^{p^r} = 1$ for some positive integer r) since

$$\varphi^k(x) = x^{p^k} = x^{p^{r+(k-r)}} = \left(x^{p^r}\right)^{p^{k-r}} = 1^{p^{k-r}} = 1.$$

Hence, since every element with order equal to some order of p belongs in a Sylow p-subgroup of G,

$$\ker \varphi^k = \bigcup_{P \in \mathrm{Syl}_p(G)} P.$$

Moreover, $\ker \varphi^k$ must be a p-subgroup of G since if not, there exists a prime $p' \neq p$ dividing $|\ker \varphi^k|$, which means by Cauchy's Theorem that $\ker \varphi^k$ contains an element of order p' (which is impossible). Hence, since $\ker \varphi^k$ is a p-subgroup of G containing a Sylow p-subgroup, by maximality of Sylow p-subgroups, $\ker \varphi^k$ must be a Sylow p-subgroup of G. Hence, G has a unique Sylow p-subgroup. And since kernels of homomorphisms are normal subgroups, this Sylow p-subgroup must be normal. [!! Complete Later !!]

Problem 2018-J-II-5. Let G denote a group of order 80. Prove that G contains a nontrivial normal subgroup.

Let G be a group order $80 = 2^4 \cdot 5$. Let n_2 and n_5 denote the number of Sylow 2- and Sylow 5-subgroups of G, respectively. By Sylow's Theorems, n_5 must be divide 2^4 and be of the form $\equiv 1 \pmod 5$. Thus, we must have

$$n_5 \in \{1, 2, 4, 8, 16\} \cap \{1, 6, 11, 16, \ldots\} = \{1, 16\}.$$

If $n_5 = 1$, then we are done. So assume $n_5 = 16$. By the prime factorization of $|G|$, each Sylow 5-subgroup of G must be of order 5, and consequently, must intersect trivially (i.e., if P, Q are distinct Sylow 5-subgroups, then $P \cap Q = \{1\}$). Each Sylow 5-subgroup contains 4 non-identity elements. This means that G contains 64 elements whose order is exactly 5. This leaves exactly 15 elements of G to have an order of the form 2^α. Since each Sylow 2-subgroup contains exactly 15 nonidentity elements of G with order equal to some power of 2, all of the aforementioned elements must be contained in exactly one Sylow 2-subgroup (i.e., $n_2 = 1$). Thus, $n_2 = 1$, and so this subgroup is normal in G. In any case, G contains a normal subgroup.

Problem 2017-J-II-6. Without using the classification of simple groups, show that every simple group of order 60 has exactly 10 subgroups of order 3. Hint: Consider the action of this group on subgroups of order 3 by conjugation.

Let G denote a simple group of order $60 = 2^2 \cdot 3 \cdot 5$, and let n_2, n_3, n_5 denote the number of Sylow 2-, 3-, and 5-subgroups. By Sylow's Theorems, one has $n_3 \in \{1, 2, 4, 5, 10, 20\} \cap \{1, 4, 7, 10, \ldots, 19, 22, \ldots\} = \{1, 4, 10\}$. Moreover, it follows that every subgroup of G of order 3 is precisely a Sylow 3-subgroup, and every Sylow-3 subgroup of G is a subgroup of order 3. Since G is simple, it has no normal subgroups, and therefore, $n_3 \neq 1$. Assume to the contrary that G has exactly 4 subgroups of order 3. Then the group action of G on the set $\mathrm{Syl}_3(G)$ of Sylow 3-subgroups by conjugation induces a homomorphism $G \to S_4$. If the kernel of this action were trivial, then this homorphism is injective. I.e., G is isomorphic to a subgroup of S_4. But $|S_4| = 24$, while $|G| = 60 > 24$, which is a contradiction. Thus, the kernel must be nontrivial. If the kernel is all of G, then the action is trivial, which forces $n_3 = 1$ (a contradiction). Thus we conclude that $n_3 = 10$.

Problem 2016-J-I-1. Let G be a finite group of order 80. Prove that G is not simple.

Let G be a group order $80 = 2^4 \cdot 5$. Let n_2 and n_5 denote the number of Sylow 2- and Sylow 5-subgroups of G, respectively. By Sylow's Theorems, n_5 must be divide 2^4 and be of the form $\equiv 1 \pmod 5$. Thus, we must have

$$n_5 \in \{1, 2, 4, 8, 16\} \cap \{1, 6, 11, 16, \ldots\} = \{1, 16\}.$$

If $n_5 = 1$, then we are done. So assume $n_5 = 16$. By the prime factorization of $|G|$, each Sylow 5-subgroup of G must be of order 5, and consequently, must intersect trivially (i.e., if P, Q are distinct Sylow 5-subgroups, then $P \cap Q = \{1\}$). Each Sylow 5-subgroup contains 4 non-identity elements. This means that G contains 64 elements whose order is exactly 5. This leaves exactly 15 elements of G to have an order of the form 2^α. Since each Sylow 2-subgroup contains exactly 15 nonidentity elements of G with order equal to some power of 2, all of the aforementioned elements must be contained in exactly one Sylow 2-subgroup (i.e., $n_2 = 1$). Thus, $n_2 = 1$, and so this subgroup is normal in G. In any case, G contains a normal subgroup, and hence is not simple.

Problem 2013-J-I-2. Prove that every group of order 72 is non-simple.

Let G denote a group of order $72 = 2^3 \cdot 3^2$, and let n_2, n_3 denote the number of Sylow 2- and 3-subgroups, respectively. By Sylow's Theorems,

$$n_3 \in \{1, 2, 4, 8\} \cap \{1, 4, 7, \ldots\} = \{1, 4\}.$$

If $n_3 = 1$, then we are done since a unique Sylow subgroup must necessarily be normal. So assume $n_3 = 4$. Let G act on the set $\mathrm{Syl}_3(G)$ of Sylow 3-subgroups by conjugation. Then this action induces a homomorphism $\varphi : G \to S_4$. Let K denote the kernel of this homomorphism. If $K = \{e\}$, then $\varphi : G \hookrightarrow S_4$ is an embedding, which is impossible since $|G| = 72 > |S_4| = 24$. So K is nontrivial. If $K = G$, then the group action is trivial, which means that for every $g \in G$ and every $P \in \mathrm{Syl}_3(G)$, $gPg^{-1} = P$. But since every Sylow 3-subgroup has to be conjugate of each other, this constraint forces $n_3 = 1$, contradicting our assumption that $n_3 = 4$. This, $\ker \varphi$ is a nontrivial, proper normal subgroup of G which means that G cannot be simple.

Problem 2009-J-I-5. Let G be a group of order 63. Prove that G is not simple (i.e., that G contains a nontrivial normal subgroup).

Let G be a group of order $63 = 3^2 \cdot 7$, and let n_7 denote the number of distinct Sylow 7-subgroups of G. By Sylow's Theorems, $n_7 \in \{1,3,9\} \cap \{1,8,15,\ldots\} = \{1\}$. Thus since there exists a single Sylow 7-subgroup, by Sylow's Theorems, this subgroup must be normal. Thus since this subgroup has order $0 < 7 < 63$, we conclude that G contains a nontrivial normal subgroup, and hence cannot be simple.

Problem 2006-A-II-5. Show that a group of order 55 is not simple.

Let G denote a group of order $55 = 5 \cdot 11$, and let n_{11} denote the number of distinct Sylow 11-subgroups of G. By Sylow's theorems $n_{11} \in \{1,5\} \cap \{1,12\ldots\} = \{1\}$ so that G contains a single Sylow-11 subgroup. By Sylow's theorems, this subgroup must be normal. Since this subgroup has order $0 < 11 < 55$, we conclude that G contains a nontrivial normal subgroup, and hence is not simple.

Problem 2001-J-I-3. Let G be a finite group and $P \subset G$ a normal Sylow p-subgroup. Show that for any group homomorphism $\varphi : G \to G$, $\varphi(P) \subset P$.

Let G be a finite group, and $P \subset G$ a normal Sylow p-subgroup. This necessarily implies that P is the unique Sylow p-subgroup of G. Therefore, every element of G with order equal to some power of p must be contained in P. Now let $\varphi : G \to G$ be a group *homo*morphism. Then for any $x \in P$, $|\varphi(x)| \mid |x| = p^{\alpha_x}$ for some positive integer α_x. So since $\varphi(x)$ has an order equal to some power of p, $\varphi(x) \in P$. Therefore, since $x \in P$ was arbitrary, we must have $\varphi(P) \subset P$.

Problem 1999-A-I-1. Let P be a Sylow p-subgroup of G, and let H be a p-subgroup of G contained in $N(P)$. Here, $N(P)$ is the normalizer of P in G. Prove H is contained in P.

Let P be a Sylow p-subgroup of G, and let H be a p-subgroup of G contained in $N(P)$, the normalizer of P in G. First, this implies that PH is a subgroup of G of order

$$|PH| = \frac{|P||H|}{|P \cap H|}.$$

In particular, it follows that PH is also a p-subgroup of G. Assume to the contrary that H is not entirely contained in P. This means that $|H|/|P \cap H| > 1$. But then $|PH| > |P|$. But this contradicts the fact that P is a *Sylow* p-subgroup of G, and therefore has maximal order amongst p-subgroups of G. Thus, by contradiction, H is entirely contained in P.

Problem D&F-4.5-13. Prove that a group of order 56 has a normal Sylow p-subgroup for some prime p dividing its order.

Let G denote a group of order $56 = 2^3 \cdot 7$, and let n_2, n_7 denote the number of Sylow 2- and 7-subgroups. By Sylow's Theorems, $n_7 \in \{1,2,4,8\} \cap \{1,8,\ldots\} = \{1,8\}$. Likewise, $n_2 \in \{1,7\} \cap \{1,3,5,7,\ldots\} = \{1,7\}$. If $n_7 = 1$, we are done since by Sylow's Theorems, a unique Sylow subgroup must necessarily be normal. So suppose $n_7 = 8$. Since each Sylow 7-subgroup has prime order 7, each pair of distinct Sylow 7-subgroups intersect trivially. This means that G contains 48 elements of order 7. If $n_2 = 7$, then each pair of disinct Sylow 2-subgroups can share at most 4 elements, so that G contains at least $8 + 6(4) = 32$ elements whose order is of the form 2^α for some positive integer α. But this means that G contains at least $32 + 48 + 1 = 81 > 56$ elements, which is a contradiction. Thus, by contradiction, $n_2 = 1$. In this case, G contains a normal Sylow 2-subgroup. Since this subgroup is of order $8 < 56$, this subgroup must be proper. The proof concludes.

Problem D&F-4.5-14. Prove that a group of order 312 has a normal Sylow p-subgroup for some prime p dividing its order.

Let G denote a group of order $312 = 2^3 \cdot 3 \cdot 13$, and let n_{13} denote the number of Sylow 13-subgroups. By Sylow's Theorems,

$$n_{13} \in \{1,2,3,4,6,8,12,24\} \cap \{1,14,27,\ldots\} = \{1\}.$$

Thus G contains a unique Sylow 13-subgroup, which must then be normal in G.

Problem D&F-4.5-15. Prove that a group of order 351 has a normal Sylow p-subgroup for some prime p dividing its order.

Let G denote a group of order $351 = 3^3 \cdot 13$, and let n_3, n_{13} denote the number of Sylow 3- and 13-subgroups in G. By Sylow's Theorems,

$$n_3 \in \{1,13\} \cap \{1,4,\ldots,\} = \{1,13\}, \qquad n_{13} \in \{1,27\}.$$

If n_{13} or n_3 is equal to one, then we are done. So assume $n_{13} = 27$ and $n_3 = 13$. Since each Sylow 13-subgroup is of prime order 13, each distinct Sylow 13-subgroup intersects trivially so that G contains exactly $12(27) = 324$ elements of order 13. On the other hand, each pair of distinct Sylow 3-subgroups may share at most 9 elements, so that the two subgroups alone contain at least 45 unique elements whose order is some power of 3. But this is impossible since $324 + 45 > |G|$. Thus G must contain either a normal Sylow 13-subgroup or a normal Sylow 3-subgroup.

Problem D&F-4.5-17. Prove that if $|G| = 105$ then G has a normal Sylow 5-subgroup and a normal Sylow 7-subgroup.

Let G denote a group of order $105 = 3 \cdot 5 \cdot 7$, and let n_3, n_5, n_7 denote the number of Sylow 3-, 5-, and 7-subgroups, respectively. By Sylow's Theorems,

$$\begin{aligned}
&n_3 \in \{1,5,7,35\} \cap \{1,4,7,\ldots\} = \{1,7\}.\\
&n_5 \in \{1,3,7,21\} \cap \{1,6,11,16,21,\ldots\} = \{1,21\}.\\
&n_7 \in \{1,3,5,15\} \cap \{1,8,15,\ldots\} = \{1,15\}.
\end{aligned}$$

First, assume to the contrary that G has 15 Sylow 7-subgroups. Since each Sylow 7-subgroup has prime order, each pair of distinct Sylow 7-subgroups intersects trivially so that G contains exactly 90 elements of order 7. Since Sylow p-subgroups intersect trivially with Sylow q-subgroups for $p \neq q$, this leaves exactly 14 elements of G to have an order of either 3 or 5. This is a contradiction: for each of the possible values of (n_3, n_5) we calculate the total number of elements of G with order either 3 or 5 to be:

(n_3, n_5)	(# Elements with Order 3, # Elements with Order 5)	Total
$(1,1)$	$(2,4)$	$2+4=6$
$(1,21)$	$(2,84)$	$2+84=86$
$(7,1)$	$(14,4)$	$14+4=18$
$(7,21)$	$(14,84)$	$14+84=98$

Thus, we must have $n_7 = 1$. Likewise, assume $n_5 = 21$. This means that G contains exactly 84 elements of order 5. Therefore, G must contain exactly 20 elements of order either 3 or 7. Since G already is known to have exactly one Sylow 7-subgroup (and thus 6 elements of order 7), G must have exactly 14 elements of order 3. Altogether, we've identified the order of every element of G: 14 elements of order 3, 6 elements of order 7, 84 elements of order 5, and the identity. This means that there exist no elements of G of composite orders. Denote by P the Sylow 7-subgroup and Q a Sylow 5-subgroup. Since $Q \leq N_G(P)$, PQ is a subgroup of G of order,

$$|PQ| = \frac{|P||Q|}{|P \cap Q|} = 7 \cdot 5 = 35.$$

Since every group of order 35 is cyclic[2], it follows that $PQ = \langle x \rangle$, where $x \in G$ is an element of order 35. But this is a contradiction. Thus, we must have $n_5 = 1$. The proof concludes.

[2]Use Sylow's Theorems again to quicky prove this!

Problem D&F-4.5-32. Let P be a Sylow p-subgroup of H and let H be a subgroup of K. If $P \trianglelefteq H$ and $H \trianglelefteq K$, prove that P is normal in K. Deduce that if $P \in \text{Syl}_p(G)$ and $H = N_G(P)$, then $N_G(H) = H$ (in words: *normalizers of Sylow p-subgroups are self-normalizing*).

Let G be a finite group, and $P \trianglelefteq H \trianglelefteq K \leq G$. Our first goal is to show that $kPk^{-1} = P$ for all $k \in K$. Since H is a normal subgroup of K, we must have that $kPk^{-1} \leq H$ for all k at the very least. But kPk^{-1} is another Sylow p-subgroup of H. Since P is a normal Sylow p-subgroup of H, it must be the unique Sylow p-subgroup so that $P = kPk^{-1}$ for all k. Thus, P is normal in K.
Now assume that $P \in \text{Syl}_p(G)$ and $H = N_G(P)$. By definition, we have $P \trianglelefteq H \trianglelefteq N_G(H) \leq G$. By our above work, one must have $P \trianglelefteq N_G(H)$. But by the definition of a normalizer of a subgroup, this constraint forces $N_G(H) \leq H$. Thus, we must have $N_G(H) = H$.

Problem D&F-4.5-33. Let P be a normal Sylow p-subgroup of G and let H be any subgroup of G. Prove that $P \cap H$ is the unique Sylow p-subgroup of H.

Let P be a normal (i.e., unique) Sylow p-subgroup of G, and let H be a fixed arbitrary subgroup of G. First, it is clear that $P \cap H$ is a p-subgroup of H: since $P \cap H \leq P$, $|P \cap H| \mid |P| = p^\alpha$, where α is some positive integer so that p^α is the largest power of p dividing $|G|$, so that $|P \cap H|$ is also a power of p. Let $|P \cap H| = p^\beta$. First, we must show that $P \cap H$ is a Sylow p-subgroup of H. Suppose p^γ is the largest power of p that divides $|H|$, where $\gamma \ngeq \beta$. Then since $H \leq N_G(P)$, PH is a subgroup of G of order

$$|PH| = \frac{|P||H|}{|P \cap H|} = p^{\alpha+\gamma-\beta}m > p^\alpha = |P|,$$

which contradicts our hypothesis that P is the Sylow p-subgroup of G. Thus, we must have $\gamma = \beta$. So since $P \cap H$ is a subgroup of H with order $p^\gamma = p^\beta$, it must be a Sylow p-subgroup of H. Now, we will show that it is normal in H; i.e., we wish to show that for any $h \in H$, $h(P \cap H)h^{-1} = P \cap H$. One direction is trivial: since P is normal in G, for any $h \in H$, $hPh^{-1} = P$, and $hHh^{-1} = H$ so that $h(P \cap H)h^{-1} \leq P \cap H$. Now let $g \in P \cap H$, and choose $g' = h^{-1}gh$ for some fixed $h \in H$. Then $g' \in h^{-1}(P \cap H)h$. By the previous inclusion, it follows that $g' \in P \cap H$. Thus, since $g = hg'h^{-1} \in h(P \cap H)h^{-1}$, $P \cap H \leq h(P \cap H)h^{-1}$. Since $h \in H$ was arbitrary, this concludes the proof that $P \cap H$ is normal in H. Therefore, $P \cap H$ is the unique Sylow p-subgroup of H.

Problem D&F-4.5-34. Let $P \in \text{Syl}_p(G)$ and assume $N \trianglelefteq G$. Use the conjugacy part of Sylow's Theorem to prove that $P \cap N$ is a Sylow p-subgroup of N. Deduce that PN/N is a Sylow p-subgroup of G/N (note that this may also be done by the Second Isomorphism Theorem — cf. Exercise 9, Section 3.3).

Let G be a finite group of order $p^\alpha M$, where $p \nmid M$ (so that p^α is the largest power of the prime integer p that divides $|G|$), $P \in \text{Syl}_p(G)$ a Sylow p-subgroup, and N a normal subgroup of G. Assume $|N| = p^\gamma m$, where $p \nmid m$ for some nonnegative integer γ. Consider the subgroup $P \cap N$. Since $P \cap N \leq P$, it follows that $P \cap N$ is a p-subgroup of N. Suppose $|P \cap N| = p^\beta$ for some positive integer β, and assume to the contrary that $\gamma > \beta$. Then since PN is a subgroup of G (because N is normal in G), $|PN| = |P||N|/|P \cap N| = p^{\alpha+\gamma-\beta}m > p^\alpha m > p^\alpha = |P|$. But this is a contradiction since by virtue of P being a Sylow p-subgroup of P, p^α is the largest power of p that divides $|G|$. Thus, $\gamma = \beta$, and we have $P \cap N$ is a Sylow p-subgroup of N. Now, by the Second Isomorphism Theorem, one has that $G/N \geq PN/N \cong P/(P \cap N)$. By our work from above, since $P \cap N$ is a p-subgroup as is P, $P/P \cap N$, and thus PN/N is a p-subgroup of G/N. In particular, $|G/N| = p^\alpha M/p^\gamma m = p^{\alpha-\gamma}(M/m)$, where $M/m \in \mathbb{N}$ by Lagrange's Theorem and $p^{\alpha-\gamma} \nmid (M/m)$. Thus, $p^{\alpha-\gamma}$ is the greatest power of p that divides $|G/N|$. But $|PN/N| = p^\alpha m/p^\gamma m = p^{\alpha-\gamma}$. Thus, PN/N must be a Sylow p-subgroup of G/N.

Problem D&F-4.5-36. Prove that if N is a normal subgroup of G then $n_p(G/N) \leq n_p(G)$.

Let N be a normal subgroup of G, so that the quotient G/N is well-defined. Let G have the prime factorization,

$$G = p^\alpha p_1^{\alpha_1} \cdots p_m^{\alpha_m}.$$

Likewise, N has the prime factorization,

$$N = p^{\widetilde{\alpha}} p_1^{\widetilde{\alpha}_1} \cdots p_m^{\widetilde{\alpha}_m},$$

where $0 \leq \widetilde{\alpha}, \widetilde{\alpha}_j \leq \alpha, \alpha_j$ for each $1 \leq j \leq m$. Note that since $N \leq G$, by Lagrange's Theorem, the prime factorization of N cannot contain any primes not contained in the prime factorization of G. [!! Complete Later !!]

3.6 Classification of Groups

Problem 2024-J-I. For distinct odd primes p and q, prove that every finite group of order $2pq$ is a semidirect product of a normal subgroup of order pq and a subgroup of order 2.

Let p, q be distinct odd primes, and let G be a group of order $2pq$. Without loss of generality, assume that $p < q$. [!! Complete Later !!]

Problem 2023-J-II-6. Classify (up to isomorphism) all groups of order 20.

Let G be a group of order $20 = 2^2 \cdot 5$. By the Sylow Theorems, G contains exactly one Sylow 5-subgroup H so that H is normal in G. Let K be a Sylow 2-subgroup of G (with order 4). By Lagrange's theorem, $H \cap K = \{1\}$, and $HK = G$ since

$$|HK| = \frac{|H||K|}{|H \cap K|} = |H||K| = |G|.$$

Therefore, it follows that $G \cong H \rtimes_\varphi K$ for some group homomorphism $\varphi : K \to \mathrm{Aut}(H)$. Since H has prime order, it is a cyclic group; in particular, $H \cong \mathbb{Z}_5$. Therefore, $\mathrm{Aut}(H) \cong (\mathbb{Z}_5)^\times \cong \mathbb{Z}_4$. On the other hand, by the classification of groups of order 4, $K \cong \mathbb{Z}_4$ or $K \cong \mathbb{Z}_2 \times \mathbb{Z}_2$. We will consider each case independently.

(Case I) Assume that $K \cong \mathbb{Z}_4$. Each homomorphism $\varphi : K \to \mathrm{Aut}(H) \cong \mathbb{Z}_4$ is uniquely determined by $\varphi(1)$, with the caveat that the order of $\varphi(1)$ divides the order of 1. Since the order of 1 is 4 in $\mathbb{Z}_4$, $\varphi(1) \in \{0, 1, 2, 3\}$, making for four total group homomorphisms.

(i) Consider the homomorphism $\varphi_0 : 1 \mapsto 0$. Then since φ_0 maps every element of K to the element $0 \in \mathbb{Z}_4$, one must have that φ_0 is the trivial homomorphism. Therefore, $H \rtimes_{\varphi_0} K$ is isomorphic to the abelian direct product $H \times K \cong \mathbb{Z}_5 \times \mathbb{Z}_4 \cong \mathbb{Z}_{20}$, with the final isomorphism following from the Chinese Remainder Theorem.

(ii) Consider the map $\alpha : \mathbb{Z}_4 \to \mathbb{Z}_4$ defined by $\alpha(x) = 3x$. Then α is easily seen to be an automorphism on $\mathbb{Z}_4$ with $\alpha : 1 \mapsto 3, 3 \mapsto 1$. This means that if $\varphi_1 : K \to \mathrm{Aut}(H)$ and $\varphi_1 : K \to \mathrm{Aut}(H)$ are the maps mapping $1 \mapsto 1$ and $1 \mapsto 3$, respectively, then $\varphi_3 = \varphi_1 \circ \alpha$. Since these two homomorphisms differ by an automorphism on the domain group K, we conclude that the nonabelian semidirect product groups $\mathbb{Z}_5 \rtimes_{\varphi_1} \mathbb{Z}_4$ and $\mathbb{Z}_5 \rtimes_{\varphi_3} \mathbb{Z}_4$ are isomorphic.

(iii) Now consider the homomorphism $\varphi_2 : 1 \mapsto 2$. Since the order of 2 in $\mathbb{Z}_4$ is 2, there cannot exist any automorphisms on $\mathbb{Z}_4$ that maps 2 to any other element of $\mathbb{Z}_4$. Moreover, the image of φ_2 has order 2, while none of the other homomorphisms have image with order 2. Thus, the nonabelian semidirect product group $H \rtimes_{\varphi_2} K$ cannot be isomorphic to $\mathbb{Z}_{20}$ nor to the nonabelian group $\mathbb{Z}_5 \rtimes_{\varphi_1} \mathbb{Z}_4$.

Therefore, for the case with $K \cong \mathbb{Z}_4$, we have found three distinct isomorphism classes of groups of order 20, one of which is abelian.

(Case 2) Assume that $K \cong \mathbb{Z}_2 \times \mathbb{Z}_2$. Each homomorphism $\varphi : K \to \mathrm{Aut}(H) \cong \mathbb{Z}_4$ is uniquely determined by $\varphi((1,0))$ and $\varphi((0,1))$ (where $(1,0), (0,1)$ are the two generators of $\mathbb{Z}_2 \times \mathbb{Z}_2$, with the caveat that the order of $\varphi(\cdot)$ divides the order of $\cdot$. Since the order of $(1,0)$ and $(0,1)$ is 2, $\varphi((1,0)), \varphi((0,1)) \in \{0, 2\}$, making for four total group homomorphisms.

(i) Consider the homomorphism $\varphi_0 : (0,1), (1,0) \mapsto 0$. Then φ_0 is the trivial homomorphim. Thus, $\mathbb{Z}_5 \rtimes_\varphi (\mathbb{Z}_2 \times \mathbb{Z}_2)$ is isomorphic to the abelian direct product $\mathbb{Z}_{10} \times \mathbb{Z}_2$.

(ii) On the other hand, since $\mathrm{Aut}(\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}) \cong GL_2(\mathbb{F}_2)$ acts transitively on the nonzero elements of K, all three of the remaining homomorphisms differ only by a precomposition by an automorphism of K. Thus, the nonabelian semidirect product groups corresponding to each such homomorphism are isomorphic to each other.

Altogether, we conclude that there are five distinct isomorphism classes of groups of order 20, 2 which are abelian.

Problem 2010-J-II-5. Classify (up to isomorphism) all groups of order 45.

Let G denote a group of order $45 = 3^2 \cdot 5$. By the Sylow theorems, if n_3 and n_5 denote the number of Sylow 3- and Sylow 5-subgroups, then

$$n_3 \in \{1,5\} \cap \{1,4,7,\dots\} = \{1\}.$$
$$n_5 \in \{1,3\} \cap \{1,6,11,\dots\} = \{1\}.$$

Thus G has a normal Sylow 3-subgroup H and a normal Sylow 5-subgroup K. Since by Lagrange's theorem, $H \cap K = \{1\}$ and $HK = G$, it follows that G is isomorphic to the abelian direct product $H \times K$. Since K has prime order 5, $K \cong \mathbb{Z}_5$. On the other hand, by classification of groups of order p^2, $H \cong \mathbb{Z}_9$ or $\mathbb{Z}_3 \times \mathbb{Z}_3$. Thus, we conclude that (up to isomorphism) there are precisely 2 groups of order 45: (1) $\mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5 \cong \mathbb{Z}_3 \times \mathbb{Z}_{15}$ (by the Chinese Remainder Theorem), and (2) $\mathbb{Z}_9 \times \mathbb{Z}_5 \cong \mathbb{Z}_{45}$ (by the Chinese Remainder Theorem).

Problem 2008-J-I-3. Classify (up to isomorphism) all groups of order 28.

Let G denote a group of order $28 = 2^2 \cdot 7$. By Sylow's theorems, G must contain a normal Sylow 7-subgroup N of order 7. Let K denote a Sylow 2-subgroup of order 4. By Lagrange's theorem, $N \cap K = \{1\}$. Therefore, since

$$|NK| = |N||K|/|N \cap K| = |N||K| = |G|,$$

one has $NK = G$. Hence, we must have $G \cong N \rtimes_\varphi K$ for some group homomorphism $\varphi : K \to \mathrm{Aut}(N)$. Since N has prime order, it is cyclic of order 7 so that $\mathrm{Aut}(N) \cong (\mathbb{Z}/7\mathbb{Z})^\times \cong \mathbb{Z}/6\mathbb{Z}$. On the other hand, by the classification of groups of order 4, either $K \cong \mathbb{Z}_4$ or $K \cong \mathbb{Z}_2 \times \mathbb{Z}_2$. Therefore, we will consider two cases.

(Case I) Suppose $K \cong \mathbb{Z}_4 = \langle 1 \rangle$. Any homomorphism $\varphi \in \mathrm{Hom}_{\mathbf{Grp}}(K, \mathrm{Aut}(N))$ is uniquely determined by $\varphi(1)$ under the caveat that the order of $\varphi(1) \in \mathbb{Z}_6 \cong \mathrm{Aut}(N)$ divides the order of $1 \in \mathbb{Z}_4$ (which is 4). Therefore, $\varphi(1) \in \{0,3\}$. The homomorphism $\varphi_0 : 1 \mapsto 0$ is the trivial homomorphism, and corresponds to the abelian direct product $G \cong N \times K \cong \mathbb{Z}_7 \times \mathbb{Z}_4 \cong \mathbb{Z}_{28}$ (with the last isomorphism following from the Chinese Remainder Theorem). On the other hand, the nontrivial isomorphism $\varphi_1 : 1 \mapsto 3$ corresponds to a nonabelian semidirect product group $N \rtimes_{\varphi_1} K$, which has the group presentation

$$\langle x, y \mid x^7 = x^4 = 1, yxy^{-1} = x^{-1} \rangle.$$

(Case 2) Suppose $K \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, with the two generators $(0,1)$ and $(1,0)$. Each of these have order 2 in K. Thus any automorphism $\varphi : K \to \mathbb{Z}/6\mathbb{Z}$ is uniquely determined by $\varphi((0,1))$ and $\varphi((1,0))$ with the caveat that the images of these generators divide 2. Hence, $\varphi((0,1)), \varphi((0,1)) \in \{0,3\}$. If φ_0 maps both genertors to 0, then φ_0 is the trivial homomorphism and corresponds to the abelian direct product $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_7 \cong \mathbb{Z}_2 \times \mathbb{Z}_{14}$ (by the Chinese Remainder Theorem). On the other hand, since $\mathrm{Aut}(\mathbb{Z}_2 \times \mathbb{Z}_2) \cong \mathrm{GL}_2(\mathbb{F}_2)$ acts transitively on the nonzero elements of K, all three of the remaining (nontrivial) homomorphisms differ by only a precomposition by an automorphism of K, and thus the nonabelian semidirect product groups corresponding to these homomorphisms are all isomorphic, and has the group presentation

$$\langle x, y, z \mid x^7 = y^2 = z^2 = 1, yz = zy, yxy^{-1} = x^{-1}, zyz^{-1} = z \rangle.$$

Thus altogether, we conclude that up to isomorphism, there exist precisely 4 groups of order 28, two which are abelian.

Problem 2004-A-I-6. Let G and H be groups of order 42 and 55, respectively. Show that the only group homomorphism from G to H is the constant homomorphism.

Let G and H be groups of order 42 and 55, respectively. Let $\varphi : G \to H$ be a group homomorphism, and let $g \in G$ be an arbitrary element. By Lagrange's theorem,

$$|g| \in \{1, 2, 3, 6, 7, 14, 21, 42\} =: A.$$

On the other hand, one must have $|\varphi(g)| \mid |g|$ and

$$|\varphi(g)| \in \{1, 5, 11, 55\} =: B.$$

But the non-identity elements of B are coprime to the non-identity elements in A so that $|\varphi(g)| = 1$. Therefore, for every $g \in G$, $\varphi(g) = 1$, which means that φ is the constant (i.e., trivial) homomorphism.

Problem 2003-A-II-6. Show that Σ_7, the group of permutations of 7 elements, does not contain any subgroup with 15 elements.

Let Σ_7 denote the group of permutations of 7 elements, and assume to the contrary that Σ_7 contains a subgroup H with $15 = 3 \cdot 5$ elements. By Sylow's theorems, H must contain a unique Sylow 5-subgroup of order 5, and a unique Sylow 3-subgroup of order 3. Thus, H contains precisely $15 - (1 + 4 + 2) = 8$ elements of order 15. In particular, Σ_7 must contain some element τ whose order is 15. Let τ be expanded as the product $\alpha_1 \alpha_2 \cdots \alpha_n$ of disjoint cycles. Then since Σ_7 is the permutation group on 7 letters, $\sum_1^n \ell_i = 7$, where for each i, ℓ_i denotes the length of α_i. On the other hand, since $|\tau| = 15$, we must have $\operatorname{lcm}_i \alpha_i = 15$. Looking at the partitions of 7, we find there is no disjoint cycle decomposition of τ that satisfies both criteria. This suggests that τ cannot have order 15, which is a contradiction! Thus, by contradiction, Σ_7 cannot contain a subgroup of order 15.

Problem 2003-J-I-6.
(a) Prove that a group of order p^2, p a prime number, is abelian.
(b) Classify groups of order p^2 up to isomorphism.

(a) Let G be a group of order p^2, where p is prime, and let $Z(G)$ denote the center of G. By Lagrange's theorem, $|Z(G)| \in \{1, p, p^2\}$. If $|Z(G)| = p^2$, then we are done. Now assume to the contrary that $|Z(G)| = p$. Then $|G/Z(G)| = |G|/|Z(G)| = p^2/p = p$. By classification of prime order groups, it follows that $G/Z(G)$ is cyclic, and therefore, G is abelian. Now suppose $|Z(G)| = 1$. By the class equation, one has
$$|G| = |Z(G)| + \sum_{i=1} [G : C_G(g_i)].$$
Each $[G : C_G(g_i)]$ must be a multiple of p. Thus taking the modulus of both sides of the above equation with respect to p, one obtains $0 \equiv 1 \pmod{p}$, which is a contradiction. Thus by contradiction, we conclude that $|Z(G)| = p^2$, and so G is abelian.
(b) Now we know that every nonidentity element of G must have order p or p^2. If G contains an element y of order p^2, then since $|\langle y \rangle| = |G|$, G must be a cyclic abelian group of order p^2; i.e., $G \cong \mathbb{Z}_{p^2}$. Now suppose every nonidentity element of G has order exactly p. Then let $x \in G$ be an element of order p, and let $y \in G \setminus \langle x \rangle$. Then $|\langle y \rangle| = p$ and $\langle x \rangle \cap \langle y \rangle = 1$. Thus since $\langle x \rangle \langle y \rangle = G$, one has that $G \cong \langle x \rangle \times \langle y \rangle \cong \mathbb{Z}_p \times \mathbb{Z}_p$.

Problem D&F-5.2-1. In each of parts (a) to (e) give the number of nonisomorphic abelian groups of the specified order — do not list the groups: (a) order 100, (b) order 576, (c) order 1155, (d) order 42875, (e) order 2704.

(a) We begin by noting that $100 = 2^2 \cdot 5^2$. The number of partitions of 2 is 2. Thus, we conclude that there are exactly 4 nonisomorphic abelian groups of order 100.
(b) We begin by noting that $576 = 2^6 \cdot 3^2$. As noted above, the number of partitions of 2 is 2, and now, the number of partitions of 6 is 11. Thus, the total number of nonisomorphic abelian groups of order 576 is 22.

(c) We begin by noting that $1155 = 3 \cdot 5 \cdot 7 \cdot 11$. Since 1155 is a squarefree integer, we conclude that there exists exactly one nonisomorphic abelian group of the specified order.

(d) We start by noting that $42875 = 5^3 \cdot 7^3$. The number of partitions of 3 is 3. Thus, there exist exactly 9 nonisomorphic abelian groups of order 42875.

(e) We start by noting that $2704 = 2^4 \cdot 13^2$. The number of partitions of 4 is 5. Thus, there exist exactly 10 nonisomorphic abelian groups of order 2704.

3.7 Module Theory

Problem 2025-J-I-5. Let R be a nonzero commutative ring with multiplicative identity. Let m and n be distinct positive integers. Prove that R^n and R^m, with their natural structure of R-modules, are not isomorphic R-modules.

Let R be a nonzero commutative ring with multiplicative identity, and assume that $R^m \cong R^n$. Let I denote a maximal ideal of R, and consider the quotient $F = R/I$, which must be a field since I is maximal. Then since $R^m \cong R^n$, tensoring on the left by F gives the isomorphism $F \otimes_R R^m \cong F \otimes_R R^n$. But as R-modules, for any k, $F \otimes_R R^k \cong F^k$. Thus we have $F^m \cong F^n$ as vector spaces over F. This means that $m = n$. Therefore, by the contrapositive, if $n \neq m$, then $R^m \ncong R^n$.

Problem 2025-J-II-3 (Modified) . Let V be a vector space of dimension n over $\mathbb{Q}$. Let $T : V \to V$ be a linear transformation with minimal polynomial $x^4 - x - 2$ over $\mathbb{Q}$. Show that n must be even.

Let V be a vector space of dimension n over $\mathbb{Q}$, and let $T : V \to V$ be a linear transformation with minimal polynomial $x^4 - x - 2$ over $\mathbb{Q}$. We first note that V is a finitely generated module over the PID $\mathbb{Q}[x]$, where x acts by the linear operator T. Thus by the Structure Theorem for Finitely Generated Modules over PIDs, we have the decomposition,

$$V \cong \bigoplus_{i=1}^{N} \frac{\mathbb{Q}[x]}{(\alpha_i(x))},$$

where each $\alpha_i(x)$ is an invariant factor of T, $\alpha_1(x) \mid \alpha_2(x) \mid \cdots \mid \alpha_N(x)$, and $\alpha_N(x)$ is the minimal polynomial of T. It is straightforward to show that $\alpha_N(x)$ is irreducible over $\mathbb{Q}$. This means that each nonconstant $\alpha_i(x)$ is an associate of $\alpha_N(x)$ (and thus has degree 4). The constant $\alpha_i(x)$ all have degree 0. Thus, since

$$n = \dim V = \sum_{i=1}^{N} \deg \alpha_i(x)$$

is divisible by 4 (and hence by 2), we conclude that V is an even-dimensional vector space.

Problem 2025-J-II-3. Let V be a vector space of dimension n over $\mathbb{Q}$. Let $T : V \to V$ be a linear transformation with minimal polynomial $x^4 - x^2 - 2$ over $\mathbb{Q}$. Show that n must be even.

Let V be a vector space of dimension n over $\mathbb{Q}$, and let $T : V \to V$ be a linear transformation with minimal polynomial $x^4 - x - 2$. We first note that V is a finitely generated module over the PID $\mathbb{Q}[x]$, where x acts by the linear operator T. Thus by the Structure Theorem for Finitely Generated Modules over PIDs, we have the decomposition,

$$V \cong \bigoplus_{i=1}^{N} \frac{\mathbb{Q}[x]}{(\alpha_i(x))},$$

where each $\alpha_i(x)$ is an invariant factor of T, $\alpha_1(x) \mid \alpha_2(x) \mid \cdots \mid \alpha_N(x)$, and $\alpha_N(x)$ is the minimal polynomial of T. We note that $\alpha_N(x) = x^4 - x^2 - 2$ is reducible over $\mathbb{Q}$ as the product $(x^2 + 1)(x^2 - 2)$ of irreducible quadratics. Thus, since

$$n = \dim V = \sum_{i=1}^{N} \deg \alpha_i(x),$$

and the degree of each direct summand is even, we conclude that n is even.

Problem 2021-A-I-5. Let R be an integral domain with field of fractions Q. Show that $Q/R \otimes_R Q/R = 0$.

Let R be an integral domain with field of fractions Q. We note that every tensor in $Q/R \otimes_R Q/R$ is a finite sum of the tensors of the form,

$$\left(\frac{a}{b} + R\right) \otimes_R \left(\frac{c}{d} + R\right),$$

where $a, b, c, d \in R$ and $b, d \neq 0$. We compute,

$$\frac{a}{b} \otimes_R \frac{c}{d} = d\left(\frac{a}{db} \otimes_R \frac{c}{d}\right) = \frac{a}{db} \otimes_R c.$$

But since $c \in R$, its representative class in Q/R is the zero class. Thus, $(a/db) \otimes c = 0$. Since every such simple tensor is identically zero, we conclude that every tensor in $Q/R \otimes_R Q/R$ is identically zero so that $Q/R \otimes_R Q/R = 0$.

Problem 2005-J-II-4. Let $A : \mathbb{R}^4 \to \mathbb{R}^4$ be a linear map such that the characteristic polynomial of A is $\chi_A(t) = t^4 + 1$. How many A-invariant subspaces are there in $\mathbb{R}^4$? (A subspace W is called A-invariant if $AW \subset W$.)

Let $A : \mathbb{R}^4 \to \mathbb{R}^4$ be a linear map such that the characteristic polynomial of A is $\chi_A(t) = t^4+1$. Since $\mathbb{R}^4$ is a finitely generated vector space, we note that $\mathbb{R}^4$ can be viewed as a finitely generated module over $\mathbb{R}[x]$, where x acts by the linear operator A. Thus by the Structure Theorem for Finitely Generated Modules over PIDs, we have the decomposition,

$$V \cong \bigoplus_{i=1}^{N} \frac{\mathbb{R}[x]}{(\alpha_i(x))},$$

where each $\alpha_i(x)$ is an invariant factor of A, $\alpha_1(x) \mid \alpha_2(x) \mid \cdots \mid \alpha_N(x)$, and $\alpha_1(x)\alpha_2(x)\cdots\alpha_N(x) = \chi_A(x) = x^4 + 1$. It is immediately straightforward to see that $\chi_A(x)$ has no real roots so that it cannot be decomposed as a produt containing a linear factor. So, we will check if $\chi_A(x)$ is equal to the product of two irreducible quadratics. Doing so, we conclude that

$$(x^4 + 1) = (x^2 + \sqrt{2}x + 1)(x^2 - \sqrt{2}x + 1).$$

Each A-invariant subspace of $\mathbb{R}^4$ corresponds to one of the direct summands above. Thus since $\chi_A(t)$ has four divisors (1, $x^2 \pm \sqrt{2}x+1$, and x^4+1), we conclude that there are exactly 4 A-invariant subspaces of $\mathbb{R}^4$.

Problem D&F-10.5-1. Suppose that

$$\begin{tikzcd} A \arrow[r, "\psi"] \arrow[d, "\alpha"] & B \arrow[r, "\varphi"] \arrow[d, "\beta"] & C \arrow[d, "\gamma"] \\ A' \arrow[r, "\psi'"] & B' \arrow[r, "\varphi'"] & C' \end{tikzcd}$$

is a commutative diagram of groups and that the rows are exact. Prove that

- **(a)** if φ and α are surjective, and β is injective then γ is injective.
- **(b)** if ψ', α, and γ are injective, then β is injective.
- **(c)** If φ, α, and γ are surjective, then β is surjective.
- **(d)** if β is injective, α and γ are surjective, then γ is injective.

(a) Let $c \in C$ so that $\gamma(c) = 0$. Since φ is surjective, there exists some $b \in B$ so that $\varphi(b) = c$. By commutativity of the diagram, $\varphi'(\beta(b)) = 0$ so that $\beta(b) \in \ker\varphi'$. By exactness of the rows, $\beta(b) \in \operatorname{img}\psi'$ so that there exists $a' \in A'$ with $\psi'(a') = \beta(b)$. Since α is surjective, there exists $a \in A$ so that $\psi'(\alpha(a)) = \beta(b)$. By commutativity of the diagram, $\beta(\psi(a)) = \beta(b)$; by injectivity of β, $b = \psi(a)$. Since $b \in \operatorname{img}\psi$, by exactness, $b \in \ker\varphi$. This means that $c = \varphi(b) = 0$. Hence, γ is injective.

(b) Let $b \in B$ so that $\beta(b) = 0$. This means that $\varphi'(\beta(b)) = 0$; i.e., $\beta(b) \in \ker \varphi'$. By commutativity of the right square, $\gamma(\varphi(b)) = 0$; since γ is injective, $\varphi(b) = 0$. I.e., $b \in \ker \varphi$; by exactness, there exists $a \in A$ such that $\psi(a) = b$. Then by commutativity of the diagram, $\psi'(\alpha(a)) = \beta(b) = 0$; since ψ' is injective, $\alpha(a) = 0$. Since α is injective, $a = 0$. Therefore, $b = \psi(a) = 0$. Hence, this concludes the proof that β is injective.

(c) Let $b' \in B'$. Since γ is surjective, there exists a $c \in C$ so that $\gamma(c) = \varphi'(b)$. Since φ is surjective, there exists a $b \in B$ such that $c = \varphi(b) \implies \gamma(\varphi(b)) = \varphi'(b)$. By commutativity of the diagram, $\varphi'(\beta(b)) = \varphi'(b)$. So consider the element $\beta(b) - b \in B'$. By the previous observation, $\beta(b) - b \in \ker \varphi'$ so that by exactness, there exists $a' \in A'$ with $\psi'(a') = \beta(b) - b'$. And by surjectiveness of α, there exists $a \in A$ with $\beta(b) - b' = \psi'(\alpha(a))$. By commutativity of the right square, $\psi'(\alpha(a)) = \beta(\psi(a))$ so that in fact, $\beta(b) - b' = \beta(\psi(a)) \implies b' = \beta(b - \psi(a))$, where $b - \psi(a) \in B$. Hence, β is surjective.

(d) Let $c \in C$ so that $\gamma(c) = 0$.

Problem D&F-10.5-2. Suppose that

$$\begin{array}{ccccccc}
A & \xrightarrow{\psi} & B & \xrightarrow{\varphi} & C & \xrightarrow{\phi} & D \\
\downarrow{\scriptstyle\alpha} & & \downarrow{\scriptstyle\beta} & & \downarrow{\scriptstyle\gamma} & & \downarrow{\scriptstyle\delta} \\
A' & \xrightarrow{\psi'} & B' & \xrightarrow{\varphi'} & C' & \xrightarrow{\phi'} & D'
\end{array}$$

is a commutative diagram of groups and that the rows are exact. Prove that

(a) if α is surjective, and β, δ are injective, then γ is injective.

(b) if δ is injective, and α, γ are surjective, then β is surjective.

(a) Let α be surjective, and β, δ be injective. Let $c \in C$ such that $\gamma(c) = 0$. Then $\phi'(\gamma(c)) = 0$. By commutativity of the right square, it then follows that $\delta(\phi(c)) = 0$; since δ is injective, $\phi(c) = 0$, which means $c \in \ker \phi$. By exactness of the first row, $c \in \operatorname{im} \varphi$. I.e., there exists some $b \in B$ so that $\varphi(b) = c$. But then, by commutativity of the diagram, $\varphi'(\beta(b)) = 0$ so that $\beta(b) \in \ker \varphi'$. By exactness of the second row, $\beta(b) \in \operatorname{im} \psi'$, which means there exists $a' \in A'$ with $\psi'(a') = \beta(b)$. Since α is surjective, there exists $a \in A$ so that $\psi'(a') = \psi'(\alpha(a)) = \beta(b)$. By commutativity of the diagram, $\beta(\psi(a)) = \beta(b)$ so that $\psi(a) = b$ (by injectivity of β). This implies $b \in \operatorname{im} \psi$. By exactness of the first row, $b \in \ker \varphi$. Hence, $c = \varphi(b) = 0$, which proves that γ is injective.

(b) Let δ be injective, and α, γ be surjective. Let $b' \in B'$. By surjectivity of γ, there exists some $c \in C$ so that $\gamma(c) = \varphi'(b')$. By exactness of the second row, $\varphi'(b') \in \ker \phi'$. By commutativity of the diagram, $\delta(\phi(c)) = 0$. Since δ is injective, $\phi(c) = 0$. By exactness of the first row, there exists some $b \in B$ so that $c = \varphi(b)$. This implies $\gamma(\varphi(b)) = \varphi'(b')$. By commutativity of the middle square, $\varphi'(\beta(b) - b') = 0$ so that, by exactness, $\beta(b) - b' \in \operatorname{im} \psi'$. Then by surjectivity of α, there exists some $a \in A$ so that $\psi'(\alpha(a)) = \beta(b) - b'$. By commutativity of the left square, $\beta(\psi(a)) = \psi'(\alpha(a)) = \beta(b) - b'$ so that $b' = \beta(b - \psi(a))$. Hence, since $b - \psi(a) \in B$, we have proven that β is surjective.

Problem D&F-10.5-3. Let P_1 and P_2 be R-modules. Prove that $P_1 \oplus P_2$ is a projective R-module if and only if both P_1 and P_2 are projective.

($\Leftarrow$) Let P_1 and P_2 be projective modules. Fix arbitrary left R-modules N, M and let $f : N \twoheadrightarrow M$ be a surjection. Let $g : P_1 \oplus P_2 \to M$ be a module homomorphism. Then by the universal property for direct sums, g is uniquely determined by the maps $(g_i)_{i=1,2}$, where for each i, $g_i = g \circ \iota_i : P_i \to M$ is a module homomorphism and $\iota_i : P_i \to P_1 \oplus P_2$ is the canonical inclusion $P_i \hookrightarrow P_1 \oplus P_2$. Since each P_i is a projective R-module, there exist module homomorphisms $h_i : P_i \to N$ such that $fh_i = g_i$. By the universal property of direct sums, there exists a unique module homomorphism $h : P_1 \oplus P_2 \to N$ such that for each i, $h \circ \iota_i = h_i$, so that in fact, $fh \circ \iota_i = g_i = g \circ \iota_i$ for each i. But by uniqueness from the universal property, it follows that $fh = g$. Hence, by definition, $P_1 \oplus P_2$ is projective.

($\Rightarrow$) Assume $P_1 \oplus P_2$ is projective. Again fix arbitrary left R-modules N and M, and let $f : N \twoheadrightarrow M$ be a surjective module homomorphism. Let $g_i : P_i \to M$ be a module homomorphism for each $i = 1, 2$. By the universal property for direct sums, $(g_i)_{i=1,2}$ determine a unique homomorphism $g : P_1 \oplus P_2 \to M$; by definition of a projective module, there exists a module homomorphism $h : P_1 \oplus P_2 \to N$ such that $fh = g$. But by the universal property for direct sums, h is uniquely determined by the family of maps $(h_i)_{i=1,2}$, where for each i, $h_i = h \circ \iota_i : P_i \to N$ (where $\iota_i : P_i \hookrightarrow P_1 \oplus P_2$ is the canonical inclusion) and $fh = g$. Then we compute $f(h \circ \iota_1) = (g \circ \iota_i) \implies fh_i = g_i$. Hence, by definition, each P_i is projective.

Problem D&F-10.5-4. Let Q_1 and Q_2 be R-modules. Prove that $Q_1 \oplus Q_2$ is an injective R-module if and only if both Q_1 and Q_2 are injective.

Let Q_1 and Q_2 be R-modules. Fix arbitrary R-modules A and B, and an injection $f : A \hookrightarrow B$. Assume that Q_1 and Q_2 are injective. Let $g : A \to Q_1 \oplus Q_2$ be a module homomorphism. By the universal property for direct sums, g is uniquely determined by the family of maps $(g_i)_{i \in I}$, where for each i, $g_i = \pi_i \circ g : A \to Q_i$ and $\pi_i : Q_1 \oplus Q_2 \to Q_i$ are the canonical projection maps. Since each Q_i is injective, there exist morphisms $h_i : B \to Q_i$ such that $h_i f = g_i$. But by the universal property for direct sums, the family of maps $(h_i)_{i \in I}$ uniquely determines a map $h : B \to Q_1 \oplus Q_2$ so that for each i, $h_i = \pi_i h$. But then we observe that

$$\pi_i g = g_i = h_i f = (\pi_i h) f = \pi_i (hf), \qquad i = 1, 2.$$

By uniqueness from the universal property for direct sums, we concude that $hf = g$, and so $Q_1 \oplus Q_2$ is injective. Now suppose $Q_1 \oplus Q_2$ are injective, and let $g_i : A \to Q_i$ be morphisms. By the universal property for direct sums, these maps uniquely determine a map $g : A \to Q_1 \oplus Q_2$ such that for each i, $g_i = \pi_i \circ g$. Since $Q_1 \oplus Q_2$ is injective, there exists a morphism $h : B \to Q_1 \oplus Q_2$ so that $hf = g$. But then, h is uniquely determined by maps $h_i : B \to Q_i$, where for each i, $h_i = \pi_i \circ h$. We then note that

$$h_i f = (\pi_i h) f = \pi_i (hf) = \pi_i g = g_i.$$

Hence, we conclude that Q_1 and Q_2 are injective.

Problem D&F-10.5-5. [!! Complete Later !!]

Problem D&F-10.5-6. Prove that the following are equivalent for a ring R:

(i) Every R-module is projective.

(ii) Every R-module is injective.

Let R be a ring. We will first prove (i) $\Rightarrow$ (ii). Let P be an R-module, A and B be arbitrary R-modules, $f : A \hookrightarrow B$ an injection, and $g : A \to P$ be an R-module homomorphism. It suffices to show there exists a homorphism $h : B \to P$ such that $hf = g$. Consider the quotient module B/A; taking $\pi : B \to B/A$ to be the canonical projection map, we obtain the short exact sequence

$$0 \to A \xrightarrow{f} B \xrightarrow{\pi} B/A \to 0.$$

Since every R-module is projective by hypothesis, B/A is a projective R-module. Therefore, the short exact sequence splits, which means that $B \cong A \oplus (B/A)$. Now let $k : B/A \to P$ be any R-module homomorphism and define the map $h : B \to P$ as follows: $h(a, c) = g(a) + k(c)$ for any $(a, c) \in A \oplus (B/A)$. We observe that

$$hf(a) = h(a, 0) = g(a) + k(0) = g(a), \qquad \forall a \in A,$$

so that $hf = g$. Hence, P is an injective R-module.

Now we will show that (ii) $\Rightarrow$ (i). Let Q be an R-module, A and B be arbitrary fixed R-modules, $f : A \twoheadrightarrow B$ a surjection, and $g : Q \to B$ be an R-module homomorphism. It suffices to prove that there exists an R-module homomorphism $h : Q \to A$ such that $fh = g$. Let $K = \ker f$, and consider the short exact sequence

$$0 \to K \xrightarrow{i} A \xrightarrow{f} B \to 0,$$

where $i : K \hookrightarrow A$ is an injective mapping. Since every R-module is injective by hypothesis (i.e., K is injective), the short exact sequence splits, which means that $A \cong B \oplus K$. Since the sequence splits, there also exists a morphism $s : B \to A$ such that $f \circ s = \mathrm{id}_B$. Let $h = sg$. Then we observe that $fh = (fs)g = \mathrm{id}_B g = g$.

Problem D&F-10.5-7. Let A be a nonzero finite abelian group.
(a) Prove that A is not a projective $\mathbb{Z}$-module.
(b) Prove that A is not an injective $\mathbb{Z}$-module.

Let A be a nonzero finite abelian group.

(a) Assume to the contrary that A is a projective $\mathbb{Z}$-module. This implies that A must be the direct summand of a free $\mathbb{Z}$-module. But $\mathbb{Z}$-modules are precisely the abelian groups. Hence, A must be the direct summand of a free abelian group. Since subgroups of free abelian groups are free abelian, A must be a free abelian group. But there exists exactly one free abelian group of finite cardinality, which is the trivial group $\{0\}$. This contradicts our hypothesis that A is nonzero. Hence, A cannot be a projective $\mathbb{Z}$-module.

(b) Now assume to the contrary that A is an injective $\mathbb{Z}$-module. By Proposition 36, A must necessarily be a divisible $\mathbb{Z}$-module. But the only finite divisible $\mathbb{Z}$-module is the trivial module $\{0\}$. This contradicts our hypothesis that A is nonzero. Hence, A cannot be an injective $\mathbb{Z}$-module.

Problem D&F-10.5-9. Assume R is commutative with 1.
(a) Prove that the tensor product of two free R-modules is free. [Use the fact that tensor products commute with direct sums.]
(b) Use (a) to prove that the tensor product of two projective R-modules is projective.

Assume R is commutative with 1.

(a) Let M_1, M_2 be fixed arbitrary free R-modules so that $M_1 \cong R^{(I)}$ and $M_2 \cong R^{(J)}$ for some index sets I, J. Since the tensor product commutes with direct sums,

$$M_1 \otimes_R M_2 \cong R^{(I)} \otimes_R R^{(J)} = \bigoplus_{i\in I} R^{(i)} \otimes_R \bigoplus_{j\in J} R^{(j)} = \bigoplus_{(i,j)\in I\times J} R^{(i,j)} = R^{(I\times J)}.$$

Hence, we conclude that $M_1 \otimes_R M_2$ is a free R-module.

(b) Now let M_1, M_2 be projective R-modules. Since every projective R-module is the direct summand of a free module, suppose there exist R-modules N_1, N_2 so that $M_1 \oplus N_1 \cong R^{(I)}$ and $M_2 \oplus N_2 \cong R^{(J)}$ for some index sets I, J. Then

$$R^{(I\times J)} \cong R^{(I)} \otimes_R R^{(J)} \cong (M_1 \oplus N_1) \otimes_R (M_2 \oplus N_2) = (M_1 \otimes_R M_2) \oplus (M_1 \otimes_R N_2) \oplus \cdots \oplus (N_1 \otimes_R N_2).$$

Since $M_1 \otimes_R M_2$ is the direct summand of a free R-module, we conclude that $M_1 \otimes_R M_2$ is a projective R-module. This concludes the claim.

Problem D&F-10.5-10. Let R and S be rings with 1 and let M and N be left R-modules. Assume also that M is an (R,S)-bimodule.
(a) For $s \in S$ and for $\varphi \in \mathrm{Hom}_R(M,N)$, define $(s\varphi) : M \to N$ by $(s\varphi)(m) = \varphi(ms)$. Prove that $s\varphi$ is a homomorphism of left R-modules, and that this action of S on $\mathrm{Hom}_R(M,N)$ makes it into a left S-module.

(a) First, we will show that $s\varphi : M \to N$ is a left R-module homomorphism. Let $m_1, m_2 \in M$, and $r \in R$. Then

$$\begin{aligned}
(s\varphi)(m_1 + m_2) &= \varphi((m_1 + m_2)s) \\
&= \varphi(m_1 s + m_2 s) \\
&= \varphi(m_1 s) + \varphi(m_2 s) = (s\varphi)(m_1) + (s\varphi)(m_2). \\
(s\varphi)(rm_1) &= \varphi(rm_1 s) \\
&= \varphi(r(m_1 s)) = r\varphi(m_1 s) = r(s\varphi)(m_1).
\end{aligned}$$

Above, we are using the fact that M is a *right* S-module when multiplying m_i by s on the right. Hence, we conclude that $(s\varphi)$ is a left R-module homomorphism. Now, we will show that this action of S on $\mathrm{Hom}_R(M,N)$ makes it a left S-module. We already have, by definition of the hom set, that $\mathrm{Hom}_R(M,N)$ is an abelian additive group. Now fix an arbitrary $x \in M$, let $s, \widetilde{s} \in S$, and $\varphi, \phi \in \mathrm{Hom}_R(M,N)$. Then we observe the following:

$$\begin{aligned}
(s(\varphi+\phi))(x) &= (\varphi+\phi)(xs) \\
&= \varphi(xs) + \phi(xs) \\
&= (s\varphi)(x) + (s\phi)(x). \\
\implies (s\cdot(\varphi+\phi)) &= (s\varphi) + (s\phi). \\
(s+\widetilde{s})\cdot\varphi(x) &= \varphi(x\cdot(s+\widetilde{s})) \\
&= \varphi(xs + x\widetilde{s}) \qquad \text{(since } M \text{ is a right } S\text{-module)} \\
&= \varphi(xs) + \varphi(x\widetilde{(s)}) = (s\varphi)(x) + (\widetilde{s}\varphi)(x) \\
\implies (s+\widetilde{s})\cdot\varphi &= (s\varphi) + (\widetilde{s}\varphi). \\
(s\widetilde{s})(\varphi)(x) &= \varphi(xs\widetilde{s}) = \varphi((xs)\widetilde{s}) \\
&= \widetilde{s}\cdot\varphi(xs) \\
&= s(\widetilde{s}\varphi)(x).
\end{aligned}$$

That the operation satisfies the unit condition is straightforward to verify. Hence, $\mathrm{Hom}_R(M,N)$ is a left S-module.

Problem D&F-10.5-12. Let A be an R-module, let I be a nonempty index set and for each $i \in I$, let B_i be an R-module. Prove the following isomorphisms of abelian groups; when R is commutative also that these are R-module isomorphisms. (Arbitrary direct sums and direct products of modules are introduced in Exercise 20 of Section 3.)

(a) $\mathrm{Hom}_R(\bigoplus_{i\in I} B_i, A) \cong \prod_{i\in I} \mathrm{Hom}_R(B_i, A)$.

(b) $\mathrm{Hom}_R(A, \prod_{i\in I} B_i) \cong \prod_{i\in I} \mathrm{Hom}_R(A, B_i)$.

(a) Let $g \in \mathrm{Hom}_R(\bigoplus_{i\in I} B_i, A)$; by the universal property for direct sums, g is uniquely determined by the family of maps $(g_i)_{i\in I}$, where for each i, $g_i = g \circ \iota_i$ and $g_i : B_i \to A$. Hence, define the map $\mathcal{F} : \mathrm{Hom}_R(\bigoplus_{i\in I} B_i, A) \to \prod_{i\in I} \mathrm{Hom}_R(B_i, A)$ as follows:

$$\mathcal{F}(\varphi) = (\varphi_i)_{i\in I} \implies \mathcal{F}(\varphi)((b_i)_{i\in I}) = ((\varphi\circ\iota_i)(b_i))_{i\in I} =: (\varphi_i(b_i))_{i\in I},$$

where $(b_i)_{i\in I} \in \bigoplus_{i\in I} B_i$. First, we will show that $\mathcal{F}$ is a group homomorphism. Let $\varphi, \phi \in \mathrm{Hom}_R(\bigoplus_{i\in I} B_i, A)$, and fix some arbitrary $(b_i)_{i\in I} \in \bigoplus_{i\in I} B_i$. Then we observe that

$$\begin{aligned}
\mathcal{F}(\varphi+\phi)((b_i)) &= ((\varphi+\phi)_i(b_i))_{i\in I} \\
&= (\varphi_i(b_i) + \phi_i(b_i))_{i\in I} \\
&= (\varphi_i(b_i))_{i\in I} + (\phi_i(b_i))_{i\in I} \\
&= \mathcal{F}(\varphi)((b_i)) + \mathcal{F}(\phi)((b_i)).
\end{aligned}$$

Hence, $\mathcal{F}$ is a group homomorphism. Now let $(\varphi_i)_{i\in I} \in \prod_{i\in I} \mathrm{Hom}_R(B_i, A)$. By the universal property for direct sums, there exists a unique morphism $\varphi : \bigoplus_{i\in I} B_i \to A$ such that for each i, $\varphi_i := \varphi \circ \iota_i : B_i \to A$. Hence, $\mathcal{F} : \varphi \mapsto (\varphi_i)_{i\in I}$ so that it is surjective. Finally, we will show that $\mathcal{F}$ is injective. Let $\varphi, \phi \in \mathrm{Hom}_R(\bigoplus_{i\in I} B_i, A)$ and assume that $\mathcal{F}(\varphi) = \mathcal{F}(\phi)$. Then for any $b \in \bigoplus_{i\in I} B_i$, we note that

$$0 = \mathcal{F}(\varphi)(b) - \mathcal{F}(\phi)(b) = \mathcal{F}(\varphi - \phi)(b),$$

so that $\varphi - \phi$ must be the trivial (i.e., zero) homomorphism. Hence, $\varphi = \phi$ so that $\mathcal{F}$ is injective. Altogether, we conclude that $\mathcal{F}$ is an isomorphism of groups.

(b) Now let $g \in \mathrm{Hom}_R(A, \prod_{i\in I} B_i)$. By the universal property for direct products, g is uniquely determined by a family of maps $(g_i)_{i\in I} : A \to B_i$ where for each i, $g_i = \pi_i \circ g$. Therefore, define the map $\mathcal{F} : \mathrm{Hom}_R(A, \prod_{i\in I} B_i) \to \prod_{i\in I} \mathrm{Hom}_R(A, B_i)$ as follows:

$$\text{for each } \varphi \in \mathrm{Hom}_R(A, \prod_{i\in I} B_i). \qquad \mathcal{F}(\varphi) = (\pi_i \circ \varphi)_{i\in I}.$$

First, we will show that $\mathcal{F}$ is a group homomorphism. Let $\varphi, \phi \in \mathrm{Hom}_R(A, \prod_{i\in I} B_i)$. Then

$$\begin{aligned}\mathcal{F}(\varphi + \phi) &= (\pi_i \circ (\varphi + \phi))_{i\in I} \\ &= (\pi_i \circ \varphi + \pi_i \circ \phi)_{i\in I} \\ &= (\pi_i \circ \varphi)_{i\in I} + (\pi_i \circ \phi)_{i\in I} = \mathcal{F}(\varphi) + \mathcal{F}(\phi).\end{aligned}$$

Now let $(\varphi_i)_{i\in I}$ be a family of maps, where for each i, $\varphi_i : A \to B_i$. Then by the universal property for direct products, there exists a unique homomorphism $\varphi : A \to \prod_{i\in I} B_i$ such that for each i, $\varphi_i = \pi_i \circ \varphi$. Hence, $\mathcal{F} : \varphi \mapsto (\varphi_i)_{i\in I}$, so that this map is surjective. Finally, generalizing the argument from (a), we see that $\mathcal{F}$ is injective. Therefore, $\mathrm{Hom}_R(A, \prod_{i\in I} B_i) \cong \prod_{i\in I} \mathrm{Hom}_R(A, B_i)$.

Problem D&F-10.5-20. Prove that the polynomial ring $R[x]$ in the indeterminate x over the commutative ring R is a flat R-module.

Let R be a commutative ring and $R[x]$ the polynomial ring over R in the indeterminate x. Consider the set $\{x^n : n \geq 0\} \subset R[x]$. It is straightforward to see that this set is linearly independent since each element in this collection has a distinct degree. Moreover, since *any* $p(x) \in R[x]$ can be written as the finite sum

$$a_0 + a_1 x + \cdots + a_n x^n$$

for some nonnegative integer n and $a_0, a_1, \ldots, a_n \in R$,

$$\mathrm{span}\{x^n : n \geq 0\} = R[x].$$

Hence, this set forms a basis for $R[x]$. This means that $R[x]$ is a free R-module:

$$R[x] = \bigoplus_{n=0}^{\infty} R \cdot x^n.$$

Since free R-modules are flat (Corollary 42), we conclude that $R[x]$ is flat.

Problem FR-1-1. Let R be a commutative ring with 1. Let I and J be ideals in R. Let M be an R-module.

(a) Describe $R/I \otimes_R M$ in terms of M, its R-submodules, and its quotient R-modules.

(b) Describe $R/I \otimes_R R/J$ as a quotient of R.

(c) Using the following short exact sequence of R-modules, describe $\mathrm{Tor}_1^R(R/I, R/J)$ in terms of ideals coming from I and J.

$$0 \longrightarrow J \longrightarrow R \longrightarrow R/J \longrightarrow 0.$$

(d) Describe the finitely generated abelian group $\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/9\mathbb{Z}, \mathbb{Z}/27\mathbb{Z})$ in the "standard form" $\mathbb{Z}/a_1\mathbb{Z} \times \cdots \times \mathbb{Z}/a_m\mathbb{Z}$ with $a_1 \mid a_2 \mid \cdots a_{m-1} \mid a_m$, i.e., give the integer m and the sequence of elementary divisors $a_1, \ldots, a_m$.

(a) We claim that $R/I \otimes_R M \cong M/IM$.

(b) We have $R/I \otimes_R R/J \cong R/(I + J)$. To prove this result, construct the map $\theta : R \to R/I \otimes_R R/J$, defined by $\theta(r) = \bar{1} \otimes \bar{r}$. It follows that this map is surjective: given any $R/I \otimes_R R/J \ni \bar{a} \otimes \bar{b} = \bar{1} \otimes \overline{ab}$. Hence, θ maps ab to this desired tensor product. Now we will compute the kernel of this map. Let $i \in I, j \in J$. Then

$$\theta(i + j) = \bar{1} \otimes (\bar{i} + \bar{j}) = \bar{1} \otimes \bar{i} + \bar{1} \otimes \bar{j} = 0,$$

since $\bar{i} = 0 \in R/I$ and $\bar{j} = 0 \in R/J$. Thus, $I + J \subseteq \ker\theta$. To show that $\ker(\theta) \subseteq I + J$, define the map

$$\psi : R/I \otimes_R R/J \to R/(I+J)$$

by

$$\psi(\bar{a} \otimes \bar{b}) = \overline{ab}.$$

This map is well defined: if $a - a' \in I$, then

$$ab - a'b = (a - a')b \in I \subseteq I + J,$$

so $\overline{ab} = \overline{a'b}$ in $R/(I+J)$. Similarly, if $b - b' \in J$, then

$$ab - ab' = a(b - b') \in J \subseteq I + J.$$

Now let $r \in \ker(\theta)$. Then

$$0 = \theta(r) = \bar{1} \otimes \bar{r}.$$

Applying ψ, we obtain

$$0 = \psi(\bar{1} \otimes \bar{r}) = \bar{r},$$

which implies that $r \in I + J$. Therefore,

$$\ker(\theta) \subseteq I + J.$$

Therefore, by the First Isomorphism Theorem, we conclude that $R/(I+J) \cong R/I \otimes_R R/J$.

(c) Consider the short exact sequence

$$0 \longrightarrow J \longrightarrow R \longrightarrow R/J \longrightarrow 0.$$

Then we have the Tor long exact sequence:

$$\cdots \to \operatorname{Tor}_1^R(R/I, R) \to \operatorname{Tor}_1^R(R/I, R/J) \to R/I \otimes_R J \xrightarrow{1\otimes\beta} R/I \otimes_R R \xrightarrow{1\otimes\gamma} R/I \otimes_R R/J \to 0.$$

We know that $\operatorname{Tor}_1^R(R/I, R) = 0$. Hence, we must have

$$\operatorname{Tor}_1^R(R/I, R/J) \cong \ker\left(R/I \otimes_R J \longrightarrow R/I \otimes_R R\right).$$

By (a), $R/I \otimes_R R \cong R/I$, and $R/I \otimes_R J \cong J/IJ$. We are interested in the kernel of the map from $J/IJ \to R/I$. But this kernel is precisely $(I \cap J)/IJ$, and so $\operatorname{Tor}_1^R(R/I, R/J) \cong (I \cap J)/IJ$.

(d) By Part (c), we then have

$$\operatorname{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/9\mathbb{Z}, \mathbb{Z}/27\mathbb{Z}) \cong \frac{(9) \cap (27)}{(243)} \cong \frac{(27)}{(243)} \cong \mathbb{Z}/9\mathbb{Z}.$$

Problem D&F-17.1-6. Let $0 \to \mathcal{A} \xrightarrow{\alpha} \mathcal{B} \xrightarrow{\beta} \mathcal{C} \to 0$ be a short exact sequence of cochain complexes. Prove that if any two of $\mathcal{A}, \mathcal{B}, \mathcal{C}$ are exact, then so is the third. [Use Theorem 2.]

Let $0 \to \mathcal{A} \xrightarrow{\alpha} \mathcal{B} \xrightarrow{\beta} \mathcal{C} \to 0$ be a short exact sequence of cochain complexes. Then there exists a corresponding long exact sequence in cohomology:

$$0 \to H^0(\mathcal{A}) \to H^0(\mathcal{B}) \to H^0(\mathcal{C}) \xrightarrow{\delta_1} H^1(\mathcal{A}) \to H^1(\mathcal{B}) \to H^1(\mathcal{C}) \xrightarrow{\delta_1} H^2(\mathcal{A}) \to \cdots.$$

First assume $\mathcal{A}$ and $\mathcal{C}$ are exact. Then $H^\bullet(\mathcal{A})$ and $H^\bullet(\mathcal{C})$ are zero. So, our long exact sequence in cohomology collapses to the series of exact segments:

$$0 \to 0 \to H^\bullet(\mathcal{B}) \to 0 \xrightarrow{\delta_1} 0.$$

Hence, we conclude that each $H^\bullet(\mathcal{B}) = 0$ so that $\mathcal{B}$ is exact. In fact, repeating this argument for the other possible combinations concludes the proof.

Problem D&F-17.1-7. Prove that a finitely generated abelian group A is free if and only if

$$\mathrm{Ext}^1(A;\mathbb{Z}) = 0.$$

Let A be a finitely generated abelian group; note that every finitely generated abelian group can be considered as a $\mathbb{Z}$-module. First assume that A is free; since every free module is projective, A is a projective $\mathbb{Z}$-module. So then by Proposition 11, $\mathrm{Ext}^n_{\mathbb{Z}}(A,\mathbb{Z}) = 0$ for all $n \geq 1$. Now assume that $\mathrm{Ext}^1_{\mathbb{Z}}(A,\mathbb{Z}) = 0$. By the Fundamental Theorem for Modules Generated over a PID, one has

$$A \cong \mathbb{Z}^r \oplus \bigoplus_{1}^{s} \mathbb{Z}/n_i\mathbb{Z}.$$

Since $\mathbb{Z}^r$ is a projective $\mathbb{Z}$-module, one has

$$0 = \mathrm{Ext}^1_{\mathbb{Z}}(A,\mathbb{Z}) \cong \bigoplus_{1}^{s} \mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Z}/n_i\mathbb{Z},\mathbb{Z}) \cong \bigoplus_{1}^{s} \mathbb{Z}/n_i\mathbb{Z}.$$

Since $s < \infty$, each n_i must be 1 so that $A \cong \mathbb{Z}^r$, and is therefore free.

Problem D&F-17.1-10.

(a) Prove that an arbitrary direct sum $\bigoplus_{i\in I} P_i$ of projective modules P_i is projective and that an arbitrary direct product $\prod_{j\in J} Q_j$ of injective modules is injective.

(b) Prove that an arbitrary direct sum of projective resolutions is again projective and use this to show $\mathrm{Ext}^R_n(\bigoplus_{i\in I} A_i, B) \cong \prod_{i\in I} \mathrm{Ext}^R_n(A_i,B)$ for any collection of R-modules A_i ($i \in I$). [cf. Exercise 12 in Section 10.5.]

(c) Prove that an arbitrary direct product of injective resolutions is an injective resolution and use this to show $\mathrm{Ext}^n_R(A,\prod_{j\in J} B_j) \cong \prod_{j\in J} \mathrm{Ext}^n_R(A,B_j)$. [cf. Exercise 12 in Section 10.5.]

(d) Prove that $\mathrm{Tor}^R_n(A,\bigoplus_{j\in J} B_j) \cong \bigoplus_{j\in J} \mathrm{Tor}^R_n(A,B_j)$.

(a) Let I be an index set and $\{P_i\}_{i\in I}$ a collection of projective R-modules. Fix arbitrary left R-modules M and N, and let $f : M \twoheadrightarrow N$ be a surjection. Further, let $g : \bigoplus_{i\in I} P_i \to N$ be a R-module homomorphism. By the universal property of direct sums, g is uniquely determined by a family of maps $(g_i)_{i\in I} : P_i \to N$. Since each P_i is projective, there exists a family of module homomorphisms $h_i : P_i \to M$ such that $fh_i = g_i$ for each i. By the universal property for direct sums, there exists a unique module homomorphism $h : \bigoplus_{i\in I} P_i \to M$ such that for each i, $h_i = h \circ \iota_i$, where $\iota_i : P_i \hookrightarrow \bigoplus_{i\in I} P_i$. But then we observe that

$$g_i = fh_i \iff g \circ \iota_i = (fh) \circ \iota_i.$$

By the uniqueness property of morphisms given by the universal property, $fh = g$. Hence, $\bigoplus_{i\in I} P_i$ is a projective R-module. Now let J be an index set and $\{Q_j\}_{j\in J}$ a collection of injective R-modules. Fix arbitrary R-modules N and M, and let $f : N \hookrightarrow M$ be an injective homomorphism. Let $g : M \to \prod_{j\in J} Q_j$ be a module homomorphism. By the universal property of direct products, g is uniquely determined by a family $(g_j)_{j\in J}$ of morphisms $g_j : M \to Q_j$ for each j, where $g_j = \pi_j \circ g$, with $\pi_j : \prod_{j\in J} Q_j \twoheadrightarrow Q_j$ being the canonical projection map. Since each Q_j is an injective R-module, there exists a morphism $h_j : N \to Q_j$ so that $h_j f = g_j$ for each j. Again invoking the universal property for direct products, the family $(h_j)_{j\in J}$ uniquely determines a map $h : N \to \prod_{j\in J} Q_j$ so that for each j, $h_j = \pi_j \circ h$. But then we observe that

$$\pi_j \circ g = g_j = h_j f = \pi_j \circ (hf).$$

By the uniquenes property of morphisms given by the universal property, we conclude that $hf = g$. Hence, we conclude that $\prod_{j\in J} Q_j$ is an injective R-module.

(b) Let I be an arbitrary index set, A any R-module, and $\{(P_\bullet^{(i)}, d_\bullet^{(i)})\}_{i\in I}$ an arbitrary collection of projective resolutions of A. Then the direct sum of these projective resolutions is defined to be the sequence $\left(\bigoplus_{i\in I} P_\bullet^{(i)}, \bigoplus_{i\in I} d_\bullet^{(i)}\right)$. Since the arbitrary direct sum of projective modules is projective (by part (a)), for each $j \in \mathbb{N}_0$, $\bigoplus_{i\in I} P_j^{(i)}$ is projective. It suffices to now show that the sequence

$$\cdots \to \bigoplus_{i\in I} P_n^{(i)} \xrightarrow{\bigoplus_{i\in I} d_n^{(i)}} \bigoplus_{i\in I} P_{n-1}^{(i)} \to \cdots \xrightarrow{\bigoplus_{i\in I} d_1^{(i)}} \bigoplus_{i\in I} P_0^{(i)} \xrightarrow{\bigoplus_{i\in I} \varepsilon^{(i)}} \bigoplus_{i\in I} A \to 0$$

is exact. But this follows immediately due to the following: for each $j \in \mathbb{N}_0$,

$$\operatorname{im} \bigoplus_{i\in I} d_{j+1}^{(i)} = \bigoplus_{i\in I} \operatorname{im} d_{j+1}^{(i)} = \bigoplus_{i\in I} \ker d_j^{(i)} = \ker \bigoplus_{i\in I} d_j^{(i)}$$

Hence, the arbitrary direct sum of projective resolutions is a projective resolution. Now, we will show that $\operatorname{Ext}_R^n(\bigoplus_{i\in I} A_i, B) \cong \prod_{i\in I} \operatorname{Ext}_R^n(A_i, B)$. Let $\{(P_\bullet^{(i)}, d_\bullet^{(i)})\}_{i\in I}$ be a collection of projective resolutions for each A_i. By our work above, $\left(\bigoplus_{i\in I} P_\bullet^{(i)}, \bigoplus_{i\in I} d_\bullet^{(i)}\right)$ is a projective resolution for $\bigoplus_{i\in I} A_i$. Then by definition,

$$\begin{aligned}\operatorname{Ext}_R^n\left(\bigoplus_{i\in I} A_i, B\right) &:= H^n\left(\operatorname{Hom}_R\left(\bigoplus_{i\in I} P_\bullet^{(i)}, B\right)\right) \cong H^n\left(\prod_{i\in I} \operatorname{Hom}_R(P_\bullet^{(i)}, B)\right) \\ &\cong \prod_{i\in I} H^n\left(\operatorname{Hom}_R(P_\bullet^{(i)}, B)\right) \cong \prod_{i\in I} \operatorname{Ext}_R^n(A_i, B),\end{aligned}$$

where in the first isomorphism, we used Exercise 12 from Section 10.5, and the second isomorphism follows from the fact that cohomology commutes with products since both kernels and images commute with products.

(c) Let I be an arbitrary index set, A any R-module, and $\{(Q_\bullet^{(i)}, d_\bullet^{(i)})\}_{i\in I}$ an arbitrary collection of injective resolutions of A. Then the direct product of these injective resolutions is defined to be the sequence $\left(\prod_{i\in I} Q_\bullet^{(i)}, \prod_{i\in I} d_\bullet^{(i)}\right)$. Since the arbitrary direct product of injective modules is injective (by part (a)), for each $j \in \mathbb{N}_0$, $\prod_{i\in I} Q_j^{(i)}$ is injective. It suffices to now show that the sequence

$$0 \to \prod_{i\in I} A \xrightarrow{\prod_{i\in I} \varepsilon} \prod_{i\in I} Q_0^{(i)} \xrightarrow{\prod_{i\in I} d_1^{(i)}} \cdots \xrightarrow{\prod_{i\in I} d_{n-1}^{(i)}} \prod_{i\in I} Q_n^{(i)} \xrightarrow{\prod_{i\in I} d_n^{(i)}} \cdots$$

is exact. But since kernels and images commute with products, by injectivity of each resolution, we observe that

$$\ker \prod_{i\in I} d_n^{(i)} \cong \prod_{i\in I} \ker d_n^{(i)} \cong \prod_{i\in I} \operatorname{image} d_{n+1}^{(i)} \cong \operatorname{image} \prod_{i\in I} d_{n+1}^{(i)},$$

and so the sequence is indeed exact. Therefore, the arbitrary direct product of injective resolutions is an injective resolution. Now, we will show that $\operatorname{Ext}_R^n(A, \prod_{j\in J} B_j) \cong \prod_{j\in J} \operatorname{Ext}_R^n(A, B_j)$. Let $\{(Q_\bullet^{(j)}, d_\bullet^{(j)})\}_{j\in J}$ be a collection of injective resolutions for each B_j. By our work above,

$$\left(\prod_{j\in J} Q_\bullet^{(j)}, \prod_{j\in J} d_\bullet^{(j)}\right)$$

is an injective resolution for $\prod_{j\in J} B_j$. Then by definition

$$\begin{aligned}\operatorname{Ext}_R^n\left(A, \prod_{j\in J} B_j\right) &:= H^n\left(\operatorname{Hom}_R\left(A, \prod_{j\in J} Q_\bullet^{(j)}\right)\right) \cong H^n\left(\prod_{j\in J} \operatorname{Hom}_R(A, Q_\bullet^{(j)})\right) \\ &\cong \prod_{j\in J} H^n\left(\operatorname{Hom}_R(A, Q_\bullet^{(j)})\right) =: \prod_{j\in J} \operatorname{Ext}_R^n\left(A, B_j\right),\end{aligned}$$

where the first isomorphism follows from Exercise 12 in Section 10.5.

(d) Let A be an arbitrary right R-module and $\{B_j\}_{j\in J}$ left R-modules. Let $(P_\bullet, d_\bullet)$ be a projective resolution for A. Then

$$\begin{aligned}\operatorname{Tor}_n^R\left(A, \bigoplus_{j\in J} B_j\right) \cong H_n\left(P_\bullet \otimes_R \bigoplus_{j\in J} B_j\right) \cong H_n\left(\bigoplus_{j\in J}(P_\bullet \otimes_R B_j)\right) \cong \bigoplus_{j\in J} H_n(P_\bullet \otimes_R B_j) \\ \cong \bigoplus_{j\in J} \operatorname{Tor}_n^R(A, B_j),\end{aligned}$$

where in the second isomorphism, we used the fact that tensor products commute with arbitrary direct sums, and in the third isomorphism, we used the fact that homology commutes with arbitrary direct sums.

Problem D&F-17.1-12. Prove Proposition 13: $\operatorname{Tor}_0^R(D,A) \cong D \otimes_R A$. [Follow the proof of Proposition 3.]

Let D be a fixed arbitrary right R-module, A a fixed arbitrary left R-module, and let $(P_\bullet, d_\bullet)$ be an arbitrary projective resolution for A. Then we have the exact sequence

$$P_1 \xrightarrow{d_1} P_0 \xrightarrow{\varepsilon} A \to 0.$$

Since tensor products is right exact, tensoring on the left by D, we obtain the exact sequence

$$D \otimes_R P_1 \xrightarrow{1\otimes d_1} D \otimes_R P_0 \xrightarrow{1\otimes \varepsilon} D \otimes_R A \to 0.$$

By exactness, we must have $(1 \otimes \varepsilon)$ surjective, and $\operatorname{im}(1 \otimes d_1) = \ker(1 \otimes \varepsilon)$. By definition of Tor,

$$\operatorname{Tor}_0^R(D,A) \cong (D \otimes_R P_0)/\operatorname{im}(1 \otimes d_1) = (D \otimes_R P_0)/\ker(1 \otimes \varepsilon) \cong D \otimes_R A,$$

where the final isomorphism comes from the First Isomorphism Theorem for Modules.

Problem D&F-17.1-13. Prove Proposition 16 characterizing flat modules.

Proposition 16 states the following:

Proposition 16. For a right R-module D the following are equivalent:
(a) D is a flat R-module.
(b) $\operatorname{Tor}_1^R(D,B) = 0$ for all left R-modules B, and
(c) $\operatorname{Tor}_n^R(D,B) = 0$ for all left R-modules B and all $n \geq 1$.

Clearly (c) implies (b). Therefore, we will show that (b) implies (a) and (a) implies (c). We start with the latter. Let D be a flat R-module, and fix an arbitrary left R-module B. Let $(P_\bullet, d_\bullet)$ be a projective resolution for B. Then tensoring by D on the right, we get the *exact* (since flat R-modules preserve exact sequences) chain complexes

$$\cdots \to D \otimes_R P_2 \xrightarrow{1\otimes d_2} D \otimes_R P_1 \xrightarrow{1\otimes d_1} D \otimes P_0 \xrightarrow{1\otimes \varepsilon} D \otimes_R B \to 0.$$

Since this complex is exact, for each $n \geq 1$,

$$\operatorname{Tor}_n^R(D,B) = \ker(1 \otimes d_n)/\operatorname{image}(1 \otimes d_{n+1}) = 0.$$

Now, we will prove that (b) implies (a). Suppose that $\operatorname{Tor}_1^R(D,B) = 0$ for all left R-modules. Consider the short exact sequence of R-modules, $0 \to L \to M \to N \to 0$. Then there exists the Tor long exact sequence

$$\begin{aligned}\cdots \to \operatorname{Tor}_2^R(D,N) \xrightarrow{\delta_1} \operatorname{Tor}_1^R(D,L) \to \operatorname{Tor}_1^R(D,M) \to \operatorname{Tor}_1^R(D,N) \\ \xrightarrow{\delta_0} D \otimes_R L \to D \otimes_R M \to D \otimes_R N \to 0.\end{aligned}$$

Applying the hypothesis, we obtain the short exact sequence

$$0 \to D \otimes_R L \to D \otimes_R M \to D \otimes_R N \to 0.$$

Hence, since for any short exact sequence of R-modules, applying the functor $D \otimes_R \_\_$ preseves short exactness, it follows that D is a flat R-module.

Problem D&F-17.1-14. Suppose $0 \to A \to B \to C \to 0$ is a short exact sequence of R-modules. Prove that if C is a flat R-module, then A is flat if and only if B is also flat. [Use the Tor long exact sequence.] Given a example to show that if A and B are flat then C need not be flat.

Let $0 \to A \to B \to C \to 0$ be a short exact sequence of R-modules, and let C be a flat R-module. First assume that B is also a flat module. Then for any R-module M, there exists a long exact sequence

$$\cdots \to \mathrm{Tor}_2^R(M,C) \xrightarrow{\delta_1} \mathrm{Tor}_1^R(M,A) \to \mathrm{Tor}_1^R(M,B)$$
$$\to \mathrm{Tor}_1^R(M,C) \to M \otimes_R A \to M \otimes_R B \to M \otimes_R C \to 0.$$

Since B and C are flat, $\mathrm{Tor}_i^R(M,B)$ and $\mathrm{Tor}_i^R(M,C)$ are zero for all $i \geq 1$. This means that we get the exact sequence $0 \to \mathrm{Tor}_1^R(M,A) \to 0$, and so $\mathrm{Tor}_1^R(M,A) = 0$. Since M was an arbitrary R-module, we conclude that A is flat. Now suppose A is a flat R-module so that for any left R-module M, $\mathrm{Tor}_1^R(M,A) = 0$. Then the above Tor long exact sequence gives us the exact sequecne

$$0 \to \mathrm{Tor}_1^R(M,B) \to 0 \qquad \text{so that } \mathrm{Tor}_1^R(M,B) = 0.$$

Since this is true for all left R-modules M, we conclude that B is flat.

Problem D&F-17.1-16. Suppose R is a P.I.D. and A and B are R-modules. If $t(B)$ denotes the torsion submodule of B show that $\mathrm{Tor}_1(A,t(B)) \cong \mathrm{Tor}_1^R(t(A),t(B))$ and deduce that $\mathrm{Tor}_1^R(A,B)$ is isomorphic to $\mathrm{Tor}_1^R(t(A),t(B))$. [Use Exercise 26 in Section 10.5 to show that $B/t(B)$ is flat over R, then use the Tor long exact sequence with $D = A$ applied to the short exact sequence $0 \to t(B) \to B \to B/t(B) \to 0$ and the remarks following Proposition 16.]

Let R be a PID, and A, B arbitrary R-modules. Let $t(B)$ denote the torsion submodule of B. First, we will show that $B/t(B)$ is a torsion-free module. Let $[b] \in B/t(b)$ be nonzero and assume to the contrary there exists some nonzero $r \in R$ such that $r[b] = [rb] = 0$. But then $rb \in t(B)$, which means there exists some nonzero $s \in R$ so that $(sr)b = s(rb) = 0$. Hence, $b \in t(b)$, which contradicts our hypothesis that $b \notin t(b)$. Hence, $B/t(B)$ is torsion-free. By Exercise 26 in Section 10.5, we conclude that $B/t(B)$ is flat over R. So now consider the short exact sequence

$$0 \to t(B) \to B \to B/t(B) \to 0.$$

Then there exists a long exact sequence

$$\cdots \to \mathrm{Tor}_2^R(A,B/t(B)) \xrightarrow{\delta_1} \mathrm{Tor}_1^R(A,t(B)) \to \mathrm{Tor}_1^R(A,B) \to \mathrm{Tor}_1^R(A,B/t(B)) \xrightarrow{\varepsilon} A \otimes_R t(B)$$
$$\to A \otimes_R B \to A \otimes_R B/t(B) \to 0.$$

Since $B/t(B)$ is flat, $\mathrm{Tor}_n^R(A,B/t(B)) = 0$ for all $n \geq 1$. And so we obtain the short exact sequence

$$0 \to \mathrm{Tor}_1^R(A,t(B)) \to \mathrm{Tor}_1^R(A,B) \to 0;$$

by exactness, we conclude that $\mathrm{Tor}_1^R(A,t(B)) \cong \mathrm{Tor}_1^R(A,B)$. Repeating this argument for the short exact sequence $0 \to t(A) \to A \to A/t(A) \to 0$, we obtain the isomorphism $\mathrm{Tor}_1^R(A,B) \cong \mathrm{Tor}_1^R(t(A),B)$. Finally, using the Tor long exact sequence with $D = t(A)$ applied to the short exact sequence $0 \to t(B) \to B \to B/t(B) \to 0$, we obtain the isomorphism $\mathrm{Tor}_1^R(t(A),t(B)) \cong \mathrm{Tor}_1^R(t(A),B)$. Hence, using our previous results, we conclude that $\mathrm{Tor}_1^R(A,B) \cong \mathrm{Tor}_1^R(t(A),t(B))$.

Problem D&F-17.1-17. Let $A = \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z} \oplus \cdots$. Prove that $\mathrm{Ext}^1(A,B) \cong (B/2B) \times (B/3B) \times (B/4B) \times \cdots$ for any abelian group B. [Use Exercise 10.] Prove that $\mathrm{Ext}^1(A,B) = 0$ if and only if B is divisible.

Define the group A as prescribed above, and let B be a fixed arbitrary abelian group. In Exercise 10, we proved that

$$\mathrm{Ext}^1(A,B) = \mathrm{Ext}^1\left(\bigoplus_{j=2}^{\infty} \mathbb{Z}/j\mathbb{Z}, B\right) \cong \prod_{j=2}^{\infty} \mathrm{Ext}^1(\mathbb{Z}/j\mathbb{Z}, B).$$

We shall work out $\mathrm{Ext}^1(\mathbb{Z}/j\mathbb{Z}, B)$ for some specified j first. Consider the projective resolution

$$0 \to \mathbb{Z} \xrightarrow{j} \mathbb{Z} \to \mathbb{Z}/j\mathbb{Z} \to 0.$$

Applying the functor $\mathrm{Hom}(-,B)$, we obtain the chain complex

$$0 \to \mathrm{Hom}(\mathbb{Z}/j\mathbb{Z}, B) \to \mathrm{Hom}(\mathbb{Z}, B) \xrightarrow{j^*} \mathrm{Hom}(\mathbb{Z}, B) \to 0 \to \cdots \to 0 \to \cdots.$$

We recall the isomorphism $\mathrm{Hom}(\mathbb{Z}, B) \cong B$ so that we have the cochain complex

$$0 \to {}_jB \to B \xrightarrow{j} B \to 0 \to \cdots \to 0 \to \cdots.$$

Hence,

$$\mathrm{Ext}^1(\mathbb{Z}/j\mathbb{Z}, B) \cong \frac{\ker(B \to 0)}{\mathrm{image}(B \xrightarrow{j} B)} \cong B/jB.$$

Therefore, we conclude that

$$\mathrm{Ext}^1(A,B) \cong \prod_{j=2}^{\infty} B/jB.$$

Now suppose B is a divisible group, which means that for every positive integer j, $jB = B$. This means that $B/jB = 0$ so that

$$\mathrm{Ext}^1(A,B) \cong \prod_{j=2}^{\infty} B/jB = \prod_{j\geq 2} 0 = 0.$$

Now suppose $\mathrm{Ext}^1(A,B) \cong 0$. This means that

$$\prod_{j\geq 2} B/jB = 0.$$

This forces every $B/jB = 0$ since if not, the product cannot be equal to zero. If $B/jB = 0$, then we immediately conclude that $B/jB = 0$ for all $j \geq 2$. Moreover, it follows trivially that $B = (1)B$. Hence, B is divisible.

Problem D&F-17.1-19. Suppose $r \neq 0$ is not a zero divisor in the commutative ring R.

- **(a)** Prove that multiplication by r gives a free resolution $0 \to R \xrightarrow{r} R \to R/rR \to 0$ of the quotient R/rR.
- **(b)** Prove that $\mathrm{Ext}_R^0(R/rR, B) = {}_rB$ is the set of elements $b \in B$ with $rb = 0$, that $\mathrm{Ext}_R^1(R/rR, B) \cong B/rB$, and that $\mathrm{Ext}_R^n(R/rR, B) = 0$ for $n \geq 2$ for every R-module B.
- **(c)** Prove that $\mathrm{Tor}_0^R(A, R/rR) = A/rA$, that $\mathrm{Tor}_1^R(A, R/rR) = {}_rA$ is the set of elements $a \in A$ with $ra = 0$, and that $\mathrm{Tor}_n^R(A, R/rR) = 0$ for $n \geq 2$ for every R-module A.

Suppose $r \neq 0$ is not a zero divisor in the commutative ring R.

(a) First, it is straightforward that seen as R-modules, R is a free R-module. Hence, it suffices to show that the sequence $0 \to R \xrightarrow{r} R \xrightarrow{\pi} R/rR \to 0$ is exact. The kernel of the map $R/rR \to 0$ is precisely R/rR. On the other hand, π being the canonical projection is surjective so that $\operatorname{image}\pi = R/rR$. Hence, the sequence is exact at R/rR. Likewise, the kernel of multiplication by r is precisely 0 since if the kernel includes any nonzero elements u, then $ru = 0$, contradicting the fact that r is not a zero divisor. On the other hand, the image of the trivial homomorphism $0 \to R$ is exactly 0. Hence, the sequence is exact at the first R. On the other hand, the kernel of π is precisely rR, while the image of multiplication of R by r is also rR. Hence, the sequence is exact at the second R. Therefore, we conclude that multiplication by r gives a free resolution $0 \to R \xrightarrow{r} R \to R/rR \to 0$ of the quotient R/rR.

(b) Since any free module is projective, the above free resolution gives a projective resolution for R/rR. Since the functor $- \otimes_R N$ is right exact, we obtain by tensoring B on the right, the exact sequence

$$R \otimes_R B \xrightarrow{r\otimes 1} R \otimes_R B \to R/rR \otimes_R B \to 0.$$

Since $R \otimes_R B \cong B$, we actually obtain the exact sequence

$$B \xrightarrow{r} B \to R/rR \otimes_R B \to 0.$$

Hence, it follows that

$$\operatorname{Ext}^0_R(R/rR, B) = \ker(B \xrightarrow{r} B) = \{b \in B : rb = 0\} =: {}_rB.$$

Now we will calculate $\operatorname{Ext}^1_R(R/rR, B)$. Applying the functor $\operatorname{Hom}_R(-, B)$ to the free resolution, we obtain the exact sequence

$$0 \to \operatorname{Hom}_R(R, B) \xrightarrow{r^*} \operatorname{Hom}_R(R, B) \to 0.$$

Since $\operatorname{Hom}_R(R, B) \cong B$ (by means of the identification $f \mapsto f(1)$), $\ker r^* = rB$. Hence,

$$\operatorname{Ext}^1_R(R/rR, B) \cong \frac{\ker 0}{\operatorname{image} r} = B/rB.$$

Since the sequence only has two nonzero components, it follows that $\operatorname{Ext}^n_R(R/rR, B) = 0$ for all $n \geq 2$.

(c) Consider again the free (projective) resolution $0 \to R \xrightarrow{r} R \to R/rR \to 0$. Applying the right-exact functor $A \otimes_R -$, we obtain the exact sequence

$$A \otimes_R R \xrightarrow{1\otimes r} A \otimes_R R \to 0 \implies A \xrightarrow{r} A \to 0.$$

Hence, we conclude from this that

$$\operatorname{Tor}^R_0(A, R/rR) = \operatorname{coker} A \xrightarrow{r} A = A/rA.$$

On the other hand, $\operatorname{Tor}^R_1(A, R/rR) = \ker A \xrightarrow{r} A = {}_rA$. And since the sequence terminates here, $\operatorname{Tor}^R_n(A, R/rR) = 0$ for all $n \geq 2$.

Problem D&F-17.1-21. Let $R = k[x, y]$ where k is a field, and let I be the ideal (x, y) in R.

(a) Let $\alpha : R \to R^2$ be the map $\alpha(r) = (yr, -xr)$ and let $\beta : R^2 \to R$ be the map $\beta((r_1, r_2)) = r_1x + r_2y$. Show that

$$0 \to R \xrightarrow{\alpha} R^2 \xrightarrow{\beta} R \to k \to 0$$

where the map $R \to R/I = k$ is the canonical projection, gives a free resolution of k as an R-module.

(b) Use the resolution in (a) to show that $\operatorname{Tor}^R_2(k, k) \cong k$.

Let $R = k[x, y]$, where k is a field, and let I be the ideal (x, y) in R.

(a) It is straightforward to see that R, R^2, R are free R-modules by definition. Hence, it suffices to show that this sequence is exact. We observe the following

$$\ker\alpha = \{r \in R : xr = 0 \text{ and } yr = 0\} = \{0\}, \quad \ker\beta = \{(r_1, r_2) \in R : r_2 y = -r_1 x\}$$
$$\ker(R \to k) = I, \qquad \ker(k \to 0) = k.$$

Hence,

(i) At the first R,

$$\operatorname{image}(0 \to R) = \{0\} = \ker\alpha,$$

so that the sequence is exact at the first R.

(ii) At R^2 we compute the following: (1) for any $(yr, -xr) \in \operatorname{image}\alpha$,

$$\beta((yr, -xr)) = (yr)x - (xr)y = (yrx) - (yrx) = 0.$$

So $\operatorname{image}\alpha \subset \ker\beta$; (2) for any $(r_1, r_2) \in \ker\beta$, $r_1 x + r_2 y = 0$ so that $x \mid r_2$ and $y \mid r_1$. I.e., there exists some $r \in R$ so that $r_2 = -xr$ and $r_1 = yr$. Hence, $\ker\beta \subseteq \operatorname{image}\alpha$. Altogether, this proves exactness at R^2.

(iii) Likewise, we can keep going to show that the whole sequence is exact. Hence, this is a free resolution of k as an R-module.

(b) From the provided free (hence, projective) resolution for k, applying the functor $k \otimes_R -$ yields the exact sequence

$$k \xrightarrow{\alpha} k^2 \xrightarrow{\beta} k \to 0$$

since $k \otimes_R R \cong k$ and $k \otimes_R R^2 \cong k^2$. So therefore, $\operatorname{Tor}_2^R(k, k) = \ker\alpha = k$.

Problem D&F-17.1-22. *(Flat Base Change for Tor)* Suppose R and S are commutative rings and $f : R \to S$ is a ring homomorphism making S into an R-module as in Example 6 following Corollary 12 in Section 10.4. Prove that if S is flat as an R-module, then $\operatorname{Tor}_n^R(A, B) \cong \operatorname{Tor}_n^S(S \otimes_R A, B)$ for all R-modules A and all S-modules B. [Show that since S is flat, tensoring an R-module projective resolution for A with S gives an S-module projective resolution of $S \otimes_R A$.]

Let R and S be commutative rings, and $f : R \to S$ a ring homomorphism that turns S into an R-module. Further assume that S is flat as an R-module. Let A, B be fixed arbitrary R- and S-modules, respectively. Let $(P_\bullet, d_\bullet)$ be a projective resolution for A. Since S is flat, the sequence $(S \otimes_R P_\bullet, 1 \otimes d_\bullet)$ is an exact sequence. Now we will show that each $S \otimes_R P_j$ is a projective S-module. Let N, M be S-modules (hence R-modules by restriction of scalars), $f : N \twoheadrightarrow M$ a surjection (hence an R-module surjection by restriction of scalars) and $g : S \otimes_R P_j \to M$ a S-module homomorphism. By the universal property for tensor products, g is uniquely determined by an R-linear map $g_2 : P_j \to M$. Since P_j is a projective R-module, there exists a morphism $h_2 : P_j \to N$ so that $fh_2 = g_2$. So define the map $h : S \otimes_R P_j \to M$ as follows: $h(s \otimes p) = sh_2(p)$. Then we check that

$$fh(s \otimes p) = f(sh_2(p)) = sf(h_2(p)) = sg_2(p) = g(s \otimes p),$$

where the final equality follows from the fact that g is S-linear. Hence, $fh = g$, which proves that $S \otimes_R P_j$ is a projective S-module for all nonnegative integers j. This means that $(S \otimes_R P_\bullet, d_\bullet)$ is a projective resolution of $S \otimes_R A$. So since $(S \otimes_R P_\bullet) \otimes_S B \cong S \otimes_R (P_\bullet \otimes_R B)$, we conclude that

$$\operatorname{Tor}_n^S(S \otimes_R A, B) \coloneqq H_n((S \otimes_R P_\bullet) \otimes_S B) = H_n(S \otimes_R (P_\bullet \otimes_R B)) = H_n((S \otimes_R P_\bullet) \otimes_R B)$$
$$\eqqcolon \operatorname{Tor}_n^R(S \otimes_R A, B).$$

Problem D&F-17.1-23. *(Localization and Tor)* Let $D^{-1}R$ be the localization of the commutative ring R with respect to the multiplicative subset D of R. Prove that localization commutes with Tor, i.e.,
$D^{-1}\mathrm{Tor}_n^R(A,B) \cong \mathrm{Tor}_n^{D^{-1}R}(D^{-1}A, D^{-1}B)$. [Use the previous exercise and the fact that $D^{-1}R$ is flat over R, cf. Proposition 42(6) in Section 15.4.]

Let R be a commutative ring, D a multiplicative subset of R, and $D^{-1}R$ the localization of R. Let $(P_\bullet, d_\bullet)$ be a projective resolution for A so that

$$\mathrm{Tor}_n^R(A,B) \coloneqq H_n(P_\bullet \otimes_R B).$$

Since localizations are exact and preserve prjective modules, $D^{-1}P_\bullet$ is a projective resolution for $D^{-1}A$. This means that

$$\mathrm{Tor}_n^{D^{-1}R}(D^{-1}A, D^{-1}B) = H_n(D^{-1}P_\bullet \otimes_R D^{-1}B).$$

Here, we note that since localization commutes with tensor products,

$$D^{-1}P_\bullet \otimes_R D^{-1}B \cong D^{-1}(P_\bullet \otimes_R B).$$

Then since localizations commute with homology, $H_n(D^{-1}(P_\bullet \otimes_R B)) \cong D^{-1}H_n(P_\bullet \otimes_R B)$. Hence,

$$\mathrm{Tor}_n^{D^{-1}R}(D^{-1}A, D^{-1}B) = H_n(D^{-1}P_\bullet \otimes_R D^{-1}B) \cong D^{-1}H_n(P_\bullet \otimes_R B) \cong D^{-1}\mathrm{Tor}_n^R(A,B).$$

This concludes the proof that localization commutes with Tor.

4 Real Analysis

4.1 Convergence of Functions

Problem 2026-J-II-2. Let c be a positive real number, and let $\{f_n(x)\}_{n\in\mathbb{N}}$ be a sequence of integrable functions from $[0,1]$ to $[-c,c]$. For the following functions on $[0,1]$,

$$g_n(x) := \int_0^x f_n(t)\,dt,$$

prove that some subsequence $(g_{n_m})_{m\in\mathbb{N}}$ converges uniformly to a c-Lipschitz function.

Define the sequence $\{g_n\}$ of functions on $[0,1]$ as given above. We begin by showing that each g_n is c-Lipschitz. Fix $\varepsilon > 0$, and let $\delta = \varepsilon/c > 0$. For any $x, y \in [0,1]$ with $x > y$ and $|x-y| < \delta$, one computes,

$$|g_n(x) - g_n(y)| \leq \int_y^x |f_n(t)|\,dt \leq \int_y^x c\,dt = c|x-y| < \varepsilon.$$

Thus we have that each g_n is c-Lipschitz, *and* the sequence $\{g_j\}$ is equicontinuous. Thus by the Arzelà-Ascoli theorem, it follows that there exists a subsequence $\{g_{n_m}\}_{m=1}^{\infty}$ that converges uniformly to a function g on $[0,1]$. We will now show that this limit is c-Lipschitz. Fix $\varepsilon > 0$, and take $x, y \in [0,1]$ with $x > y$. Then

$$\begin{aligned}
|g(x) - g(y)| &= |g(x) - g_{n_m}(x) + g_{n_m}(x) - g_{n_m}(y) + g_{n_m}(y) - g(y)| \\
&\leq |g(x) - g_{n_m}(x)| + |g_{n_m}(x) - g_{n_m}(y)| + |g_{n_m}(y) - g(y)| \\
&\leq |g(x) - g_{n_m}(x)| + c|x-y| + |g_{n_m}(y) - g(y)|.
\end{aligned}$$

By uniform convergence of the subsequence, in the limit $m \to \infty$, one has $|g(x) - g_{n_m}(x)|, |g(y) - g_{n_m}(y)| \to 0$. Thus, we conclude that $|g(x) - g(y)| \leq c|x-y|$ so that g is c-Lipschitz.

Problem 2025-J-I-2. Let $\{f_n\}_{n\geq 1}$ be a sequence of Lebesgue measurable functions on $[0,1]$. Suppose that

$$\int_0^1 f_n^2\,dm \leq \frac{1}{n^2}$$

for all $n \geq 1$. Prove that f_n converges to 0 a.e. on $[0,1]$.

Let $\{f_n\}_{n\geq 1}$ be a sequence of Lebesgue measurable functions on $[0,1]$ such that for each $n \geq 1$,

$$\int_0^1 f_n^2\,dm \leq \frac{1}{n^2}.$$

To show that $f_n \to 0$ a.e., it suffices to show that $f_n^2 \to 0$ a.e., and thus $\sum_1^\infty f_n^2$ converges a.e. We note that $\sum_1^N f_n^2 \to \sum_1^\infty f_n^2$ a.e., and the sequence of such partial sums is increasing. By the Monotone Convergence Theorem, one has the following:

$$\int_0^1 \sum_1^\infty f_n^2\,dm = \int_0^1 \lim_{N\to\infty} \sum_1^N f_n^2\,dm = \lim_{N\to\infty} \int_0^1 \sum_1^N f_n^2\,dm = \sum_1^\infty \int_0^1 f_n^2\,dm \leq \sum_1^\infty \frac{1}{n^2} < \infty.$$

Thus, $\sum_1^\infty f_n^2 < \infty$ a.e. on $[0,1]$ so that the sum converges a.e. Since the tail of a convergent series must tend to zero a.e., we must have $f_n^2 \to 0$ a.e., and thus $f_n \to 0$ a.e.

Problem 2004-A-II-1. Let $\{f_n\}_{n=1}^{\infty}$ be a sequence of real valued absolutely continuous functions defined on $[0,1]$. Assume there exist constants c_0 and c_1 such that

$$\sup_{0\leq x\leq 1} |f_n(x)| \leq c_0 \qquad \text{and} \qquad \int_0^1 |f_n'(x)|^2\,dx \leq c_1$$

for $n = 1, 2, 3, \ldots$. Prove that $\{f_n\}_{n=1}^{\infty}$ has a subsequence which converges uniformly on $[0,1]$.

Let $\{f_n\}_{n=1}^{\infty}$ be a sequence of real-valued absolutely continuous functions defined on $[0,1]$ with the above properties. Since $[0,1]$ is compact, by the Arzelà-Ascoli theorem, it suffices to show that the sequence is uniformly bounded and equicontinuous; the first condition on the sequence already proves uniform boundedness. Therefore, we will prove that the sequence is equicontinuous. Let $\varepsilon > 0$, and let $\delta = \varepsilon^2/c_1$. Then for any $x, y \in [0,1]$ with $x > y$ and all $j \geq 1$, one has

$$\begin{aligned} |f_j(x) - f_j(y)| &= \left| \int_y^x f_j'(x)\, dx \right| \leq \int_y^x |f_j'(x)|\, dx \\ &\leq \left(\int_y^x |f_j'(x)|^2\, dx \right)^{1/2} \left(\int_y^x dx \right)^{1/2} && \text{(by Cauchy-Schwarz Inequality)} \\ &\leq c_1^{1/2} \cdot |x-y|^{1/2} < \varepsilon. \end{aligned}$$

Thus it follows that $\{f_j\}_{j\geq 1}$ is uniformly continuous. By Arzelà-Ascoli, we conclude that the sequence has a subsequence which converges uniformly on $[0,1]$.

Problem 2021-J-II-2. A real-valued function $f : [0,1] \to \mathbb{R}$ is said to be Hölder continuous of order α if there is a constant $C > 0$ such that $|f(x) - f(y)| \leq C|x-y|^\alpha$ for all $x, y \in [0,1]$. Define

$$\|f\|_\alpha = \max_{x\in[0,1]} |f(x)| + \sup_{x,y\in[0,1]} \frac{|f(y) - f(x)|}{|x-y|^\alpha}.$$

Show that for $\alpha \in (0,1)$, the set of functions

$$S_\alpha = \{f \in C([0,1]) : \|f\|_\alpha \leq 1\}$$

is a compact subset of $C([0,1])$ in the topology of uniform convergence.

Fix some order $\alpha \in (0,1)$ and define the set of functions,

$$S_\alpha = \{f \in C([0,1]) : \|f\|_\alpha \leq 1\}.$$

By the Arzelà-Ascoli theorem, S_α is a compact subset of $C([0,1])$ if and only if it is closed, bounded, and equicontinuous under the topology of uniform convergence. Boundedness is immediately clear: for all $f \in S_\alpha$, one has

$$\|f\|_\infty = \max_{x\in[0,1]} |f(x)| \leq \|f\|_\alpha = 1,$$

where the upper bound of 1 is independent of the individual function f. Now we will show that S_α is closed under the supremum norm; we must show that if a sequence $\{f_n\} \subset S_\alpha$ converges uniformly to a function f, then $\|f\|_\alpha \leq 1$. We begin by showing first that f is Hölder continuous of exponent α. For any $x, y \in [0,1]$, one has

$$\begin{aligned} |f(x) - f(y)| &\leq |f(x) - f_n(x)| + |f_n(x) - f_n(y)| + |f_n(y) - f(y)| \\ &\leq |f(x) - f_n(x)| + C|x-y|^\alpha + |f(y) - f_n(y)| && \text{(since each } f_n \text{ is Hölder continuous)} \\ &\xrightarrow{n\to\infty} C|x-y|^\alpha && \text{(by uniform convergence).} \end{aligned}$$

Now, we must show that $\|f\|_\alpha \leq 1$. We will do this by showing that the Hölder norm is lower semicontinuous. For each distinct pair $x, y \in [0,1]$, the functional $\rho_{x,y} : C([0,1]) \to [0,\infty)$ given by

$$\rho_{x,y}(g) := \frac{|g(x) - g(y)|}{|x-y|^\alpha}$$

is continuous under the topology of uniform convergence (hence lower semicontinuous). Since the supremum of lower semicontinuous functions is lower semicontinuous, the Hölder seminorm $[\,\cdot\,]_\alpha$

must be lower semicontinuous. Thus since $\|\cdot\|_\infty$ is continuous (hence, lower semicontinuous), one has that the Hölder norm is lower semicontinuous. This means that

$$\|f\|_\alpha \le \liminf_{n\to\infty} \|f_n\|_\alpha \le 1.$$

Thus, S_α is closed. Finally, we will show equicontinuity. Let $\varepsilon > 0$. and take $\delta = \varepsilon^{1/\alpha}$. Then for any $x, y \in [0,1]$ with $|x-y| < \delta$, one has that for all n,

$$|f_n(x) - f_n(y)| \le |x-y|^\alpha < \varepsilon,$$

since $[f_n]_\alpha \le 1$ (which means we can take $C = 1$) for all n.

Problem 2019-A-I-3. For every positive integer n, let $T_n : X \to Y$ be a linear transformation with X, Y normed vector spaces. Suppose that $\|T_n\| \le 1$. Let $\{x_\alpha\}$ be a dense subset of X. Suppose that for any α,

$$\lim_{n\to\infty} T_n(x_\alpha) = y_\alpha.$$

for some $y_\alpha \in Y$. Suppose further that if $x = \lim_i x_{\alpha_i}$, then y_{α_i} converges to some $y_x \in Y$. Show that

$$\lim_{n\to\infty} T_n(x) = y_x.$$

Fix some arbitrary $x \in X$, and let $\varepsilon > 0$. By the density of $\{x_\alpha\}$ in X, there exists a subsequence $\{x_{\alpha_i}\}_{i=1}^\infty$ such that the following holds: there exists a positive integer I such that for all $i \ge I$, $\|x_{\alpha_i} - x\|_X \le \varepsilon/3$ (i.e., $x_{\alpha_i} \to x$). By the hypotheses, for each $i \ge 1$, there exists a corresponding $y_{\alpha_i} \in Y$ and a positive integer N_i so that for all $n \ge N_i$, $\|T_n x_{\alpha_i} - y_{\alpha_i}\|_Y \le \varepsilon/3$. By the final hypothesis, there exists some $y_x \in Y$ and some positive integer J such that for all $i \ge J$, $\|y_{\alpha_i} - y_x\|_Y \le \varepsilon/3$. Set $M = \max\{I, J\}$. Then for all $n \ge N_M$, one has

$$\begin{aligned}
\|T_n x - y_x\|_Y &= \|T_n x - T_n x_{\alpha_M} + T_n x_{\alpha_M} - y_{\alpha_M} + y_{\alpha_M} - y_x\|_Y \\
&\le \|T_n(x - x_{\alpha_M})\|_Y + \|T_n x_{\alpha_M} - y_{\alpha_M}\|_Y + \|y_{\alpha_M} - y_x\|_Y \\
&\le \|x - x_{\alpha_M}\|_Y + \|T_n x_{\alpha_M} - y_{\alpha_M}\|_Y + \|y_{\alpha_M} - y_x\|_Y \qquad (\text{since } \|T_n\| \le 1) \\
&\le 3 \cdot \frac{\varepsilon}{3} = \varepsilon.
\end{aligned}$$

Thus, since ε was arbitrary, we conclude that $T_n x \to y_x$ under the norm on Y.

Problem 2015-J-II-5. Let E be a measurable subset of the real numbers whose Lebesgue measure is finite. Let $\chi_E : \mathbb{R} \to \{0,1\}$ be the characteristic function of E (which equals 1 on the set E, and 0 otherwise). Show that $\chi_E(x-t)$ converges to $\chi_E(x)$ in L^1 as $t \to 0$.

Let E be a finite Lebesgue measurable subset of $\mathbb{R}$, and let χ_E denote the characteristic function of E. We wish to show that

$$\int_{\mathbb{R}} |\chi_E(x) - \chi_E(x-t)|\, dx \xrightarrow{t\to 0} 0.$$

We begin by studying the function $\chi_E(x) - \chi_E(x-t)$. We note the following:

$$\chi_E(x) - \chi_E(x-t) = \begin{cases} 1, & x \in E \cap (E+t)^c \\ 0, & x \in E \cap (E+t) \text{ or } x \in E^c \cap (E+t)^c \\ -1, & x \in (E+t) \cap E^c \end{cases}$$

Therefore,

$$\int_{\mathbb{R}} |\chi_E(x) - \chi_E(x-t)|\, dx = \int_{E\cap(E+t)^c} dx + \int_{(E+c)\cap E^c} dx = \mu(E \cap (E+t)^c) + \mu((E+t) \cap E^c).$$

Hence, we need to show that each of the terms on the right side converges to zero as $t \to 0$. [!! Complete Later !!]

4.2 L^p Spaces

Problem 2025-A-II-4.

(a) Characterize L^1-functions $f : \mathbb{R} \to \mathbb{R}$ such that $f * f = f$.

(b) The same question for L^2-functions.

(a) Let $f \in L^1(\mathbb{R})$ so that $f * f = f$. Taking the Fourier transform of both sides, one has

$$\hat{f} = (f * f)\hat{} = \hat{f} \cdot \hat{f} = (\hat{f})^2,$$

so that at each point $\xi \in \mathbb{R}$, either $\hat{f}(\xi) = 0$ or $\hat{f}(\xi) = 1$. Thus, $\hat{f}$ is the indicator function of some measurable subset $E \subset \mathbb{R}$. We recall that the Fourier transform maps integrable functions to the space of continuous functions $C_0(\mathbb{R})$ that vanish at infinity. By the vanishing criteria, E must be a proper subset of $\mathbb{R}$, and by continuity, E has to be the empty set. Thus in fact, $\hat{f} \equiv 0$. By the Fourier Inversion theorem, it follows that $f = 0$ a.e.

(b) Now let $f \in (L^1 \cap L^2)(\mathbb{R})$ so that $f = f * f$. As above, taking the Fourier transform of both sides forces $\hat{f}(\xi) = 0$ or $\hat{f}(\xi) = 1$ at each $\xi \in \mathbb{R}$. Therefore, $\hat{f}$ must be the characteristic function of some measurable subset $E \subset \mathbb{R}$. By Plancherel's Theorem, since $f \in L^2$, $\hat{f} \in L^2$. This means that

$$\infty > \int_{\mathbb{R}} |\hat{f}(\xi)|^2 \, d\xi = \int_{\mathbb{R}} \chi_E^2 \, d\xi = \int_{\mathbb{R}} \chi_E \, d\xi = m(E).$$

Thus $\hat{f} = \chi_E$ for some finitely-measured subset $E \subset \mathbb{R}$. By the Fourier Inversion Theorem, it follows that

$$f(x) = \mathcal{F}^{-1}\{\chi_E\} = \int_E e^{2\pi i x \xi} \, d\xi,$$

for some finite-measured subset $E \subseteq \mathbb{R}$.

Problem 2024-J-I-6. Let f and g be Lebesgue-measurable functions on $\mathbb{R}$. Define the convolution

$$(f * g)(x) = \int_{\mathbb{R}} f(x - y) g(y) \, dy$$

for all x such that the integral exists. Prove that if $f \in L^p(\mathbb{R})$ and $g \in L^q(\mathbb{R})$ with $p, q \in (1, \infty)$ satisfying $\frac{1}{p} + \frac{1}{q} = 1$, then $f * g$ is a bounded continuous function on $\mathbb{R}$.

Let f and g be Lebesgue-measurable functions on $\mathbb{R}$, and define their convolution as given above. Further assume that $f \in L^p(\mathbb{R})$ and $g \in L^q(\mathbb{R})$ with $(p, q) \in (1, \infty)$ satisfying $\frac{1}{p} + \frac{1}{q} = 1$. First, we will show that $f * g$ is bounded. For any $x \in \mathbb{R}$, by Hölder's continuity, one has

$$|(f * g)(x)| = \left| \int_{\mathbb{R}} f(x - y) g(y) \, dy \right| \leq \int_{\mathbb{R}} |f(x - y)||g(y)| \, dy \leq \|f(x - \cdot)\|_p \|g\|_q = \|f\|_p \|g\|_q < \infty,$$

where the final equality follows from translation invariance of the L^p-norm. Since the bound is independent of x, we conclude that $f * g$ is bounded. To show that $f * g$ is continuous on $\mathbb{R}$, we observe the following:

$$\begin{aligned} |(f * g)(x + h) - (f * g)(x)| &\leq \int_{\mathbb{R}} |f(x + h - y) - f(x - y)||g(y)| \, dy \\ &= \int_{\mathbb{R}} |f(t + h) - f(t)||g(x - t)| \, dt \\ &\leq \|\tau_h f - f\|_p \|g(x - \cdot)\|_q = \|\tau_h f - f\|_p \|g\|_q \xrightarrow{h \to 0} 0, \end{aligned}$$

where the final line follows from continuity of the translation operator. Thus the convolution is (uniformly) continuous.

Problem 2019-J-I-4. Let (X,μ) be a measure space and let $f \in L^1(X,\mu) \cap L^\infty(X,\mu)$. Show that $f \in L^q(X,\mu)$ for all $q > 1$ and
$$\|f\|_\infty = \lim_{q\to\infty} \|f\|_q .$$

Let (X,μ) be a measure space, and let $f \in L^1(X,\mu) \cap L^\infty(X,\mu)$. Fix any arbitrary $q > 1$. Then

$$\|f\|_{L^q}^q = \int_X |f|^q \, d\mu = \int_X |f| \cdot |f|^{q-1} \, d\mu \le \int_X |f| \cdot \|f\|_{L^\infty}^{q-1} = \|f\|_{L^1} \cdot \|f\|_{L^\infty}^{q-1} < \infty.$$

Thus since $\|f\|_{L^q} < \infty$, we conclude that $f \in L^q(X,\mu)$. From the above result, one has

$$\|f\|_{L^q} \le \|f\|_{L^1}^{1/q} \cdot \|f\|_{L^\infty}^{1-(1/q)},$$

so that $\lim_{q\to\infty} \|f\|_{L^q} \le \|f\|_{L^\infty}$. Now fix $\varepsilon > 0$, and let $E = \{x \in X : |f(x)| \ge \|f\|_\infty - \varepsilon\}$. By definition of the essential supremum, E must have positive measure. On the other hand, since $f \in L^\infty$, E must have finite measure. Then one has

$$\|f\|_q^q \ge \int_E |f(x)|^q \, d\mu \ge \int_E (\|f\|_\infty - \varepsilon)^q \, d\mu \ge \mu(E) \cdot (\|f\|_\infty - \varepsilon)^q.$$

Taking the qth roots, one has
$$\|f\|_q \ge \mu(E)^{1/q} \cdot (\|f\|_\infty - \varepsilon).$$

Since the left side is independent of ε, passing $\varepsilon \to 0$ gives $\|f\|_q \ge \mu(E)^{1/q} \cdot \|f\|_\infty$ so that $\lim_{q\to\infty} \|f\|_q \ge \|f\|_\infty$. This concludes the proof.

Problem 2017-J-II-2. Suppose $f : [0,1] \to \mathbb{R}$ is measurable. Suppose further that for all $g \in L^2([0,1])$ we have that $fg \in L^2([0,1])$. Show f is in $L^\infty([0,1])$.

Suppose $f : [0,1] \to \mathbb{R}$ is measurable. Define the functional $T : L^2([0,1]) \to L^2([0,1])$ by $Tg := fg$ for any $g \in L^2([0,1])$. It is immediately clear that this operator is linear. We will now show that T is a closed linear operator. Let $\{g_j\}_{j=1}^\infty \subset L^2([0,1])$ be a sequence of square-integrable functions so that $g_j \to 0$ under the L^2 norm. Further assume that Tg_j converges to some $y \in L^2([0,1])$ under the L^2-norm. Since convergence under the L^2 norm implies pointwise a.e. convergence of a subsequence, there exists a subsequence $\{g_{j_k}\}_{k=1}^\infty$ so that $g_{j_k} \to 0$ pointwise a.e. Since $Tg_{j_k} \to y$ under the L^2 norm, there exists a further subsequence $\{g_{j_{k_l}}\}_{l=1}^\infty$ so that $Tg_{j_{k_l}}$ converges to y pointwise a.e. and by construction, $g_{j_{k_l}}$ converges to 0 pointwise a.e. By definition of T, it then follows that $f(x)g_{j_{k_l}}(x)$ converges to 0 pointwise a.e. so that $y = 0$ a.e. Thus in fact, we have shown that the graph of T, $\Gamma(T)$, is closed. By the closed graph theorem, it follows that T is continuous, and thus bounded. Therefore, we conclude that f must be bounded, and hence, must be in $L^\infty([0,1])$.

Problem F-6-1-2. Prove Theorem 6.8.

Theorem 6.8 states the following

6.8 Theorem.

a. If f and g are measurable functions on X, then $\|fg\|_1 \le \|f\|_1 \|g\|_\infty$. If $f \in L^1$ and $g \in L^\infty$, $\|fg\|_1 = \|f\|_1 \|g\|_\infty$ iff $|g(x)| = \|g\|_\infty$ a.e. on the set where $f(x) \ne 0$.
b. $\|\cdot\|_\infty$ is a norm on L^∞.
c. $\|f_n - f\|_\infty \to 0$ iff there exists $E \in \mathcal{M}$ such that $\mu(E^c) = 0$ and $f_n \to f$ uniformly on E.
d. L^∞ is a Banach space.
e. Simple functions are dense in L^∞.

a. Let f, g be measurable functions on X. Since $|g(x)| \leq \|g\|_\infty$ a.e., one has

$$\|fg\|_1 = \int_X |fg| \, d\mu \leq \int_X |f| \, \|g\|_\infty \, d\mu =: \|f\|_1 \|g\|_\infty .$$

Now suppose $|g(x)| = \|g\|_\infty$ a.e. on the set $E = \{x : f(x) \neq 0\}$. Then

$$\|fg\|_1 = \int_X |fg| \, d\mu = \int_E |f||g| \, d\mu = \int_E |f| \, \|g\|_\infty \, d\mu = \|f\|_1 \|g\|_\infty .$$

Now suppose $\|fg\|_1 = \|f\|_1 \|g\|_\infty$ on E. I.e., one has

$$\int_X |fg| \, d\mu = \int_E |f||g| \, d\mu = \int_E |f| \, \|g\|_\infty \, d\mu \implies \int_E |f|(|g| - \|g\|_\infty) \, d\mu = 0.$$

We note that on E, $|f| \ngeq 0$. Moreover, $|g| - \|g\|_\infty \leq 0$ on E. Thus, the integral is zero if and only if $|g| - \|g\|_\infty = 0$ a.e. on E so that $|g(x)| = \|g\|_\infty$ a.e. on E.

b. Recall that we defined the following

$$\|f\|_\infty = \inf\{a \geq 0 : |f(x)| \leq a \text{ a.e.}\}.$$

Thus it follows immediately that for any $f \in L^\infty$, $\|f\|_\infty \geq 0$. Now suppose $f(x) = 0$ a.e. Then $|f(x)| = 0$ a.e. so that $\|f\|_\infty = 0$. On the other hand, if $\|f\|_\infty = 0$, then $0 \leq |f(x)| \leq 0$ for a.e. x so that $f(x) = 0$ a.e. Now we will show that the essential supremum is homogeneous. If $\lambda = 0$, the proof is immediate. So suppose $\lambda \neq 0$. Let $\mathcal{A} = \{a \geq 0 : |f(x)| \leq a \text{ a.e.}\}$, and let $\mathcal{B} = \{b \geq 0 : |\lambda f(x)| \leq b \text{ a.e.}\}$. Suppose $|f(x)| \leq a$ a.e. Then it follows that for any $\lambda \in \mathbb{C} \setminus \{0\}$, $|\lambda f(x)| = |\lambda||f(x)| \leq |\lambda| a$ a.e. This shows that $|\lambda|\mathcal{A} \subset \mathcal{B}$ (which means that $|\lambda| \|f\|_\infty = |\lambda| \inf \mathcal{A} \geq \|\lambda f\|_\infty = \inf \mathcal{B}$). Now suppose $|\lambda f(x)| \leq b$ a.e. so that $|f(x)| \leq b/|\lambda|$ a.e. This means that $(1/|\lambda|)\mathcal{B} \subset \mathcal{A}$ so that $(1/|\lambda|) \|\lambda f\|_\infty \geq \|f\|_\infty \iff \|\lambda f\|_\infty \geq |\lambda| \|f\|_\infty$. Thus, one has $\|\lambda f\|_\infty = |\lambda| \|f\|_\infty$. Now we will show that the triangle inequality holds. Let $f, g \in L^\infty$. Further define the sets $\mathcal{A} = \{a \geq 0 : |f(x) + g(x)| \leq a \text{ a.e.}\}$, $\mathcal{F} = \{b \geq 0 : |f(x)| \leq b \text{ a.e.}\}$, and $\mathcal{G} = \{c \geq 0 : |g(x)| \leq c \text{ a.e.}\}$. Let $b \in \mathcal{F}, c \in \mathcal{G}$. Then a.e. x,

$$|f(x) + g(x)| \leq |f(x)| + |g(x)| \leq b + c,$$

so that $\mathcal{F} + \mathcal{G} \subset \mathcal{A}$. Taking the infimum of both sides, one has $\|f + g\|_\infty = \inf \mathcal{A} \leq \inf \mathcal{F} + \inf \mathcal{G} = \|f\|_\infty + \|g\|_\infty$. Thus, the proof that $\|\cdot\|_\infty$ is a norm concludes.

c. Suppose there exists $E \in \mathcal{M}$ so that $\mu(E^c) = 0$ and $f_n \to f$ uniformly on E. I.e., we have that $f_n \to f$ uniformly a.e. This means that for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ large enough so that for all $n \geq N$ and almost all $x \in X$, $|f_n(x) - f(x)| \leq \varepsilon$. By definition of the essential supremum, it follows that for all $n \geq N$, $\|f_n - f\|_\infty \leq \varepsilon$. Thus we conclude that $\|f_n - f\|_\infty \to 0$. Now suppose that $\|f_n - f\|_\infty \to 0$. This means that for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ large enough so that for all $n \geq N$, $\|f_n - f\|_\infty \leq \varepsilon$. For each fixed $n \geq N$, let Z_n denote the set $\{x : |f_n(x) - f(x)| > \|f_n - f\|_\infty\}$; note that $\mu(Z_n) = 0$. Let $E_n = X \setminus Z_n$. One has that $f_n \to f$ uniformly on $E := \bigcap_{n \geq N} E_n$. On the other hand,

$$\mu\left(\left(\bigcap_{n \geq N} E_n\right)^c\right) = \mu\left(\bigcup_{n \geq N} Z_n\right) = 0,$$

by countable subadditivity.

d. Let $\{f_n\}_{n \geq 1} \subseteq L^\infty(X)$ be a Cauchy sequence. It suffices to show that $\{f_n\}$ converges to some $f \in L^\infty$. by definition of Cauchy sequences, for every $\varepsilon > 0$, there exists a positive integer N so that for all $n, m \geq N$, $\|f_n - f_m\|_\infty \leq \varepsilon$. For each pair $(n, m) \geq (N, N)$, let $E_{n,m}$ denote the set $\{x : |f_n(x) - f_m(x)| > \|f_n - f_m\|_\infty\}$, and let $E = \bigcup_{n,m=1}^\infty E_{n,m}$. By countable subadditivity, one has $\mu(E) = 0$. On the other hand, for every $x \in E^c$, the sequence $\{f_n(x)\}_{n=1}^\infty$ is a Cauchy sequence in $\mathbb{C}$ and thus converges to some value. Define the function $f : X \to \mathbb{C}$ as follows

$$f(x) = \begin{cases} \lim_{n \to \infty} f_n(x), & x \in E^c \\ 0, & x \in E. \end{cases}$$

First, we must show that $f \in L^\infty$. Since $\{f_n\}$ is a Cauchy sequence, it follows that $\{\|f_n\|_\infty\}_{n\geq 1}$ is a Cauchy sequence in $\mathbb{R}$, and thus must be bounded above by some finite constant $M > 0$. Therefore, for every $x \in E^c$, one has

$$|f(x)| = \lim_{n\to\infty} |f_n(x)| \leq \lim_{n\to\infty} \|f_n\|_\infty = M,$$

so that $\|f\|_\infty \leq M < \infty$. Thus, $f \in L^\infty$. Now we need to show that $f_n \to f$ under the L^∞ norm. Let $\varepsilon > 0$. Let $N \in \mathbb{N}$ so that for all $n, m \geq N$, $\|f_n - f_m\| < \varepsilon$. Then for every $x \in E^c$,

$$|f_n(x) - f(x)| = \lim_{m\to\infty} |f_n(x) - f_m(x)| \leq \varepsilon.$$

Since this inequality holds for all $n \geq n$, $x \in E^c$, we conclude that $\|f_n - f\|_\infty \leq \varepsilon$, and thus $f_n \to f$ under the L^∞-norm.

e. First, we will start by showing that simple functions are contained in L^∞. To do so, we will look at *characteristic functions*. Since for every $x \in X$, $|\chi_E(x)| \leq 1$ for any measurable subset $E \subseteq X$, it follows that χ_E is bounded everywhere on X, and hence, is contained in L^∞. Thus, it follows that every simple function is contained in L^∞. Now we will show that the set of simple functions in L^∞ is dense in this space. Fix an arbitrary $f \in L^\infty$. By Theorem 2.10, there exists a sequence $\{\phi_n\}$ of simple functions such that $0 \leq |\phi_1| \leq |\phi_2| \leq \cdots \leq |f|$, $\phi_n \to f$ pointwise, and $\phi_n \to f$ uniformly on any set on which f is bounded. Since f is essentially bounded, there exists some measurable set $E \in \mathcal{M}$ so that $\mu(E^c) = 0$ and f is bounded on E. Thus, $\phi_n \to f$ uniformly on E. But by part (c), it follows that $\|\phi_n - f\|_\infty \to \infty$; i.e., for every $\varepsilon > 0$, there exists some positive integer N so that for all $n \geq N$, $\|\phi_n - f\|_\infty \leq \varepsilon$. Thus, this proves density of simple functions in L^∞.

Problem F-6-1-3. If $1 \leq p < r \leq \infty$, $L^p \cap L^r$ is a Banach space with norm $\|f\| = \|f\|_p + \|f\|_r$, and if $p < q < r$, the inclusion map $L^p \cap L^r \to L^q$ is continuous.

Fix $1 \leq p < r \leq \infty$, and define the map $\|\cdot\| : L^p \cap L^r \to [0, \infty)$ as given above. First, we will begin by showing that this map is a well-defined norm. Let $f \in L^p \cap L^r$, and assume $\|f\| = \|f\|_p + \|f\|_r = 0$. Then since $\|f\|_p, \|f\|_r \geq 0$, one must have $\|f\|_p = 0$ and $\|f\|_r = 0$. But by positive definiteness of the L^p norms, one must have $f = 0$ a.e. Now we will show homogeneity. Let $\lambda \in \mathbb{C}$. Then

$$\begin{aligned}\|\lambda f\| &= \|\lambda f\|_p + \|\lambda f\|_r \\ &= |\lambda|(\|f\|_p + \|f\|_r) \\ &= |\lambda|\,\|f\|.\end{aligned}$$

Finally, we will show that the triangle inequality is satisfied. Let $f, g \in L^p \cap L^r$. Then

$$\begin{aligned}\|f + g\| &= \|f + g\|_p + \|f + g\|_r \\ &\leq \|f\|_p + \|g\|_p + \|f\|_r + \|g\|_r \\ &= \|f\| + \|g\|.\end{aligned}$$

Thus $\|\cdot\|$ is a norm on $L^p \cap L^r$. Now we will show that $L^p \cap L^r$ is a Banach space under this norm. Let $\{f_n\}_{n\geq 1} \subset L^p \cap L^r$ be a Cauchy sequence. Then by construction of $\|\cdot\|$, one has that $\{f_n\}$ is a Cauchy sequence in L^p and in L^r. Thus since each of these spaces are Banach, $f_n \to f$ in L^p and $f_n \to g$ in L^r. Our first task is to show that $f = g$ a.e. Since $\{f_n\}$ converges to f in L^p, there exists a subsequence $\{f_{n_k}\}$ that converges to f pointwise a.e. But since $\{f_{n_k}\} \subset \{f_n\}$ and the latter converges to g in L^r, there exists a further subsequence $\{f_{n_{k_j}}\}$ that converges pointwise to g a.e. But since $\{f_{n_k}\} \to f$ pointwise a.e., every subsequence must also converge to f pointwise a.e. Thus by uniqueness of limits, we must have $f = g$ a.e. This shows that $f, g \in L^p \cap L^r$. Thus,

$$\|f_n - f\| = \|f_n - f\|_p + \|f_n - f\|_r \xrightarrow{n\to\infty} 0.$$

This concludes the proof that $(L^p \cap L^r, \|\cdot\|)$ is Banach. Now let $p < q < r$. By Proposition 6.10, one has $L^p \cap L^r \subset L^q$ so that the inclusion $\iota : L^p \cap L^r \hookrightarrow L^q$ is well-defined. To show that ι is continuous, it suffices to show that it is bounded. By Proposition 6.10 and Young's inequality (in that order), one has

$$\|\iota(f)\|_q \le \|f\|_p^\lambda \|f\|_r^{1-\lambda} \le \lambda \|f\|_p + (1-\lambda)\|f\|_r \le \|f\|_p + \|f\|_r =: \|f\|,$$

where λ is defined by the relation $q^{-1} = \lambda p^{-1} + (1-\lambda) q^{-1}$ and $\lambda \in (0,1)$. Thus, since $f \in L^p \cap L^r$ was arbitrary, ι is bounded, and thus continuous.

Problem F-6-1-7. If $f \in L^p \cap L^\infty$ for some $p < \infty$, so that $f \in L^q$ for all $q > p$, then $\|f\|_\infty = \lim_{q\to\infty} \|f\|_q$.

Fix some $f \in L^p \cap L^\infty$ for some $p < \infty$. By Proposition 6.10, one has that $f \in L^q$ for all $q > p$. Let $\varepsilon > 0$, and let $A = \{x : |f(x)| > \|f\|_\infty - \varepsilon\}$; by definition of the essential supremum, A has positive measure; note that since $f \in L^q$, A must have *finite* measure. Then one has

$$\|f\|_q^q = \int_X |f(x)|^q \, d\mu \ge \int_A |f(x)|^q \, d\mu \ge \mu(A)(\|f\|_\infty - \varepsilon)^q.$$

Since both sides are finite, one has $\|f\|_q \ge \mu(A)^{1/q}(\|f\|_\infty - \varepsilon)$. Since $\varepsilon > 0$ was arbitrary, one has $\|f\|_q \ge \mu(A)^{1/q} \|f\|_\infty$. Thus taking the limit $q \to \infty$, one has

$$\lim_{q\to\infty} \|f\|_q \ge \|f\|_\infty.$$

On the other hand, set $\lambda \in (0,1)$ so that $q^{-1} = \lambda p^{-1}$. Then by Proposition 6.10, one has

$$\|f\|_q \le \|f\|_p^\lambda \|f\|_\infty^{1-\lambda} = \|f\|_p^{p/q} \|f\|_\infty^{1-p/q}.$$

Thus, taking the limit $q \to \infty$ on both sides yields,

$$\lim_{q\to\infty} \|f\|_q \le \lim_{q\to\infty} \|f\|_q^{p/q} \|f\|_\infty^{1-p/q} = \|f\|_\infty.$$

Thus the proof concludes.

Problem F-6-1-10. Suppose $1 \le p < \infty$. If $f_n, f \in L^p$, and $f_n \to f$ a.e., then $\|f_n - f\|_p \to 0$ iff $\|f_n\|_p \to \|f\|_p$. (Use Exercise 20 in §2.3.)

Suppose $1 \le p < \infty$, $f_n, f \in L^p$, and $f_n \to f$ a.e. Exercise 20 in §2.3 states the following

> **(Generalized Dominated Convergence Theorem)** If $f_n, g_n, f, g \in L^1$, $f_n \to f$ and $g_n \to g$ a.e., $|f_n| \le g_n$, and $\int g_n \to \int g$, then $\int f_n \to \int f$.

First assume that $\|f_n - f\|_p \to 0$. Then one has

$$\|f_n\|_p \le \|f_n - f\|_p + \|f\|_p.$$

Thus taking the limit as $n \to \infty$ on both sides, one has

$$\lim_{n\to\infty} \|f_n\|_p = \lim_{n\to\infty} \|f_n - f\|_p + \|f\|_p = \|f\|_p.$$

Now suppose $\|f_n\|_p \to \|f\|_p$. We will use Exercise 20 from §2.3. Since $f_n, f \in L^p$, one has that $|f_n - f|^p, 0, 2^{p-1}(|f_n|^p + |f|^p), 2^p|f|^p \in L^1$, $|f_n - f|^p \le 2^{p-1}(|f_n|^p + |f|^p)$ for all n, and $\int 2^{p-1}(|f_n|^p + |f|^p) \to \int 2^p|f|^p$ (by the hypothesis). Thus by the Generalized Dominated Convergence Theorem, we conclude that

$$\|f_n - f\|_p^p = \int |f_n - f|^p \to \int 0 = 0 \implies \|f_n - f\|_p \to 0.$$

Problem F-6-1-14. If $g \in L^\infty$, the operator T defined by $Tf = fg$ is bounded on L^p for $1 \le p \le \infty$. Its operator norm is at most $\|g\|_\infty$, with equality if μ is semifinite.

Fix an arbitrary $g \in L^\infty$, $1 \le p \le \infty$, and define the operator T on L^p by $Tf := fg$. We begin by showing that for any $f \in L^p$, $fg \in L^p$. We compute (for $1 \le p < \infty$),

$$\|fg\|_p^p = \int_X |f|^p |g|^p \, d\mu \le \int_X \|g\|_\infty^p |f|^p \, d\mu = \|g\|_\infty^p \|f\|_p^p < \infty \implies \|fg\|_p < \infty.$$

The case for $p = \infty$ can be shown trivially. Thus, one has that for any $f \in L^p$,

$$\|Tf\|_p = \|fg\|_p \le \|g\|_\infty \|f\|_p,$$

so that T is a bounded operator. Indeed we see that the operator norm of T is at most $\|g\|_\infty$. Now assume that μ is semifinite. Let $\varepsilon > 0$, and first assume that $1 \le p < \infty$. By semifiniteness, there exists a measurable set $E \in \mathcal{M}$ of finite positive measure so that $|g(x)| > \|g\|_\infty - \varepsilon$ on E. Then one has that for all $f \in L^p$,

$$\|fg\|_p^p = \int_X |f|^p |g|^p \, d\mu \ge \int_E |f|^p (\|g\|_\infty - \varepsilon)^p \, d\mu.$$

Let

$$f = \frac{\chi_E \operatorname{sgn}(g)}{\mu(E)^{1/p}} \implies \|f\|_p = 1.$$

For this particular choice of f, one has

$$\|Tf\|_p^p \ge (\|g\|_\infty - \varepsilon)^p \implies \|Tf\|_p \ge \|g\|_\infty - \varepsilon.$$

Since this is true for all $\varepsilon > 0$, $\|Tf\|_p \ge \|g\|_\infty$. Thus in fact, one has

$$\|T\| = \sup_{\|h\|_p \le 1} \|Th\| \ge \|Tf\|_p \ge \|g\|_\infty.$$

Therefore, in this case, we conclude that the operator norm of T is exactly $\|g\|_\infty$. Now consider the case $p = \infty$, and let $f = \chi_E \operatorname{sgn}(g)$. Then, one has

$$|f(x)g(x)| = |g(x)| > \|g\|_\infty - \varepsilon \implies \|fg\|_\infty \ge \|g\|_\infty - \varepsilon.$$

Therefore, one has

$$\|T\| = \sup_{\|h\|_\infty \le 1} \|Th\|_\infty \ge \|fg\|_\infty \ge \|g\|_\infty - \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, we conclude that $\|T\| \ge \|g\|_\infty$. Thus the proof concludes in both cases by using the upper bound we proved above.

Problem F-6-1-4. If $1 \le p < r \le \infty$, $L^p + L^r$ is a Banach space with norm $\|f\| = \inf\{\|g\|_p + \|h\|_r : f = g + h\}$, and if $p < q < r$, the inclusion map $L^q \hookrightarrow L^p + L^r$ is continuous.

Let $1 \le p < r \le \infty$, and let $\|\cdot\| : L^p + L^r \to [0, \infty)$ be the map defined by

$$\|\cdot\| := \inf\{\|g\|_p + \|h\|_r : \cdot = g + h\}.$$

First, we will show that the above map is a norm. We begin by showing positive definiteness. Suppose $f \in L^p + L^r$ with $f = 0$ a.e. Then the sum $0 + 0$ is a decomposition of f into a sum of an L^p function with an L^r function; i.e., $f = 0 + 0$ a.e. But then

$$\|f\| = \inf\{\|g\|_p + \|h\|_r : f = g + h\} \le \|0\|_p + \|0\|_r = 0 + 0 = 0.$$

Now assume $\|f\| = 0$. By definition of infimum, for every $j \geq 1$, there exists a decomposition $g_j + h_j$ of f so that $\|g_j\|_p + \|h_j\|_r \leq j^{-1}$. In particular, one has $\|g_j\|_p \leq j^{-1}$ and $\|h_j\|_r \leq j^{-1}$. Consider the sequence $\{g_j\}_{j=1}^{\infty}$. By the above reasoning, it follows that the sequence converges to $0 = \|0\|_p$. Thus since $\{g_j\}$ converges to 0 under the L^p norm, it follows that there exists a subsequence $\{g_{j_k}\}_{k=1}^{\infty}$ that converges to 0 pointwise a.e. Let $\{h_{j_k}\}_{k=1}^{\infty}$ be the corresponding subsequence. Then since $\{h_j\} \to 0$ under the L^r norm, $\{h_{j_k}\}_{k=1}^{\infty}$ converges to 0 under the L^r norm. Thus there exists a subsequence $\{h_{j_{k_m}}\}_{m=1}^{\infty}$ that converges pointwise to 0 a.e. By construction, $\{g_{j_{k_m}}\}_{m=1}^{\infty}$ also converges pointwise to 0 a.e. Thus, since $f = g_{j_{k_m}} + h_{j_{k_m}}$, taking the limit $m \to \infty$ yields that $f = 0$ a.e. Thus, we conclude that $\|\cdot\|$ is positive definite. Now we will show that $\|\cdot\|$ is homogeneous. If $\lambda \in \mathbb{C}$ is zero, the proof follows trivially. Thus let $\lambda \in \mathbb{C} \setminus \{0\}$, $\mathcal{A} = \{\|g\|_p + \|h\|_r : f = g + h\}$ and $\mathcal{B} = \{\|i\|_p + \|j\|_r : \lambda f = i + j\}$. If $g + h$ is a decomposition of f, then $\lambda g + \lambda h$ is a decomposition of λf. Thus, we have $|\lambda|\mathcal{A} \subseteq \mathcal{B}$ so that $\|\lambda f\| = \inf \mathcal{B} \leq |\lambda| \inf \mathcal{A} = |\lambda| \|f\|$. On the other hand, suppose $i + j$ is a decomposition of λf. Then $(1/\lambda)(i + j)$ is a decomposition of f so that $1/|\lambda|\mathcal{B} \subseteq \mathcal{A}$, which implies $\|f\| \leq \frac{1}{|\lambda|} \inf \mathcal{B} = \frac{1}{|\lambda|} \|\lambda f\| \implies \|\lambda f\| \geq |\lambda| \|f\|$. Thus we conclude that $\|\lambda f\| = |\lambda| \|f\|$ so that $\|\cdot\|$ is homogeneous. Finally, we will show that the triangle inequality is satisfied. Let $f_1, f_2 \in L^p + L^r$, and define the following sets

$$\begin{aligned}
\mathcal{A} &= \{\|g_1\|_p + \|h_1\|_r : f_1 = g_1 + h_1\} \\
\mathcal{B} &= \{\|g_2\|_p + \|h_2\|_r : f_2 = g_2 + h_2\} \\
\mathcal{C} &= \{\|g_3\|_p + \|h_3\|_r : f_1 + f_2 = g_3 + h_3\}.
\end{aligned}$$

Let $g_1 + h_1$ and $g_2 + h_2$ be decompositions of f_1 and f_2, respectively. Then $(g_1 + g_2) + (h_1 + h_2)$ is a decomposition of $f + g$ so that $\mathcal{A} + \mathcal{B} \subset \mathcal{C}$. This implies that

$$\|f + g\| = \inf \mathcal{C} \leq \inf(\mathcal{A} + \mathcal{B}) = \inf \mathcal{A} + \inf \mathcal{B} = \|f\| + \|g\|.$$

Thus, we conclude that $\|\cdot\|$ is indeed a valid norm on $L^p + L^r$. Now we need to show that $(L^p + L^r, \|\cdot\|)$ is Banach. Let $\{f_n\}_{n=1}^{\infty}$ be a Cauchy sequence under this norm.

4.3 Baire Category Theorem

Problem 2024-J-II-5. Let P be the vector space over $\mathbb{R}$ of (finite degree) polynomials in the variable $x \in (-\infty, \infty)$. Show that P cannot be a Banach space with respect to any norm, that is, if $\|\cdot\|$ is some norm on P, then P is not complete under this norm. Hint: you may use the Baire category theorem.

Let P be the vector space over $\mathbb{R}$ of (finite degree) polynomials in the variable $x \in (-\infty, \infty)$. Let $\|\cdot\|$ be a norm on P, and assume to the contrary that P is complete under this norm. We can write

$$P = \bigcup_{n \geq 1} P_n,$$

where for each n, P_n is the subset consisting of all polynomials with degree at most n. By the Baire Category Theorem, there must exist at least one $n_0 \in \mathbb{N}$ so that $\overline{P}_{n_0}$ has nonempty interior. Since P_{n_0} is a finite-dimensional subspace of a Banach space, it must be closed: $\overline{P}_{n_0} = P_{n_0}$. So since P_{n_0} has nonempty interior, there exists some $p \in P_{n_0}$ and $R > 0$ so that $B_R(p) \Subset P_{n_0}$. Fix an arbitrary polynomial $q \in P$. Then define

$$r = p + \frac{(q - p)R}{\|q - p\|} \in \overline{B}_R(p) \subset P_{n_0}.$$

But then since

$$q = p + \frac{\|q - p\|}{R}(r - p) \in P_{n_0},$$

one has that $q \in P_{n_0}$. Thus since any arbitrary polynomial in P is contained in P_{n_0}, we are led to conclude that $P = P_{n_0}$. But this is impossible! Thus, by contradiction, P cannot be complete under $\|\cdot\|$.

4.4 Distribution Theory

Problem F-9-1-1. Suppose that $f_1, f_2, \ldots$, and f are in $L^1_{\text{loc}}(U)$. The conditions in (a) and (b) below imply that $f_n \to f$ in $\mathcal{D}'(U)$, but the condition in (c) does not.

a. $f_n \in L^p(U)$ $(1 \le p \le \infty)$ and $f_n \to f$ in the L^p norm or weakly in L^p.

b. For all n, $|f_n| \le g$ for some $g \in L^1_{\text{loc}}(U)$, and $f_n \to f$ a.e.

c. $f_n \to f$ pointwise.

a. First assume that $f_n \in L^p(U)$, and further assume that $f_n \to f$ in the L^p norm. Fix an arbitrary test function $\phi \in C_c^\infty(U)$; let K be a compact subset of U containing the support of ϕ. Then one has

$$\begin{aligned}\left|\int_U f_n\phi - \int_U f\phi\right| &= \left|\int_K (f_n - f)\phi\right| \\ &\le \int_K |f_n - f||\phi| \\ &\le \|f_n - f\|_p \|\phi\|_q \qquad \text{(by Hölder's Inequality)} \xrightarrow{n\to\infty} 0,\end{aligned}$$

so that by arbitrariness of ϕ, $f_n \to f$ in $\mathcal{D}'(U)$. Now suppose that $f_n \to f$ weakly in L^p for $1 \le p < \infty$. I.e., if q is the conjugate exponent of p, then $\int_U f_n g \to \int_U fg$ for all $g \in L^q$. We know that $C_c^\infty(U) \subset L^q(U)$. Thus it follows immediately that for every $\phi \in C_c^\infty(U)$, $\int_U f_n\phi \to \int_U f\phi$ so that $f_n \to f$ in $\mathcal{D}'(U)$.

b. Assume that $f_n \to f$ pointwise a.e. and assume there exists some $g \in L^1_{\text{loc}}(U)$ so that $|f_n| \le g$ for all n. Fix some test function $\phi \in C_c^\infty(U)$ and let $K \subset U$ be a compact subset that contains the support of ϕ. Then since each $f_n, g \in L^1(K)$, one has by the dominated convergence theorem that

$$\lim_{n\to\infty} \int_U f_n\phi = \lim_{n\to\infty} \int_K f_n\phi = \int_K \lim_{n\to\infty} f_n\phi = \int_K f\phi = \int_U f\phi.$$

Thus, $f_n \to f$ in $\mathcal{D}'(U)$.

c. Consider the sequence $\{n\chi_{(0,1/n)}\}_{n\in\mathbb{N}}$. Pointwise, the sequence converges to the zero distribution a.e., but as distributions, the sequence converges to the Dirac delta distribution supported at the origin, which is not identically zero. Thus $f_n \to f$ pointwise does not guarantee convergence on the level of distributions.

Problem F-9-1-2. The product rule for derivatives is valid for products of smooth functions and distributions.

Let $U \subseteq \mathbb{R}^n$ be an open domain, $\ell \in \mathcal{D}'(U)$, and fix arbitrary $f, \phi \in C_c^\infty(U) = \mathcal{D}(U)$. We will solve this problem in two steps.
(Step 1): Let $|\alpha| = 1$; there exists some $i \in \{1, 2, \ldots, n\}$ so that $D^\alpha = \partial_i$. In particular, we must have

$$\begin{aligned}\langle \partial_i(f\ell), \phi\rangle &= -\langle f\ell, \partial_i\phi\rangle = -\langle \ell, f\partial_i\phi\rangle = -\langle \ell, \partial_i(f\phi)\rangle + \langle \ell, \partial_i(f)\phi\rangle \\ &= \langle f\partial_i\ell, \phi\rangle + \langle \partial_i(f)\ell, \phi\rangle = \langle f\,\partial_i\ell + \partial_i f\,\ell, \phi\rangle.\end{aligned} \tag{1}$$

(Step 2): So now consider the case when $|\alpha| = m$ for some $m > 1$. Then there exist indices $\{i_j\}_{j=1}^m \subset$

$\{1, \ldots, n\}$ so that $D^\alpha = \partial_{i_1} \cdots \partial_{i_m}$. Then,

$$\begin{aligned}
\langle \partial_{i_1} \cdots \partial_{i_m}(f\ell), \phi \rangle &= (-1)^m \langle f\ell, \partial_{i_1} \cdots \partial_{i_m} \phi \rangle \\
&= (-1)^m \langle \ell, f\partial_{i_1} \cdots \partial_{i_m} \phi \rangle \\
&= (-1)^m \langle \ell, f\partial_{i_1} \psi \rangle & (\psi \coloneqq \partial_{i_2} \cdots \partial_{i_m} \phi) \\
&= (-1)^{m+1} \langle f\, \partial_{i_1} \ell + \partial_{i_1} f\ \ell, \partial_{i_2} \varphi \rangle & (\varphi \coloneqq \partial_{i_3} \cdots \partial_{i_m} \phi) \\
&= (-1)^{m+1} \left[\langle \partial_{i_1} \ell, f\partial_{i_2} \varphi \rangle + \langle \ell, (\partial_{i_1} f) \partial_{i_2} \varphi \rangle \right] \\
&= (-1)^{m+2} \left[\langle f\partial_{i_1} \partial_{i_2} \ell, \partial_{i_3} \Phi \rangle + \langle (\partial_{i_2} f)(\partial_{i_1} \ell), \partial_{i_3} \Phi \rangle \right. \\
&\qquad \left. + \langle (\partial_{i_1 f})(\partial_{i_2} \ell), \partial_{i_3} \Phi \rangle + \langle (\partial_{i_1} \partial_{i_2} f)\ell, \partial_{i_3} \Phi \rangle \right] \\
&\ \ \vdots \\
&= \sum_{|\beta| \le m} \binom{\alpha}{\beta} \langle D^\beta(f) D^{\alpha-\beta} \ell, \phi \rangle,
\end{aligned} \tag{2}$$

where $\Phi \coloneqq \partial_{i_4} \cdots \partial_{i_m} \phi$, and we repeated the same types of calculation to get to the last line. Hence, since ϕ was arbitrary, we conclude that

$$D^\alpha(f\ell) = \sum_{|\beta| \le m} \binom{\alpha}{\beta} (D^\beta f)(D^{\alpha-\beta} \ell)$$

Problem F-9-1-3. On $\mathbb{R}$, if $\psi \in C^\infty$ then $\psi\delta^{(k)} = \sum_0^k (-1)^j \binom{k}{j} \psi^{(j)}(0) \delta^{(k-j)}$, where the superscripts denote derivatives.

Let $\psi \in C^\infty(\mathbb{R})$, and let $\delta \coloneqq \delta_0 \in \mathcal{D}'(\mathbb{R})$ be the Dirac delta function. Let us fix arbitrary $\phi \in C_c^\infty(\mathbb{R})$. For some $k \ge 1$, we compute

$$\begin{aligned}
\langle \psi \partial^k \delta, \phi \rangle &= (-1)^k \langle \phi, \sum_{j=0}^k \binom{k}{j} \psi^{(j)} \phi^{(k'-j)} \rangle \\
&= (-1)^k \sum_{j=0}^k \binom{k}{j} \psi^{(j)}(0) \phi^{(k-j)}(0) \\
&= \sum_{j=0}^k \binom{k}{j} \psi^{(j)}(0) (-1)^k (-1)^{k-j} \langle \delta^{(k-j)}, \phi \rangle \\
&= \sum_{j=0}^k (-1)^j \binom{k}{j} \psi^{(j)}(0) \langle \delta^{(k-j)}, \phi \rangle.
\end{aligned} \tag{3}$$

Since ϕ was arbitrary, we conclude that

$$\psi\delta^{(k)} = \sum_{j=0}^k (-1)^j \binom{k}{j} \psi^{(j)}(0) \delta^{(k-j)}.$$

Problem F-9-1-5. Suppose that f is continuously differentiable on $\mathbb{R}$ except at $x_1, \ldots, x_m$, where f has jump discontinuities, and that its pointwise derivative df/dx (defined except at the x_j's is in $L^1_{\text{loc}}(\mathbb{R})$. Then the distribution derivative f' of f is given by $f' = (df/dx) + \sum_1^m [f(x_j+) - f(x_j-)] \tau_{x_j} \delta$.

Suppose that f is continuously differentiable on $\mathbb{R}$, except at the points $x_1, \ldots, x_m$, where f has jump discontinuities. Let df/dx, defined except at the x_j's denote its pointwise derivative. Let f' denote the distribution derivative of f. Fix an arbitrary test function $\phi \in C_c^\infty(\mathbb{R})$, and let $[x_0, x_{m+1}]$ be a compact set so that $\operatorname{supp}\phi \subsetneq K$, where $x_0 < x_1 < x_2 < \cdots < x_{m+1}$. We begin by looking at $\langle f, \phi' \rangle := \int_{\mathbb{R}} f(x)\phi'(x)\, dx$. We have the following:

$$\begin{aligned}
\langle f, \phi' \rangle &= \int_{\mathbb{R}} f(x)\phi'(x)\, dx = \int_{x_0}^{x_{m+1}} f(x)\phi'(x)\, dx \\
&= \sum_{j=0}^{m} \int_{x_j}^{x_{j+1}} f(x)\phi'(x)\, dx \\
&= \sum_{j=0}^{m} \left[f(x)\phi(x)\Big|_{x_j}^{x_{j+1}} - \int_{x_j}^{x_{j+1}} \frac{df}{dx}\phi(x)\, dx \right] \qquad \text{(by integration by parts)} \\
&= \sum_{j=0}^{m} [f(x_{j+1}-)\phi(x_{j+1}) - f(x_j+)\phi(x_j)] - \sum_{j=0}^{m} \int_{x_j}^{x_{j+1}} \frac{df}{dx}\phi(x)\, dx.
\end{aligned}$$

Telescoping the sum and using the fact that $\phi(x_{m+1}) = \phi(x_0) = 0$ since $\operatorname{supp}\phi \subsetneq [x_0, x_{m+1}]$, one has

$$\begin{aligned}
\langle f, \phi' \rangle &= -\left\langle \sum_{j=1}^{m} [f(x_j+) - f(x_j-)]\tau_{x_j}\delta, \phi \right\rangle - \left\langle \frac{df}{dx}, \phi \right\rangle \\
&= -\left\langle \frac{df}{dx} + \sum_{j=1}^{m} [f(x_j+) - f(x_j-)]\tau_{x_j}\delta, \phi \right\rangle.
\end{aligned}$$

By definition of the distribution derivative, we conclude that the distribution derivative of f is given by,

$$f' = \frac{df}{dx} + \sum_{j=1}^{m} [f(x_j+) - f(x_j-)]\tau_{x_j}\delta.$$

Problem F-9-1-6. If f is absolutely continuous on compact subsets of an interval $U \subset \mathbb{R}$, the distribution derivative $f' \in \mathcal{D}'(U)$ coincides with the pointwise (a.e.-defined) derivative of f.

Let f be absolutely continuous on compact subsets of an interval $U \subset \mathbb{R}$ (in particular, since f is continuous on compact subsets of U, one has $f \in L^1_{\text{loc}}(U)$). Let f' denote the distribution derivative of f, and let df/dx be the pointwise (a.e.-defined) derivative of f. Fix an arbitrary test function $\phi \in C_c^\infty$ and let $K = [a, b]$ be a compact subset of U so that $\operatorname{supp}\phi \subsetneq [a, b]$. Then one has

$$\langle f, \phi' \rangle = \int_a^b f(x)\phi'(x)\, dx = f(x)\phi(x)\Big|_a^b - \int_a^b \frac{df}{dx}\phi(x)\, dx,$$

where the second equality follows from integration by parts. Since $\operatorname{supp}\phi \subsetneq [a, b]$, one has $\phi(a) = \phi(b) = 0$. Thus the first term vanishes. It remains to be seen that df/dx is locally integrable. This follows immediately: since f is absolutely continuous, its pointwise a.e.-defined derivative must be integrable, and hence, must be contained in $L^1_{\text{loc}}(U)$ so that $df/dx \in \mathcal{D}'(U)$. Therefore, by definition of the distribution derivative, for every test function $\phi \in C_c^\infty(U)$, one has

$$\langle f', \phi \rangle = -\langle f, \phi' \rangle = \left\langle \frac{df}{dx}, \phi \right\rangle,$$

whence $f' = df/dx$ a.e.

5 Complex Analysis

5.1 Holomorphic Functions

Problem 2026-J-II-4. Let U be an open subset of $\mathbb{C}$. Let $(f_j(z))$ be a collection of holomorphic functions on U that converges uniformly on every compact subset. Carefully prove that the pointwise limit is holomorphic.

We will use Morera's theorem for this proof. Recall the statement of Morera's Theorem:

> **Theorem. (Morera)** A continuous, complex-valued function f defined on an open set D in the complex plane that satisfies
>
> $$\oint_\gamma f(z)\,dz = 0$$
>
> for every closed piecewise differentiable curve γ in D is holomorphic on D.

Assume the hypotheses, and define f on U to be $f(z) = \lim_{j\to\infty} f_j(z)$ for all $z \in U$. Since each f_j is holomorphic (and thus continuous), and the uniform limit of continuous functions is continuous, f is continuous. Then for any closed piecewise differentiable curve γ in U,

$$\oint_\gamma f(z)\,dz = \oint_\gamma \lim_{j\to\infty} f_j(z)\,dz = \lim_{j\to\infty}\oint_\gamma f_j(z)\,dz = \lim_{j\to\infty} 0 = 0,$$

where the limit comes out of the integral due to uniform on compacta convergence of the f_j, and holomorphicity of each f_j implies $\oint_\gamma f_j(z)\,dz = 0$. Thus, we conclude by Morera's Theorem that f is holomorphic in U.

5.2 Rouché's Theorem

Problem 2023-A-II-6. Find the number of solutions (counting multiplicity) to $z^8 - 5z^6 + 2z^3 - z - 1 = 0$ that lie inside the unit disk.

Let $p(z) = z^8 - 5z^7 + 2z^3 - z - 1$, and let $g(z) = -5z^6 + 2z^3$, $\Delta \coloneqq \{z \in \mathbb{C} : |z| \le 1\}$. It is straightforward to check that neither $p(z)$ nor $g(z)$ have any zeros on $\partial\Delta$. Indeed, on $\partial\Delta$, one has

$$\begin{aligned}|p(z) - g(z)| &= |z^8 - z - 1| \le |z|^8 + |z| + |1| \\ &= 3 < 7 = |g(z)|.\end{aligned}$$

By Rouché's Theorem, it follows that $p(z)$ has the same number of zeros as $g(z)$ in Δ. We can write $g(z) = z^3(-5z^3 + 2)$. Thus, since $g(z)$ has six roots (counting multiplicity) inside the unit disk, we conclude that $p(z)$ also has only six roots inside the unit disk.

Problem 2020-A-II-1. How many roots (counted with multiplicity) does the function

$$g(z) = 6z^3 + e^z + 1$$

have in the unit disk $|z| < 1$?

Let $g(z) = 6z^3 + e^z + 1$, $p(z) = 6z^3$, and $\Delta \coloneqq \{z \in \mathbb{C} : |z| \le 1\}$. Then it is straightforward to see that neither $p(z)$ nor $g(z)$ has any roots in $\partial\Delta$. Thus, since on $\partial\Delta$ one has

$$|g(z) - p(z)| = |e^z + 1| \le |e^z| + 1 \le e + 1 < 6 = |p(z)|.$$

Thus, by Rouché's Theorem, $g(z)$ has the same number of roots as $p(z)$ inside the unit disk (counting multiplicity). Thus, since $p(z)$ has 3 roots inside Δ, we conclude that $g(z)$ also has 3 roots inside the unit disk.

Problem 2018-A-I-3. Show that if $c > 1$, then the function

$$f(z) = ze^{c-z} - 1$$

has precisely one root in $\Delta = \{|z| < 1\}$, and this root is real and positive.

Let $c > 1$, and define $f(z) = ze^{c-z} - 1$. Define the function $g(z) = ze^{c-z}$. Then on $\partial\Delta$, one has

$$|f(z) - g(z)| = 1 < e^{c-1} = |g(z)|,$$

so that by Rouché's Theorem, $f(z)$ has the same number of zeros as $g(z)$ inside Δ. Since the exponential function is never zero, and the polynomial z is zero exactly at the origin (with multiplicity one), $g(z)$ has exactly one zero inside the unit disk. Thus, $f(z)$ also has only one zero inside the unit disk. Now consider the restriction $f|_{[0,1]}$ given by $f(x) = xe^{c-x} - 1$, for $x \in [0,1]$. By elementary calculus, $f(x)$ is continuous on $[0,1]$, and $f(0) = -1$, $f(1) = e^{c-1} - 1 > 0$ since $c > 1$. Thus, by the Intermediate Value Theorem, there exists some $x_0 \in (0,1)$ so that $f(x_0) = 0$. Thus, x_0 is a root of f in Δ. Since f has exactly one root inside the unit disk, and $x_0 > 0$, the proof concludes.

Problem 2015-A-I-3. Let f be a holomorphic function defined on a neighborhood of the closed unit disk $\overline{\mathbb{D}} := \{z \in \mathbb{C} : |z| \leq 1\}$, and assume that $|f(z)| < 1$ for all $|z| = 1$. Determine the number of fixed points of f in $\overline{\mathbb{D}}$.

Let $f(z)$ be a holomorphic function defined on a neighborhood of the closed unit disk $\overline{\mathbb{D}}$ with the aforementioned properties. The number of fixed points of f is equivalent to the number of roots of the holomorphic function $f(z) - z$ in $\overline{\mathbb{D}}$. Indeed, by Rouché's theorem, since for all $z \in \partial\overline{\mathbb{D}}$,

$$|(f(z) - z) - (-z)| = |f(z)| < 1 = |-z|,$$

$f(z) - z$ has the same number of roots as $-z$ inside the closed unit disk. Since the latter has exactly one zero inside the closed unit disk, $f(z) - z$ has exactly one zero inside the closed unit disk. Thus, $f(z)$ has exactly one fixed point inside $\overline{\mathbb{D}}$.

Problem 2012-A-I-4. Find, with proof, the precise number of zeros of the complex polynomial $p(z) = z^9 - 2z^6 + z^2 - 8z - 2$ inside the annulus $1 < |z| < 2$.

Let $p(z) = z^9 - 2z^6 + z^2 - 8z - 2$. By Rouché's theorem, the number of zeros inside the annulus is equal to the difference of the number of zeros contained in the outer disk $|z| < 2$ and the number of zeros contained in the inner disk $|z| < 1$ (on a quick note, it is important and straightforward to check that $p(z)$ has no roots in $|z| = 1$ nor $|z| = 2$).
FINDING NUMBER OF ROOTS IN $|z| < 2$:
Let $g(z) = z^9$. Then on $|z| = 2$,

$$\begin{aligned} |-2z^6 + z^2 - 8z + 2| &\leq 2|z|^6 + \cdots + 2 \\ &= 2^7 + 2^2 + 2^4 + 2 < 2^9 = |g(z)|. \end{aligned}$$

Thus, by Rouché's theorem, $p(z)$ has the same number of zeros inside $|z| < 2$ as does $g(z)$ (which is 9).
FINDING NUMBER OF ROOTS IN $|z| < 1$: Let $g(z) = -8z$. Then on $|z| = 1$,

$$\begin{aligned} |p(z) - g(z)| &\leq |z|^9 + 2|z|^6 + |z|^2 + 2 \\ &= 1 + 2 + 1 + 2 = 6 < 8 = |g(z)|. \end{aligned}$$

Thus, by Rouché's Theorem, $p(z)$ has the same number of zeros inside $|z| < 1$ as does $g(z)$ (which is 1).
Therefore, we conclude that $p(z)$ has only eight zeros inside the annulus.

6 Partial Differential Equations

6.1 Introduction

Problem E-1-1. Classify each of the partial differential equations in §1.2 as follows:
(a) Is the PDE linear, semilinear, quasilinear or fully nonlinear?
(b) What is the order of the PDE?

Name	Equation	Classification	Order
Laplace's Equation	$\Delta u = 0$	Linear	2
Helmholt's (Eigenvalue) Equation)	$-\Delta u = \lambda u$	Linear	2
Linear Transport Equation	$u_t + \sum_{i=1}^{n} b^i u_{x_i} = 0$	Linear	1
Liouville's Equation	$u_t - \sum_{i=1}^{n} (b^i u)_{x_i} = 0$	Linear	1
Heat (or diffusion) Equation	$u_t - \Delta u = 0$	Linear	2
Schrödinger's Equation	$iu_t + \Delta u = 0$	Linear	2
Kolmogorov's Equation	$u_t - \sum_{i,j=1}^{n} a^{ij} u_{x_i x_j} + \sum_{i=1}^{n} b^i u_{x_i} = 0$	Linear	2
Fokker-Planck Equation	$u_t - \sum_{i,j=1}^{n} (a^{ij} u)_{x_i x_j} - \sum_{i=1}^{n} (b^i u)_{x_i} = 0$	Linear	2
Wave Equation	$u_{tt} - \Delta u = 0$	Linear	2
Klein-Gordon Equation	$u_{tt} - \Delta u + m^2 u = 0$	Linear	2
Telegraph Equation	$u_{tt} + 2du_t - u_{xx} = 0$	Linear	2
General Wave Equation	$u_{tt} - \sum_{i,j=1}^{n} a^{ij} u_{x_i x_j} + \sum_{i=1}^{n} b^i u_{x_i} = 0$	Linear	2
Airy's Equation	$u_t + u_{xxx} = 0$	Linear	3
Beam Equation	$u_{tt} + u_{xxxx} = 0$	Linear	4

Name	Equation	Classification	Order
Eikonal Equation	$\|Du\| = 1$	Fully Nonlinear	1
Nonlinear Poisson Equation	$-\Delta u = f(u)$	Semilinear	2
p-Laplacian Equation	$\operatorname{div}(\|Du\|^{p-2} Du) = 0$	Quasilinear	2
Minimal Surface Equation	$\operatorname{div}\left(\frac{Du}{(1+\|Du\|^2)^{1/2}}\right) = 0$	Quasilinear	2
Monge-Ampère Equation	$\det(D^2 u) = f$	Fully Nonlinear	2
Hamilton-Jacobi Equation	$u_t + H(Du, x) = 0$	Quasilinear	1
Scalar Conservation Law	$u_t + \operatorname{div}\mathbf{F}(u) = 0$	Fully Nonlinear	1
Inviscid Burger's Equation	$u_t + uu_x = 0$	Quasilinear	1
Scalar Reaction-Diffusion Equation	$u_t - \Delta u = f(u)$	Semilinear	2
Porous Medium Equation	$u_t - \Delta(u^\gamma) = 0$	Quasilinear	2
Nonlinear Wave Equation	$u_{tt} - \Delta u + f(u) = 0$	Semilinear	2
Korteweg-de Vries (KdV) Equation	$u_t + uu_x + u_{xxx} = 0$	Semilinear	3
Nonlinear Schrödinger Equation	$iu_t + \Delta u = f(\|u\|^2)u$	Semilinear	2

Problem E-1-2. Let k be a positive integer. Show that a smooth function defined in $\mathbb{R}^n$ has in

general
$$\binom{n+k-1}{k} = \binom{n+k-1}{n-1}$$
distinct partial derivatives of order k.
(Hint: This is the number of ways of inserting $n-1$ dividers $|$ within a row of k symbols $\circ$: for example $\circ\circ \,||\, \circ\circ\circ \,|\, \circ \,|\, \circ\circ\circ \,||\, \circ\circ\circ\circ \,|$. Explain why each such pattern corresponds to precisely one of the partial derivatives of order k.)

Let $k \geq 1$ be a positive integer, and let $u : \mathbb{R}^n \to \mathbb{R}$ be a smooth function. We want to compute the total number of distinct partial derivatives of u of the form

$$\frac{\partial^k u}{\partial x_{i_1} \partial x_{i_2} \cdots \partial x_{i_k}} \qquad (1 \leq i_1, \ldots, i_k \leq n).$$

Problem E-1-4. Prove *Leibniz's Formula:*

$$D^\alpha(uv) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} D^\beta u D^{\alpha-\beta} v,$$

where $u, v : \mathbb{R}^n \to \mathbb{R}$ are smooth, $\binom{\alpha}{\beta} = \frac{\alpha!}{\beta!(\alpha-\beta)!}$, and $\beta \leq \alpha$ means $\beta_i \leq \alpha_i$ ($i = 1, \ldots, n$).

We can prove Leibniz's formula using induction. First, we consider the case where $|\alpha| = 1$, and assume that $D^\alpha = \partial_{x_i}$ for some positive integer $1 \leq i \leq n$ (i.e., $\alpha = (0, \ldots, 1, \ldots, 0)$ where the 1 is in the i^{th} place). We note that
$$\mathcal{A} := \{\beta \in \mathbb{N}^n : \beta \leq \alpha\} = \{(0, \ldots, 0), \alpha\}.$$
Then by the usual product rule for one variable, one has

$$D^\alpha(uv) = \frac{\partial}{\partial x_i}(uv) = (u)\left(\frac{\partial v}{\partial x^i}\right) + \left(\frac{\partial u}{\partial x_i}\right)(v) = \sum_{\beta \in \mathcal{A}} \binom{\alpha}{\beta} D^\beta u D^{\alpha-\beta} v.$$

Now assume that the result holds for all $|\alpha| = k$ for some $k \geq 1$. We now consider D^α, where $|\alpha| = k+1$. There exists a $(k+1)$-tuple $(i_1, \ldots, i_{k+1})$ so that $D^\alpha = \partial_{x_{i_1}} \cdots \partial_{x_{i_k}}$. Let γ be the k-tuple so that $D^\gamma = \partial_{x_2} \cdots \partial_{x_{k+1}}$. Then by the induction hypothesis, one has

$$D^\gamma(uv) = \sum_{\beta \leq \gamma} \binom{\gamma}{\beta} D^\beta u D^{\gamma-\beta} v,$$

so that

$$\begin{aligned} D^\alpha(uv) &= \sum_{\beta \leq \gamma} \binom{\gamma}{\beta} \partial_{x_{i_1}} (D^\beta u D^{\gamma-\beta} v) \\ &= \sum_{\beta \leq \gamma} \binom{\gamma}{\beta} \left[\partial_{x_{i_1}} (D^\beta u) D^{\gamma-\beta} v + D^\beta u \partial_{x_{i_1}} (D^{\gamma-\beta} v)\right]. \end{aligned}$$

Now we note that $\partial_{x_{i_1}} D^{\gamma-\beta} v = D^{\alpha-\beta} v$. On the other hand, we can write $\partial_{x_{i_1}} (D^\beta u) D^{\gamma-\beta} v$ as

$$\partial_{x_{i_1}} (\partial^\beta u) D^{\gamma-\beta} v = D^{\beta+e_{i_1}} u \cdot D^{\alpha-(\beta+e_{i_1})} v.$$

7 Notes

7.1 Differential Geometry and Topology

7.1.1 De Rham Cohomology

- **Def. (de Rham Cohomology Group)** Let M be a smooth manifold with or without boundary, and let p be a nonnegative positive integer. Define the following spaces:

$$\mathcal{Z}^p(M) = \ker\left(d : \Omega^p(M) \to \Omega^{p+1}(M)\right) = \{\text{closed } p\text{-forms on } M\},$$
$$\mathcal{B}^p(M) = \operatorname{im}\left(d : \Omega^{p-1}(M) \to \Omega^p(M)\right) = \{\text{exact } p\text{-forms on } M\}.$$

  Then w define the *de Rham cohomology group in degree p* to be the quotient vector space

$$H^p_{\mathrm{dR}}(M) \coloneqq \frac{\mathcal{Z}^p(M)}{\mathcal{B}^p(M)}.$$

- **Def. (Cohomologous Closed Form)** Let M be a smooth manifold with or without boundary, and let ω, ω' be closed p-forms on M. Then ω and ω' are said to be *cohomologous* if and only if $\omega = \omega + \beta$, where β is an exact p-form. Then $[\omega] = [\omega']$ in $H^p_{\mathrm{dR}}(M)$.

- **Prop. (Induced Cohomology Groups)** For any smooth map $F : M \to N$ between smooth manifolds with or without boundary, the pullback $F^* : \Omega^p(N) \to \Omega^p(M)$ carries $\mathcal{Z}^p(N)$ and $\mathcal{B}^p(N)$ into $\mathcal{Z}^p(M)$ and $\mathcal{B}^p(M)$, respectively. Thus it descends into a linear map from $H^p_{\mathrm{dR}}(N)$ to $H^p_{\mathrm{dR}}(M)$.

- **Prop. (Homotopic Maps and De Rham Cohomology)** Suppose M and N are smooth manifolds with or without boundary, and let $F, G : M \to N$ are homotopic smooth maps. For every p, the induced cohomology maps $F^*, G^* : H^p_{\mathrm{dR}}(N) \to H^p_{\mathrm{dR}}(M)$ are equal.

- **Thm. (Homotopy Invariance of De Rham Cohomology)** If M and N are homotopy equivalent smooth manifolds with or without boundary, then $H^p_{\mathrm{dR}}(M) \cong H^p_{\mathrm{dR}}(N)$ for each p. The isomorphisms are induced by any smooth homotopy equivalence $F : M \to N$.

- **Cor. (Topological Invariance of De Rham Cohomology)** The de Rham cohomology groups are topological invariants: if M and N are homeomorphic smoth manifolds with or without boundary, then their de Rham cohomology groups are isomorphic.

- **Def. (Contractible Topological Space)** A topological space X is said to be *contractible* if the identity map of X is homotopic to a constant map.

- **Thm. (Contractible Manifolds)** If M is a contractible smooth manifold with or without boundary, then $H^p_{\mathrm{dR}}(M) = 0$ for $p \geq 1$.

  Proof. Let M be a contractible smooth manifold. Then there exists some $q \in M$ so that id_M is homotopic to the constant map $c_q : M \to M$ sending all of M to $\{q\}$. Let $\iota_q : \{q\} \hookrightarrow M$ denote the inclusion map. Then since $c_q \circ \iota_q = \mathrm{Id}_M = \iota_q \circ c_q$, ι_q is a homotopy equivalence. This means that $H^p_{\mathrm{dR}}(\{q\}) \cong H^p_{\mathrm{dR}}(M)$ for each p. But $H^p_{\mathrm{dR}}(\{q\}) \cong 0$ for each $p \geq 1$ since $\{q\}$ is a 0-manifold. Thus the proof concludes. □

- **Thm. (Poincaré Lemma)** If U is a star-shaped open subset of $\mathbb{R}^n$ or $\mathbb{H}^n$ then $H^p_{\mathrm{dR}}(U) = 0$ for $p \geq 1$.

  Proof. Let U be star-shaped with respect to $c \in U$. Then the straight-line homotopy $H(x, t) = c + t(x - c)$ shows that U is contractible. Thus by the above theorem, $H^p_{\mathrm{dR}}(U) = 0$ for all $p \geq 1$. □

- **Cor. (Local Exactness of Closed Forms)** Let M be a smooth manifold with or without boundary. Each point of M has a neighborhood on which every closed form is exact.

Proof. Every point of M has a neighborhood diffeomorphic to an open ball in $\mathbb{R}^n$ or an open half-ball in $\mathbb{H}^n$, each of which is star-shaped. Thus by the Poincaré Lemma and the diffeomorphism invariance of the de Rham cohomology, the proof concludes. □

- **Thm. (First Cohomology and the Fundamental Group)** Suppose M is a connected smooth manifold. For each $q \in M$, the linear map $\Phi : H^1_{\mathrm{dR}}(M) \to \mathrm{Hom}(\pi_1(M, q), \mathbb{R})$ is well-defined and injective (in particular, an isomorphism).
- **Cor. (Finite Fundamental Group and First Cohomology)** If M is a connected smooth manifold with finite fundamental group, then $H^1_{\mathrm{dR}}(M) = 0$.

  Proof. Let M be a connected smooth manifold with finite fundamental group. By the above theorem, one has that $H^1_{\mathrm{dR}}(M) \cong \mathrm{Hom}(\pi_1(M), \mathbb{R})$. Since there is no nontrivial homomorphism from a finite group to $\mathbb{R}$, it follows that $\mathrm{Hom}(\pi_1(M), \mathbb{R})$ is trivial. Thus the proof concludes. □

Exercise 17.19. A group Γ is called a ***torsion group*** if for each $g \in \Gamma$, there exists an integer k such that $g^k = 1$. Show that if M is a connected smooth manifold whose fundamental group is a torsion group, then $H^1_{\mathrm{dR}}(M) = 0$.

Let M be connected smooth manifold whose fundamental group $\pi_1(M)$ is a torsion group. By the above theorem, one has $H^1_{\mathrm{dR}}(M) \cong \mathrm{Hom}(\pi_1(M), \mathbb{R})$. Suppose $\varphi : \pi_1(M) \to \mathbb{R}$ is a homomorphism. Let $g \in \pi_1(M)$ be an arbitrary nonidentity element, and let k be a positive integer so that $g^k = 1$. By the property of group homomorphisms,

$$0 = \varphi(e) = \varphi(g^k) = k \cdot \varphi(g).$$

Since k is a positive integer and $\mathbb{R}$ is an integral domain, we conclude that $\varphi(g) = 0$. But since g was an arbitrary non-identity element of the fundamental group, we conclude that φ is the trivial homomorphism. Thus in fact, $\mathrm{Hom}(\pi_1(M), \mathbb{R}) = 0$ so that $H^1_{\mathrm{dR}}(M) = 0$.

- **Thm. (Mayer-Vietoris)** Let M be a smooth manifold with or without boundary, and let U, V be open subsets of M whose union is M. For each p, there is a linear map $\delta : H^p_{\mathrm{dR}}(U \cap V) \to H^{p+1}_{\mathrm{dR}}(M)$ such that the following sequence called the ***Mayer-Vietoris sequence*** for the open cover $\{U, V\}$ is exact:

$$\cdots \xrightarrow{\delta} H^p_{\mathrm{dR}}(M) \xrightarrow{k^* \oplus \ell^*} H^p_{\mathrm{dR}}(U) \oplus H^p_{\mathrm{dR}}(V) \xrightarrow{i^* - j^*} H^p_{\mathrm{dR}}(U \cap V) \xrightarrow{\delta} H^{p+1}_{\mathrm{dR}}(M) \xrightarrow{k^* \oplus \ell^*} \cdots,$$

  where $i : U \cap V \hookrightarrow U$, $k : U \hookrightarrow M$, $\ell : V \hookrightarrow M$, and $j : U \cap V \hookrightarrow V$ are inclusions.
- **Thm. (Cohomology of Spheres)** For $n \geq 1$, the de Rham cohomology groups of S^n are

$$H^p_{\mathrm{dR}}(S^n) \cong \begin{cases} \mathbb{R}, & \text{if } p = 0 \text{ or } p = n, \\ 0, & \text{if } 0 < p < n. \end{cases}$$

  Proof. Since S^n is connected, $H^0_{\mathrm{dR}}(S^n) \cong \mathbb{R}$. Thus we only need to prove the above result for $p \geq 1$. We will do so by induction. For $n = 1$, since any orientation form on S^1 has nonzero integral, they must all be non-exact. Thus $\dim H^1_{\mathrm{dR}}(S^1) \geq 1$. On the other hand, there is an injective linear map from $H^1_{\mathrm{dR}}(S^1) \to \mathrm{Hom}(\pi_1(S^n, 1), \mathbb{R})$, which is 1-dimensional. Thus $H^1_{\mathrm{dR}}(S^1)$ has dimension exactly 1, and is spanned by the cohomology class of any orientation form.

  Now suppose $n \geq 2$, and assume by induction that the theorem is true for S^{n-1}. Since S^n is simply connected, $H^1_{\mathrm{dR}}(S^n) = 0$. For $p > 1$, we use the Mayer-Vietoris theorem as follows. Let N and S be the north and south poles in S^n, respectively, and let $U = S^n \setminus \{S\}$ and $V = S^n \setminus \{N\}$. By stereographic projection, both U and V are diffeomorphic to $\mathbb{R}^n$ so that $U \cap V$ is diffeomorphic to $\mathbb{R}^n \smallsetminus \{0\}$. Part of the Mayer-Vietoris sequence for $\{U, V\}$ reads

$$H^{p-1}_{\mathrm{dR}}(U) \oplus H^{p-1}_{\mathrm{dR}}(V) \to H^{p-1}_{\mathrm{dR}}(U \cap V) \to H^{p}_{\mathrm{dR}}(S^n) \to H^{p}_{\mathrm{dR}}(U) \oplus H^{p}_{\mathrm{dR}}(V).$$

Since U and V are diffeomorphic to $\mathbb{R}^n$, the groups on both ends are trivial when $p > 1$, which implies that $H^p_{\mathrm{dR}}(S^n) \cong H^{p-1}_{\mathrm{dR}}(U \cap V)$. Moreover, $U \cap V$ is diffeomorphic to $\mathbb{R}^n \smallsetminus \{0\}$ and therefore homotopy equivalent to $\mathbb{S}^{n-1}$. Thus, we conclude that $H^p_{\mathrm{dR}}(S^n) \cong H^{p-1}_{\mathrm{dR}}(S^{n-1})$ for $p > 1$. Therefore, the above result follows by induction. As in the $n = 1$ case, any smooth orientation form on $\mathbb{S}^n$ determines a nonzero cohomology class, which therefore spans $H^n_{\mathrm{dR}}(S^n)$. □

- **Cor. (Cohomology of Punctured Euclidean Space)** Suppose $n \geq 2$ and $x \in \mathbb{R}^n$. Let $M = \mathbb{R}^n \smallsetminus \{x\}$. The only nontrivial de Rham groups of M are $H^0_{\mathrm{dR}}(M)$ and $H^{n-1}_{\mathrm{dR}}(M)$, both of which are 1-dimensional. A closed $(n-1)$-form η on M is exact if and only if $\int_S \eta = 0$ for some (and hence every) $(n-1)$-dimensional sphere $S \subseteq M$ centered at x.

  Proof. Let $S \subseteq M$ be any $(n-1)$-dimensional sphere centered a x. Because inclusion $\iota : S \hookrightarrow M$ is a homotopy equivalence, $i^* : H^p_{\mathrm{dR}}(M) \to H^p_{\mathrm{dR}}(S)$ is an isomorphism for each p, so the assertion aout the dimension of $H^p_{\mathrm{dR}}(M)$ follows from the cohomology of spheres. If η is a closed $(n-1)$-form on M, it follows that η is exact if and only if $\iota^*\eta$ is exact on S, which in turn is true if and only if $\int_S \eta = \int_S \iota^*\eta = 0$ by Exercise 17.22. □

- **Cor. (Punctured Open Set)** Suppose $n \geq 2$, $U \subseteq \mathbb{R}^n$ is any open subset, and $x \in U$. Then $H^{n-1}_{\mathrm{dR}}(U \smallsetminus \{x\}) \neq 0$.

- **Lem. (Poincaré Lemma with Compact Support)** Let $n \geq p \geq 1$, and suppose ω is a compactly supported closed p-form on $\mathbb{R}^n$. If $p = n$, suppose in addition that $\int_{\mathbb{R}^n} \omega = 0$. Then there exists a compactly supported smooth $(p-1)$-form η on $\mathbb{R}^n$ such that $\omega = d\eta$.

  Proof. Let $n \geq p \geq 1$, and suppose ω is a compactly supported closed p-form on $\mathbb{R}^n$. By the Poincaré Lemma, $H^k_{\mathrm{dR}}(\mathbb{R}^n) = 0$ for all $k \geq 1$ so that ω is automatically exact. Thus, it suffices to show that ω is the exterior derivative of a *compactly supported* form. Our proof of this lemma will be split into a few different cases.

  Case 1: Suppose $n = p = 1$. Assume $\int_{\mathbb{R}^n} \omega = 0$. Then we can write $\omega = f\ dx$ for some smooth, compactly supported function $f \in C_c^\infty(\mathbb{R})$. Define $F : \mathbb{R} \to \mathbb{R}$ by

  $$F(x) = \int_{-\infty}^{x} f(t)\ dt.$$

  By the fundamental theorem of calculus, $dF = F'\ dx = f\ dx = \omega$. Thus it suffices to show that F is compactly supported. Let $R > 0$ so that $\operatorname{supp} f \subset [-R, R]$. When $x < -R$, by construction, $F(x) = 0$. When $x > R$, since $\int_{\mathbb{R}} \omega = 0$, one has

  $$F(x) = \int_{-\infty}^{x} f(t)\ dt = \int_{-\infty}^{\infty} f(t)\ dt = 0.$$

  Thus, $\operatorname{supp} F \subset [-R, R]$. Since F is compactly supported and $\omega = dF$, the proof concludes.

  Case 2: Now assume $n \geq 2$, and let $B, B' \subseteq \mathbb{R}^n$ be open balls centered at the origin such that $\operatorname{supp} \omega \subseteq B \subseteq \overline{B} \subseteq B'$. By the ordinary Poincaré Lemma, there exists a smooth (but not necessarily compactly supported) $(p-1)$-form η_0 on $\mathbb{R}^n$ such that $d\eta_0 = \omega$. In particular, $d\eta_0 = 0$ on $\mathbb{R}^n \smallsetminus \overline{B}$. Consider the further three cases:

  > Case A: $p = 1$. In this case, η_0 must be a smooth function. Since $\mathbb{R}^n \smallsetminus \overline{B}$ is connected when $n \geq 2$, it follows that η_0 is equal to a constant c on $\mathbb{R}^n \smallsetminus \overline{B}$. Let $\eta = \eta_0 - c$. Then $\eta = \eta_0 - c$ is compactly supported and satisfies $d\eta = \omega$ as claimed.

CASE B: $1 < p < n$. The restriction of η_0 to $\mathbb{R}^n \setminus \overline{B}$ is a closed $(p-1)$-form. Since $H^{p-1}_{\text{dR}}(\mathbb{R}^n \setminus \overline{B}) = 0$, there is a smooth $(p-2)$-form γ on $\mathbb{R}^n \setminus \overline{B}$ such that $d\gamma = \eta_0$ there. Let ψ be a smooth bump function that is supported in $\mathbb{R}^n \setminus \overline{B}$ and equal to 1 on $\mathbb{R}^n \setminus B'$. Then $\eta = \eta_0 - d(\psi\gamma)$ is smooth on all of $\mathbb{R}^n$ and satisfies $d\eta = d\eta_0 = \omega$. Because $d(\psi\gamma) = d\gamma = \eta_0$ on $\mathbb{R}^n \setminus B'$, η is compactly supported.

CASE C: $p = n$. In this case, we cannot use the same argument as in Case B since $H^{n-1}_{\text{dR}}(\mathbb{R}^n \setminus \overline{B}) \neq 0$. However, the restriction of η_0 to $\mathbb{R}^n \setminus \overline{B}$ is exact provided its integral is zero over some sphere centered at the origin and cntained in $\mathbb{R}^n \setminus \overline{B}$. By Stokes' Theorem,

$$0 = \int_{\mathbb{R}^n} \omega = \int_{\overline{B}'} \omega = \int_{\overline{B}'} d\eta_0 = \int_{\partial B'} \eta_0.$$

Thus η_0 is exact on $\mathbb{R}^n \setminus \overline{B}$, and the proof proceeds exactly as in Case B.

□

- **Def. (pth Compactly Supported de Rham Cohomology Group)** Let M be a smooth manifold with or without boundary, and let $\Omega^p_c(M)$ denote the vector space of compactly supported smooth p-forms on M. The p^{th} compactly supported de Rham cohomology group of M is the quotient space

$$H^p_c(M) := \frac{\ker\left(d : \Omega^p_c(M) \to \Omega^{p+1}_c(M)\right)}{\operatorname{im}\left(\Omega^{p-1}_c(M) \to \Omega^p_c(M)\right)}.$$

- **Thm. (Compactly Supported Cohomology of $\mathbb{R}^n$)** For $n \geq 1$, the compactly supported de Rham cohomology groups of $\mathbb{R}^n$ are

$$H^p_c(\mathbb{R}^n) = \begin{cases} 0, & \text{if } 0 \leq p < n, \\ \mathbb{R}, & \text{if } p = n. \end{cases}$$

Exercise 17.29. Prove this theorem.

Fix some $n \geq 1$, and let $M = \mathbb{R}^n$. For all $1 \leq p < n$, by the Poincaré Lemma with compact support, every compactly supported closed p-form on $\mathbb{R}^n$ is the exterior derivative of some compactly supported smooth $(p-1)$-form on $\mathbb{R}^n$. Thus for all $1 \leq p < n$, $H^p_c(M) = 0$. Now let $p = 0$, and $\omega \in \Omega^0_c(M)$. Since ω is closed, $d\omega$ is identically zero on $\mathbb{R}^n$. Since $\mathbb{R}^n$ is connected and ω is actually just a smooth function on $\mathbb{R}^n$, ω must be a constant everywhere on $\mathbb{R}^n$. But since ω also has to be compactly supported, $\omega = 0$. Thus since every compactly supported closed smooth 0-form on $\mathbb{R}^n$ is identically zero, $H^0_c(M) = 0$. We have proven the first half of the theorem. Now suppose $p = n$. First assume $n = 1$. Let $\psi(x)$ be a smooth, non-negative bump function supported in some interval $B = [a, b]$ so that $\int_{\mathbb{R}} \psi(x)\, dx > 0$, and let $\omega = \psi(x)\, dx$. Let f be a smooth function on $\mathbb{R}$ such that $\omega = df$. In particular, we must then have

$$f(x) = \int_{-\infty}^{x} \psi(t)\, dt.$$

For any $x > b$,

$$f(x) = \int_{-\infty}^{x} \psi(t)\, dt = \int_{-\infty}^{\infty} \psi(t)\, dt = C > 0,$$

which means f is not compactly supported. Thus, $\dim H^1_c(\mathbb{R}) \geq 1$ since we have shown the existence of a closed, non-exact, compactly supported 1-form. [!! Complete Later !!]

- **Thm. (Top Cohomology, Orientable Compact Support Case)** If M is a connected oriented smooth n-manifold, then the integration map $I : H^n_c(M) \to \mathbb{R}$ is an isomorphism so that $H^n_c(M)$ is 1-dimensional.

- **Thm. (Top Cohomology, Orientable Compact Case)** If M is a compact connected orientable smooth n-manifold, then $H^n_{\mathrm{dR}}(M)$ is 1-dimensional, and is spanned by the cohomology class of any smooth orientation form.
- **Thm. (Top Cohomology, Orientable Noncompact Case)** If M is a noncompact connected orientable smooth n-manifold, then $H^n_{\mathrm{dR}}(M) = 0$.
- **Lem. (Nonorientability and Cohomology)** Suppose M is a connected nonorientable smooth manifold and $\hat{\pi} : \hat{M} \to M$ is its orientation covering. For each p, the induced cohomology maps $\hat{\pi}^* : H^p_{\mathrm{dR}}(M) \to H^p_{\mathrm{dR}}(\hat{M})$ and $\hat{\pi}^* : H^p_c(M) \to H^p_c(\hat{M})$ are injective.
- **Thm. (Top Cohomology, Nonorientable Case)** If M is a connected nonorientable smooth n-manifold, then $H^n_c(M) = 0$ and $H^n_{\mathrm{dR}}(M) = 0$.

Problem LSM-17-5. For each $n \geq 1$, compute the de Rham cohomology groups of $\mathbb{R}^n \setminus \{e_1, -e_1\}$; and for each nonzero cohomology group, give specific differential forms whose cohomology classes form a basis.

We will work through this problem through several cases.
Case 1: First suppose that $n = 1$, and fix $e_1 \in \mathbb{R}$. Then $\mathbb{R} \setminus \{e_1, -e_1\}$ is the disjoint union of three connected components M_1, M_2, M_3 so that $H^1_{\mathrm{dR}}(M) \cong \mathbb{R}^3$. Next, since each connected component is contractible,

$$H^1_{\mathrm{dR}}(M) \cong \prod_1^3 H^1_{\mathrm{dR}}(M_j) = 0.$$

Case 2: Now let $n \geq 2$. Since M is connected, $H^0_{\mathrm{dR}}(M) = \mathbb{R}$. Since M is connected, noncompact, and orientable, $H^n_{\mathrm{dR}}(M) = 0$. Thus it suffices to compute the de Rham cohomology groups for $1 \leq p \leq n-1$. We first focus on $1 < p < n-1$. Let $e_1 = (1, 0, \ldots, 1)$, and define the open subsets $U = \{x \in \mathbb{R}^n : x_1 > -0.5\} \setminus \{e_1\}$ and $V = \{x \in \mathbb{R}^n : x_1 < 0.5\} \setminus \{e_2\}$. U and V are both homotopy equivalent to S^{n-1} and $U \cap V$ is contractible. By the Mayer-Vietoris Theorem, we have the exact sequence

$$\cdots \to H^{p-1}_{dR}(U) \oplus H^{p-1}_{dR}(V) \to H^{p-1}_{dR}(U \cap V) \to H^p_{dR}(M) \to H^p_{dR}(U) \oplus H^p_{dR}(V) \to H^p_{dR}(U \cap V) \to \cdots,$$

which then becomes

$$\cdots \to 0 \to 0 \to H^p_{dR}(M) \to 0 \to 0 \to \cdots.$$

Thus for $1 < p < n-1$, one has $H^p_{\mathrm{dR}}(M) = 0$. Now let $p = n-1$. Then Mayer-Vietoris gives us the exact sequence

$$\cdots \to H^{n-2}_{\mathrm{dR}}(U \cap V) \to H^{n-1}_{\mathrm{dR}}(M) \to H^{n-1}_{\mathrm{dR}}(U) \oplus H^{n-1}_{\mathrm{dR}}(V) \to H^{n-1}_{\mathrm{dR}}(U \cap V) \to \cdots.$$

For $n > 2$, the above exact sequence becomes

$$\cdots \to 0 \to H^{n-1}_{\mathrm{dR}}(M) \to \mathbb{R}^2 \to 0 \to \cdots,$$

so that $H^{n-1}_{\mathrm{dR}}(M) \cong \mathbb{R}^2$. Finally, we consider the case where $p = 1$. By the Mayer-Vietoris Theorem, we have the exact sequence

$$\cdots \to H^0_{\mathrm{dR}}(U) \oplus H^0_{\mathrm{dR}}(V) \to H^0_{\mathrm{dR}}(U \cap V) \to H^1_{\mathrm{dR}}(M) \to H^1_{\mathrm{dR}}(U) \oplus H^1_{\mathrm{dR}}(V) \to H^1_{\mathrm{dR}}(U \cap V) \to \cdots,$$

which becomes

$$\cdots \to \mathbb{R}^2 \to \mathbb{R} \to H^1_{\mathrm{dR}}(M) \to H^1_{\mathrm{dR}}(U) \oplus H^1_{\mathrm{dR}}(V) \to 0 \to \cdots.$$

The map $\mathbb{R}^2 \to \mathbb{R}$ is defined by $(f, g) \mapsto f|_{U \cap V} - g|_{U \cap V}$. But since $H^0_{\mathrm{dR}}(V), H^0_{\mathrm{dR}}(V) \cong \mathbb{R}$, we can treat the elements in the direct sum $H^0_{\mathrm{dR}}(U) \oplus H^0_{\mathrm{dR}}(V)$ simply as pairs of real numbers (c_1, c_2). Thus the map is surjective. By the First Isomorphism Theorem,

$$\mathrm{Im}(\mathbb{R} \to H^1_{\mathrm{dR}}(M)) \cong \mathbb{R} / \ker(\mathbb{R} \to H^1_{\mathrm{dR}}(M)).$$

But $\ker(\mathbb{R} \to H^1_{\mathrm{dR}}(M)) = \mathrm{Im}(\mathbb{R}^2 \to \mathbb{R}) = \mathbb{R}$. Thus, $\mathrm{Im}(\mathbb{R} \to H^1_{\mathrm{dR}}(M)) = 0$. This means that we have the exact sequence

$$0 \to H^1_{\mathrm{dR}}(M) \to H^1_{\mathrm{dR}}(U) \oplus H^1_{\mathrm{dR}}(V) \to 0,$$

so that $H^1_{\mathrm{dR}}(M) \cong H^1_{\mathrm{dR}}(U) \oplus H^1_{\mathrm{dR}}(V)$. Thus, by using the cohomology of spheres, the computations conclude.

Problem LSM-17-6. Let M be a connected smooth manifold of dimension $n \geq 3$. For any $x \in M$ and $0 \leq p \leq n-2$, prove that the map $H^p_{\mathrm{dR}}(M) \to H^p_{\mathrm{dR}}(M \smallsetminus \{x\})$ induced by inclusion $M \smallsetminus \{x\} \hookrightarrow M$ is an isomorphism. Prove that the same is true for $p = n-1$ if M is compact and orientable. [Hint: use the Mayer-Vietoris theorem. The cases $p = 0$, $p = 1$, and $p = n-1$ require special handling.]

Let M be a connected smooth manifold of dimension $n \geq 3$. Fix an arbitrary $x \in M$. Let us denote the following: (1) $U = M \smallsetminus \{x\}$, (2) V an open neighborhood of x in M, (3) $i : U \cap V \hookrightarrow U$, (4) $j : U \cap V \hookrightarrow V$, (5) $k : U \hookrightarrow M$, and $l : V \hookrightarrow M$. We will work through the proof through several different cases.
CASE 1: $1 < p \leq n-2$, $n > 3$. For each such p, the Mayer-Vietoris theorem gives us the exact sequence

$$\cdots \to H^{p-1}_{\mathrm{dR}}(U \cap V) \to H^p_{\mathrm{dR}}(M) \to H^p_{\mathrm{dR}}(U) \oplus H^p_{\mathrm{dR}}(V) \to H^p_{\mathrm{dR}}(U \cap V) \to \cdots.$$

Since $U \cap V$ is homotopy equivalent to S^{n-1}, for $1 < p \leq n-2$ (with $n > 3$), $H^p_{\mathrm{dR}}(U \cap V), H^{p-1}_{\mathrm{dR}}(U \cap V) \cong 0$. Moreover, since V is contractible, $H^p_{\mathrm{dR}}(V) \cong 0$. Thus, we have the short exact segment

$$0 \to H^p_{\mathrm{dR}}(M) \xrightarrow{k^* \oplus l^*} H^p_{\mathrm{dR}}(U) \oplus 0 \to 0,$$

so that $H^p_{\mathrm{dR}}(M) \cong H^p_{\mathrm{dR}}(U) \oplus 0$. Thus $k^* \oplus l^*$ is an isomorphism, which then implies that k^* is an isomorphism since l^* is the zero map.
CASE 2: Now we will look at the case $p = 0$ for all $n \geq 3$. The Mayer-Vietoris theorem gives us the following exact sequence

$$0 \to H^0_{\mathrm{dR}}(M) \xrightarrow{k^* \oplus l^*} H^0_{\mathrm{dR}}(U) \oplus H^0_{\mathrm{dR}}(V) \xrightarrow{i^* - j^*} H^0_{\mathrm{dR}}(U \cap V) \xrightarrow{\delta} H^1_{\mathrm{dR}}(M) \to \cdots.$$

In particular, the above sequence becomes,

$$0 \to \mathbb{R} \xrightarrow{k^* \oplus l^*} \mathbb{R}^2 \xrightarrow{i^* - j^*} \mathbb{R} \xrightarrow{\delta} H^1_{\mathrm{dR}}(M) \to \cdots.$$

In particular, we already know that $k^* \oplus l^*$ is injective by exactness. On the other hand, $i^* - j^*$ is the map that maps $(a, b) \mapsto a - b$, and therefore is surjective. By exactness, $\ker \delta = \mathrm{Im}(i^* - j^*) = \mathbb{R}$ so that δ is the zero map. So we actually have the short exact sequence

$$0 \to \mathbb{R} \xrightarrow{k^* \oplus l^*} \mathbb{R}^2 \xrightarrow{i^* - j^*} \mathbb{R} \to 0.$$

Since $k^* \oplus l^* : c \mapsto (k^* c, l^* c) = (c, c)$, $k^* : c \mapsto c$. Since $H^0_{\mathrm{dR}}(M) \cong H^0_{\mathrm{dR}}(U)$ and k^* maps each $c \in \mathbb{R}$ to itself, k^* is the identity map, and thus is an isomorphism.
CASE 3: Now we will look at the case $p = 1$ (which also includes the case $n = 3$). The Mayer-Vietoris Theorem gives us the following exact sequence

$$\cdots \to H^0_{\mathrm{dR}}(U \cap V) \xrightarrow{\delta} H^1_{\mathrm{dR}}(M) \xrightarrow{k^* \oplus l^*} H^1_{\mathrm{dR}}(U) \oplus H^1_{\mathrm{dR}}(V) \xrightarrow{i^* - j^*} H^1_{\mathrm{dR}}(U \cap V) \xrightarrow{\delta} H^2_{\mathrm{dR}}(M) \to \cdots.$$

In particular, one has

$$\cdots \to \mathbb{R} \xrightarrow{\delta} H^1_{\mathrm{dR}}(M) \xrightarrow{k^* \oplus l^*} H^1_{\mathrm{dR}}(U) \oplus 0 \xrightarrow{i^* - j^*} 0 \xrightarrow{\delta} H^2_{\mathrm{dR}}(M) \to \cdots.$$

But in our previous work, we showed that δ was the zero map. So in fact, we must have the short exact segment

$$0 \xrightarrow{\delta} H^1_{\mathrm{dR}}(M) \xrightarrow{k^* \oplus l^*} H^1_{\mathrm{dR}}(U) \oplus 0 \xrightarrow{i^* - j^*} 0,$$

so that $k^* \oplus l^*$ is an isomorphism, and therefore, k^* is an isomorphism.
CASE 4: Now assume that M is compact and orientable. We will look at the case $p = n-1$. The Mayer-Vietoris Theorem gives us the following exact sequence

$$\cdots \to H^{n-2}_{\mathrm{dR}}(U \cap V) \xrightarrow{\delta} H^{n-1}_{\mathrm{dR}}(M) \xrightarrow{k^* \oplus l^*} H^{n-1}_{\mathrm{dR}}(U) \oplus H^{n-1}_{\mathrm{dR}}(V) \xrightarrow{i^* - j^*} H^{n-1}_{\mathrm{dR}}(U \cap V)$$
$$\xrightarrow{\delta} H^n_{\mathrm{dR}}(M) \to H^n_{\mathrm{dR}}(U) \oplus H^n_{\mathrm{dR}}(V) \to \cdots.$$

In particular, using the information we have, the above short exact segment can be written as the following:

$$\cdots \to 0 \xrightarrow{\delta} H^{n-1}_{\mathrm{dR}}(M) \xrightarrow{k^*\oplus l^*} H^{n-1}_{\mathrm{dR}}(U) \oplus 0 \xrightarrow{i^*-j^*} \mathbb{R} \xrightarrow{\delta} \mathbb{R} \xrightarrow{k^*\oplus l^*} 0,$$

where the penultimate term follows from the compactness, connectedness, and orientability of M, and the last term follows from the fact that U is a noncompact, connected, orientable smooth n-manifold. Thus, we have that $\delta : \mathbb{R} \to \mathbb{R}$ is surjective. Since δ is a surjection between equal dimensional vector spaces, δ is an isomorphism. Thus, $0 = \ker\delta = \mathrm{Im}(i^* - j^*)$. Therefore, we have the short exact segment $\cdots \to 0 \to H^{n-1}_{\mathrm{dR}}(M) \to H^{n-1}_{\mathrm{dR}}(U) \oplus 0 \to 0$, whence we conclude that k^* is an isomorphism.

Problem LSM-17-7. Let M_1, M_2 be connected smooth manifolds of dimension $n \geq 3$, and let $M_1 \# M_2$ denote their smooth connected sum (Example 9.31). Prove that $H^p_{\mathrm{dR}}(M_1\#M_2) \cong H^p_{\mathrm{dR}}(M_1) \oplus H^p_{\mathrm{dR}}(M_2)$ for $0 < p < n-1$. Prove that the same is true for $p = n-1$ if M_1 and M_2 are both compact and orientable. [Hint: use Problems 9-12 and 17-6.]

Let M_1, M_2 be connected smooth manifolds of dimension $n \geq 3$, and let M denote their smooth connected sum. Fix $p_1 \in M_1$ and $p_2 \in M_2$. By Problem 9-12, there exist open subsets $\widetilde{M}_1, \widetilde{M}_2 \subseteq M$ diffeomorphic to $M_1 \setminus \{p_1\}$ and $M_2 \setminus \{p_2\}$, respectively, such that $\widetilde{M}_1 \cup \widetilde{M}_2 = M$ and $\widetilde{M}_1 \cap \widetilde{M}_2$ is diffeomorphic to $(-1,1) \times S^{n-1}$.
Next, for each $0 \leq p < n-1$, $H^p_{\mathrm{dR}}(\widetilde{M}_j) \cong H^p_{\mathrm{dR}}(M_j/\{p_j\}) \cong H^p_{\mathrm{dR}}(M_j)$ for $j = 1,2$, where the first isomorphism follows by diffeomorphism invariance of the de Rham cohomology group, and the second isomorphism follows from Problem LSM-17-6. Now we will examine a few different cases.
CASE 1: Suppose $1 < p \leq n-2, n > 3$. The Mayer-Vietoris theorem gives us the following exact sequence,

$$\cdots \to H^{p-1}_{\mathrm{dR}}(\widetilde{M}_1 \cap \widetilde{M}_2) \to H^p_{\mathrm{dR}}(M) \to H^p_{\mathrm{dR}}(\widetilde{M}_1) \oplus H^p_{\mathrm{dR}}(\widetilde{M}_2) \to H^p_{\mathrm{dR}}(\widetilde{M}_1 \cap \widetilde{M}_2) \to \cdots,$$

which then begets the exact sequence,

$$\cdots \xrightarrow{i^*-j^*} H^{p-1}_{\mathrm{dR}}((-1,1)\times S^{n-1}) \xrightarrow{\delta} H^p_{\mathrm{dR}}(M) \xrightarrow{k^*\oplus l^*} H^p_{\mathrm{dR}}(M_1) \oplus H^p_{\mathrm{dR}}(M_2) \xrightarrow{i^*-j^*} H^p_{\mathrm{dR}}((-1,1)\times S^{n-1}) \xrightarrow{\delta} \cdots.$$

Since $(-1,1) \times S^{n-1}$ is homotopy equivalent to S^{n-1}, for $1 < p \leq n-2$,

$$H^{p-1}_{\mathrm{dR}}((-1,1)\times S^{n-1}), H^p_{\mathrm{dR}}((-1,1)\times S^{n-1}) \cong 0.$$

Thus, we have the short exact sequence

$$0 \xrightarrow{\delta} H^p_{\mathrm{dR}}(M) \xrightarrow{k^*\oplus l^*} H^p_{\mathrm{dR}}(M_1) \oplus H^p_{\mathrm{dR}}(M_2) \xrightarrow{i^*-j^*} 0,$$

CASE 2: Now consider the case $p = 1$ (which also includes the case $n = 3$). From the Mayer-Vietoris theorem, we have the following exact sequence:

$$\begin{aligned}\cdots \to H^0_{\mathrm{dR}}(M_1) \oplus H^0_{\mathrm{dR}}(M_2) \xrightarrow{i^*-j^*} H^0_{\mathrm{dR}}(\widetilde{M}_1 \cap \widetilde{M}_2) \xrightarrow{\delta} H^1_{\mathrm{dR}}(M) \xrightarrow{k^*\oplus l^*} H^1_{\mathrm{dR}}(M_1) \oplus H^1_{\mathrm{dR}}(M_2)& \\ \xrightarrow{i^*-j^*} H^1_{\mathrm{dR}}(\widetilde{M}_1 \cap \widetilde{M}_2) \xrightarrow{\delta} H^2_{\mathrm{dR}}(M) \to \cdots,&\end{aligned}$$

which begets the following exact sequence

$$\cdots \to \mathbb{R}^2 \xrightarrow{i^*-j^*} \mathbb{R} \xrightarrow{\delta} H^1_{\mathrm{dR}}(M) \xrightarrow{k^*\oplus l^*} H^1_{\mathrm{dR}}(M_1) \oplus H^1_{\mathrm{dR}}(M_2) \xrightarrow{i^*-j^*} 0 \to \cdots.$$

Here $i^* - j^* : (c,d) \in \mathbb{R}^2 \mapsto c - d$ so that $i^* - j^*$ is surjective. By exactness, $\ker\delta = \mathbb{R}$ so that δ is the zero map. Thus we actually have the exact sequence

$$0 \to H^1_{\mathrm{dR}}(M) \xrightarrow{k^*\oplus l^*} H^1_{\mathrm{dR}}(M_1) \oplus H^1_{\mathrm{dR}}(M_2) \xrightarrow{i^*-j^*} 0,$$

Case 3: Now assume both M_1 and M_2 are compact and orientable, and let $p = n-1$. Then by Problems LSM-9-12 and LSM-17-6, one has for $j = 1, 2$: $H^{n-1}_{\mathrm{dR}}(\widetilde{M}_j) \cong H^{n-1}_{\mathrm{dR}}(M_j)$. Thus the Mayer-Vietoris theorem gives the following exact sequence:

$$\cdots \to H^{n-2}_{\mathrm{dR}}(\widetilde{M}_1 \cap \widetilde{M}_2) \xrightarrow{\delta} H^{n-1}_{\mathrm{dR}}(M) \xrightarrow{k^* \oplus l^*} H^{n-1}_{\mathrm{dR}}(M_1) \oplus H^{n-1}_{\mathrm{dR}}(M_2) \xrightarrow{i^* - j^*} H^{n-1}_{\mathrm{dR}}(\widetilde{M}_1 \cap \widetilde{M}_2)$$
$$\xrightarrow{\delta} H^{n}_{\mathrm{dR}}(M) \xrightarrow{k^* \oplus l^*} H^{n}_{\mathrm{dR}}(\widetilde{M}_1) \oplus H^{n}_{\mathrm{dR}}(\widetilde{M}_2) \to \cdots.$$

Using the information we have, we obtain the following exact sequence:

$$\cdots \to 0 \xrightarrow{\delta} H^{n-1}_{\mathrm{dR}}(M) \xrightarrow{k^* \oplus l^*} H^{n-1}_{\mathrm{dR}}(M_1) \oplus H^{n-1}_{\mathrm{dR}}(M_2) \xrightarrow{i^* - j^*} \mathbb{R} \xrightarrow{\delta} \mathbb{R} \xrightarrow{k^* \oplus l^*} 0,$$

where we used the facts that the top de Rham cohomology of a compact connected orientable manifold is one-dimensional, $M_1 \# M_2$ is compact connected and orientable, and $\widetilde{M}_1, \widetilde{M}_2$ are both noncompact and orientable so that their top de Rham cohomology groups are trivial. By exactness, δ must be surjective; since it is a surjective linear map between equal dimensional vector spaces, it must also be injective. Thus by exactness, $i^* - j^*$ is the zero map. Thus, we have the exact sequence

$$0 \xrightarrow{\delta} H^{n-1}_{\mathrm{dR}}(M) \xrightarrow{k^* \oplus l^*} H^{n-1}_{\mathrm{dR}}(M_1) \oplus H^{n-1}_{\mathrm{dR}}(M_2) \to 0,$$

whence $H^{n-1}_{\mathrm{dR}}(M) \cong H^{n-1}_{\mathrm{dR}}(M_1) \oplus H^{n-1}_{\mathrm{dR}}(M_2)$.
The proof concludes.

7.2 Partial Differential Equations

7.2.1 Introduction

- **Def. (k^{th}-Order Partial Differential Equation)** Fix an integer $k \geq 1$, and let U denote an open subset of $\mathbb{R}^n$. An expression of the form

$$F(D^k u(x), D^{k-1}u(x), \ldots, Du(x), u(x), x) = 0, \qquad (x \in U),$$

where $F : \mathbb{R}^{n^k} \times \mathbb{R}^{n^{k-1}} \times \cdots \times \mathbb{R}^n \times \mathbb{R} \times U \to \mathbb{R}$ is given and $u : U \to \mathbb{R}$ is an unknown function, is called a *k-th order partial differential equation* (PDE).

- **Def. (Classification of PDE Types)** Consider a k-th order PDE of the above form. Then:
  (i) the PDE is *linear* iff it has the form

$$\sum_{|\alpha| \leq k} a_\alpha(x) D^\alpha u(x) = f(x),$$

  for given functions $a_\alpha(x)$ ($|\alpha| \leq k$), f. The linear PDE is *homogeneous* if $f \equiv 0$;
  (ii) the PDE is *semilinear* if it has the form

$$\sum_{|\alpha|=k} a_\alpha(x) D^\alpha u + a_0(D^{k-1}u, \ldots, Du, u, x) = 0;$$

  (iii) the PDE is *quasilinear* if it has the form

$$\sum_{|\alpha|=k} a_\alpha(D^{k-1}u, \ldots, Du, u, x) D^\alpha u + a_0(D^{k-1}u, \ldots, Du, u, x) = 0;$$

  (iv) the PDE is *fully nonlinear* if it depends nonlinearly upon the highest order derivatives.

- **Def. (k-th Order System of PDE)** Fi an integer $k \geq 1$, and let U denote an open subset of $\mathbb{R}^n$. An expression of the form

$$\mathbf{F}(D^k \mathbf{u}(x), D^{k-1}\mathbf{u}(x), \ldots, D\mathbf{u}(x), \mathbf{u}(x), x) = \mathbf{0}, \qquad (x \in U),$$

where $\mathbf{F} : \mathbb{R}^{mn^k} \times \mathbb{R}^{mn^{k-1}} \times \cdots \times \mathbb{R}^{mn} \times \mathbb{R}^m \times U \to \mathbb{R}^m$ is given and $\mathbf{u} : U \to \mathbb{R}^m, \mathbf{u} = (u^1, \ldots, u^m)$ is the unknown, is called a *k-th order system of PDE*.

- **Def. (Well-Posed PDE)** A PDE is said to be *well-posed* iff:
  (i) the PDE problem has a solution;
  (ii) the solution is unique;
  (iii) the solution depends *continuously* on the data given in the problem (i.e., "a small change in the initial data only changes the solution by a small amount").

- **Def. (Classical Solution to a k-th Order PDE)** A function u is called a *classical solution* to a k-th order partial differential equation problem if and only if u satisfies the PDE and $u \in C^k(U)$. While higher regularity is desirable, this could be too much to ask for, in general (e.g., the scalar conservation law).

7.2.2 Four Important Linear Partial Differential Equations

Transport Equation

- **Def. (Transport Equation with Constant Coefficients)** The transport equation with constant coefficients is the PDE,

$$u_t + b \cdot Du = 0 \qquad \text{in } \mathbb{R}^n \times (0, \infty),$$

where $b = (b_1, \ldots, b_n) \in \mathbb{R}^n$ is fixed, $u = u(x,t) : \mathbb{R}^n \times [0, \infty) \to \mathbb{R}$ is the unknown, and $Du = D_x u = (u_{x_1}, \ldots, u_{x_n})$ denotes the gradient of u with respect to the spatial variables x.

- **Def. (Initial-Value Problem)** The initial-value problem for the transport equation with constant coefficients is given by

$$\begin{cases} u_t + b \cdot Du = 0 & \text{in } \mathbb{R}^n \times (0, \infty) \\ u = g & \text{on } \mathbb{R}^n \times \{t = 0\}, \end{cases}$$

  where $b \in \mathbb{R}^n$ and $g : \mathbb{R}^n \to \mathbb{R}$ are known, and the problem is to compute the unknown function g.

- **Prop. (Solving Transport Equation with Constant Coefficients)** To determine which functions u solve the transport equation, we begin by assuming we have *some* solution u and try to compute this. First fix some point $(x, t) \in \mathbb{R}^n \times (0, \infty)$ and define

$$z(s) := u(x + sb, t + s) \qquad (s \in \mathbb{R}).$$

  Then we compute

$$z(s) = Du(x + sb, t + s) \cdot b + u_t(x + sb, t + s) = 0.$$

  I.e., $z(\cdot)$ is a constant function of s. Consequently, for each point (x, t), u is constant on the line through (x, t) with the direction $(b, 1) \in \mathbb{R}^{n+1}$. Therefore, if we know the value of u at any point on each such line we know its value everywhere in $\mathbb{R}^n \times (0, \infty)$. Thus given (x, t) as above, the line through (x, t) with direction $(b, 1)$ is represented parametrically by $(x+sb, t+s)$ $(s \in \mathbb{R})$. This line hits the plane $\Gamma := \mathbb{R}^n \times \{t = 0\}$ when $s = -t$, at the point $(x - tb, 0)$. Since u is constant on this line, and $u(x - tb, 0) = g(x - tb)$, we deduce that

$$u(x, t) = g(x - tb) \qquad (x \in \mathbb{R}^n, t \geq 0).$$

  So *if* the transport problem with constant coefficients has a sufficiently regular solution u, it must be given by the above equation. Conversely, if $g \in C^1$, then u defined by above is certainly a solution of the transport equation. If $g \notin C^1$, then the transport equation cannot have any C^1 solution. But we will declare $u(x, t) = g(x - tb)$ to be a *weak solution* of the transport equation.

- **Prop. (Solving Nonhomogeneous Transport Equation with Constant Coefficients)** Consider the associated nonhomogeneous problem

$$\begin{cases} u_t + b \cdot Du = f & \text{in } \mathbb{R}^n \times (0, \infty) \\ u = g, & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

  As before, fix $(x, t) \in \mathbb{R}^{n+1}$, and set $z(s) = u(x + sb, t + s)$ for $s \in \mathbb{R}$. Then

$$z(s) = Du(x + sb, t + s) \cdot b + u_t(x + sb, t + s) = f(x + sb, t + s).$$

  Consequently,

$$\begin{aligned} u(x, t) - g(x - tb) = z(0) - z(-t) &= \int_{-t}^{0} z(s)\, ds \\ &= \int_{-t}^{0} f(x + sb, t + s)\, ds \\ &= \int_{0}^{t} f(x + (s - t)b, s)\, ds. \end{aligned}$$

  This means that

$$u(x, t) = g(x - tb) + \int_0^t f(x + (s - t)b, s)\, ds \quad (x \in \mathbb{R}^n, t \geq 0)$$

  solves the initial-value problem.

Laplace's Equation

- **Def. (Laplace's and Poisson's Equations)** Fix an open subset $U \subseteq \mathbb{R}^n$, and let the unknown function be $u : \overline{U} \to \mathbb{R}$, $u = u(x)$ ($x \in U$). Also fix some function $f : U \to \mathbb{R}$. Then:
  (1) (LAPLACE'S EQUATION) $\Delta u = 0$;
  (2) (POISSON'S EQUATION) $\Delta u = f$;

  where $\Delta u \coloneqq \sum_{i=1}^n u_{x_i x_i}$ is the *Laplacian of u.*

- **Def. (Harmonic Function)** a C^2 function u satisfying Laplace's equation.

- **Prop. (Derivation of Laplace's Equation)** Let $U^{\text{open}} \subseteq \mathbb{R}^n$ be a fixed open subset, V any smooth subregion within U, and let u denote the density of some quantity (e.g., a chemical concentration) in equilibrium. Then the next flux of u through ∂V is zero (equilibrium means net flux is zero):
$$\int_{\partial V} \mathbf{F} \cdot \boldsymbol{\nu} \, dS = 0,$$
  where $\mathbf{F}$ denotes the flux density and $\boldsymbol{\nu}$ denotes the unit outer normal field. By the Gauss-Green Theorem, one has
$$\int_V \operatorname{div}\mathbf{F} \, dx = \int_{\partial V} \mathbf{F} \cdot \boldsymbol{\nu} \, ds = 0,$$
  so that
$$\operatorname{div}\mathbf{F} = 0$$
  since V was arbitrary. Assume $\mathbf{F} = -aDu$ ($a > 0$). Then substituting into the above result, we have that
$$\Delta u = \operatorname{div}(Du) = 0.$$

- **Prop. (Derivation of Fundamental Solution to Laplace's Equation)** Our strategy to study Laplace's equation is to identify explicit solutions, then exploit linearity to assemble more complicated solutions. In particular, we will restrict attention to functions with certain symmetry properties. Thus, we will begin by looking for radial solutions (i.e., functions of $r = |x|$) since the Laplace equation is rotation invariant. Let $u(x) = v(r)$, where $r = |x| = (x_1^2 + \cdots + x_n^2)^{1/2}$ and v is to be selected (if possible) so that $\Delta u = 0$. First, we observe that for $i = 1, \ldots, n$,
$$\frac{\partial r}{\partial x_i} = \frac{1}{2}(x_1^2 + \cdots + x_n^2)^{-1/2} 2x_i = \frac{x_i}{r} \qquad (x \neq 0).$$
  Therefore, we must have
$$u_{x_i} = v'(r)\frac{x_i}{r} \implies u_{x_i x_i} = v''(r)\frac{x_i^2}{r^2} + v'(r)\left(\frac{1}{r} - \frac{x_i^2}{r^3}\right).$$
  Therefore,
$$\begin{aligned} \Delta u = \sum_{i=1}^n u_{x_i x_i} &= \frac{v''(r)}{r^2}\sum_{i=1}^n x_i^2 + \frac{v'(r)}{r}\sum_{i=1}^n 1 - \frac{v'(r)}{r^3}\sum_{i=1}^n x_i^2 \\ &= v''(r) + \frac{nv'(r)}{r} - \frac{v'(r)}{r} = v''(r) + \frac{n-1}{r}v'(r). \end{aligned}$$
  Thus, $\Delta u = 0$ if and only if
$$v''(r) + \frac{n-1}{r}v'(r) = 0.$$
  Suppose $v' \neq 0$. Then we have
$$\log(v')' = \frac{v''}{v'} = \frac{n-1}{r},$$

so that $v'(r) = ar^{1-n}$ for some constant a. Consequently, if $r > 0$, we have

$$v(r) = \begin{cases} b\log r + c & (n=2) \\ br^{2-n} + c & (n \geq 3) \end{cases},$$

where b and c are constants.

- **Def. (Fundamental Solutions of Laplace's Equation)** The function

$$\Phi(x) = \begin{cases} -\frac{1}{2\pi}\log|x| & (n=2) \\ \frac{1}{n(n-2)\alpha(n)}\frac{1}{|x|^{n-2}} & (n \geq 3), \end{cases}$$

defined for $x \in \mathbb{R}^n \setminus \{0\}$, is the fundamental solution of Laplace's equation, where $\alpha(n)$ denotes the volume of the unit ball in $\mathbb{R}^n$. Note that the fundamental solution is radial as it only depends on $|x|$.

- **Prop. (Estimates for the Fundamental Solution)** Let $\Phi(x)$ denote the fundamental solution to Laplace's equation. Then, one has the following estimates:

$$|D\Phi(x)| \leq \frac{C}{|x|^{n-1}} \qquad \text{and} \qquad |D^2\Phi(x)| \leq \frac{C}{|x|^n} \qquad (x \neq 0),$$

for some constant $C > 0$.

- **Obs. (Naive Solution to Poisson's Equation)** By construction, the function $x \mapsto \Phi(x)$ is harmonic for $x \neq 0$. Since shifting the origin to the new point $y \in \mathbb{R}^n$ leaves Laplace's equation intact, the map $x \mapsto \Phi(x-y)$ is aso harmonic as a function of x, for $x \neq y$. Let $f : \mathbb{R}^n \to \mathbb{R}$. Then it follows that the mapping $x \mapsto \Phi(x-y)f(y)$ $(x \neq y)$ is harmonic for each point $y \in \mathbb{R}^n$, and thus so is the sum of finitely many such expressions built for different points y. Thus *one might think* that the convolution

$$u(x) = \int_{\mathbb{R}^n} \Phi(x-y)f(y)\,dy = \begin{cases} -\dfrac{1}{2\pi}\displaystyle\int_{\mathbb{R}^2} \log(|x-y|)f(y)\,dy & (n=2) \\ \dfrac{1}{n(n-2)\alpha(n)}\displaystyle\int_{\mathbb{R}^n} \frac{f(y)}{|x-y|^{n-2}}\,dy & (n \geq 3) \end{cases}$$

will solve Laplace's equation. But this is not the case. The major obstacle is differentiation under the integral sign. In particular, this is because $D^2\Phi(x-y)$ is not integrable at $y = x$ so that differentiation through the integral sign is unjustified.

- **Prop. (Solving the Poisson Equation)** Let u be defined as

$$u(x) = \int_{\mathbb{R}^n} \Phi(x-y)f(y)\,dy,$$

where $f \in C_c^2(\mathbb{R}^n)$. Then
(i) $u \in C^2(\mathbb{R}^n)$;
(ii) $-\Delta u = f$ in $\mathbb{R}^n$.

Consequently, the above equation does give a formula for a solution of Poisson's equation in $\mathbb{R}^n$.

Proof. The proof of this result will be split into four steps. In the first step, we will establish that $u \in C^2(\mathbb{R}^n)$; in the second, we will compute Δu on some small ball since Φ blows up at the origin; in the third, we will continue working through Δu over small balls; finally, in the fourth step, we will take limits as our balls shrink to the origin.

(1) Remember that convolutions are symmetric (i.e., $f * g = g * y$). Thus,

$$u(x) = \int_{\mathbb{R}^n} \Phi(x-y)f(y)\,dy = \int_{\mathbb{R}^n} \Phi(y)f(x-y)\,dy.$$

From this, if $h \neq 0$ and $e_i = (0,\ldots,1,\ldots,0)$ (the 1 in the i^{th}-slot), then

$$\frac{u(x+he_i)-u(x)}{h} = \int_{\mathbb{R}^n} \Phi(y)\left[\frac{f(x+he_i-y)-f(x-y)}{h}\right]dy.$$

Since $f \in C_c^2(\mathbb{R}^n)$,

$$\frac{f(x+he_i-y)-f(x-y)}{h} \xrightarrow{h\to 0} f_{x_i}(x-y) \text{ uniformly on } \mathbb{R}^n.$$

Therefore, since the limit can pass inside the integral sign,

$$u_{x_i}(x) = \int_{\mathbb{R}^n} \Phi(y)f_{x_i}(x-y)\,dy. \qquad (i=1,\ldots,n).$$

Likewise,

$$u_{x_ix_j}(x) = \int_{\mathbb{R}^n} \Phi(y)f_{x_ix_j}(x-y)\,dy \qquad (i,j=1,\ldots,n).$$

Since the integral on the right side is continuous in x, we conclude that $u \in C^2(\mathbb{R}^n)$.

(2) Now we turn to the second result. Since Φ blows up at 0, we will need to isolate this singularity inside a small ball. Fix $\varepsilon > 0$. Then

$$\Delta u(x) = \int_{B(0,\varepsilon)} \Phi(y)\Delta_x f(x-y)\,dy + \int_{\mathbb{R}^n\setminus B(0,\varepsilon)} \Phi(y)\Delta_x f(x-y)\,dy =: I_\varepsilon + J_\varepsilon.$$

Let us see what happens with I_ε and J_ε. We first have[3],

$$\begin{aligned}|I_\varepsilon| &= \left|\int_{B(0,\varepsilon)} \Phi(y)\Delta_x f(x-y)\,dy\right| \leq \int_{B(0,\varepsilon)} |\Phi(y)|\,|\Delta_x f(x-y)|\,dy \\ &\leq C\left\|D^2 f\right\|_{L^\infty(\mathbb{R}^n)} \int_{B(0,\varepsilon)} |\Phi(y)|\,dy \leq \begin{cases} C\varepsilon^2|\log\varepsilon| & (n=2) \\ C\varepsilon^2 & (n\geq 3)\end{cases}\end{aligned}$$

Therefore, as $\varepsilon \to 0$, $I_\varepsilon \to 0$. Now we shift to J_ε. Using integration by parts, one has

$$\begin{aligned}J_\varepsilon &= \int_{\mathbb{R}^n\setminus B(0,\varepsilon)} \Phi(y)\Delta_y f(x-y)\,dy \\ &= \int_{\mathbb{R}^n\setminus B(0,\varepsilon)} D\Phi(y)\cdot D_y f(x-y)\,dy + \int_{\partial B(0,\varepsilon)} \Phi(y)\frac{\partial f}{\partial \nu}(x-y)\,dS(y) \\ &=: K_\varepsilon + L_\varepsilon.\end{aligned}$$

where ν denotes the *inward* pointing unit normal along $\partial B(0,\varepsilon)$. We check

$$|L_\varepsilon| \leq \|Df\|_{L^\infty(\mathbb{R}^n)} \int_{\partial B(0,\varepsilon)} |\Phi(y)|\,dS(y) \leq \begin{cases} C\varepsilon|\log(\varepsilon)| & (n=2) \\ C\varepsilon & (n\geq 3).\end{cases}$$

Thus, $L_\varepsilon \to 0$ as $\varepsilon \to 0$.

[3] Recall that $\left\|D^2 f\right\|_{L^\infty} = \max_{1\leq i,j\leq n}\left\|\partial_{x_ix_j} f\right\|_{L^\infty}$. On the other hand, one has (using polar coordinates),

$$\begin{aligned}\int_{B(0,\varepsilon)} |\log|x||\,dx &= \int_0^\varepsilon \int_{\partial B(0,\varepsilon)} r^{n-1}|\log(r)|\,dS\,dr = C'\int_0^\varepsilon r^{n-1}|\log(r)|\,dr \\ &= C'\cdot\varepsilon^n(-n^{-1}\log\varepsilon + n^{-2}) \leq C\cdot\varepsilon^n|\log\varepsilon| \leq C\varepsilon^2|\log\varepsilon|,\end{aligned}$$

where without loss of generality, we assume that $0<\varepsilon<1$. This covers the $n=2$ case. For $n\geq 3$, one has,

$$\int_{B(0,\varepsilon)} |x|^{2-n}\,dx = \int_0^\varepsilon \int_{\partial B(0,\varepsilon)} r^{n-1}r^{2-n}\,dS\,dr = C'\cdot\int_0^\varepsilon r\,dr = C\varepsilon^2.$$

(3) Now we investigate K_ε. If we can show that $K_\varepsilon \to -f(x)$ as $\varepsilon \to 0$, then we will be done. Using integration by parts, one has

$$\begin{aligned} K_\varepsilon &= \int_{\mathbb{R}^n \setminus B(0,\varepsilon)} \Delta\Phi(y) f(x-y)\, dy - \int_{\partial B(0,\varepsilon)} \frac{\partial \Phi}{\partial \nu}(y) f(x-y)\, dS(y) \\ &= -\int_{\partial B(0,\varepsilon)} \frac{\partial \Phi}{\partial \nu}(y) f(x-y)\, dS(y), \end{aligned}$$

since Φ is harmonic away from the origin. By definition of Φ, one has

$$D\Phi(y) = -\frac{1}{n\alpha(n)} \frac{y}{|y|^n} \qquad (y \neq 0)$$

and

$$\nu = -\frac{y}{|y|} = -\frac{y}{\varepsilon} \qquad (y \in \partial B(0,\varepsilon)).$$

Consequently,

$$\frac{\partial \Phi}{\partial \nu}(y) = \nu \cdot D\Phi(y) = \frac{1}{n\alpha(n)\varepsilon^{n-1}} \qquad (\in \partial B(0,\varepsilon)).$$

But $n\alpha(n)\varepsilon^{n-1}$ is the surface area of the sphere $\partial B(0,\varepsilon)$. Thus, we have

$$\begin{aligned} K_\varepsilon &= -\frac{1}{n\alpha(n)\varepsilon^{n-1}} \int_{\partial B(0,\varepsilon)} f(x-y)\, dS(y) \\ &= -⨍_{\partial B(x,\varepsilon)} f(y)\, dS(y) \xrightarrow{\varepsilon \to 0} -f(x) \end{aligned}$$

(4) Therefore, combining all of the above computations, one has that in the limit $\varepsilon \to 0$, $-\Delta u \to f(x)$.

□

- **Thm. (Mean-Value Formulas for Laplace's Equation)** Fix an open subset $U \subseteq \mathbb{R}^n$. If $u \in C^2(U)$ is harmonic, then

$$u(x) = ⨍_{\partial B(x,r)} u\, dS = ⨍_{B(x,r)} u\, dy,$$

for each ball $B(x,r) \subset U$. I.e., the average of f over the sphere $\partial B(x,r)$ is equal to the average of f over the ball $B(x,r)$ for all such balls completely contained in U, and is equal to $u(x)$.

Proof. Our proof will consist of two steps: (1) in the first step, we will show that the first equality is true; (2) in the second step, we will show that the second equality is true.

(1) Define

$$\phi(r) := ⨍_{\partial B(x,r)} u(y)\, dS(y) = ⨍_{\partial B(0,1)} u(x+rz)\, dS(z).$$

Then differentiating with respect to r,

$$\phi'(r) = ⨍_{\partial B(0,1)} Du(x+rz) \cdot z\, dS(z).$$

Consequently, by Green's formulas[4],

$$\begin{aligned}
\phi'(r) &= \fint_{\partial B(x,r)} Du(y) \cdot \frac{y-x}{r} \cdot dS(z) \qquad ((y-x)/r \text{ is the unit normal}) \\
&= \fint_{\partial B(x,r)} \frac{\partial u}{\partial \nu}\, dS(y) \\
&= \frac{r}{n} \fint_{B(x,r)} \Delta u(y)\, dy = 0,
\end{aligned}$$

since u is harmonic in U and $B(x,r) \subset U$. Hence, since ϕ is constant,

$$\phi(r) = \lim_{t\to 0} \phi(t) = \lim_{t \to 0} \fint_{\partial B(x,t)} u(y)\, dS(y) = u(x).$$

(2) Using polar coordinates (see, e.g., Theorem 2.49 from Folland), one has

$$\begin{aligned}
\int_{B(x,r)} u\, dy &= \int_0^r \left(\int_{\partial B(x,s)} u\, dS \right) ds \\
&= u(x) \cdot \int_0^r n\alpha(n) s^{n-1}\, ds = \alpha(n) r^n u(x) \\
\implies u(x) &= \frac{1}{\alpha(n) r^n} \int_{B(x,r)} u\, dy =: \fint_{B(x,r)} u\, dy.
\end{aligned}$$

□

- **Thm. (Converse to Mean-Value Property)** If $u \in C^2(U)$ satisfies

$$u(x) = \fint_{\partial B(x,r)} u\, dS$$

for each ball $B(x,r) \subset U$, then u is harmonic.

Proof. Assume to the contrary that u is not harmonic (i.e., assume $\Delta u \not\equiv 0$). Then there exists some ball $B(x,r)$ so that $\Delta u > 0$ within this ball. Define

$$\phi(r) = \fint_{\partial B(x,r)} u(y)\, dS(y).$$

Then by Green's formulas,

$$0 = \phi'(r) = \frac{r}{n} \fint_{\partial B(x,r)} \Delta u(y)\, dy > 0,$$

which is a contradiction. □

- **Thm. (Strong Maximum Principle)** Suppose $u \in C^2(U) \cap C(\bar{U})$ is harmonic within U.
  (i) $\max_{\bar{U}} u = \max_{\partial U} u$.
  (ii) If U is connected and there exists $x_0 \in U$ so that

$$u(x_0) = \max_{\bar{U}} u,$$

then u is constant within U.

[4] By Green's Formulas,

$$\fint_{\partial B(x,r)} \frac{\partial u}{\partial \nu}\, dS(y) = \frac{1}{|\partial B(x,r)|} \int_{B(x,r)} \Delta u\, dy = \frac{|B(x,r)|}{|\partial B(x,r)|} \fint_{B(x,r)} \Delta u\, dy = \frac{r}{n} \fint_{B(x,r)} \Delta u(y)\, dy.$$

The first result is the *maximum principle* for Laplace's equation, while the second result is the *strong maximum principle*. The first asserts that if u has a maximum in U, it must also be the maximum of u on the closure of U; the second asserts that if u achieves its maximum in U, then it is constant in U. Replacing u by $-u$ gives us similar assertions with the minimum instead of maximum.

Proof. We will first prove the second statement. The first statement follows from the first.

(ii) Consider the set $\{x \in U : u(x) = M\}$, where $u(x_0) = M = \max_{\bar{U}} u$. Note that the set is nonempty as it contains (at least) the point x_0. By the mean-value property,

$$M = u_{x_0} = \fint_{B(x_0,r)} u\,dy \leq M,$$

where $0 < r < \text{dist}(x_0, \partial U)$. Equality holds if and only if $u \equiv M$ with $B(x_0, r)$ so that $u(y) = M$ for all $y \in B(x_0, r)$. This proves that the set $\{x \in U : u(x) = M\}$ is open in U. By continuity of M, $\{x \in U : u(x) = M\}$ is closed in U. Thus, since U is connected, $\{x \in U : u(x) = M\} = U$. Therefore, u is constant in U.

□

- **Cor. (Positivity Principle)** Let U be a connected open subset of $\mathbb{R}^n$, and assume $u \in C^2(U) \cap C(\bar{U})$ satisfies

$$\begin{cases} \Delta u = 0 & \text{in } U \\ u = g & \text{on } \partial U, \end{cases}$$

where $g \geq 0$. Then u is positive everywhere in U if g is positive somewhere on ∂U.

Proof. Since u satisfies Laplace's equation in U, it follows that u is harmonic in U so that the strong maximum/minimum principles apply to u. By the weak minimum principle, for every $x \in \bar{U}$, $u(x) \geq \min_{\partial U} u = \min_{\partial U} g \geq 0$ so that $g(x)$ is nonnegative everywhere. Define the subset

$$A = \{z \in U : u(z) > 0\}.$$

If A is empty, then $u(x) = 0$ everywhere on U. Thus by continuity of u on $\bar{U}$, $u \equiv 0$ on ∂U so that $g \equiv 0$ on ∂U. But this contradicts our hypothesis that g is positive somewhere on ∂U. Therefore, A is nonempty. By continuity of u, A is open in U. Thus it suffices to show that A is relatively closed in U. Let $\{z_k\} \subset A$ be a sequence that converges to some $z \in U$. We must show that $z \in A$. Since $u(z_k) > 0$ for all k, $u(z) \geq 0$. Assume to the contrary that $u(z) = 0$. Then u achieves its minimum at z so that by the strong minimum principle, u must be zero everywhere in U, which is a contradiction. Thus by contradiction, $u(z) > 0$. I.e., $z \in A$, which completes the proof that A is relatively closed in U. Thus since U is connected, we conclude that $A = U$. □

- **Thm. (Uniqueness of Solutions to Laplace Equation)** Let $g \in C(\partial U)$, $f \in C(U)$. Then there exists at most one solution $u \in C^2(U) \cap C(\bar{U})$ of the boundary-value problem

$$\begin{cases} -\Delta u = f & \text{in } U \\ u = g & \text{on } \partial U. \end{cases}$$

Proof. Let u and $\widetilde{u}$ satisfy the above boundary-value problem. Then we consider the function $w = \pm(u - \widetilde{u})$. Since $(u - \widetilde{u})|_{\partial U} = 0$, by the weak maximum principle, $\max_{\bar{U}} w = 0$. Likewise, by the weak minimum principle, $\min_{\bar{U}} w = 0$. Thus, $w = 0$ everywhere on U so that $u = \widetilde{u}$ everywhere in U. □

- **Thm. (Smoothness of Harmonic Functions)** If $u \in C(U)$ satisfies the mean-value property for each $B(x, r) \subset U$, then $u \in C^\infty(U)$. Note that u may not necessarily be smooth, or even continuous, up to ∂U.

Proof. We will prove this result using mollifiers, and showing that u is equivalent to a smooth function locally. Let η be a standard mollifier (recall that η is a radial function). Set $u^\varepsilon = \eta_\varepsilon * u$ in $U_\varepsilon = \{x \in U : \mathrm{dist}(x, \partial U) > \varepsilon\}$. Note that $u^\varepsilon \in C^\infty(U_\varepsilon)$. Fix some $x \in U^\varepsilon$. Then

$$\begin{aligned}
u^\varepsilon(x) &= \int_U \eta_\varepsilon(x-y)u(y)\,dy \\
&= \frac{1}{\varepsilon^n}\int_{B(x,\varepsilon)} \eta\left(\frac{|x-y|}{\varepsilon}\right)u(y)\,dy \\
&= \frac{1}{\varepsilon^n}\int_0^\varepsilon \eta\left(\frac{r}{\varepsilon}\right)\left(\int_{\partial B(x,r)} u\,dS\right)dr \qquad \text{(Polar Coordinates)} \\
&= \frac{1}{\varepsilon^n}u(x)\cdot\int_0^\varepsilon \eta\left(\frac{r}{\varepsilon}\right)n\alpha(n)r^{n-1}\,dr \qquad \text{(Mean-Value Property)} \\
&= u(x)\int_{B(0,\varepsilon)} \eta_\varepsilon\,dy = u(x) \qquad \text{(Property of Standard Mollifiers).}
\end{aligned}$$

Therefore, $u^\varepsilon \equiv u$ in U_ε so that $u \in C^\infty(U_\varepsilon)$ for every $\varepsilon > 0$. This proves that u is smooth on U. □

- **Thm. (Estimates on Derivatives of Harmonic Functions)** Assume u is harmonic in U. Then

$$|D^\alpha u(x_0)| \le \frac{C_k}{r^{n+k}}\|u\|_{L^1(B(x_0,r))}$$

for each ball $B(x_0,r) \subset U$ and each multiindex α of order $|\alpha| = k$. Here

$$C_0 = \frac{1}{\alpha(n)}, C_k = \frac{(2^{n+1}nk)^k}{\alpha(n)} \qquad (k = 1, \ldots).$$

Proof. The proof of these results will follow from induction. We will first focus on the base cases $k = 0, 1$, and then work through the general case $k \ge 2$.

(1) First assume $k = 0$. Since u is harmonic, it satisfies the mean-value property so that for all $x_0 \in U$ and each ball $B(x_0,r) \subset U$, one has

$$u(x_0) = \fint_{B(x_0,r)} u\,dy \implies |u(x_0)| \le \frac{1}{\alpha(n)r^n}\int_{B(x_0,r)}|u(y)|\,dy \le \frac{1}{\alpha(n)r^n}\|u\|_{L^1(B(x_0,r))}$$

Now suppose $k = 1$. By differentiating Laplace's equation, we observe that u_{x_i} $(i = 1, \ldots, n)$ is harmonic for each i. This means that each u_{x_i} satisfies the mean-value property. Therefore,

$$\begin{aligned}
|u_{x_i}(x_0)| &= \left|\fint_{B(x_0,r/2)} u_{x_i}\,dx\right| \\
&= \left|\frac{2^n}{\alpha(n)r^n}\int_{\partial B(x_0,r/2)} u\nu_i\,dS\right| \\
&\le \frac{2n}{r}\|u\|_{L^\infty(\partial B(x_0,r/2))}.
\end{aligned}$$

Now if $x \in \partial B(x_0,r/2)$, then $B(x,r/2) \subset B(x_0,r) \subset U$, and so by the case for $k = 0$,

$$|u(x)| \le \frac{1}{\alpha(n)}\left(\frac{2}{r}\right)^n\|u\|_{L^1(B(x_0,r))} \implies \|u\|_{L^\infty(\partial B(x_0,r/2))} \le \frac{1}{\alpha(n)}\frac{2^n}{r^n}\|u\|_{L^1(B(x_0,r))}.$$

Therefore, combining the above inequalities, one has that if $|\alpha| = 1$, then

$$|D^\alpha u(x_0)| \le \frac{2n}{r}\frac{1}{\alpha}\frac{2^n}{r^n}\|u\|_{L^1(B(x_0,r))} = \frac{2^{n+1}n}{\alpha(n)r^{n+1}}\|u\|_{L^1(B(x_0,r))}.$$

This proves the case for $k = 1$.

(2) Now we will prove the general case for $k \geq 2$. Assume that the estimates are valid for all balls in U and each multiindex of order less than or equal to $k-1$. Fix $B(x_0, r) \subset U$ and let α be a multiindex with $|\alpha| = k$. Then $D^\alpha u = (D^\beta u)_{x_i}$ for some $i \in \{1, \ldots, n\}$, $|\beta| = k-1$. By repeatedly differentiating Laplace's equation, we can show that $D^\alpha u$ is harmonic. Thus by the mean-value formula, one has

$$\begin{aligned} |D^\alpha u(x_0)| &= \left| \fint_{B(x_0, r/k)} D^\alpha u \, dx \right| \\ &= \left| \frac{k^n}{\alpha(n) r^n} \int_{\partial B(x_0, r/k)} D^\beta u \nu_i \, dS \right| \\ &\leq \frac{nk}{r} \left\| D^\beta u \right\|_{L^\infty(\partial B(x_0, r/k))}. \end{aligned}$$

If $x \in \partial B(x_0, r/k)$, then $B(x, r/k) \subset B(x_0, r) \subset U$. Therefore, by the estimates for $k-1$, one has

$$|D^\beta u(x)| \leq \frac{(2^{n+1} n(k-1))^{k-1}}{\alpha(n) \left(\frac{k-1}{k} r\right)^{n+k-1}} \|u\|_{L^1(B(x_0, r))}.$$

Thus combining the two estimates gets us,

$$|D^\alpha u(x_0)| \leq \frac{(2^{n+1} nk)^k}{\alpha(n) r^{n+k}} \|u\|_{L^1(B(x_0, r))}.$$

Thus by induction, the estimate holds for all $k \geq 1$.

□

- **Thm. (Liouville's Theorem)** Suppose $u : \mathbb{R}^n \to \mathbb{R}$ is harmonic and bounded. Then u is constant.

Proof. Fix $x_0 \in \mathbb{R}^n$, $r > 0$, and let $B(x_0, r)$ be the ball of radius r centered at x_0. To show that u is constant, we just need to show that $Du \equiv 0$; connectedness of $\mathbb{R}^n$ will then show that u is constant everywhere on $\mathbb{R}^n$. By the above estimates, one has

$$\begin{aligned} |Du(x_0)| &\leq \frac{2^{n+1} n}{\alpha(n) r^{n+1}} \|u\|_{L^1(B(x_0, r))} \\ &= \frac{2^{n+1} n}{\alpha(n) r^{n+1}} \int_{B(x_0, r)} |u(x)| \, dx \\ &\leq \frac{2^{n+1} n}{\alpha(n) r^{n+1}} \int_{B(x_0, r)} \|u\|_{L^\infty(B(x_0, r))} \, dx \\ &= \frac{2^{n+1} n}{r} \|u\|_{L^\infty(B(x_0, r))} \leq \frac{2^{n+1} n}{r} \|u\|_{L^\infty(\mathbb{R}^n)} \xrightarrow{r \to \infty} 0. \end{aligned}$$

Therefore, $Du \equiv 0$; by connectedness of $\mathbb{R}^n$, we conclude that u is constant. □

- **Thm. (Representation Formula)** Let $f \in C_c^2(\mathbb{R}^n)$ $n \geq 3$. Then any bounded solution of $-\Delta u = f$ in $\mathbb{R}^n$ has the form

$$u(x) = \int_{\mathbb{R}^n} \Phi(x - y) f(y) \, dy + C \qquad (x \in \mathbb{R}^n)$$

for some constant C.

Proof. For $n \geq 3$, $\Phi(x) \to 0$ as $|x| \to \infty$. Therefore, $\widetilde{u}(x) = \int_{\mathbb{R}^n} \Phi(x-y) f(y) \, dy$ is a bounded solution of $-\Delta u = f$ in $\mathbb{R}^n$. If u is another solution, $w = u - \widetilde{u}$ is constant according to Liouville's Theorem. As a side remark, for $n = 2$, $\Phi(x)$ is unbounded as $|x| \to \infty$, so that the above form may also be unbounded. □

- **Thm. (Analyticity of Harmonic Functions)** Assume u is harmonic in U. Then u is analytic in U.

  Proof. Fix any point $x_0 \in U$. Our goal is to show that u can be represented by a convergent power series in some neighborhood of x_0. Our proof will be split into multiple steps.

  (1) Let $r = \frac{1}{4}\operatorname{dist}(x_0, \partial U)$. Then since $u \in C^2(U)$, $u \in L^1(B(x_0, 2r))$ so that the quantity

  $$M = \frac{1}{\alpha(n)r^n}\|u\|_{L^1(B(x_0,2r))} < \infty.$$

  (2) Since $B(x,r) \subset B(x_0, 2r) \subset U$ for each $x \in B(x_0, r)$, one has

  $$\begin{aligned} |D^\alpha u(x)| &\le \left(\frac{2^{n+1}n|\alpha|}{\alpha(n)}\right)^{|\alpha|} \cdot \frac{1}{r^{n+|\alpha|}}\|u\|_{L^1(B(x_0,2r)} \\ &= \left(\frac{2^{n+1}n}{r}\right)^{|\alpha|}\frac{1}{\alpha(n)^{|\alpha|}r^n}\|u\|_{L^1(B(x_0,2r))}|\alpha|^{|\alpha|} \\ &= C'M\cdot\left(\frac{2^{n+1}n}{r}\right)^{|\alpha|}|\alpha|^{|\alpha|} \\ \implies \|D^\alpha u\|_{L^\infty(B(x_0,r))} &\le C'\cdot M\left(\frac{2^{n+1}n}{r}\right)^{|\alpha|}|\alpha|^{|\alpha|}, \end{aligned}$$

  where C' is some constant that depends solely on n. Then since $k^k/k! < e^k$ for all positive integers k, one has

  $$|\alpha|^{|\alpha|} \le e^{|\alpha|}|\alpha|!$$

  for all multiindices α. By the multinomial theorem, one has

  $$n^k = (1 + \cdots + 1)^k = \sum_{|\alpha|=k}\frac{|\alpha|!}{\alpha!},$$

  so that $|\alpha|! \le n^{|\alpha|}\alpha!$. Thus,

  $$\|D^\alpha u\|_{L^\infty(B(x_0,r))} \le CM\left(\frac{2^{n+1}n^2e}{r}\right)^{|\alpha|}\alpha!,$$

  where C is some constant that depends on n.

  (3) The Taylor series for u at x_0 is

  $$\sum_\alpha \frac{D^\alpha u(x_0)}{\alpha!}(x - x_0)^\alpha,$$

  where the sum is taken over all multiindices α. We will now show that this power series converges provided $|x - x_0| < \frac{r}{2^{n+2}n^3e}$. To show this, it suffices to show that the remainder term tends to zero. I.e., we begin by computing,

  $$\begin{aligned} R_N(x) &:= u(x) - \sum_{k=0}^{N-1}\sum_{|\alpha|=k}\frac{D^\alpha u(x_0)}{\alpha!}(x - x_0)^\alpha \\ &= \sum_{|\alpha|=N}\frac{D^\alpha u(x_0 + t(x - x_0))(x - x_0)^\alpha}{\alpha!}, \end{aligned}$$

  for some $0 \le t \le 1$ depending on x, where we obtain the above formula by writing out the first N terms and the error in ther Tayor expansion about 0 for the function of one variable

$g(t) = u(x_0 + t(x - x_0))$ at $t = 1$. So using the estimates above and from (2), we have

$$\begin{aligned}
|R_N(x)| &\leq CM \sum_{|\alpha|=N} \left(\frac{2^{n+1}n^2 e}{r}\right)^N \left(\frac{r}{2^{n+2}n^3 e}\right)^N \\
&= CM \sum_{|\alpha|=N} \frac{1}{(2n)^N} \leq CMn^N \frac{1}{(2n)^N} = \frac{CM}{2^N} \xrightarrow{N\to\infty} 0.
\end{aligned}$$

Thus the proof concludes.

□

- **Thm. (Harnack's Inequality)** For each connected open set $V \Subset U$ (i.e., $V \subset \overline{V} \subset U$ with $\overline{V}$ compact), there exists a positive constant C, depending only on V such that

$$\sup_V u \leq C \inf_V u,$$

for all nonnegative harmonic functions in U.

Proof. [!! Complete Later !!] □

- **Thm. (Representation Formula for Solution to Poisson's Equation using Green's Function)** Assume now that $U \subset \mathbb{R}^n$ is open, bounded, and ∂U is C^1. If $u \in C^2(\bar{U})$ solves the boundary-value problem

$$\begin{cases} -\Delta u = f & \text{in } U \\ u = g & \text{on } \partial U, \end{cases}$$

then

$$u(x) = -\int_{\partial U} g(y) \frac{\partial G}{\partial \nu}(x,y)\, dS(y) + \int_U f(y) G(x,y)\, dy \qquad (x \in U),$$

where $G = G(x,y)$ is Green's function for the given domain U.

Proof. Suppose $u \in C^2(\overline{U})$ is an arbitrary function. Fix $u \in U$, and choose $\varepsilon > 0$ so small so that $B(x,\varepsilon) \subset U$, and apply Green's formula on the region $V_\varepsilon \coloneqq U \setminus B(x,\varepsilon)$ to $u(y)$ and $\Phi(y-x)$. We compute

$$\int_{V_\varepsilon} u(y)\Delta\Phi(y-x) - \Phi(y-x)\Delta u(y)\, dy = \int_{\partial V_\varepsilon} u(y)\frac{\partial \Phi}{\partial \nu}(y-x) - \Phi(y-x)\frac{\partial u}{\partial \nu}(y)\, dS(y),$$

where ν denotes the outer unit normal vector on ∂V_ε. For $x \neq y$, $\Delta\Phi(x-y) = 0$. Moreover[5],

$$\left| \int_{\partial B(x,\varepsilon)} \Phi(y-x)\frac{\partial u}{\partial \nu}(y)\, dS(y) \right| \leq C\varepsilon^{n-1} \max_{\partial B(0,\varepsilon)} |\Phi| = o(1) \text{ as } \varepsilon \to 0.$$

Furthermore,

$$\int_{\partial B(x,\varepsilon)} u(y)\frac{\partial \Phi}{\partial \nu}(y-x)\, dS(y) = ⨍_{\partial B(x,\varepsilon)} u(y)\, dS(y) \to u(x)$$

as $\varepsilon \to 0$. Thus in the limit $\varepsilon \to 0$,

$$u(x) = \int_{\partial U} \Phi(y-x)\frac{\partial u}{\partial \nu}(y) - u(y)\frac{\partial \Phi}{\partial \nu}(y-x)\, dS(y) - \int_U \Phi(y-x)\Delta u(y)\, dy.$$

This integral is valid for any $x \in U$ and $u \in C^2(\bar{U})$. If we knew the values of Δu inside U and the values of u, $\partial u/\partial \nu$ along ∂U, then we could use the above formula to solve for $u(x)$. However, we do not know $\partial u/\partial \nu$. Thus we must remove this term from the above result. For a fixed x, we introduce the corrector function $\phi^x(y)$ which solves the boundary value problem

$$\begin{cases} \Delta\phi^x = 0 & \text{in } U \\ \phi^x = \Phi(y-x) & \text{in } \partial U. \end{cases}$$

Applying Green's Formula, one obtains

$$\begin{aligned} -\int_U \phi^x(y)\Delta u(y)\, dy &= \int_{\partial U} u(y)\frac{\partial \phi^x}{\partial \nu}(y) - \phi^x(y)\frac{\partial u}{\partial \nu}(y)\, dS(y) \\ &= \int_{\partial U} u(y)\frac{\partial \phi^x}{\partial \nu}(y) - \Phi(y-x)\frac{\partial u}{\partial \nu}(y)\, dS(y). \end{aligned}$$

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$$\begin{aligned} \left| \int_{\partial B(x,\varepsilon)} \Phi(y-x)\frac{\partial u}{\partial \nu}(y)\, dS(y) \right| &\leq \int_{\partial B(x,\varepsilon)} |\Phi(y-x)| \left| \frac{\partial u}{\partial \nu}(y) \right| dS(y) = \int_{\partial B(0,\varepsilon)} |\Phi(t)| \left| \frac{\partial u}{\partial \nu}(t+x) \right| dS(t) \\ &\leq \int_{\partial B(0,\varepsilon)} \|\Phi\|_{L^\infty(\partial B(0,\varepsilon))} \left\| \frac{\partial u}{\partial \nu}(x+\cdot) \right\|_{L^\infty(\partial B(0,\varepsilon))} dS(t) = \int_{\partial B(0,\varepsilon)} \|\Phi\|_{L^\infty(\partial B(0,\varepsilon))} \left\| \frac{\partial u}{\partial \nu} \right\|_{L^\infty(\partial B(x,\varepsilon))} dS(t) \\ &= \|\Phi\|_{L^\infty(\partial B(0,\varepsilon))} \left\| \frac{\partial u}{\partial \nu} \right\|_{L^\infty(\partial B(x,\varepsilon))} n\alpha(n)\varepsilon^{n-1}. \end{aligned}$$

So now suppose that $G(x,y) := \Phi(y-x) - \phi^x(y)$ $(x,y \in U, x \neq y)$ (i.e., *Green's function for the region* U). Using this terminology and adding the above result to the original expansion for u, one obtains

$$u(x) = -\int_{\partial U} u(y) \frac{\partial G}{\partial \nu}(x,y)\, dS(y) - \int_U G(x,y)\Delta u(y)\, dy \qquad (x \in U),$$

where $\partial G/\partial \nu(x,y) = D_y G(x,y) \cdot \boldsymbol{\nu}(y)$ is the outer normal derivative of G with respect to the variable y. Since $\partial u/\partial \nu$ does not appear in this fixed equation, the corrector function ϕ^x was chosen correctly. Suppose now that $u \in C^2(\bar{U})$ satisfies the above boundary-value problem. Then making the necessary substitutions, one obtains the representation formula given above. □

- **Thm. (Symmetry of Green's Function)** For all $x, y \in U$, $x \neq y$, we have $G(y,x) = G(x,y)$.
- **[!! Fill in Green's Functions for Different Geometries !!]**
- **Thm. (Uniqueness of Solution to Poisson Problem; Energy Method)** Consider the boundary-value problem

$$\begin{cases} -\Delta u = f & \text{in } U \\ u = g & \text{on } \partial U, \end{cases}$$

where U is open, bounded, and ∂U is C^1. Then there exists at most one solution $u \in C^2(\bar{U})$ of the above problem.

Proof. Assume $\tilde{u}$ is another solution to the given boundary-value problem. Set $w = u - \tilde{u}$. Then $\Delta w = 0$ in U, which means that by integration by parts,

$$0 = -\int_U w\Delta w\, dx = \int_U |Dw|^2\, dx.$$

Since $|Dw|^2 \geq 0$, it follows that $Dw \equiv 0$ in U. Since $w = 0$ on ∂U, we deduce that $w \equiv 0$ in U so that $u \equiv \tilde{u}$. □

- **Def. (Energy Functional for Laplace's Equation)** Define the *energy functional*,

$$I[w] := \int_U \frac{1}{2}|Dw|^2 - wf\, dx,$$

where w belongs to the *admissible set*

$$\mathcal{A} := \{w \in C^2(\bar{U}) : w = g \text{ on } \partial U\}.$$

- **Thm. (Dirichlet's Principle)** Assume $u \in C^2(\bar{U})$ solves the Poisson boundary-value problem. Then

$$I[u] = \min_{w \in \mathcal{A}} I[w].$$

Conversely, if $u \in \mathcal{A}$ satisfies the above condition, then u solves the Poisson boundary-value problem. I.e., a solution of the Poisson boundary-value problem can be characterized as the minimizer of the energy functional.

Proof. The proof will proceed in two steps. In the first, we will show that $I[u]$ is indeed the minimum of $I[w]$; in the second, we will show that if u is the minimizer of the energy functional, then it solves Poisson's equation.

(1) [!! Complete Later !!]

□

Heat Equation

- **Def. ((Non)homogeneous Heat Equation)** The *(non)homogeneous heat equation* is the following:
$$u_t - \Delta u = 0\ (f),$$
subject to appropriate initial and boundary conditions, where $t > 0$ and $x \in U^{\text{open}} \subseteq \mathbb{R}^n$. The Laplacian Δ is taken with respect to the spatial variables, and in the second case, f is some given function $f : U \times [0, \infty) \to \mathbb{R}$.

- **Obs. (Physical Interpretation of the Heat Equation)** The heat equation describes the time evolution of the density u of some quantity. If $V \subset U$ is any smooth subregion, the rate of change of the total quantity within V equals the negative of the net flux through ∂V:
$$\frac{d}{dt}\int_V u\, dx = -\int_{\partial V} \mathbf{F} \cdot \boldsymbol{\nu}\, dS,$$
where $\mathbf{F}$ is the flux density. Since V was arbitrary, one has $u_t = -\text{div}\mathbf{F}$. If one assumes that $\mathbf{F} = -aDu$ ($a > 0$), then we obtain the PDE
$$u_t = a\,\text{div}(Du) = a\Delta u,$$
which for $a = 1$ is the heat equation.

- **Prop. (Deriving the Fundamental Solution to the Heat Equation)** We begin our study of the heat equation by trying to find *specific* solutions. We begin by looking for solutions u that are *invariant under dilation scaling*. I.e., we look for solutions u so that for all $\lambda > 0$, $x \in \mathbb{R}^n$, and $t > 0$,
$$u(x, t) = \lambda^\alpha u(\lambda^\beta x, \lambda t),$$
where the constants α and β must be found. This leads to us looking for solutions u having the special structure,
$$u(x, t) = \frac{1}{t^\alpha} v\left(\frac{x}{t^\beta}\right) \qquad (x \in \mathbb{R}^n, t > 0).$$

7.2.3 Sobolev Spaces

- **Def. (Lipschitz Continuous Functions)** Assume $U^{\text{open}} \subseteq \mathbb{R}^n$ is open, and let $0 < \gamma \leq 1$. $u : U \to \mathbb{R}$ is said to be *Lipschitz continuous* if and only if

$$|u(x) - u(y)| \leq C|x - y| \qquad (x, y \in U),$$

for some constant C. In particular, it follows that u is uniformly continuous.

- **Def. (Hölder Continuous Functions)** A function $u : U \to \mathbb{R}$ is said to be *Hölder continuous with exponent* γ if and only if

$$|u(x) - u(y)| \leq C|x - y|^\gamma \qquad (x, y \in U),$$

for some constant C and $0 < \gamma \leq 1$.

- **Lem. (Constant Hölder Continuous Functions)** If $u : U \to \mathbb{R}$ is Hölder continuous with exponent $\gamma > 1$, where U is connected, then u is constant.

Proof. Let U be a connected open subset of $\mathbb{R}^n$ and assume that $u : U \to \mathbb{R}$ is Hölder continuous with exponent $\gamma > 1$. Fix an arbitrary $x \in U$, and let e_i be a basis vector of $\mathbb{R}^n$. For $h \in \mathbb{R}$ small enough so that $x + he_i \in U$, one has by Hölder continuity

$$|u(x + he_i) - u(x)| \leq C|h|^\gamma \implies 0 \leq \left|\frac{u(x + he_i) - u(x)}{h}\right| \leq C|h|^{\gamma - 1}.$$

Since $\gamma - 1 > 0$, the RHS tends to zero. On the other hand, since u is differentiable, the middle term tends to $D_{e_i}u(x)$. Thus since e_i and x were arbitrary, one has that $Du(x) = 0$ for all $x \in U$. Thus since U is connected, we conclude that $Du \equiv 0$, which means that u is constant. □

- **Def. (Hölder Norms)** We define the following:
  (i) (Uniform Norm) If $u : U \to \mathbb{R}$ is bounded and continuous, we write

$$\|u\|_{C(\bar{U})} \coloneqq \sup_{x \in U} |u(x)|.$$

  (ii) The γ^{th}-Hölder seminorm of $u : U \to \mathbb{R}$ is

$$[u]_{C^{0,\gamma}(\bar{U})} \coloneqq \sup_{\substack{x,y \in U \\ x \neq y}} \left\{\frac{|u(x) - u(y)|}{|x - y|^\gamma}\right\},$$

  and the γth-Hölder norm is

$$\|u\|_{C^{0,\gamma}(\bar{U})} \coloneqq \|u\|_{C(\bar{U})} + [u]_{C^{0,\gamma}(\bar{U})}.$$

- **Def. (Hölder Space, $C^{k,\gamma}(\bar{U})$)** Let $k \geq 1$ be a positive integer. Then we define

$$C^{k,\gamma}(\bar{U}) \coloneqq \left\{u \in C^k(\bar{U}) : \|u\|_{C^{k,\gamma}(\bar{U})} \coloneqq \sum_{|\alpha| \leq k} \|D^\alpha u\|_{C(\bar{U})} + \sum_{|\alpha| = k} [D^\alpha u]_{C^{0,\gamma}(\bar{U})} < \infty\right\}.$$

- **Thm. (Hölder and Banach Spaces)** The space of functions $C^{k,\gamma}(\bar{U})$ is a Banach space under the aforementioned norm.

- **Def. (Test Function)** A function $f \in C_c^\infty(U)$.

- **Def. (Weak Partial Derivatives)** Suppose $u, v \in L^1_{\text{loc}}(U)$ and α is a multiindex. We say that α is the α^{th}-weak partial derivative of u, written

$$D^\alpha u = v,$$

if and only if

$$\int_U u D^\alpha \phi \, dx = (-1)^{|\alpha|} \int_U v\phi \, dx,$$

for all test functions $\phi \in C_c^\infty(U)$.

- **Obs. (Motivation for Weak Derivatives)** Suppose $u \in C^1(U)$. Then if $\phi \in C_c^\infty(U)$, by the integration by parts formula,

$$\int_U u\phi_{x_i} \, dx = -\int_U u_{x_i}\phi \, dx \qquad (i = 1, \dots, n).$$

Note that there are no boundary terms since ϕ has compact support in U and vanishes near ∂U. More generally, if k is a positive integer, $u \in C^k(U)$, and $\alpha = (\alpha_1, \dots, \alpha_n)$ is a multiindex of order $|\alpha| = \alpha_1 + \cdots + \alpha_n = k$, then

$$\int_U u D^\alpha \phi \, dx = (-1)^{|\alpha|} \int_U D^\alpha u\phi \, dx,$$

obtained by induction. We want to see if this formula is true even if $u \notin C^k(U)$. The left side is defined regardless of the regularity of u so long as it is locally integrable. Thus, we replace the definition of $D^\alpha u$ as suggested above.

- **Lem. (Uniqueness of Weak Derivatives)** A weak α^{th}-partial derivative of u, if it exists, is uniquely defined up to a set of measure zero.

Proof. Assume that $v, \widetilde{v} \in L^1_{\text{loc}}(U)$ satisfy

$$\int_U u D^\alpha \phi \, dx = (-1)^{|\alpha|} \int_U v\phi \, dx = (-1)^{|\alpha|} \int_U \widetilde{v}\phi \, dx$$

for all $\phi \in C_c^\infty(U)$. Then

$$\int_U (v - \widetilde{v})\phi \, dx = 0$$

for all $\phi \in C_c^\infty(U)$, whence $v - \widetilde{v} = 0$ a.e. □

- **Cor. (Weak Derivative in $\mathbb{R}$)** Let $n = 1$, $U = (0, 2)$, and

$$u(x) =$$

- **Def. (Sobolev Space)** Fix $1 \leq p \leq \infty$ and let k be a nonnegative integer. Then we define the *Sobolev space*

$$W^{k,p}(U) := \left\{ u : U \to \mathbb{R} \middle| u \in L^1_{\text{loc}}(U) \text{ and } \forall |\alpha| \leq k,\ D^\alpha u \begin{array}{c}\text{exists in the weak sense}\\ \text{and belongs to } L^p(U)\end{array} \right\}$$

7.3 Semidirect Products

- **Ex. 1 (Classification of Groups of Order pq)** Consider the finite group G of order pq, with p, q distinct primes and without loss of generality, $p < q$. By Sylow's theorems, it follows that G contains a unique (hence normal) Sylow q-subgroup Q. Let P denote a Sylow p-subgroup. Then it follows that $G \cong Q \rtimes_\varphi P$, where $\varphi : P \to \mathrm{Aut}(Q)$ is a group homomorphism. Since P and Q are both of prime order, both subgroups are cyclic. Moreover, $\mathrm{Aut}(Q) \cong (\mathbb{Z}/q\mathbb{Z})^\times \cong \mathbb{Z}/(q-1)\mathbb{Z}$. We consider two cases.

  (Case I) First consider the case $p \nmid (q-1)$. Then as in Problem 2024-J-I, the only homomorphism from Q to $\mathrm{Aut}(P)$ is the trivial homomorphism. Therefore, it follows that in this case, the only possible semidirect product is the direct product. I.e., $G \cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z} \cong \mathbb{Z}/(pq)\mathbb{Z}$. Thus up to isomorphism, there exists exactly one group of order pq, which is cyclic and abelian.

  (Case II) Now consider the case $p \mid (q-1)$. Let $P = \langle y \rangle$. Since $\mathrm{Aut}(Q)$ is cyclic of order $q-1$, and $p \mid (q-1)$, by Cauchy's theorem, $\mathrm{Aut}(Q)$ contains a unique subgroup of order p, $\langle \gamma \rangle$. Then any homomorphism $\varphi : P \to \mathrm{Aut}(Q)$ must map y to a generator of $\langle \gamma \rangle$. Thus in fact, there are p homomorphisms: $\varphi_i : y \mapsto \gamma^i$. Note that when $i = 0$, φ_0 corresponds to the trivial homomorphism so that $G \cong Q \rtimes_{\varphi_0} P \cong Q \times P \cong \mathbb{Z}/(pq)\mathbb{Z}$. On the other hand, all of the other homomorphisms are isomorphic since for every $i > 0$, there exists some generator y_i of P so that $p_i(y_i) = \gamma$. Thus up to isomorphism, there exist exactly 2 groups of order pq with $p < q$; one is cyclic and abelian, and the other is nonabelian.

- **Ex. 2 (Classification of Groups of Order 30)** Let G be a group of order $30 = 2 \cdot 3 \cdot 5$. We begin by showing that at least one of n_3 and n_5 must be one. By Sylow's theorems, one has

$$n_3 \in \{1, 2, 5, 10\} \cap \{1, 4, 7, 10, \ldots\} = \{1, 10\}.$$
$$n_5 \in \{1, 2, 3, 6\} \cap \{1, 11, 16, 21, \ldots\} = \{1, 6\}.$$

  Suppose $n_3 = 10$ and $n_5 = 6$. Since each Sylow 5-subgroup and each Sylow 3-subgroup has prime order, they must all intersect trivially. Moreover, each Sylow 5-subgroup has trivial intersection with each Sylow 3-subgroup by Lagrange's theorem. Thus, there are 30 elements of G with order 5, and 20 elements of G with order 3. This means that $|G| > 30 + 20 = 50 > 30 = |G|$, which is a contradiction. Thus G must have a normal Sylow 3-subgroup or a normal Sylow 5-subgroup. In any case, it then follows that G must have a subgroup H of order 50. Since H has index 2 in G, it must be normal and cyclic in G. Therefore, denoting a Sylow 2-subgroup of G by K, $G \cong H \rtimes_\varphi K$ for some homomorphism $\varphi : K \to \mathrm{Aut}(H)$. Then since $H \cong \mathbb{Z}_{15}$, one has

$$\mathrm{Aut}(\mathbb{Z}_{15}) \cong (\mathbb{Z}/15\mathbb{Z})^\times \cong (\mathbb{Z}/5\mathbb{Z})^\times \times (\mathbb{Z}/3\mathbb{Z})^\times \qquad \text{(Chinese Remainder Theorem)}$$

  Thus, $\mathrm{Aut}(\mathbb{Z}_{15}) \cong \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.